ANSWERS TO ODD-NUMBERED EXERCISES

Chapter 1

SECTION 1.1, pp. 29–31

1. D: $(-\infty, \infty)$, R: $[1, \infty)$ **3.** D: $[-2, \infty)$, R: $[0, \infty)$

5. D: $(-\infty, 3) \cup (3, \infty)$, R: $(-\infty, 0) \cup (0, \infty)$

7. **(a)** Not a function of x because some values of x have two values of y

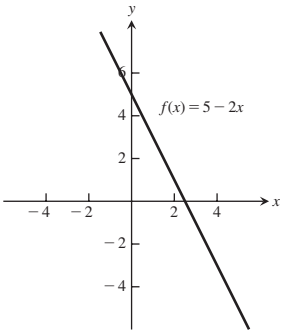
(b) A function of x because for every x there is only one possible y

9. $A = \dfrac{\sqrt{3}}{4}x^2$, $p = 3x$

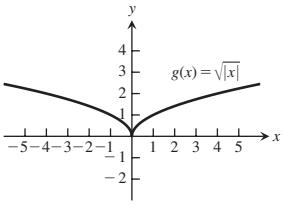
11. $x = \dfrac{d}{\sqrt{3}}$, $A = 2d^2$, $V = \dfrac{d^3}{3\sqrt{3}}$

13. $L = \dfrac{\sqrt{20x^2 - 20x + 25}}{4}$

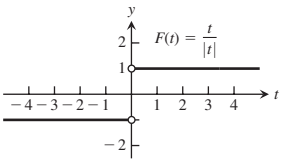
15. $(-\infty, \infty)$



17. $(-\infty, \infty)$

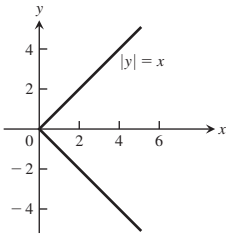


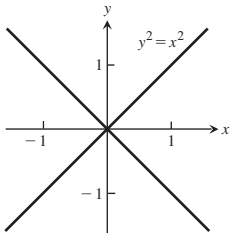
19. $(-\infty, 0) \cup (0, \infty)$



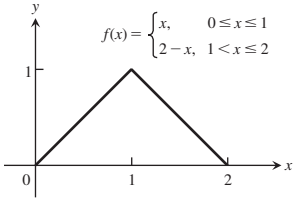
21. $(-\infty, -5) \cup (-5, -3] \cup [3, 5) \cup (5, \infty)$

23. **(a)** For each positive value of x, there are two values of y. **(b)** For each value of $x \neq 0$, there are two values of y.

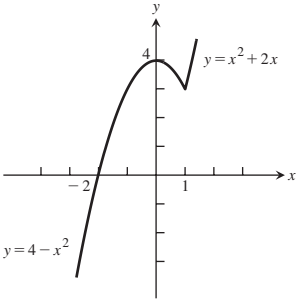




25.



27.



29. **(a)** $f(x) = \begin{cases} x, & 0 \le x \le 1 \\ -x + 2, & 1 < x \le 2 \end{cases}$

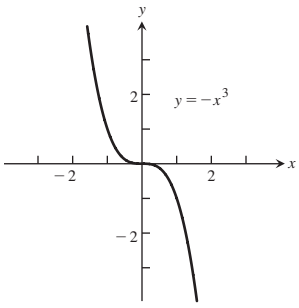
(b) $f(x) = \begin{cases} 2, & 0 \le x < 1 \\ 0, & 1 \le x < 2 \\ 2, & 2 \le x < 3 \\ 0, & 3 \le x \le 4 \end{cases}$

31. **(a)** $f(x) = \begin{cases} -x, & -1 \le x < 0 \\ 1, & 0 < x \le 1 \\ -\frac{1}{2}x + \frac{3}{2}, & 1 < x < 3 \end{cases}$

(b) $f(x) = \begin{cases} \frac{1}{2}x, & -2 \le x \le 0 \\ -2x + 2, & 0 < x \le 1 \\ -1, & 1 < x \le 3 \end{cases}$

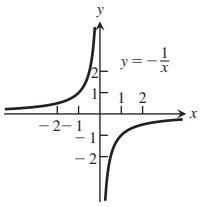
33. **(a)** $0 \le x < 1$ **(b)** $-1 < x \le 0$ **35.** Yes

37. Symmetric about the origin



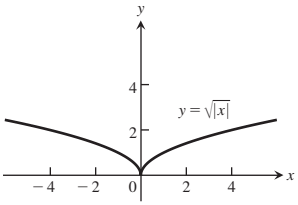
Dec. $-\infty < x < \infty$

39. Symmetric about the origin



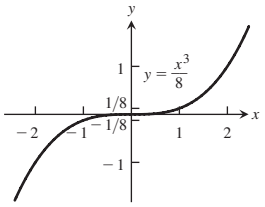
Inc. $-\infty < x < 0$ and $0 < x < \infty$

41. Symmetric about the y-axis



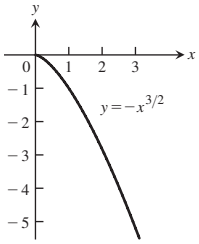
Dec. $-\infty < x \le 0$;
Inc. $0 \le x < \infty$

43. Symmetric about the origin



Inc. $-\infty < x < \infty$

45. No symmetry



Dec. $0 \le x < \infty$

47. Even **49.** Even **51.** Odd **53.** Even

55. Neither **57.** Neither **59.** Odd **61.** Even

63. $t = 180$ **65.** $s = 2.4$ **67.** $V = x(14 - 2x)(22 - 2x)$

69. **(a)** h **(b)** f **(c)** g **71.** **(a)** $(-2, 0) \cup (4, \infty)$

75. $C = 5(2 + \sqrt{2})\,h$

SECTION 1.2, pp. 36–39

1. $D_f : -\infty < x < \infty$, $D_g : x \geq 1$, $R_f : -\infty < y < \infty$, $R_g : y \geq 0$, $D_{f+g} = D_{f \cdot g} = D_g$, $R_{f+g} : y \geq 1$, $R_{f \cdot g} : y \geq 0$

3. $D_f : -\infty < x < \infty$, $D_g : -\infty < x < \infty$, $R_f : y = 2$, $R_g : y \geq 1$, $D_{f/g} : -\infty < x < \infty$, $R_{f/g} : 0 < y \leq 2$, $D_{g/f} : -\infty < x < \infty$, $R_{g/f} : y \geq 1/2$

5. **(a)** 2 **(b)** 22 **(c)** $x^2 + 2$ **(d)** $x^2 + 10x + 22$ **(e)** 5 **(f)** -2 **(g)** $x + 10$ **(h)** $x^4 - 6x^2 + 6$

7. $13 - 3x$ **9.** $\sqrt{\dfrac{5x+1}{4x+1}}$

11. **(a)** $f(g(x))$ **(b)** $j(g(x))$ **(c)** $g(g(x))$ **(d)** $j(j(x))$ **(e)** $g(h(f(x)))$ **(f)** $h(j(f(x)))$

13.

	$g(x)$	$f(x)$	$(f \circ g)(x)$
(a)	$x - 7$	$\sqrt{x}$	$\sqrt{x-7}$
(b)	$x + 2$	$3x$	$3x + 6$
(c)	x^2	$\sqrt{x-5}$	$\sqrt{x^2-5}$
(d)	$\dfrac{x}{x-1}$	$\dfrac{x}{x-1}$	x
(e)	$\dfrac{1}{x-1}$	$1 + \dfrac{1}{x}$	x
(f)	$\dfrac{1}{x}$	$\dfrac{1}{x}$	x

15. **(a)** 1 **(b)** 2 **(c)** -2 **(d)** 0 **(e)** -1 **(f)** 0

17. **(a)** $f(g(x)) = \sqrt{\dfrac{1}{x} + 1}$, $g(f(x)) = \dfrac{1}{\sqrt{x+1}}$

(b) $D_{f \circ g} = (-\infty, -1] \cup (0, \infty)$, $D_{g \circ f} = (-1, \infty)$

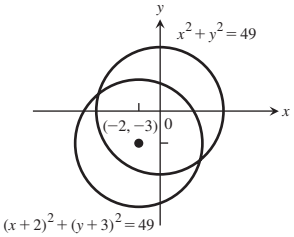
(c) $R_{f \circ g} = [0, 1) \cup (1, \infty)$, $R_{g \circ f} = (0, \infty)$

19. $g(x) = \dfrac{2x}{x-1}$ **21.** $V(t) = 4t^2 - 8t + 6$

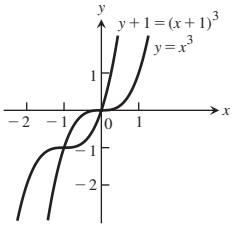
23. **(a)** $y = -(x+7)^2$ **(b)** $y = -(x-4)^2$

25. **(a)** Position 4 **(b)** Position 1 **(c)** Position 2 **(d)** Position 3

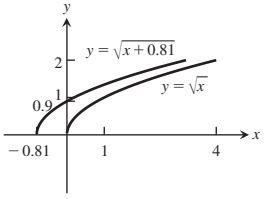
27. $(x+2)^2 + (y+3)^2 = 49$



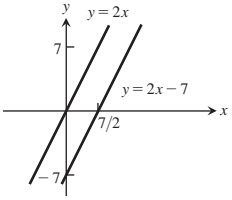
29. $y + 1 = (x+1)^3$



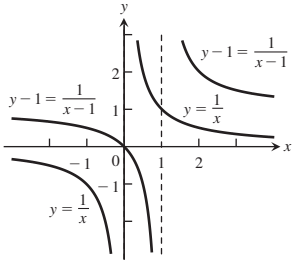
31. $y = \sqrt{x + 0.81}$



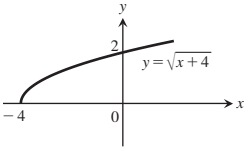
33. $y = 2x$



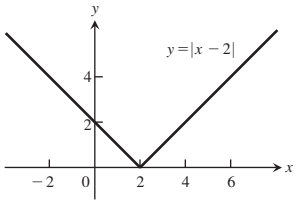
35. $y - 1 = \dfrac{1}{x-1}$



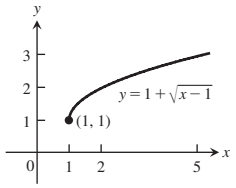
37.



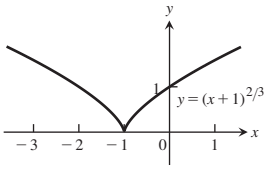
39.



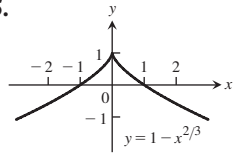
41.



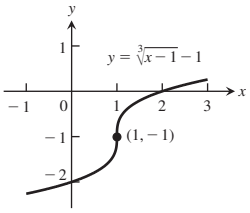
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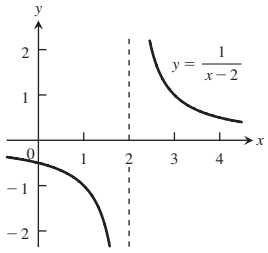
45.



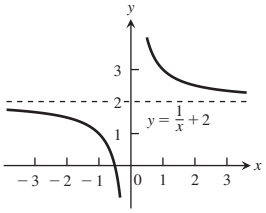
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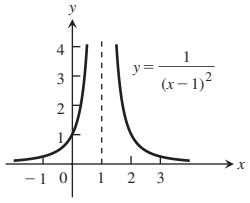
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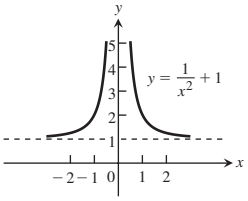
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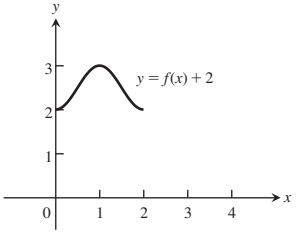
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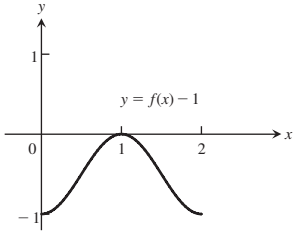


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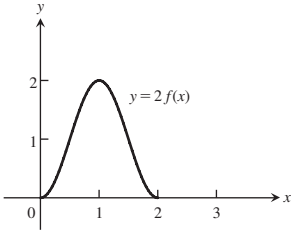


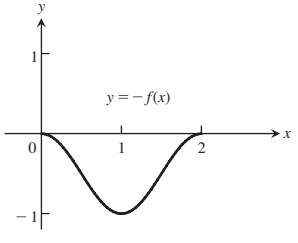
57. **(a)** $D : [0, 2]$, $R : [2, 3]$ **(b)** $D : [0, 2]$, $R : [-1, 0]$



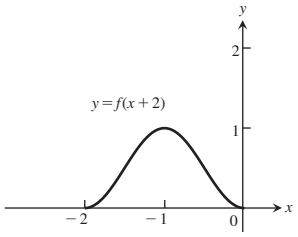


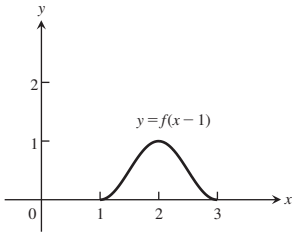
(c) $D : [0, 2]$, $R : [0, 2]$ **(d)** $D : [0, 2]$, $R : [-1, 0]$



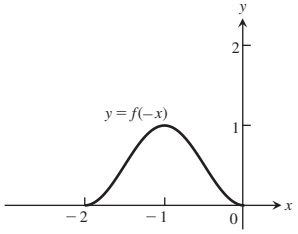


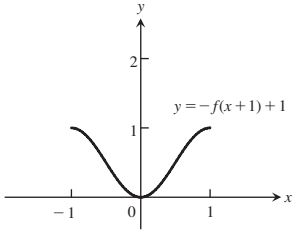
(e) $D:[-2,0]$, $R:[0,1]$ **(f)** $D:[1,3]$, $R:[0,1]$





(g) $D:[-2,0]$, $R:[0,1]$ **(h)** $D:[-1,1]$, $R:[0,1]$

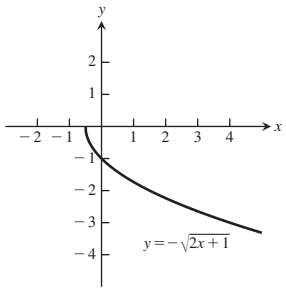




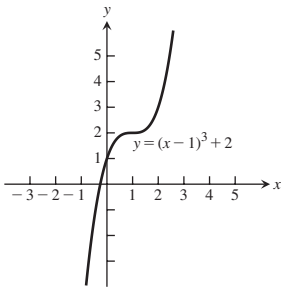
59. $y = 3x^2 - 3$ **61.** $y = \frac{1}{2} + \frac{1}{2x^2}$ **63.** $y = \sqrt{4x+1}$

65. $y = \sqrt{4 - \frac{x^2}{4}}$ **67.** $y = 1 - 27x^3$

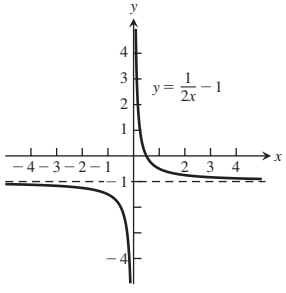
69.



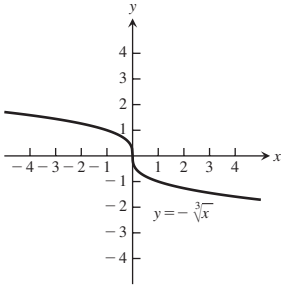
71.



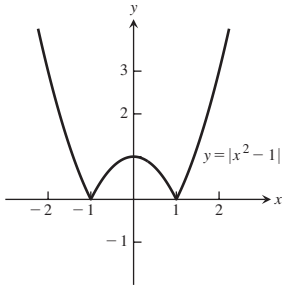
73.



75.



77.



79. **(a)** Odd **(b)** Odd **(c)** Odd **(d)** Even **(e)** Even **(f)** Even **(g)** Even **(h)** Even **(i)** Odd

SECTION 1.3, pp. 45–47

1. **(a)** 8π m **(b)** $\frac{55\pi}{9}$ m **3.** 8.4 cm (or 84 mm)

5.

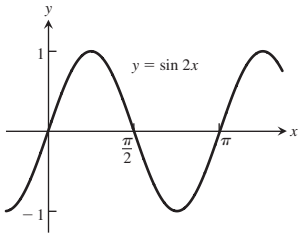
θ	$-\pi$	$-2\pi/3$	0	$\pi/2$	$3\pi/4$
$\sin\theta$	0	$-\frac{\sqrt{3}}{2}$	0	1	$\frac{1}{\sqrt{2}}$
$\cos\theta$	-1	$-\frac{1}{2}$	1	0	$-\frac{1}{\sqrt{2}}$
$\tan\theta$	0	$\sqrt{3}$	0	UND	-1
$\cot\theta$	UND	$\frac{1}{\sqrt{3}}$	UND	0	-1
$\sec\theta$	-1	-2	1	UND	$-\sqrt{2}$
$\csc\theta$	UND	$-\frac{2}{\sqrt{3}}$	UND	1	$\sqrt{2}$

7. $\cos x = -4/5$, $\tan x = -3/4$

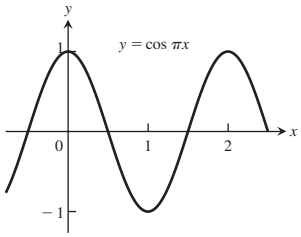
9. $\sin x = -\frac{\sqrt{8}}{3}$, $\tan x = -\sqrt{8}$

11. $\sin x = -\frac{1}{\sqrt{5}}$, $\cos x = -\frac{2}{\sqrt{5}}$

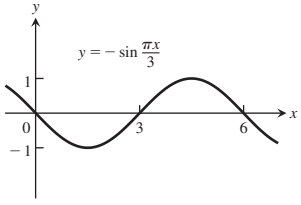
13. Period π



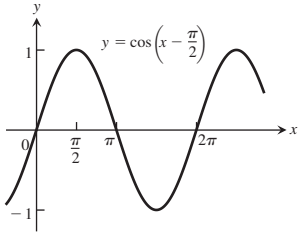
15. Period 2



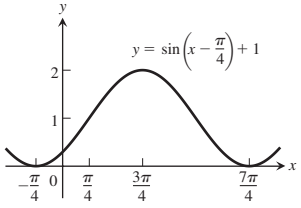
17. Period 6



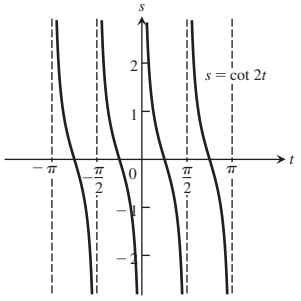
19. Period 2π



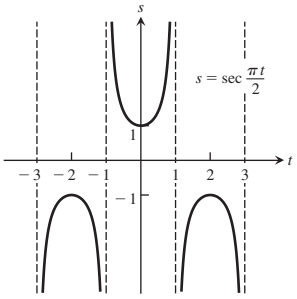
21. Period 2π



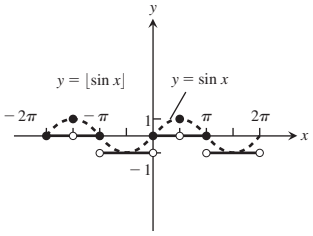
23. Period $\pi/2$, symmetric about the origin



25. Period 4, symmetric about the y-axis



29. $D: (-\infty, \infty)$, $R: y = -1, 0, 1$



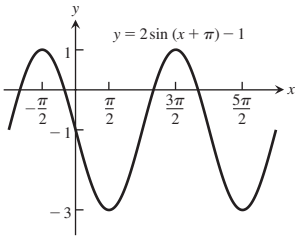
39. $-\cos x$ **41.** $-\cos x$ **43.** $\dfrac{\sqrt{6}+\sqrt{2}}{4}$ **45.** $\dfrac{\sqrt{2}+\sqrt{6}}{4}$

47. $\dfrac{2+\sqrt{2}}{4}$ **49.** $\dfrac{2-\sqrt{3}}{4}$ **51.** $\dfrac{\pi}{3}, \dfrac{2\pi}{3}, \dfrac{4\pi}{3}, \dfrac{5\pi}{3}$

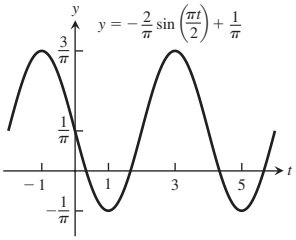
53. $\dfrac{\pi}{6}, \dfrac{\pi}{2}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$ **59.** $\sqrt{7} \approx 2.65$ **63.** $a = 1.464$

65. $r = \dfrac{\alpha \sin\theta}{1 - \sin\theta}$

67. $A = 2, B = 2\pi, C = -\pi, D = -1$



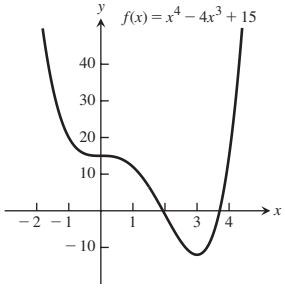
69. $A = -\dfrac{2}{\pi}, B = 4, C = 0, D = \dfrac{1}{\pi}$



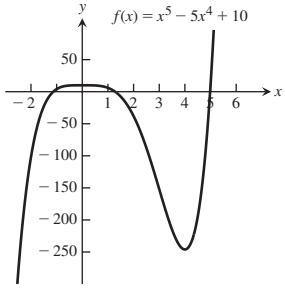
SECTION 1.4, p. 51

1. d **3.** d

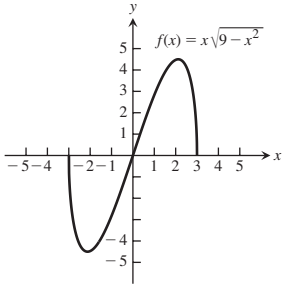
5. $[-3, 5]$ by $[-15, 40]$



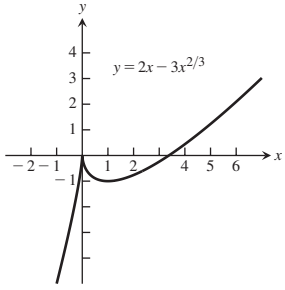
7. $[-3, 6]$ by $[-250, 50]$



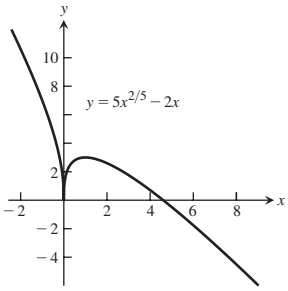
9. $[-5, 5]$ by $[-6, 6]$



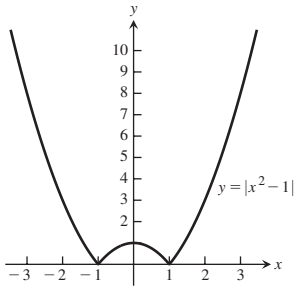
11. $[-2, 6]$ by $[-5, 4]$



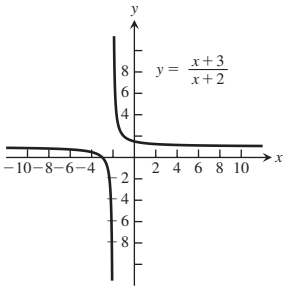
13. $[-2, 8]$ by $[-5, 10]$



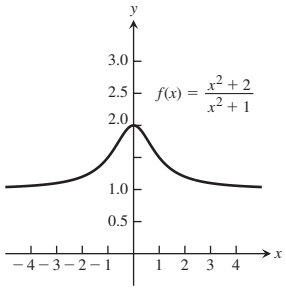
15. $[-3, 3]$ by $[0, 10]$



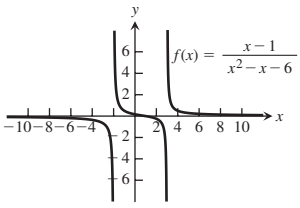
17. $[-10, 10]$ by $[-10, 10]$



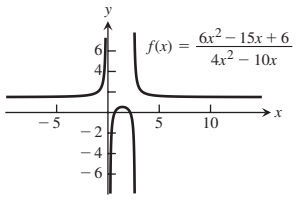
19. $[-4, 4]$ by $[0, 3]$



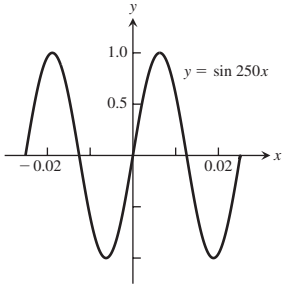
21. $[-10, 10]$ by $[-6, 6]$



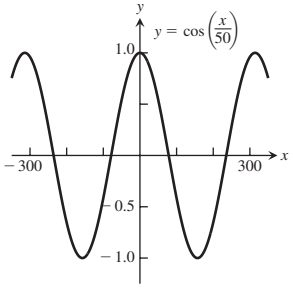
23. $[-6, 10]$ by $[-6, 6]$



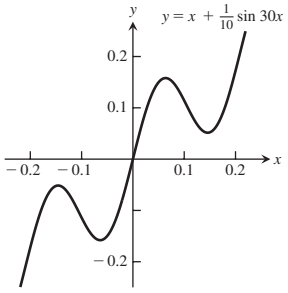
25. $\left[-\dfrac{\pi}{125}, \dfrac{\pi}{125}\right]$ by $[-1.25, 1.25]$



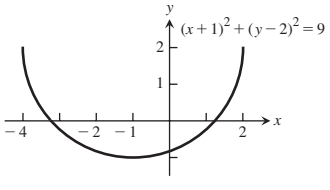
27. $[-100\pi, 100\pi]$ by $[-1.25, 1.25]$



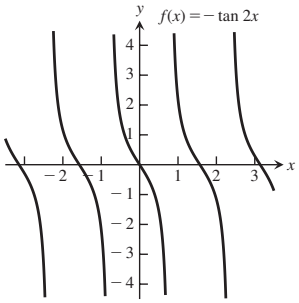
29. $\left[-\dfrac{\pi}{15}, \dfrac{\pi}{15}\right]$ by $[-0.25, 0.25]$



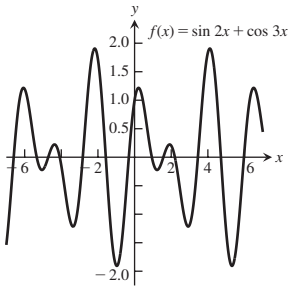
31.



33.



35.



PRACTICE EXERCISES, pp. 52–53

1. $A = \pi r^2, C = 2\pi r, A = \dfrac{C^2}{4\pi}$ **3.** $x = \tan\theta, y = \tan^2\theta$

5. Origin **7.** Neither **9.** Even **11.** Even

13. Odd **15.** Neither

17. **(a)** Even **(b)** Odd **(c)** Odd **(d)** Even **(e)** Even

19. **(a)** Domain: all reals **(b)** Range: $[-2, \infty)$

21. **(a)** Domain: $[-4, 4]$ **(b)** Range: $[0, 4]$

23. **(a)** Domain: all reals **(b)** Range: $(-3, \infty)$

25. **(a)** Domain: all reals **(b)** Range: $[-3, 1]$

27. **(a)** Domain: all reals **(b)** Range: $[0, 2]$

29. **(a)** Domain: $(-\infty, -1]$ and $[3, \infty)$ **(b)** Range: $(-\infty, 5]$

31. **(a)** Domain: $(-\infty, 0)$ and $(0, \infty)$ **(b)** Range: $[-4, 4]$

33. **(a)** Increasing **(b)** Neither **(c)** Decreasing **(d)** Increasing

35. **(a)** Domain: $[-4, 4]$ **(b)** Range: $[0, 2]$

37. $f(x) = \begin{cases} 1 - x, & 0 \le x < 1 \\ 2 - x, & 1 \le x \le 2 \end{cases}$

39. **(a)** 1 **(b)** $\dfrac{1}{\sqrt{2.5}} = \sqrt{\dfrac{2}{5}}$ **(c)** $x, x \neq 0$

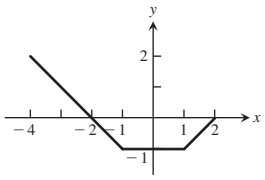
(d) $\dfrac{1}{\sqrt{1/\sqrt{x+2}+2}}$

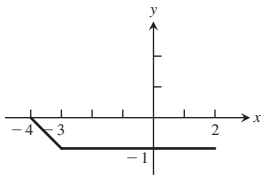
41. **(a)** $(f \circ g)(x) = -x, x \ge -2, (g \circ f)(x) = \sqrt{4 - x^2}$

(b) Domain $(f \circ g)$: $[-2, \infty)$, domain $(g \circ f)$: $[-2, 2]$

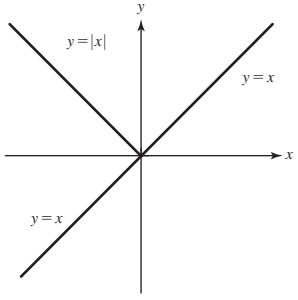
(c) Range $(f \circ g)$: $(-\infty, 2]$, range $(g \circ f)$: $[0, 2]$

43.





45. Replace the portion for $x < 0$ with the mirror image of the portion for $x > 0$ to make the new graph symmetric with respect to the y-axis.



47. Reflects the portion for $y < 0$ across the x-axis

49. Reflects the portion for $y < 0$ across the x-axis

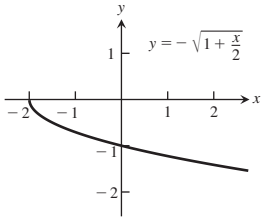
51. Adds the mirror image of the portion for $x > 0$ to make the new graph symmetric with respect to the y-axis

53. **(a)** $y = g(x - 3) + \dfrac{1}{2}$ **(b)** $y = g\left(x + \dfrac{2}{3}\right) - 2$

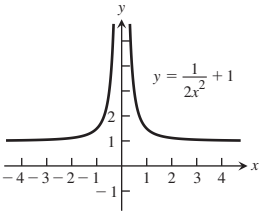
(c) $y = g(-x)$ **(d)** $y = -g(x)$ **(e)** $y = 5g(x)$

(f) $y = g(5x)$

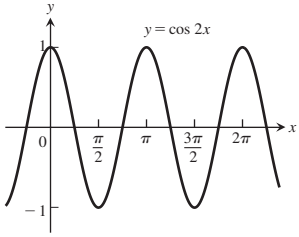
55.



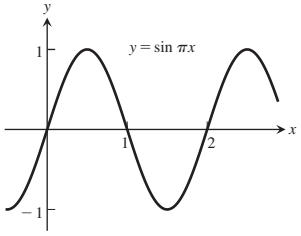
57.



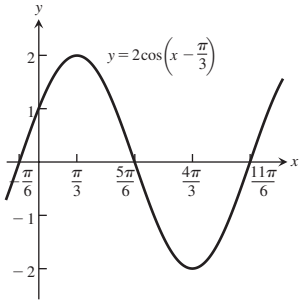
59. Period π



61. Period 2



63.



65. **(a)** $a = 1 \quad b = \sqrt{3}$ **(b)** $a = 2\sqrt{3}/3 \quad c = 4\sqrt{3}/3$

67. **(a)** $a = \dfrac{b}{\tan B}$ **(b)** $c = \dfrac{a}{\sin A}$

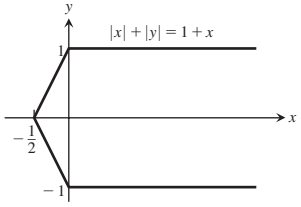
69. ≈ 16.98 m **71.** **(b)** 4π

ADDITIONAL AND ADVANCED EXERCISES, pp. 53–54

1. Yes. For instance: $f(x) = 1/x$ and $g(x) = 1/x$, or $f(x) = 2x$ and $g(x) = x/2$, or $f(x) = e^x$ and $g(x) = \ln x$.

3. If $f(x)$ is odd, then $g(x) = f(x) - 2$ is not odd. Nor is $g(x)$ even, unless $f(x) = 0$ for all x. If f is even, then $g(x) = f(x) - 2$ is also even.

5.



Chapter 2

SECTION 2.1, pp. 61–63

1. **(a)** 19 **(b)** 1

3. **(a)** $-\frac{4}{\pi}$ **(b)** $-\frac{3\sqrt{3}}{\pi}$ **5.** 1

7. **(a)** 4 **(b)** $y = 4x - 9$

9. **(a)** 2 **(b)** $y = 2x - 7$

11. **(a)** 12 **(b)** $y = 12x - 16$

13. **(a)** -9 **(b)** $y = -9x - 2$

15. **(a)** $-1/4$ **(b)** $y = -x/4 - 1$

17. **(a)** $1/4$ **(b)** $y = x/4 + 1$

19. Your estimates may not completely agree with these.

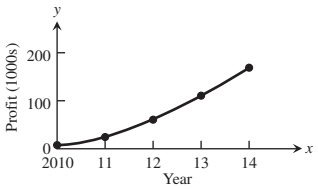
(a)

PQ_1	PQ_2	PQ_3	PQ_4
43	46	49	50

The appropriate units are m/s.

(b) ≈50 m/s or 180 km/h

21. **(a)**



(b) ≈$56,000/year

(c) ≈$42,000/year

23. **(a)** 0.414213, 0.449489, $\left(\sqrt{1+h}-1\right)/h$ **(b)** $g(x) = \sqrt{x}$

$1+h$	1.1	1.01	1.001	1.0001
$\sqrt{1+h}$	1.04880	1.004987	1.0004998	1.0000499
$\left(\sqrt{1+h}-1\right)/h$	0.4880	0.4987	0.4998	0.499

1.00001	1.000001
1.000005	1.0000005
0.5	0.5

(c) 0.5 **(d)** 0.5

25. **(a)** 15 km/h, 3.3 km/h, 10 km/h **(b)** 10 km/h, 0 km/h, 4 km/h

(c) 20 km/h when $t = 3.5$ h

SECTION 2.2, pp. 71–74

1. **(a)** Does not exist. As x approaches 1 from the right, $g(x)$ approaches 0. As x approaches 1 from the left, $g(x)$ approaches 1. There is no single number L that all the values $g(x)$ get arbitrarily close to as $x \to 1$.

(b) 1 **(c)** 0 **(d)** $1/2$

3. **(a)** True **(b)** True **(c)** False **(d)** False **(e)** False **(f)** True **(g)** True **(h)** False **(i)** True **(j)** True **(k)** False

5. As x approaches 0 from the left, $x/|x|$ approaches -1. As x approaches 0 from the right, $x/|x|$ approaches 1. There is no single number L that the function values all get arbitrarily close to as $x \to 0$.

7. Nothing can be said. **9.** No; no; no **11.** -4 **13.** -8

15. 3 **17.** $-25/2$ **19.** 16 **21.** $3/2$ **23.** $1/10$

25. -7 **27.** $3/2$ **29.** $-1/2$ **31.** -1 **33.** $4/3$

35. $1/6$ **37.** 4 **39.** $1/2$ **41.** $3/2$ **43.** -1

45. 1 **47.** $1/3$ **49.** $\sqrt{4-\pi}$

51. **(a)** Quotient Rule **(b)** Difference and Power Rules

(c) Sum and Constant Multiple Rules

53. **(a)** -10 **(b)** -20 **(c)** -1 **(d)** $5/7$

55. **(a)** 4 **(b)** -21 **(c)** -12 **(d)** $-7/3$

57. 2 **59.** 3 **61.** $1/\left(2\sqrt{7}\right)$ **63.** $\sqrt{5}$

65. **(a)** The limit is 1.

67. **(a)** $f(x) = (x^2 - 9)/(x + 3)$

x	-3.1	-3.01	-3.001	-3.0001	-3.00001	-3.000001
$f(x)$	-6.1	-6.01	-6.001	-6.0001	-6.00001	-6.000001

x	-2.9	-2.99	-2.999	-2.9999	-2.99999	-2.999999
$f(x)$	-5.9	-5.99	-5.999	-5.9999	-5.99999	-5.999999

(c) $\lim_{x \to -3} f(x) = -6$

69. **(a)** $G(x) = (x + 6)/(x^2 + 4x - 12)$

x	-5.9	-5.99	-5.999	-5.9999
$G(x)$	$-.126582$	$-.1251564$	$-.1250156$	$-.1250015$

-5.99999	-5.999999
$-.1250001$	$-.1250000$

x	-6.1	-6.01	-6.001	-6.0001
$G(x)$	$-.123456$	$-.124843$	$-.124984$	$-.124998$

-6.00001	-6.000001
$-.124999$	$-.124999$

(c) $\lim_{x \to -6} G(x) = -1/8 = -0.125$

71. **(a)** $f(x) = (x^2 - 1)/(|x| - 1)$

x	-1.1	-1.01	-1.001	-1.0001	-1.00001	-1.000001
$f(x)$	2.1	2.01	2.001	2.0001	2.00001	2.000001

x	$-.9$	$-.99$	$-.999$	$-.9999$	$-.99999$	$-.999999$
$f(x)$	1.9	1.99	1.999	1.9999	1.99999	1.999999

(c) $\lim_{x \to -1} f(x) = 2$

73. **(a)** $g(\theta) = (\sin\theta)/\theta$

θ	.1	.01	.001	.0001	.00001	.000001
$g(\theta)$	.998334	.999983	.999999	.999999	.999999	.999999

θ	$-.1$	$-.01$	$-.001$	$-.0001$	$-.00001$	$-.000001$
$g(\theta)$	.998334	.999983	.999999	.999999	.999999	.999999

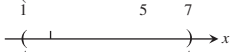
$\lim_{\theta \to 0} g(\theta) = 1$

75. $c = 0, 1, -1$; the limit is 0 at $c = 0$, and 1 at $c = 1, -1$.

77. 7 **79.** **(a)** 5 **(b)** 5 **81.** **(a)** 0 **(b)** 0

SECTION 2.3, pp. 79–82

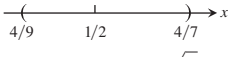
1. $\delta = 2$



3. $\delta = 1/2$



5. $\delta = 1/18$



7. $\delta = 0.1$ **9.** $\delta = 7/16$ **11.** $\delta = \sqrt{5} - 2$

13. $\delta = 0.36$ **15.** $(3.99, 4.01)$, $\delta = 0.01$

17. $(-0.19, 0.21)$, $\delta = 0.19$ **19.** $(3, 15)$, $\delta = 5$

21. $(10/3, 5)$, $\delta = 2/3$

23. $\left(-\sqrt{4.5}, -\sqrt{3.5}\right)$, $\delta = \sqrt{4.5} - 2 \approx 0.12$

25. $\left(\sqrt{15}, \sqrt{17}\right)$, $\delta = \sqrt{17} - 4 \approx 0.12$

27. $\left(2 - \frac{0.03}{m}, 2 + \frac{0.03}{m}\right), \quad \delta = \frac{0.03}{m}$

29. $\left(\frac{1}{2} - \frac{c}{m}, \frac{c}{m} + \frac{1}{2}\right), \quad \delta = \frac{c}{m}$

31. $L = -3, \quad \delta = 0.01$ **33.** $L = 4, \quad \delta = 0.05$

35. $L = 4, \quad \delta = 0.75$

55. $[\,8.7332, 8.7476\,]$. To be safe, the left endpoint was rounded up and the right endpoint rounded down.

59. The limit does not exist as x approaches 3.

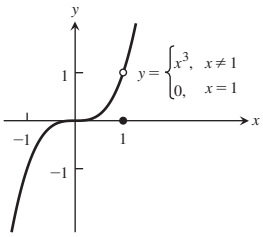
SECTION 2.4, pp. 88–90

1. **(a)** True **(b)** True **(c)** False **(d)** True **(e)** True **(f)** True **(g)** False **(h)** False **(i)** False **(j)** False **(k)** True **(l)** False

3. **(a)** 2, 1 **(b)** No, $\lim_{x\to 2^+} f(x) \neq \lim_{x\to 2^-} f(x)$ **(c)** 3, 3 **(d)** Yes, 3

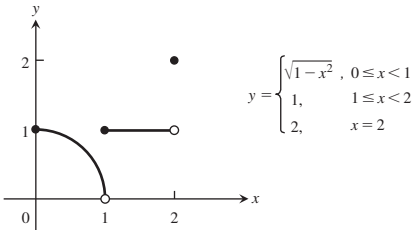
5. **(a)** No **(b)** Yes, 0 **(c)** No

7. **(a)**



(b) 1, 1 **(c)** Yes, 1

9. **(a)** $D: 0 \le x \le 2, R: 0 < y \le 1$ and $y = 2$ **(b)** $[\,0, 1) \cup (1, 2\,]$ **(c)** $x = 2$ **(d)** $x = 0$



11. $\sqrt{3}$ **13.** 1 **15.** $2/\sqrt{5}$ **17.** **(a)** 1 **(b)** -1

19. **(a)** 1 **(b)** -1 **21.** **(a)** 1 **(b)** $2/3$ **23.** 1 **25.** $3/4$

27. 2 **29.** $1/2$ **31.** 2 **33.** 0 **35.** 1 **37.** $1/2$

39. 0 **41.** $3/8$ **43.** 3 **45.** 0

51. $\delta = \varepsilon^2, \lim_{x\to 5^+} \sqrt{x - 5} = 0$

55. **(a)** 400 **(b)** 399 **(c)** The limit does not exist.

SECTION 2.5, pp. 99–101

1. No; not defined at $x = 2$

3. Continuous **5.** **(a)** Yes **(b)** Yes **(c)** Yes **(d)** Yes

7. **(a)** No **(b)** No **9.** 0

11. 1, nonremovable; 0, removable **13.** All x except $x = 2$

15. All x except $x = 3, x = 1$ **17.** All x

19. All x except $x = 0$ **21.** All x except $n\pi/2$, n any integer

23. All x except $n\pi/2$, n an odd integer **25.** All $x \ge -3/2$

27. All x **29.** All x **31.** 0; continuous at $x = \pi$

33. 1; continuous at $y = 1$ **35.** $\sqrt{2}/2$; continuous at $t = 0$

37. $g(3) = 6$ **39.** $f(1) = 3/2$ **41.** $a = 4/3$

43. $a = -2, 3$ **45.** $a = 5/2, b = -1/2$

65. $x \approx 1.8794, -1.5321, -0.3473$ **67.** $x \approx 1.7549$

69. $x \approx 0.7391$

SECTION 2.6, pp. 111–114

1. **(a)** 0 **(b)** -2 **(c)** 2 **(d)** Does not exist **(e)** -1 **(f)** ∞ **(g)** Does not exist **(h)** 1 **(i)** 0

3. **(a)** -3 **(b)** -3 **5.** **(a)** $1/2$ **(b)** $1/2$ **7.** **(a)** $-5/3$ **(b)** $-5/3$ **9.** 0 **11.** -1 **13.** **(a)** $2/5$ **(b)** $2/5$

15. **(a)** 0 **(b)** 0 **17.** **(a)** 7 **(b)** 7 **19.** **(a)** 0 **(b)** 0

21. **(a)** ∞ **(b)** ∞ **23.** 2 **25.** ∞ **27.** 0 **29.** 1

31. ∞ **33.** 1 **35.** $1/2$ **37.** ∞ **39.** $-\infty$

41. $-\infty$ **43.** ∞ **45.** **(a)** ∞ **(b)** $-\infty$ **47.** ∞

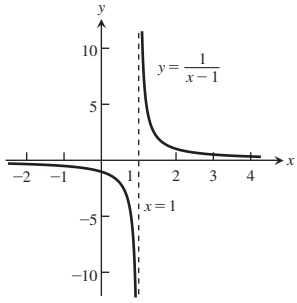
49. ∞ **51.** $-\infty$ **53.** **(a)** ∞ **(b)** $-\infty$ **(c)** $-\infty$ **(d)** ∞

55. **(a)** $-\infty$ **(b)** ∞ **(c)** 0 **(d)** $3/2$

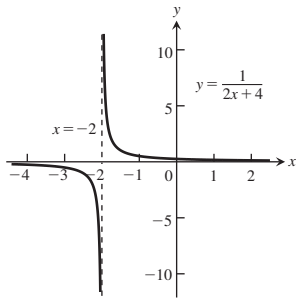
57. **(a)** $-\infty$ **(b)** $1/4$ **(c)** $1/4$ **(d)** $1/4$ **(e)** It will be $-\infty$.

59. **(a)** $-\infty$ **(b)** ∞ **61.** **(a)** ∞ **(b)** ∞ **(c)** ∞ **(d)** ∞

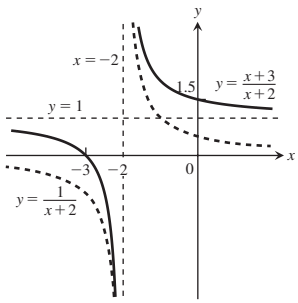
63.



65.



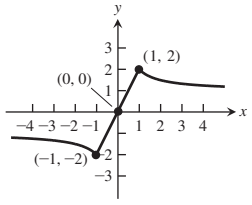
67.



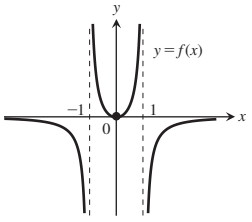
69. Domain: $(-\infty, \infty)$, Range: $[\,4, 7)$

71. Domain: $(-\infty, 0)$ and $(0, \infty)$, Range: $(-\infty, -1)$ and $(1, \infty)$

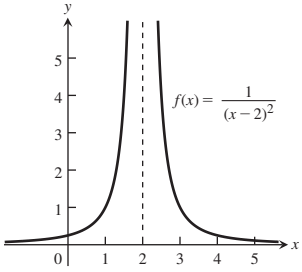
73. Here is one possibility.



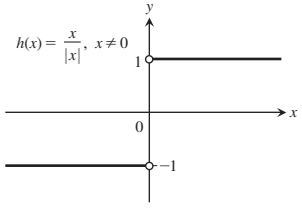
75. Here is one possibility.



77. Here is one possibility.



79. Here is one possibility.



83. At most one **85.** 0 **87.** $-3/4$ **89.** $5/2$

97. (a) For every positive real number B there exists a corresponding number $\delta > 0$ such that for all x

$$c - \delta < x < c \implies f(x) > B.$$

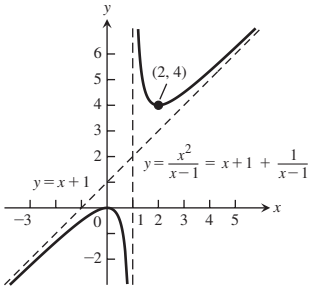
(b) For every negative real number $-B$ there exists a corresponding number $\delta > 0$ such that for all x

$$c < x < c + \delta \implies f(x) < -B.$$

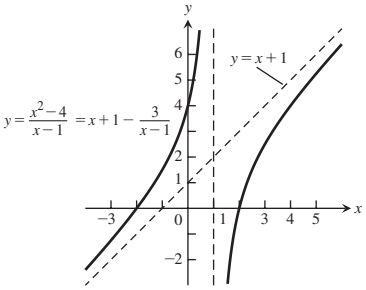
(c) For every negative real number $-B$ there exists a corresponding number $\delta > 0$ such that for all x

$$c - \delta < x < c \implies f(x) < -B.$$

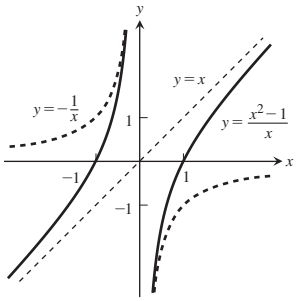
103.



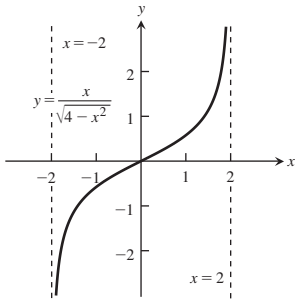
105.



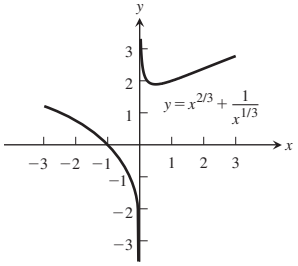
107.



109.



111.



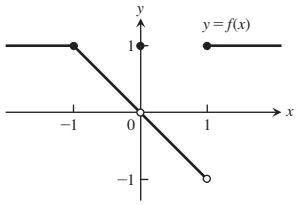
113. At ∞: ∞, at $-\infty$: 0

PRACTICE EXERCISES, pp. 115–116

1. At $x = -1$: $\lim_{x\to -1^-} f(x) = \lim_{x\to -1^+} f(x) = 1$, so $\lim_{x\to -1} f(x) = 1 = f(-1)$; continuous at $x = -1$

At $x = 0$: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = 0$, so $\lim_{x\to 0} f(x) = 0$. However, $f(0) \neq 0$, so f is discontinuous at $x = 0$. The discontinuity can be removed by redefining $f(0)$ to be 0.

At $x = 1$: $\lim_{x\to 1^-} f(x) = -1$ and $\lim_{x\to 1^+} f(x) = 1$, so $\lim_{x\to 1} f(x)$ does not exist. The function is discontinuous at $x = 1$, and the discontinuity is not removable.



3. (a) -21 **(b)** 49 **(c)** 0 **(d)** 1 **(e)** 1 **(f)** 7 **(g)** -7 **(h)** $-\frac{1}{7}$ **5.** 4

7. (a) $(-\infty, +\infty)$ **(b)** $[0, \infty)$ **(c)** $(-\infty, 0)$ and $(0, \infty)$ **(d)** $(0, \infty)$

9. (a) Does not exist **(b)** 0 **11.** $\frac{1}{2}$ **13.** $2x$ **15.** $-\frac{1}{4}$

17. $2/3$ **19.** $2/\pi$ **21.** 1 **23.** 4 **25.** 2 **27.** 0

31. No in both cases, because $\lim_{x\to 1} f(x)$ does not exist, and $\lim_{x\to -1} f(x)$ does not exist.

33. Yes, f does have a continuous extension, to $a = 1$ with $f(1) = 4/3$.

35. No **37.** $2/5$ **39.** 0 **41.** $-\infty$ **43.** 0 **45.** 1

47. (a) $x = 3$ **(b)** $x = 1$ **(c)** $x = -4$

49. Domain: $[-4, 2)$ and $(2, 4]$, Range: $(-\infty, \infty)$

ADDITIONAL AND ADVANCED EXERCISES, pp. 116–119

1. 0; the left-hand limit was taken because the function is undefined for $v > c$.

3. $17.5 < t < 22.5$; within 2.5°C **11. (a)** B **(b)** A **(c)** A **(d)** A

19. (a) $\lim_{a\to 0} r_+(a) = 0.5$, $\lim_{a\to -1^+} r_+(a) = 1$

(b) $\lim_{a\to 0} r_-(a)$ does not exist, $\lim_{a\to -1^+} r_-(a) = 1$

23. 0 **25.** 1 **27.** 4 **29.** $y = 2x$ **31.** $y = x, y = -x$

35. $-4/3$

37. (a) Domain: $\{1, 1/2, 1/3, 1/4 \ldots\}$

(b) The domain intersects (a, b) if $a < 0$ and $b > 0$.

(c) 0

39. (a) Domain: $(-\infty, -1/\pi] \cup [-1/(2\pi), -1/(3\pi)] \cup [-1/(4\pi), -1/(5\pi)] \cup \cdots \cup [1/(5\pi), 1/(4\pi)] \cup [1/(3\pi), 1/(2\pi)] \cup [1/\pi, \infty)$

(b) The domain intersects any open interval (a, b) containing 0 because $1/(n\pi) < b$ for large enough n.

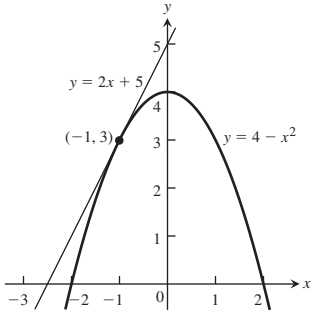
(c) 0

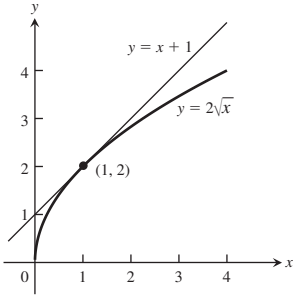
Chapter 3

SECTION 3.1, pp. 122–124

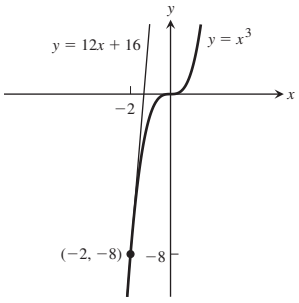
1. P_1: $m_1 = 1$, P_2: $m_2 = 5$ **3.** P_1: $m_1 = 5/2$, P_2: $m_2 = -1/2$

5. $y = 2x + 5$ **7.** $y = x + 1$





9. $y = 12x + 16$



11. $m = 4, y - 5 = 4(x - 2)$
13. $m = -2, y - 3 = -2(x - 3)$
15. $m = 12, y - 8 = 12(t - 2)$
17. $m = \frac{1}{4}, y - 2 = \frac{1}{4}(x - 4)$
19. $m = -1$ **21.** $m = -1/4$
23. **(a)** It is the rate of change of the number of cells when $t = 5$. The units are the number of cells per hour.
(b) $P'(3)$ because the slope of the curve is greater there.
(c) $51.72 \approx 52$ cells/h
25. $(-2,-5)$ **27.** $y = -(x + 1), y = -(x - 3)$
29. 19.6 m/s **31.** 6π **35.** Yes **37.** Yes
39. **(a)** Nowhere **41.** **(a)** At $x = 0$ **43.** **(a)** Nowhere
45. **(a)** At $x = 1$ **47.** **(a)** At $x = 0$

SECTION 3.2, pp. 129–133

1. $-2x, 6, 0, -2$ **3.** $-\frac{2}{t^3}, 2, -\frac{1}{4}, -\frac{2}{3\sqrt{3}}$

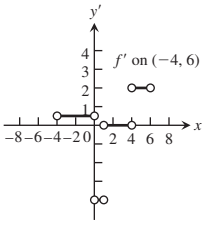
5. $\frac{3}{2\sqrt{3\theta}}, \frac{3}{2\sqrt{3}}, \frac{1}{2}, \frac{3}{2\sqrt{2}}$ **7.** $6x^2$ **9.** $\frac{1}{(2t + 1)^2}$

11. $\frac{3}{2}q^{1/2}$ **13.** $1 - \frac{9}{x^2}, 0$ **15.** $3t^2 - 2t, 5$

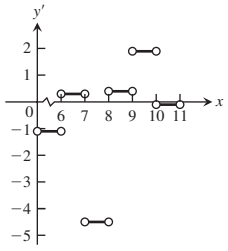
17. $\frac{-4}{(x - 2)\sqrt{x - 2}}, \quad y - 4 = -\frac{1}{2}(x - 6)$ **19.** 6

21. $1/8$ **23.** $\frac{-1}{(x + 2)^2}$ **25.** $\frac{-1}{(x - 1)^2}$ **27.** (b) **29.** (d)

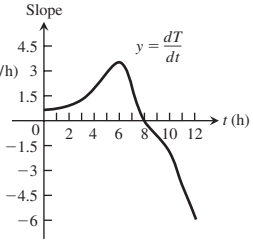
31. **(a)** $x = 0, 1, 4$
(b)



33.



35. **(a)** **i)** 0.77 °C/h **ii)** 1.43 °C/h
iii) 0 °C/h **iv)** −1.88 °C/h
(b) 3.63 °C/h at 12 P.M., −4 °C/h at 6 P.M.
(c)



37. Since $\lim_{h \to 0^+} \frac{f(0 + h) - f(0)}{h} = 1$
while $\lim_{h \to 0^-} \frac{f(0 + h) - f(0)}{h} = 0,$
$f'(0) = \lim_{h \to 0} \frac{f(0 + h) - f(0)}{h}$ does not exist and $f(x)$ is not differentiable at $x = 0$.
39. Since $\lim_{h \to 0^+} \frac{f(1 + h) - f(1)}{h} = 2$ while
$\lim_{h \to 0^-} \frac{f(1 + h) - f(1)}{h} = \frac{1}{2}, \quad f'(1) = \lim_{h \to 0} \frac{f(1 + h) - f(1)}{h}$
does not exist and $f(x)$ is not differentiable at $x = 1$.
41. Since $f(x)$ is not continuous at $x = 0$, $f(x)$ is not differentiable at $x = 0$.
43. Since $\lim_{h \to 0^+} \frac{f(0 + h) - f(0)}{h} = 3$ while
$\lim_{h \to 0^-} \frac{f(0 + h) - f(0)}{h} = 0$, f is not differentiable at $x = 0$.
45. **(a)** $-3 \le x \le 2$ **(b)** None **(c)** None
47. **(a)** $-3 \le x < 0, 0 < x \le 3$ **(b)** None **(c)** $x = 0$
49. **(a)** $-1 \le x < 0, 0 < x \le 2$ **(b)** $x = 0$ **(c)** None

SECTION 3.3, pp. 139–141

1. $\frac{dy}{dx} = -2x, \frac{d^2y}{dx^2} = -2$

3. $\frac{ds}{dt} = 15t^2 - 15t^4, \frac{d^2s}{dt^2} = 30t - 60t^3$

5. $\frac{dy}{dx} = 4x^2 - 1, \frac{d^2y}{dx^2} = 8x$

7. $\frac{dw}{dz} = -\frac{6}{z^3} + \frac{1}{z^2}, \frac{d^2w}{dz^2} = \frac{18}{z^4} - \frac{2}{z^3}$

9. $\frac{dy}{dx} = 12x - 10 + 10x^{-3}, \frac{d^2y}{dx^2} = 12 - 30x^{-4}$

11. $\frac{dr}{ds} = \frac{-2}{3s^3} + \frac{5}{2s^2}, \frac{d^2r}{ds^2} = \frac{2}{s^4} - \frac{5}{s^3}$

13. $y' = -5x^4 + 12x^2 - 2x - 3$

15. $y' = 3x^2 + 10x + 2 - \frac{1}{x^2}$ **17.** $y' = \frac{-19}{(3x - 2)^2}$

19. $g'(x) = \frac{x^2 + x + 4}{(x + 0.5)^2}$ **21.** $\frac{dv}{dt} = \frac{t^2 - 2t - 1}{(1 + t^2)^2}$

23. $f'(s) = \frac{1}{\sqrt{s}(\sqrt{s} + 1)^2}$ **25.** $v' = -\frac{1}{x^2} + 2x^{-3/2}$

27. $y' = \frac{-4x^3 - 3x^2 + 1}{(x^2 - 1)^2(x^2 + x + 1)^2}$

29. $y' = 2x^3 - 3x - 1, y'' = 6x^2 - 3, y''' = 12x, y^{(4)} = 12,$
$y^{(n)} = 0$ for $n \ge 5$
31. $y' = 3x^2 + 8x + 1, y'' = 6x + 8, y''' = 6, y^{(n)} = 0$ for $n \ge 4$
33. $y' = 2x - 7x^{-2}, y'' = 2 + 14x^{-3}$

35. $\frac{dr}{d\theta} = 3\theta^{-4}, \frac{d^2r}{d\theta^2} = -12\theta^{-5}$ **37.** $\frac{dw}{dz} = -z^{-2} - 1, \frac{d^2w}{dz^2} = 2z^{-3}$

39. **(a)** 13 **(b)** −7 **(c)** 7/25 **(d)** 20

41. **(a)** $y = -\frac{x}{8} + \frac{5}{4}$ **(b)** $m = -4$ at $(0, 1)$
(c) $y = 8x - 15, y = 8x + 17$
43. $y = 4x, y = 2$ **45.** $a = 1, b = 1, c = 0$
47. $(2, 4)$ **49.** $(0, 0), (4, 2)$ **51.** $y = -16x + 24$

53. **(a)** $y = 2x + 2$ **(c)** $(2, 6)$ **55.** 50 **57.** $a = -3$

59. $P'(x) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + 2a_2 x + a_1$

61. The Product Rule is then the Constant Multiple Rule, so the latter is a special case of the Product Rule.

63. **(a)** $\dfrac{d}{dx}(uvw) = uvw' + uv'w + u'vw$

(b) $\dfrac{d}{dx}(u_1u_2u_3u_4) = u_1u_2u_3u_4' + u_1u_2u_3'u_4 + u_1u_2'u_3u_4 + u_1'u_2u_3u_4$

(c) $\dfrac{d}{dx}(u_1 \cdots u_n) = u_1u_2 \cdots u_{n-1}u_n' + u_1u_2 \cdots u_{n-2}u_{n-1}'u_n + \cdots + u_1'u_2 \cdots u_n$

65. $\dfrac{dP}{dV} = -\dfrac{nRT}{(V-nb)^2} + \dfrac{2an^2}{V^3}$

SECTION 3.4, pp. 148–152

1. **(a)** -2 m, -1 m/s
(b) 3 m/s, 1 m/s; 2 m/s^2, 2 m/s^2
(c) Changes direction at $t = 3/2$ s

3. **(a)** -9 m, -3 m/s
(b) 3 m/s, 12 m/s; 6 m/s^2, -12 m/s^2
(c) No change in direction

5. **(a)** -20 m, -5 m/s
(b) 45 m/s, $(1/5)$ m/s; 140 m/s^2, $(4/25)$ m/s^2
(c) No change in direction

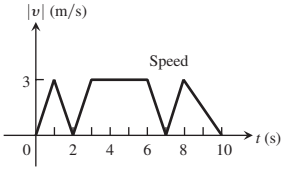
7. **(a)** $a(1) = -6$ m/s^2, $a(3) = 6$ m/s^2
(b) $v(2) = 3$ m/s **(c)** 6 m

9. Mars: ≈ 7.5 s, Jupiter: ≈ 1.2 s

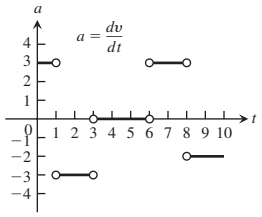
11. $g_s = 0.75$ m/s^2

13. **(a)** $v = -9.8t$, $|v| = 9.8t$ m/s, $a = -9.8$ m/s^2
(b) $t \approx 3.4$ s
(c) $v \approx -33.1$ m/s

15. **(a)** $t = 2, t = 7$ **(b)** $3 \le t \le 6$
(c)



(d)



17. **(a)** 57 m/s **(b)** 2 s **(c)** 8 s, 0 m/s
(d) 10.8 s, 27 m/s **(e)** 2.8 s
(f) Greatest acceleration happens 2 s after launch
(g) Constant acceleration between 2 and 10.8 s, -10 m/s^2

19. **(a)** $\frac{4}{7}$ s, 280 cm/s **(b)** 560 cm/s, 980 cm/s^2
(c) 29.75 flashes/s

21. C = position, A = velocity, B = acceleration

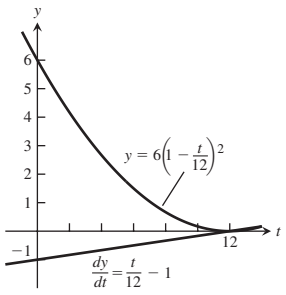
23. **(a)** \$110/machine **(b)** \$80 **(c)** \$79.90

25. **(a)** $b'(0) = 10^4$ bacteria/h **(b)** $b'(5) = 0$ bacteria/h
(c) $b'(10) = -10^4$ bacteria/h

27. **(a)** $\dfrac{dy}{dt} = \dfrac{t}{12} - 1$

(b) The largest value of $\dfrac{dy}{dt}$ is 0 m/h when $t = 12$ and the smallest value of $\dfrac{dy}{dt}$ is -1 m/h when $t = 0$.

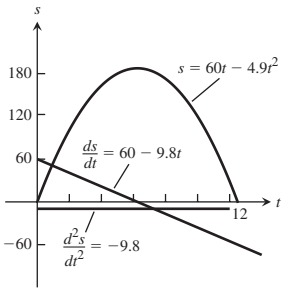
(c)



29. 0.846 m, 1.482 m, the additional number of meters required to stop the car if speed is increased by 1 km/h

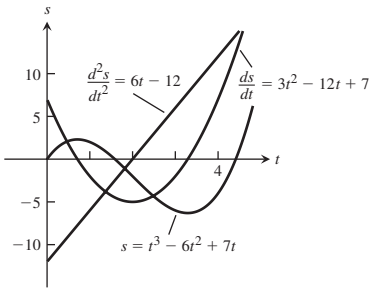
31. $t = 25$ s, $D = \dfrac{6250}{9}$ m

33.



(a) $v = 0$ when $t = 6.12$ s
(b) $v > 0$ when $0 \le t < 6.12 \Rightarrow$ the object moves up; $v < 0$ when $6.12 < t \le 12.24 \Rightarrow$ the object moves down.
(c) The object changes direction at $t = 6.12$ s.
(d) The object speeds up on $(6.12, 12.24]$ and slows down on $[0, 6.12)$.
(e) The object is moving fastest at the endpoints $t = 0$ and $t = 12.24$ when it is traveling 60 m/s. It's moving slowest at $t = 6.12$ when the speed is 0.
(f) When $t = 6.12$ the object is $s = 183.6$ m from the origin and farthest away.

35.



(a) $v = 0$ when $t = \dfrac{6 \pm \sqrt{15}}{3}$ s

(b) $v < 0$ when $\dfrac{6 - \sqrt{15}}{3} < t < \dfrac{6 + \sqrt{15}}{3} \Rightarrow$ the object moves left; $v > 0$ when $0 \le t < \dfrac{6 - \sqrt{15}}{3}$ or $\dfrac{6 + \sqrt{15}}{3} < t \le 4 \Rightarrow$ the object moves right.

(c) The object changes direction at $t = \dfrac{6 \pm \sqrt{15}}{3}$ s.

(d) The object speeds up on $\left(\frac{6-\sqrt{15}}{3}, 2\right) \cup \left(\frac{6+\sqrt{15}}{3}, 4\right]$ and slows down on $\left[0, \frac{6-\sqrt{15}}{3}\right) \cup \left(2, \frac{6+\sqrt{15}}{3}\right)$.

(e) The object is moving fastest at $t = 0$ and $t = 4$ when it is moving 7 units/s and slowest at $t = \frac{6 \pm \sqrt{15}}{3}$ s.

(f) When $t = \frac{6+\sqrt{15}}{3}$ the object is at position $s \approx -6.303$ units and farthest from the origin.

SECTION 3.5, pp. 156–158

1. $-10 - 3\sin x$ **3.** $2x\cos x - x^2 \sin x$

5. $-\csc x \cot x - \frac{2}{\sqrt{x}}$ **7.** $\sin x \sec^2 x + \sin x$

9. $\sec x + x \sec x \tan x - \frac{1}{x^2}$ **11.** $\frac{-\csc^2 x}{(1+\cot x)^2}$

13. $4\tan x \sec x - \csc^2 x$ **15.** 0

17. $3x^2 \sin x \cos x + x^3 \cos^2 x - x^3 \sin^2 x$

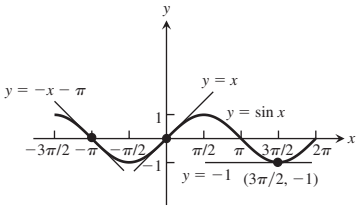
19. $\sec^2 t - 1$ **21.** $\frac{-2\csc t \cot t}{(1-\csc t)^2}$ **23.** $-\theta(\theta\cos\theta + 2\sin\theta)$

25. $\sec\theta\csc\theta(\tan\theta - \cot\theta) = \sec^2\theta - \csc^2\theta$ **27.** $\sec^2 q$

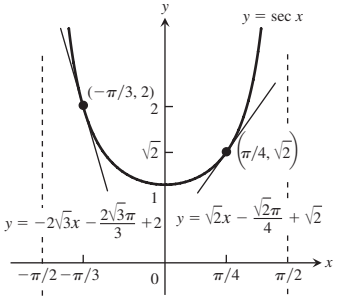
29. $\sec^2 q$ **31.** $\frac{q^3\cos q - q^2 \sin q - q\cos q - \sin q}{(q^2-1)^2}$

33. **(a)** $2\csc^3 x - \csc x$ **(b)** $2\sec^3 x - \sec x$

35.

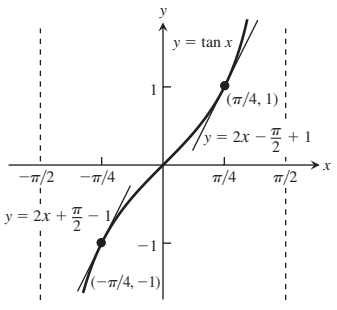


37.



39. Yes, at $x = \pi$ **41.** No **43.** Yes, at $x = 0, \pi$, and 2π

45. $\left(-\frac{\pi}{4}, -1\right); \left(\frac{\pi}{4}, 1\right)$



47. **(a)** $y = -x + \pi/2 + 2$ **(b)** $y = 4 - \sqrt{3}$

49. 0 **51.** $\sqrt{3}/2$ **53.** -1 **55.** 0

57. $-\sqrt{2}$ m/s, $\sqrt{2}$ m/s, $\sqrt{2}$ m/s^2, $\sqrt{2}$ m/s^3

59. $c = 9$ **61.** **(a)** $\sin x$ **(b)** $3\cos x - \sin x$ **(c)** $73\sin x + x\cos x$

63. **(a)** **i)** 10 cm **ii)** 5 cm **iii)** $-5\sqrt{2} \approx -7.1$ cm
(b) **i)** 0 cm/s **ii)** $-5\sqrt{3} \approx -8.7$ cm/s
iii) $-5\sqrt{2} \approx -7.1$ cm/s

SECTION 3.6, pp. 163–166

1. $12x^3$ **3.** $3\cos(3x+1)$ **5.** $\frac{\cos x}{2\sqrt{\sin x}}$

7. $2\pi x \sec^2(\pi x^2)$

9. With $u = (2x+1)$, $y = u^5$: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = 5u^4 \cdot 2 = 10(2x+1)^4$

11. With $u = (1 - (x/7))$, $y = u^{-7}$: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = -7u^{-8} \cdot \left(-\frac{1}{7}\right) = \left(1 - \frac{x}{7}\right)^{-8}$

13. With $u = ((x^2/8) + x - (1/x))$, $y = u^4$: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = 4u^3 \cdot \left(\frac{x}{4} + 1 + \frac{1}{x^2}\right) = 4\left(\frac{x^2}{8} + x - \frac{1}{x}\right)^3\left(\frac{x}{4} + 1 + \frac{1}{x^2}\right)$

15. With $u = \tan x$, $y = \sec u$: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = (\sec u \tan u)(\sec^2 x) = \sec(\tan x)\tan(\tan x)\sec^2 x$

17. With $u = \tan x$, $y = u^3$: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = 3u^2\sec^2 x = 3\tan^2 x(\sec^2 x)$

19. $-\frac{1}{2\sqrt{3-t}}$ **21.** $\frac{4}{\pi}(\cos 3t - \sin 5t)$ **23.** $\frac{\csc\theta}{\cot\theta + \csc\theta}$

25. $2x\sin^4 x + 4x^2\sin^3 x\cos x + \cos^{-2} x + 2x\cos^{-3} x\sin x$

27. $(3x-2)^5 - \frac{1}{x^3\left(4 - \frac{1}{2x^2}\right)^2}$ **29.** $\frac{(4x+3)^3(4x+7)}{(x+1)^4}$

31. $\sqrt{x}\sec^2\left(2\sqrt{x}\right) + \tan\left(2\sqrt{x}\right)$ **33.** $\frac{x\sec x\tan x + \sec x}{2\sqrt{7 + x\sec x}}$

35. $\frac{2\sin\theta}{(1+\cos\theta)^2}$ **37.** $-2\sin(\theta^2)\sin 2\theta + 2\theta\cos(2\theta)\cos(\theta^2)$

39. $\left(\frac{t+2}{2(t+1)^{3/2}}\right)\cos\left(\frac{t}{\sqrt{t+1}}\right)$

41. $2\pi\sin(\pi t - 2)\cos(\pi t - 2)$ **43.** $\frac{8\sin(2t)}{(1+\cos 2t)^5}$

45. $10t^{10}\tan^9 t\sec^2 t + 10t^9\tan^{10} t$

47. $\frac{-3t^6(t^2+4)}{(t^3-4t)^4}$ **49.** $-2\cos(\cos(2t-5))(\sin(2t-5))$

51. $\left(1 + \tan^4\left(\frac{t}{12}\right)\right)^2\left(\tan^3\left(\frac{t}{12}\right)\sec^2\left(\frac{t}{12}\right)\right)$

53. $-\frac{t\sin(t^2)}{\sqrt{1+\cos(t^2)}}$ **55.** $6\tan(\sin^3 t)\sec^2(\sin^3 t)\sin^2 t\cos t$

57. $3(2t^2-5)^3(18t^2-5)$ **59.** $\frac{6}{x^3}\left(1 + \frac{1}{x}\right)\left(1 + \frac{2}{x}\right)$

61. $2\csc^2(3x-1)\cot(3x-1)$ **63.** $16(2x+1)^2(5x+1)$

65. $f'(x) = 0$ for $x = 1, 4$; $f''(x) = 0$ for $x = 2, 4$

67. $5/2$ **69.** $-\pi/4$ **71.** 0 **73.** -5

75. **(a)** $2/3$ **(b)** $2\pi + 5$ **(c)** $15 - 8\pi$ **(d)** $37/6$ **(e)** -1 **(f)** $\sqrt{2}/24$ **(g)** $5/32$ **(h)** $-5/\left(3\sqrt{17}\right)$

77. 5 **79.** **(a)** 1 **(b)** 1 **81.** $y = 1 - 4x$

83. **(a)** $y = \pi x + 2 - \pi$ **(b)** $\pi/2$

85. It multiplies the velocity, acceleration, and jerk by 2, 4, and 8, respectively.

87. $v(6) = \frac{2}{5}\text{m/s}, a(6) = -\frac{4}{125}\text{m/s}^2$

SECTION 3.7, pp. 169–171

1. $\dfrac{-2xy - y^2}{x^2 + 2xy}$ **3.** $\dfrac{1 - 2y}{2x + 2y - 1}$

5. $\dfrac{-2x^3 + 3x^2y - xy^2 + x}{x^2y - x^3 + y}$ **7.** $\dfrac{1}{y(x+1)^2}$ **9.** $\cos y \cot y$

11. $\dfrac{-\cos^2(xy) - y}{x}$ **13.** $\dfrac{-y^2}{y\sin\left(\frac{1}{y}\right) - \cos\left(\frac{1}{y}\right) + xy}$

15. $-\dfrac{\sqrt{r}}{\sqrt{\theta}}$ **17.** $\dfrac{-r}{\theta}$ **19.** $y' = -\dfrac{x}{y}, y'' = \dfrac{-y^2 - x^2}{y^3}$

21. $\dfrac{dy}{dx} = \dfrac{x+1}{y}, \dfrac{d^2y}{dx^2} = \dfrac{x^2 + 2x}{y^3}$

23. $y' = \dfrac{\sqrt{y}}{\sqrt{y}+1}, y'' = \dfrac{1}{2\left(\sqrt{y}+1\right)^3}$

25. $y' = \dfrac{3x^2}{1 - \cos y}, y'' = \dfrac{6x(1 - \cos y)^2 - 9x^4 \sin y}{(1 - \cos y)^3}$

27. -2 **29.** $(-2, 1) : m = -1, (-2, -1) : m = 1$

31. **(a)** $y = \frac{7}{4}x - \frac{1}{2}$ **(b)** $y = -\frac{4}{7}x + \frac{29}{7}$

33. **(a)** $y = 3x + 6$ **(b)** $y = -\frac{1}{3}x + \frac{8}{3}$

35. **(a)** $y = \frac{6}{7}x + \frac{6}{7}$ **(b)** $y = -\frac{7}{6}x - \frac{7}{6}$

37. **(a)** $y = -\frac{\pi}{2}x + \pi$ **(b)** $y = \frac{2}{\pi}x - \frac{2}{\pi} + \frac{\pi}{2}$

39. **(a)** $y = 2\pi x - 2\pi$ **(b)** $y = -\frac{x}{2\pi} + \frac{1}{2\pi}$

41. Points: $\left(-\sqrt{7}, 0\right)$ and $\left(\sqrt{7}, 0\right)$, Slope: -2

43. $m = -1$ at $\left(\frac{\sqrt{3}}{4}, \frac{\sqrt{3}}{2}\right)$, $m = \sqrt{3}$ at $\left(\frac{\sqrt{3}}{4}, \frac{1}{2}\right)$

45. $(-3, 2) : m = -\frac{27}{8}; (-3, -2) : m = \frac{27}{8}; (3, 2) : m = \frac{27}{8}$; $(3, -2) : m = -\frac{27}{8}$

47. $(3, -1)$

53. $\dfrac{dy}{dx} = -\dfrac{y^3 + 2xy}{x^2 + 3xy^2}, \dfrac{dx}{dy} = -\dfrac{x^2 + 3xy^2}{y^3 + 2xy}, \dfrac{dx}{dy} = \dfrac{1}{dy/dx}$

SECTION 3.8, pp. 176–180

1. $\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$ **3.** 10 **5.** -6 **7.** $-3/2$

9. $31/13$ **11.** **(a)** $-180\ \text{m}^2/\text{min}$ **(b)** $-135\ \text{m}^3/\text{min}$

13. **(a)** $\dfrac{dV}{dt} = \pi r^2 \dfrac{dh}{dt}$ **(b)** $\dfrac{dV}{dt} = 2\pi hr \dfrac{dr}{dt}$

(c) $\dfrac{dV}{dt} = \pi r^2 \dfrac{dh}{dt} + 2\pi hr \dfrac{dr}{dt}$

15. **(a)** $1\ \text{volt/s}$ **(b)** $-\frac{1}{3}\ \text{amp/s}$

(c) $\dfrac{dR}{dt} = \dfrac{1}{I}\left(\dfrac{dV}{dt} - \dfrac{V}{I}\dfrac{dI}{dt}\right)$

(d) $3/2$ ohms/s, R is increasing.

17. **(a)** $\dfrac{ds}{dt} = \dfrac{x}{\sqrt{x^2 + y^2}}\dfrac{dx}{dt}$

(b) $\dfrac{ds}{dt} = \dfrac{x}{\sqrt{x^2 + y^2}}\dfrac{dx}{dt} + \dfrac{y}{\sqrt{x^2 + y^2}}\dfrac{dy}{dt}$

(c) $\dfrac{dx}{dt} = -\dfrac{y}{x}\dfrac{dy}{dt}$

19. **(a)** $\dfrac{dA}{dt} = \frac{1}{2}ab\cos\theta\dfrac{d\theta}{dt}$

(b) $\dfrac{dA}{dt} = \frac{1}{2}ab\cos\theta\dfrac{d\theta}{dt} + \frac{1}{2}b\sin\theta\dfrac{da}{dt}$

(c) $\dfrac{dA}{dt} = \frac{1}{2}ab\cos\theta\dfrac{d\theta}{dt} + \frac{1}{2}b\sin\theta\dfrac{da}{dt} + \frac{1}{2}a\sin\theta\dfrac{db}{dt}$

21. **(a)** $14\ \text{cm}^2/\text{s}$, increasing **(b)** $0\ \text{cm/s}$, constant **(c)** $-14/13\ \text{cm/s}$, decreasing

23. **(a)** $-3.6\ \text{m/s}$ **(b)** $-5.355\ \text{m}^2/\text{s}$ **(c)** $-1\ \text{rad/s}$

25. $6\ \text{m/s}$

27. **(a)** $\dfrac{dh}{dt} = 11.19\ \text{cm/min}$ **(b)** $\dfrac{dr}{dt} = 14.92\ \text{cm/min}$

29. **(a)** $\frac{-1}{24\pi}\ \text{m/min}$ **(b)** $r = \sqrt{26y - y^2}\ \text{m}$

(c) $\dfrac{dr}{dt} = -\dfrac{5}{288\pi}\ \text{m/min}$

31. $1\ \text{m/min}, 40\pi\ \text{m}^2/\text{min}$ **33.** $3.16\ \text{m/s}$

35. Increasing at $466/1681\ L/\text{min}^2$

37. $-5\ \text{m/s}$ **39.** $-441\ \text{m/s}$

41. $\frac{5}{72\pi}\ \text{cm/min}, \frac{10}{3}\text{cm}^2/\text{min}$

43. **(a)** $-10/\sqrt{13} \approx -2.774\ \text{m/s}$

(b) $d\theta_1/dt = 5/39\ \text{rad/s}, d\theta_2/dt = -5/39\ \text{rad/s}$

(c) $d\theta_1/dt = 1/6\ \text{rad/s}, d\theta_2/dt = -1/6\ \text{rad/s}$

45. $-5.5\ \text{deg/min}$ **47.** $12\pi\ \text{km/min}$

SECTION 3.9, pp. 189–191

1. $L(x) = 10x - 13$ **3.** $L(x) = 2$ **5.** $L(x) = x - \pi$

7. $2x$ **9.** $-x - 5$ **11.** $\frac{1}{12}x + \frac{4}{3}$

13. $f(0) = 1$. Also, $f'(x) = k(1 + x)^{k-1}$, so $f'(0) = k$. This means the linearization at $x = 0$ is $L(x) = 1 + kx$.

15. **(a)** 1.01 **(b)** 1.003

17. $\left(3x^2 - \dfrac{3}{2\sqrt{x}}\right)dx$ **19.** $\dfrac{2 - 2x^2}{(1 + x^2)^2}\,dx$

21. $\dfrac{1 - y}{3\sqrt{y} + x}\,dx$ **23.** $\dfrac{5}{2\sqrt{x}}\cos\left(5\sqrt{x}\right)dx$

25. $(4x^2)\sec^2\left(\dfrac{x^3}{3}\right)dx$

27. $\dfrac{3}{\sqrt{x}}\left(\csc\left(1 - 2\sqrt{x}\right)\cot\left(1 - 2\sqrt{x}\right)\right)dx$

29. **(a)** 0.41 **(b)** 0.4 **(c)** 0.01

31. **(a)** 0.231 **(b)** 0.2 **(c)** 0.031

33. **(a)** $-1/3$ **(b)** $-2/5$ **(c)** $1/15$

35. $dV = 4\pi r_0^2\, dr$ **37.** $dS = 12x_0\, dx$ **39.** $dV = 2\pi r_0 h\, dr$

41. **(a)** 0.08π m^2 **(b)** 2% **43.** $dV \approx 565.5$ cm^3

45. **(a)** 2% **(b)** 4% **47.** $\frac{1}{3}$% **49.** 3%

51. The ratio equals 37.52, so a change in the acceleration of gravity on the moon has about 38 times the effect that a change of the same magnitude has on Earth.

53. Increase $V \approx 40\%$

55. **(a)** **i)** $b_0 = f(a)$ **ii)** $b_1 = f'(a)$ **iii)** $b_2 = \dfrac{f''(a)}{2}$

(b) $Q(x) = 1 + x + x^2$ **(d)** $Q(x) = 1 - (x - 1) + (x - 1)^2$

(e) $Q(x) = 1 + \dfrac{x}{2} - \dfrac{x^2}{8}$

(f) The linearization of any differentiable function $u(x)$ at $x = a$ is $L(x) = u(a) + u'(a)(x - a) = b_0 + b_1(x - a)$, where b_0 and b_1 are the coefficients of the constant and linear terms of the quadratic approximation. Thus, the linearization for $f(x)$ at $x = 0$ is $1 + x$; the linearization for $g(x)$ at $x = 1$ is $1 - (x - 1)$ or $2 - x$; and the linearization for $h(x)$ at $x = 0$ is $1 + \dfrac{x}{2}$.

PRACTICE EXERCISES, pp. 192–197

1. $5x^4 - 0.25x + 0.25$ **3.** $3x(x - 2)$

5. $2(x + 1)(2x^2 + 4x + 1)$

7. $3(\theta^2 + \sec\theta + 1)^2(2\theta + \sec\theta\tan\theta)$

9. $\dfrac{1}{2\sqrt{t}\left(1 + \sqrt{t}\right)^2}$ **11.** $2\sec^2 x \tan x$

13. $8\cos^3(1 - 2t)\sin(1 - 2t)$ **15.** $5(\sec t)(\sec t + \tan t)^5$

17. $\dfrac{\theta\cos\theta + \sin\theta}{\sqrt{2\theta\sin\theta}}$ **19.** $\dfrac{\cos\sqrt{2\theta}}{\sqrt{2\theta}}$

21. $x\csc\left(\dfrac{2}{x}\right) + \csc\left(\dfrac{2}{x}\right)\cot\left(\dfrac{2}{x}\right)$

23. $\dfrac{1}{2}x^{1/2}\sec(2x)^2\left[16\tan(2x)^2 - x^{-2}\right]$

25. $-10x\csc^2(x^2)$ **27.** $8x^3\sin(2x^2)\cos(2x^2) + 2x\sin^2(2x^2)$

29. $\dfrac{-(t + 1)}{8t^3}$ **31.** $\dfrac{1 - x}{(x + 1)^3}$ **33.** $\dfrac{-1}{2x^2\left(1 + \dfrac{1}{x}\right)^{1/2}}$

35. $\dfrac{-2\sin\theta}{(\cos\theta - 1)^2}$ **37.** $3\sqrt{2x + 1}$ **39.** $-9\left[\dfrac{5x + \cos 2x}{(5x^2 + \sin 2x)^{5/2}}\right]$

41. $-\dfrac{y + 2}{x + 3}$ **43.** $\dfrac{-3x^2 - 4y + 2}{4x - 4y^{1/3}}$ **45.** $-\dfrac{y}{x}$

47. $\dfrac{1}{2y(x + 1)^2}$ **49.** $\dfrac{dp}{dq} = \dfrac{6q - 4p}{3p^2 + 4q}$

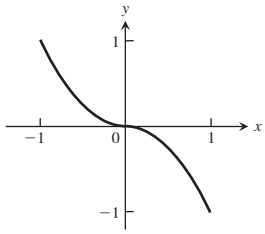
51. $\dfrac{dr}{ds} = (2r - 1)(\tan 2s)$

53. **(a)** $\dfrac{d^2y}{dx^2} = \dfrac{-2xy^3 - 2x^4}{y^5}$ **(b)** $\dfrac{d^2y}{dx^2} = \dfrac{-2xy^2 - 1}{x^4y^3}$

55. **(a)** 7 **(b)** -2 **(c)** $5/12$ **(d)** $1/4$ **(e)** 12 **(f)** $9/2$ **(g)** $3/4$

57. 0 **59.** $\sqrt{3}$ **61.** $-\dfrac{1}{2}$ **63.** $\dfrac{-2}{(2t + 1)^2}$

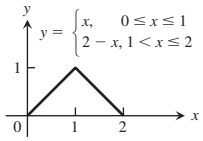
65. **(a)**



$$f(x) = \begin{cases} x^2, & -1 \le x < 0 \\ -x^2, & 0 \le x < 1 \end{cases}$$

(b) Yes **(c)** Yes

67. **(a)**



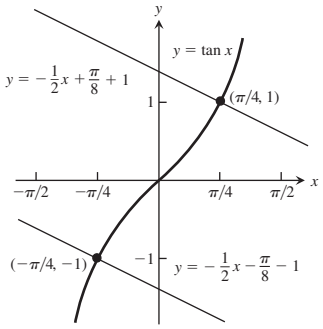
(b) Yes **(c)** No

69. $\left(\dfrac{5}{2}, \dfrac{9}{4}\right)$ and $\left(\dfrac{3}{2}, -\dfrac{1}{4}\right)$

71. $(-1, 27)$ and $(2, 0)$

73. **(a)** $(-2, 16)$, $(3, 11)$ **(b)** $(0, 20)$, $(1, 7)$

75.



77. $\dfrac{1}{4}$ **79.** 4

81. Tangent: $y = -\dfrac{1}{4}x + \dfrac{9}{4}$, normal: $y = 4x - 2$

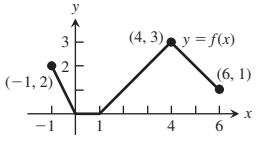
83. Tangent: $y = 2x - 4$, normal: $y = -\dfrac{1}{2}x + \dfrac{7}{2}$

85. Tangent: $y = -\dfrac{5}{4}x + 6$, normal: $y = \dfrac{4}{5}x - \dfrac{11}{5}$

87. $(1, 1)$: $m = -\dfrac{1}{2}$; $(1, -1)$: m not defined

89. B = graph of f, A = graph of f'

91.



93. **(a)** 0, 0 **(b)** 1700 rabbits, $\approx$1400 rabbits

95. -1 **97.** $1/2$ **99.** 4 **101.** 1

103. To make g continuous at the origin, define $g(0) = 1$.

105. **(a)** $\dfrac{dS}{dt} = (4\pi r + 2\pi h)\dfrac{dr}{dt}$

(b) $\dfrac{dS}{dt} = 2\pi r\dfrac{dh}{dt}$

(c) $\dfrac{dS}{dt} = (4\pi r + 2\pi h)\dfrac{dr}{dt} + 2\pi r\dfrac{dh}{dt}$

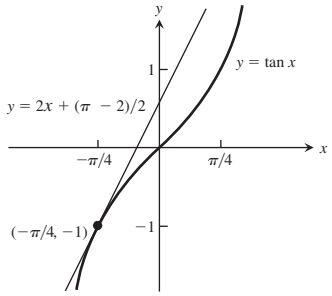
(d) $\dfrac{dr}{dt} = -\dfrac{r}{2r + h}\dfrac{dh}{dt}$

107. $-40\ \text{m}^2/\text{s}$ **109.** $0.02\ \text{ohm}/\text{s}$ **111.** $2\ \text{m}/\text{s}$

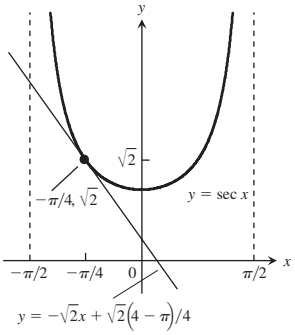
113. **(a)** $r = \frac{2}{5}h$ **(b)** $-\frac{5}{16\pi}\ \text{m}/\text{min}$

115. **(a)** $\frac{3}{5}\ \text{km}/\text{s}$ or $600\ \text{m}/\text{s}$ **(b)** $\frac{18}{\pi}\ \text{rpm}$

117. **(a)** $L(x) = 2x + \frac{\pi - 2}{2}$



(b) $L(x) = -\sqrt{2}x + \frac{\sqrt{2}(4 - \pi)}{4}$



119. $L(x) = 1.5x + 0.5$ **121.** $dS = \frac{\pi r h_0}{\sqrt{r^2 + h_0^2}}dh$

123. **(a)** 4% **(b)** 8% **(c)** 12%

ADDITIONAL AND ADVANCED EXERCISES, pp. 197–199

1. **(a)** $\sin 2\theta = 2\sin\theta\cos\theta$; $2\cos 2\theta = 2\sin\theta(-\sin\theta) + \cos\theta(2\cos\theta)$; $2\cos 2\theta = -2\sin^2\theta + 2\cos^2\theta$; $\cos 2\theta = \cos^2\theta - \sin^2\theta$

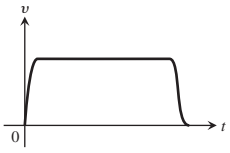
(b) $\cos 2\theta = \cos^2\theta - \sin^2\theta$; $-2\sin 2\theta = 2\cos\theta(-\sin\theta) - 2\sin\theta(\cos\theta)$; $\sin 2\theta = \cos\theta\sin\theta + \sin\theta\cos\theta$; $\sin 2\theta = 2\sin\theta\cos\theta$

3. **(a)** $a = 1, b = 0, c = -\frac{1}{2}$ **(b)** $b = \cos a, c = \sin a$

5. $h = -4, k = \frac{9}{2}, a = \frac{5\sqrt{5}}{2}$

7. **(a)** 0.09y **(b)** Increasing at 1% per year

9. Answers will vary. Here is one possibility.



11. **(a)** 2 s, 19.6 m/s **(b)** 12.25 s, 120 m

15. **(a)** $m = -\frac{b}{\pi}$ **(b)** $m = -1, b = \pi$

17. **(a)** $a = \frac{3}{4}, b = \frac{9}{4}$ **19.** f odd $\Rightarrow f'$ is even

23. h' is defined but not continuous at $x = 0$; k' is defined *and* continuous at $x = 0$.

25. $\frac{43}{75}\ \text{rad}/\text{s}$

29. **(a)** 0.248 m **(b)** 0.02015 s
(c) It will lose about 28.3 min/day.

Chapter 4

SECTION 4.1, pp. 206–208

1. Absolute minimum at $x = c_2$; absolute maximum at $x = b$

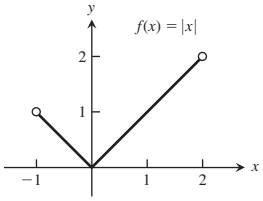
3. Absolute maximum at $x = c$; no absolute minimum

5. Absolute minimum at $x = a$; absolute maximum at $x = c$

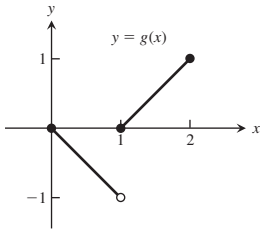
7. No absolute minimum; no absolute maximum

9. Absolute maximum at (0, 5) **11.** (c) **13.** (d)

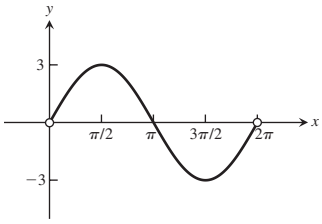
15. Absolute minimum at $x = 0$; no absolute maximum



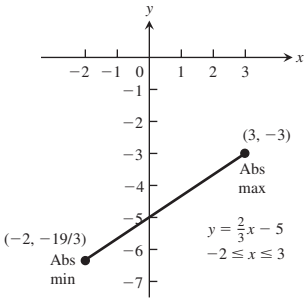
17. Absolute maximum at $x = 2$; no absolute minimum



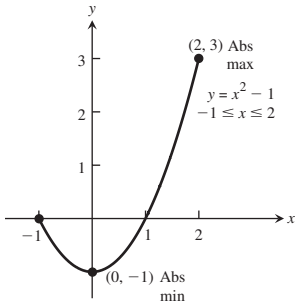
19. Absolute maximum at $x = \pi/2$; absolute minimum at $x = 3\pi/2$



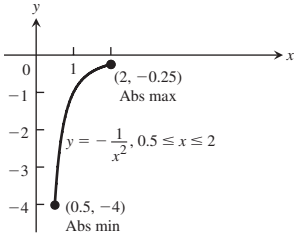
21. Absolute maximum: -3; absolute minimum: $-19/3$



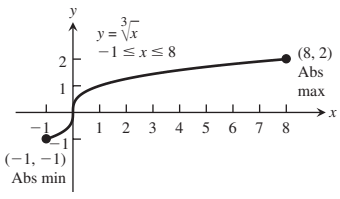
23. Absolute maximum: 3; absolute minimum: -1



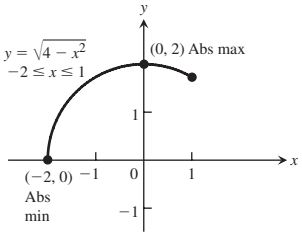
25. Absolute maximum: -0.25; absolute minimum: -4



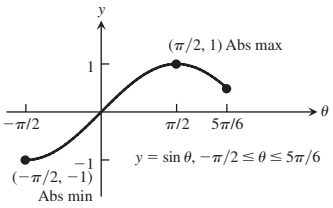
27. Absolute maximum: 2; absolute minimum: -1



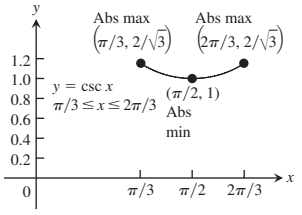
29. Absolute maximum: 2; absolute minimum: 0



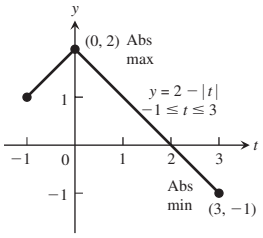
31. Absolute maximum: 1; absolute minimum: -1



33. Absolute maximum: $2/\sqrt{3}$; absolute minimum: 1



35. Absolute maximum: 2; absolute minimum: -1



37. Increasing on $(0, 8)$, decreasing on $(-1, 0)$; absolute maximum: 16 at $x = 8$; absolute minimum: 0 at $x = 0$

39. Increasing on $(-32, 1)$; absolute maximum: 1 at $\theta = 1$; absolute minimum: -8 at $\theta = -32$

41. $x = 3$

43. $x = 1, x = 4$

45. $x = 1$

47. $x = 0$ and $x = 4$

49. $x = 2$ and $x = -4$

51.

Critical point or endpoint	Derivative	Extremum	Value
$x = -\frac{4}{5}$	0	Local max	$\frac{12}{25}10^{1/3} \approx 1.034$
$x = 0$	Undefined	Local min	0

53.

Critical point or endpoint	Derivative	Extremum	Value
$x = -2$	Undefined	Local max	0
$x = -\sqrt{2}$	0	Minimum	-2
$x = \sqrt{2}$	0	Maximum	2
$x = 2$	Undefined	Local min	0

55.

Critical point or endpoint	Derivative	Extremum	Value
$x = 1$	Undefined	Minimum	2

57.

Critical point or endpoint	Derivative	Extremum	Value
$x = -1$	0	Maximum	5
$x = 1$	Undefined	Local min	1
$x = 3$	0	Maximum	5

59. **(a)** No

(b) The derivative is defined and nonzero for $x \neq 2$. Also, $f(2) = 0$ and $f(x) > 0$ for all $x \neq 2$.

(c) No, because $(-\infty, \infty)$ is not a closed interval.

(d) The answers are the same as parts (a) and (b), with 2 replaced by a.

61. y is increasing on $(-\infty, \infty)$ and so has no extrema.

63. Yes

65. g assumes a local maximum at $-c$.

67. **(a)** Maximum value is 144 at $x = 2$.

(b) The largest volume of the box is 144 cubic units, and it occurs when $x = 2$.

69. $\frac{v_0^2}{2g} + s_0$

71. Maximum value is 11 at $x = 5$; minimum value is 5 on the interval $[-3, 2]$; local maximum at $(-5, 9)$.

73. Maximum value is 5 on the interval $[3, \infty)$; minimum value is -5 on the interval $(-\infty, -2]$.

SECTION 4.2, pp. 213–215

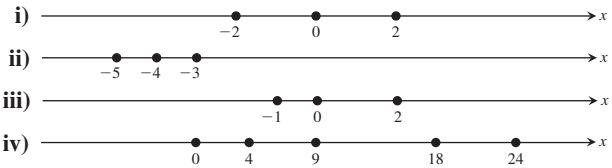
1. $1/2$ **3.** 1

5. $\frac{1}{3}\left(1 + \sqrt{7}\right) \approx 1.22, \frac{1}{3}\left(1 - \sqrt{7}\right) \approx -0.549$

7. Does not; f is not differentiable at the interior domain point $x = 0$.

9. Does **11.** Does not; f is not differentiable at $x = -1$.

15. **(a)**



27. Yes **29.** **(a)** 4 **(b)** 3 **(c)** 3

31. **(a)** $\frac{x^2}{2} + C$ **(b)** $\frac{x^3}{3} + C$ **(c)** $\frac{x^4}{4} + C$

33. **(a)** $\frac{1}{x} + C$ **(b)** $x + \frac{1}{x} + C$ **(c)** $5x - \frac{1}{x} + C$

35. **(a)** $-\frac{1}{2}\cos 2t + C$ **(b)** $2\sin\frac{t}{2} + C$

(c) $-\frac{1}{2}\cos 2t + 2\sin\frac{t}{2} + C$

37. $f(x) = x^2 - x$ **39.** $r(\theta) = 8\theta + \cot\theta - 2\pi - 1$

41. $s = 4.9t^2 + 5t + 10$ **43.** $s = \frac{1 - \cos(\pi t)}{\pi}$

45. $s = 16t^2 + 20t + 5$ **47.** $s = \sin(2t) - 3$

49. If $T(t)$ is the temperature of the thermometer at time t, then $T(0) = -19\,^\circ\text{C}$ and $T(14) = 100\,^\circ\text{C}$. From the Mean Value Theorem, there exists a $0 < t_0 < 14$ such that $\frac{T(14) - T(0)}{14 - 0} = 8.5\,^\circ\text{C/s} = T'(t_0)$, the rate at which the temperature was changing at $t = t_0$ as measured by the rising mercury on the thermometer.

51. Because its average speed was approximately 7.667 knots, and by the Mean Value Theorem, it must have been going that speed at least once during the trip.

55. The conclusion of the Mean Value Theorem yields

$$\frac{\frac{1}{b}-\frac{1}{a}}{b-a}=-\frac{1}{c^2}\Rightarrow c^2\left(\frac{a-b}{ab}\right)=a-b\Rightarrow c=\sqrt{ab}.$$

59. $f(x)$ must be zero at least once between a and b by the Intermediate Value Theorem. Now suppose that $f(x)$ is zero twice between a and b. Then, by the Mean Value Theorem, $f'(x)$ would have to be zero at least once between the two zeros of $f(x)$, but this can't be true since we are given that $f'(x)\neq 0$ on this interval. Therefore, $f(x)$ is zero once and only once between a and b.

69. $1.09999 \le f(0.1) \le 1.1$

SECTION 4.3, pp. 219–220

1. (a) 0, 1
(b) Increasing on $(-\infty, 0)$ and $(1, \infty)$; decreasing on $(0, 1)$
(c) Local maximum at $x = 0$; local minimum at $x = 1$

3. (a) -2, 1
(b) Increasing on $(-2, 1)$ and $(1, \infty)$; decreasing on $(-\infty, -2)$
(c) No local maximum; local minimum at $x = -2$

5. (a) -2, 1, 3
(b) Increasing on $(-2, 1)$ and $(3, \infty)$; decreasing on $(-\infty, -2)$ and $(1, 3)$
(c) Local maximum at $x = 1$; local minimum at $x = -2, 3$

7. (a) 0, 1
(b) Increasing on $(-\infty, -2)$ and $(1, \infty)$; decreasing on $(-2, 0)$ and $(0, 1)$
(c) Local minimum at $x = 1$

9. (a) -2, 2
(b) Increasing on $(-\infty, -2)$ and $(2, \infty)$; decreasing on $(-2, 0)$ and $(0, 2)$
(c) Local maximum at $x = -2$; local minimum at $x = 2$

11. (a) -2, 0
(b) Increasing on $(-\infty, -2)$ and $(0, \infty)$; decreasing on $(-2, 0)$
(c) Local maximum at $x = -2$; local minimum at $x = 0$

13. (a) $\frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}$
(b) Increasing on $\left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$; decreasing on $\left(0, \frac{\pi}{2}\right)$, $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$, and $\left(\frac{4\pi}{3}, 2\pi\right)$
(c) Local maximum at $x = 0$ and $x = \frac{4\pi}{3}$; local minimum at $x = \frac{2\pi}{3}$ and $x = 2\pi$

15. (a) Increasing on $(-2, 0)$ and $(2, 4)$; decreasing on $(-4, -2)$ and $(0, 2)$
(b) Absolute maximum at $(-4, 2)$; local maximum at $(0, 1)$ and $(4, -1)$; absolute minimum at $(2, -3)$; local minimum at $(-2, 0)$

17. (a) Increasing on $(-4, -1)$, $(1/2, 2)$, and $(2, 4)$; decreasing on $(-1, 1/2)$
(b) Absolute maximum at $(4, 3)$; local maximum at $(-1, 2)$ and $(2, 1)$; no absolute minimum; local minimum at $(-4, -1)$ and $(1/2, -1)$

19. (a) Increasing on $(-\infty, -1.5)$; decreasing on $(-1.5, \infty)$
(b) Local maximum: 5.25 at $t = -1.5$; absolute maximum: 5.25 at $t = -1.5$

21. (a) Decreasing on $(-\infty, 0)$; increasing on $(0, 4/3)$; decreasing on $(4/3, \infty)$
(b) Local minimum at $x = 0$ $(0, 0)$; local maximum at $x = 4/3$ $(4/3, 32/27)$; no absolute extrema

23. (a) Decreasing on $(-\infty, 0)$; increasing on $(0, 1/2)$; decreasing on $(1/2, \infty)$
(b) Local minimum at $\theta = 0$ $(0, 0)$; local maximum at $\theta = 1/2$ $(1/2, 1/4)$; no absolute extrema

25. (a) Increasing on $(-\infty, \infty)$; never decreasing
(b) No local extrema; no absolute extrema

27. (a) Increasing on $(-2, 0)$ and $(2, \infty)$; decreasing on $(-\infty, -2)$ and $(0, 2)$
(b) Local maximum: 16 at $x = 0$; local minimum: 0 at $x = \pm 2$; no absolute maximum; absolute minimum: 0 at $x = \pm 2$

29. (a) Increasing on $(-\infty, -1)$; decreasing on $(-1, 0)$; increasing on $(0, 1)$; decreasing on $(1, \infty)$
(b) Local maximum: 0.5 at $x = \pm 1$; local minimum: 0 at $x = 0$; absolute maximum: $1/2$ at $x = \pm 1$; no absolute minimum

31. (a) Increasing on $(10, \infty)$; decreasing on $(1, 10)$
(b) Local maximum: 1 at $x = 1$; local minimum: -8 at $x = 10$; absolute minimum: -8 at $x = 10$

33. (a) Decreasing on $\left(-2\sqrt{2}, -2\right)$; increasing on $(-2, 2)$; decreasing on $\left(2, 2\sqrt{2}\right)$
(b) Local minima: $g(-2) = -4$, $g\left(2\sqrt{2}\right) = 0$; local maxima: $g\left(-2\sqrt{2}\right) = 0$, $g(2) = 4$; absolute maximum: 4 at $x = 2$; absolute minimum: -4 at $x = -2$

35. (a) Increasing on $(-\infty, 1)$; decreasing when $1 < x < 2$, decreasing when $2 < x < 3$; discontinuous at $x = 2$; increasing on $(3, \infty)$
(b) Local minimum at $x = 3$ $(3, 6)$; local maximum at $x = 1$ $(1, 2)$; no absolute extrema

37. (a) Increasing on $(-2, 0)$ and $(0, \infty)$; decreasing on $(-\infty, -2)$
(b) Local minimum: $-6\sqrt[3]{2}$ at $x = -2$; no absolute maximum; absolute minimum: $-6\sqrt[3]{2}$ at $x = -2$

39. (a) Increasing on $\left(-\infty, -2/\sqrt{7}\right)$ and $\left(2/\sqrt{7}, \infty\right)$; decreasing on $\left(-2/\sqrt{7}, 0\right)$ and $\left(0, 2/\sqrt{7}\right)$
(b) Local maximum: $24\sqrt[3]{2}/7^{7/6} \approx 3.12$ at $x = -2/\sqrt{7}$; local minimum: $-24\sqrt[3]{2}/7^{7/6} \approx -3.12$ at $x = 2/\sqrt{7}$; no absolute extrema

41. (a) Local maximum: 1 at $x = 1$; local minimum: 0 at $x = 2$
(b) Absolute maximum: 1 at $x = 1$; no absolute minimum

43. (a) Local maximum: 1 at $x = 1$; local minimum: 0 at $x = 2$
(b) No absolute maximum; absolute minimum: 0 at $x = 2$

45. (a) Local maxima: -9 at $t = -3$ and 16 at $t = 2$; local minimum: -16 at $t = -2$
(b) Absolute maximum: 16 at $t = 2$; no absolute minimum

47. (a) Local minimum: 0 at $x = 0$
(b) No absolute maximum; absolute minimum: 0 at $x = 0$

49. (a) Local maximum: 5 at $x = 0$; local minimum: 0 at $x = -5$ and $x = 5$
(b) Absolute maximum: 5 at $x = 0$; absolute minimum: 0 at $x = -5$ and $x = 5$

51. (a) Local maximum: 2 at $x = 0$;
local minimum: $\dfrac{\sqrt{3}}{4\sqrt{3}-6}$ at $x = 2-\sqrt{3}$
(b) No absolute maximum; an absolute minimum at $x = 2-\sqrt{3}$

53. (a) Local maximum: 1 at $x = \pi/4$;
local maximum: 0 at $x = \pi$;
local minimum: 0 at $x = 0$;
local minimum: -1 at $x = 3\pi/4$

55. Local maximum: 2 at $x = \pi/6$;
local maximum: $\sqrt{3}$ at $x = 2\pi$;
local minimum: -2 at $x = 7\pi/6$;
local minimum: $\sqrt{3}$ at $x = 0$

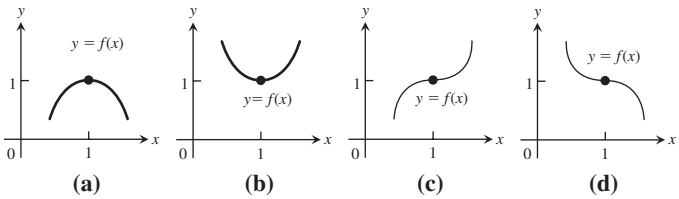
57. (a) Local minimum: $(\pi/3) - \sqrt{3}$ at $x = 2\pi/3$;
local maximum: 0 at $x = 0$;
local maximum: π at $x = 2\pi$

59. (a) Local minimum: 0 at $x = \pi/4$

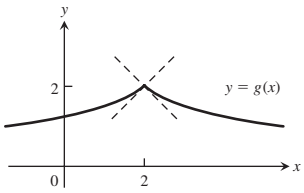
61. Local minimum at $x = 1$; no local maximum

63. Local maximum: 3 at $\theta = 0$;
local minimum: -3 at $\theta = 2\pi$

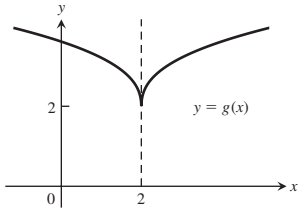
65.



67. (a)



(b)



71. $a = -2, b = 4$

SECTION 4.4, pp. 228–232

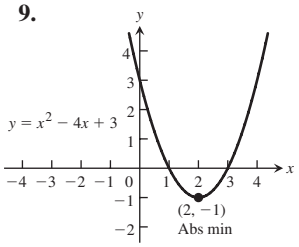
1. Local maximum: $3/2$ at $x = -1$; local minimum: -3 at $x = 2$; point of inflection at $(1/2, -3/4)$; rising on $(-\infty, -1)$ and $(2, \infty)$; falling on $(-1, 2)$; concave up on $(1/2, \infty)$; concave down on $(-\infty, 1/2)$

3. Local maximum: $3/4$ at $x = 0$; local minimum: 0 at $x = \pm 1$; points of inflection at $\left(-\sqrt{3}, \dfrac{3\sqrt[3]{4}}{4}\right)$ and $\left(\sqrt{3}, \dfrac{3\sqrt[3]{4}}{4}\right)$; rising on $(-1, 0)$ and $(1, \infty)$; falling on $(-\infty, -1)$ and $(0, 1)$; concave up on $\left(-\infty, -\sqrt{3}\right)$ and $\left(\sqrt{3}, \infty\right)$; concave down on $\left(-\sqrt{3}, \sqrt{3}\right)$

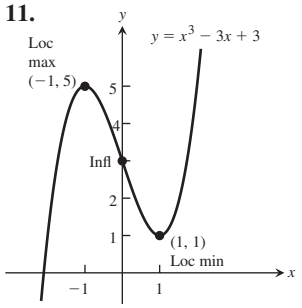
5. Local maxima: $\dfrac{-2\pi}{3} + \dfrac{\sqrt{3}}{2}$ at $x = -2\pi/3$, $\dfrac{\pi}{3} + \dfrac{\sqrt{3}}{2}$ at $x = \pi/3$; local minima: $-\dfrac{\pi}{3} - \dfrac{\sqrt{3}}{2}$ at $x = -\pi/3$, $\dfrac{2\pi}{3} - \dfrac{\sqrt{3}}{2}$ at $x = 2\pi/3$; points of inflection at $(-\pi/2, -\pi/2)$, $(0, 0)$, and $(\pi/2, \pi/2)$; rising on $(-\pi/3, \pi/3)$; falling on $(-2\pi/3, -\pi/3)$ and $(\pi/3, 2\pi/3)$; concave up on $(-\pi/2, 0)$ and $(\pi/2, 2\pi/3)$; concave down on $(-2\pi/3, -\pi/2)$ and $(0, \pi/2)$

7. Local maxima: 1 at $x = -\pi/2$ and $x = \pi/2$, 0 at $x = -2\pi$ and $x = 2\pi$; local minima: -1 at $x = -3\pi/2$ and $x = 3\pi/2$, 0 at $x = 0$; points of inflection at $(-\pi, 0)$ and $(\pi, 0)$; rising on $(-3\pi/2, -\pi/2)$, $(0, \pi/2)$, and $(3\pi/2, 2\pi)$; falling on $(-2\pi, -3\pi/2)$, $(-\pi/2, 0)$, and $(\pi/2, 3\pi/2)$; concave up on $(-2\pi, -\pi)$ and $(\pi, 2\pi)$; concave down on $(-\pi, 0)$ and $(0, \pi)$

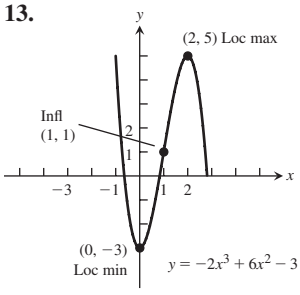
9.



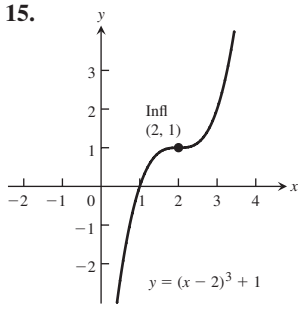
11.



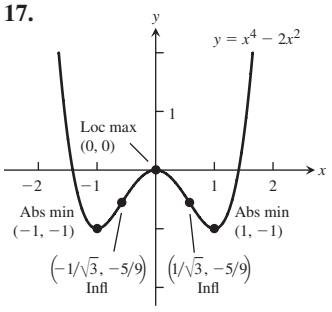
13.



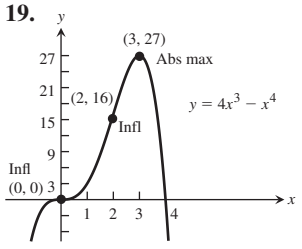
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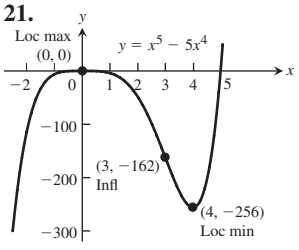
17.



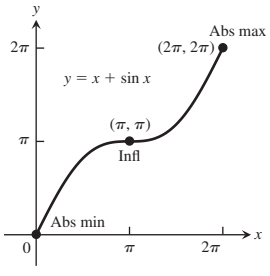
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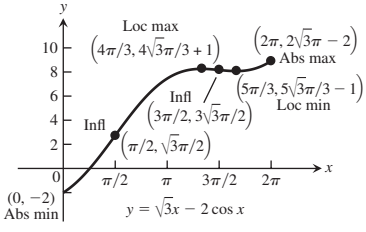
21.



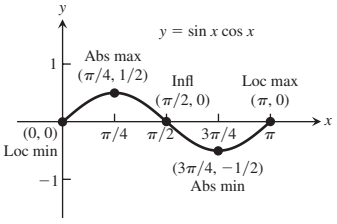
23.



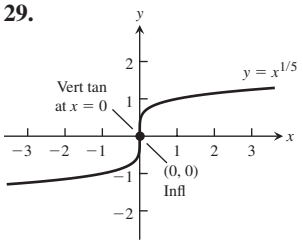
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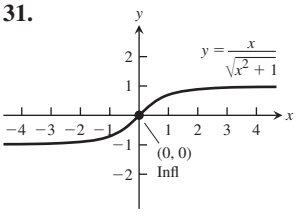
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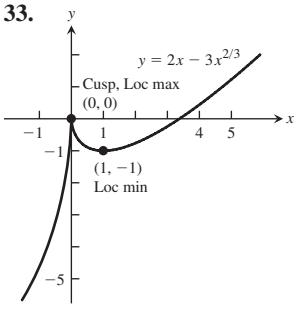
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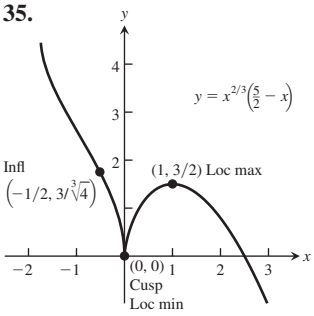
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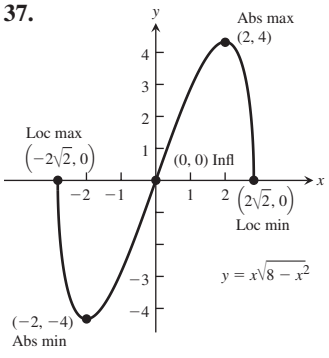
33.



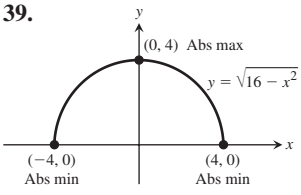
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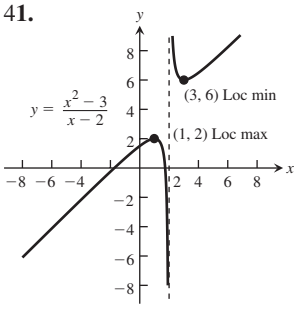
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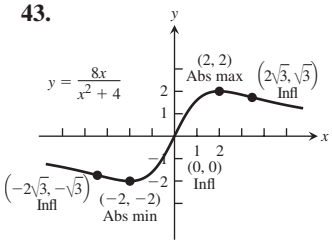
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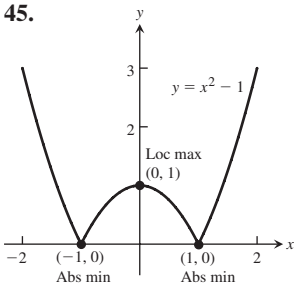
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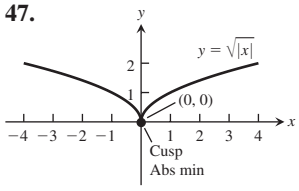
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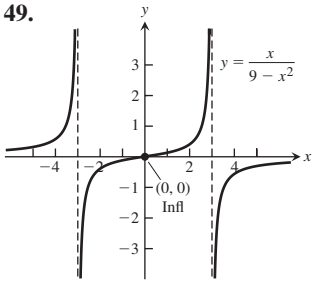
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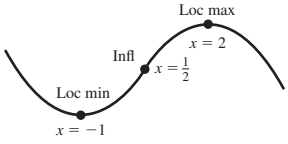
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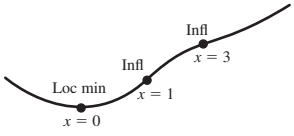
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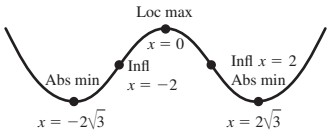
51. $y'' = 1 - 2x$



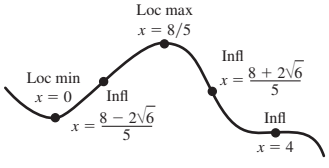
53. $y'' = 3(x - 3)(x - 1)$



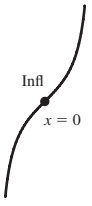
55. $y'' = 3(x - 2)(x + 2)$



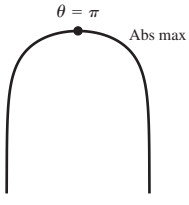
57. $y'' = 4(4 - x)(5x^2 - 16x + 8)$



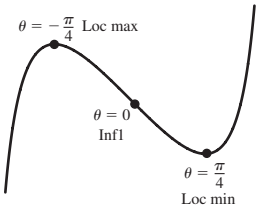
59. $y'' = 2 \sec^2 x \tan x$



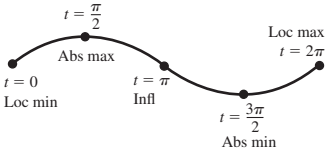
61. $y'' = -\frac{1}{2}\csc^2\frac{\theta}{2}$, $0 < \theta < 2\pi$



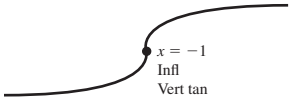
63. $y'' = 2\tan\theta\sec^2\theta,\ -\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$



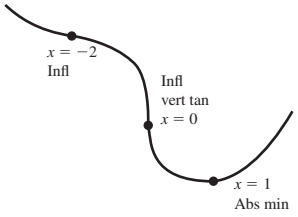
65. $y'' = -\sin t,\ 0 \le t \le 2\pi$



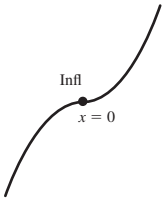
67. $y'' = -\dfrac{2}{3}(x+1)^{-5/3}$



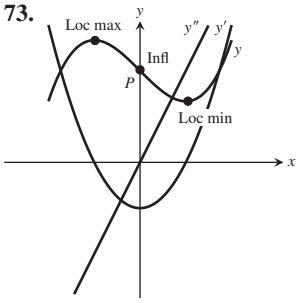
69. $y'' = \dfrac{1}{3}x^{-2/3} + \dfrac{2}{3}x^{-5/3}$



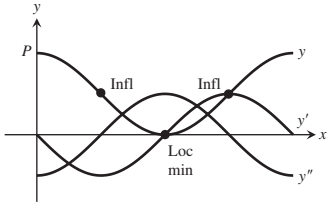
71. $y'' = \begin{cases} -2, & x < 0 \\ 2, & x > 0 \end{cases}$



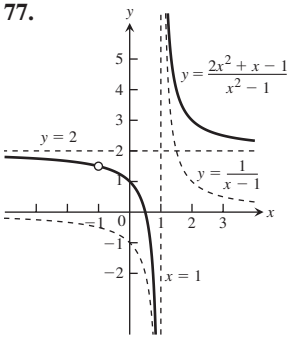
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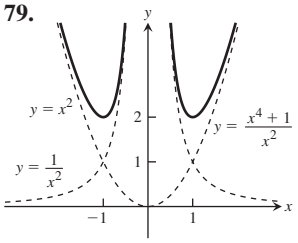
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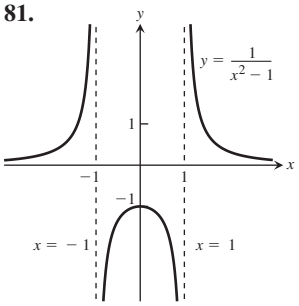
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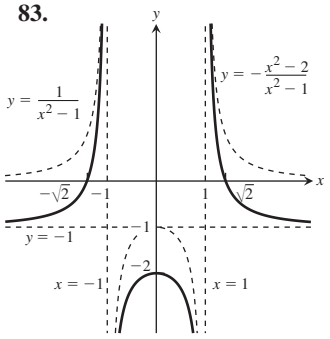
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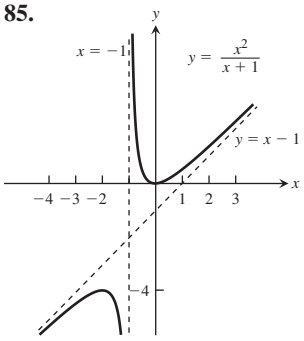
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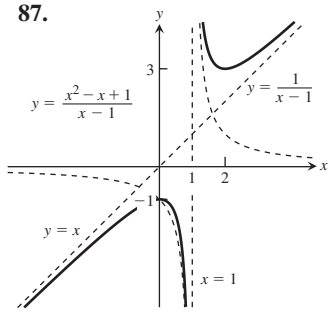
83.



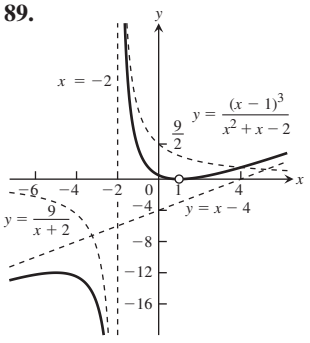
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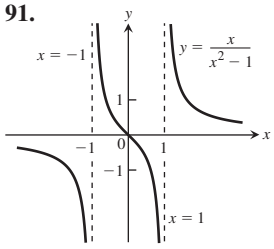
87.



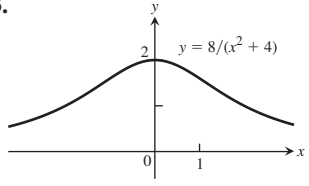
89.



91.



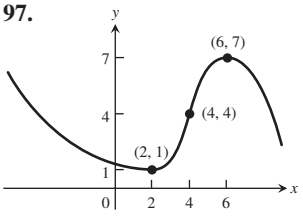
93.



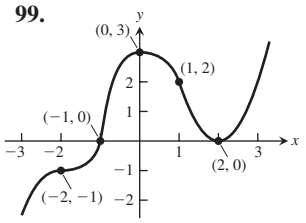
95.

Point	y'	y''
P	$-$	$+$
Q	$+$	0
R	$+$	$-$
S	0	$-$
T	$-$	$-$

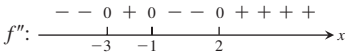
97.



99.

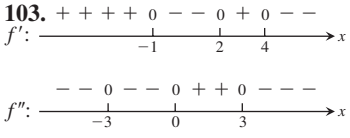


101.



There are points of inflection at $x = -3$, $x = -1$, and $x = 2$.

103.



There are local maxima at $x = -1$ and $x = 4$. There is a local minimum at $x = 2$. There are points of inflection at $x = 0$ and $x = 3$.

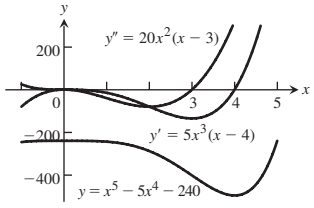
105. **(a)** Towards origin: $0 \le t < 2$ and $6 \le t \le 10$; away from origin: $2 \le t \le 6$ and $10 \le t \le 15$
(b) $t = 2, t = 6, t = 10$
(c) $t = 5, t = 7, t = 13$
(d) Positive: $5 \le t \le 7$, $13 \le t \le 15$; negative: $0 \le t \le 5$, $7 \le t \le 13$

107. ≈60 thousand units

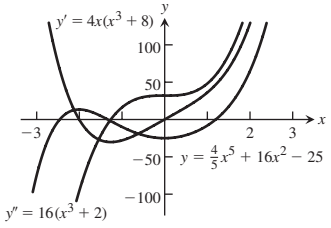
109. Local minimum at $x = 2$; inflection points at $x = 1$ and $x = 5/3$

111. $-1, 2$ **113.** $b = -3$ **119.** $a = 1, b = 3, c = 9$

121. The zeros of $y' = 0$ and $y'' = 0$ are extrema and points of inflection, respectively. Inflection at $x = 3$, local maximum at $x = 0$, local minimum at $x = 4$.



123. The zeros of $y' = 0$ and $y'' = 0$ are extrema and points of inflection, respectively. Inflection at $x = -\sqrt[3]{2}$; local maximum at $x = -2$; local minimum at $x = 0$.



SECTION 4.5, pp. 238–244

1. 16 cm, 4 cm by 4 cm

3. **(a)** $(x, 1 - x)$ **(b)** $A(x) = 2x(1 - x)$
(c) $\frac{1}{2}$ square units, 1 by $\frac{1}{2}$

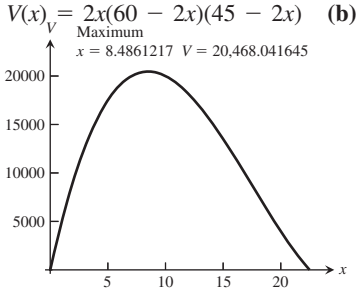
5. $14 \times 35 \times 5$ cm, 2450 cm^3

7. 80,000 m^2; 400 m by 200 m

9. **(a)** The optimum dimensions of the tank are 2 m on the base edges and 1 m deep.
(b) Minimizing the surface area of the tank minimizes its weight for a given wall thickness. The thickness of the steel walls would likely be determined by other considerations such as structural requirements.

11. 45×22.5 cm **13.** $\frac{\pi}{2}$ **15.** $h : r = 8 : \pi$

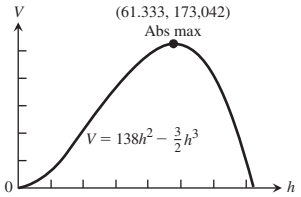
17. **(a)** $V(x) = 2x(60 - 2x)(45 - 2x)$ **(b)** Domain: (0, 22.5)



(c) Maximum volume ≈ 20,468 cm^3 when $x \approx 8.49$ cm
(d) $V'(x) = 24x^2 - 840x + 5400$, so the critical point is at $x = \dfrac{35 - 5\sqrt{13}}{2}$ which confirms the result in part (c).
(e) $x = 12.5$ cm or $x = 5$ cm

19. ≈2418.40 cm^3

21. **(a)** $h = 184/3$, $w = 46$
(b)



23. If r is the radius of the hemisphere, h the height of the cylinder, and V the volume, then $r = \left(\dfrac{3V}{8\pi}\right)^{1/3}$ and $h = \left(\dfrac{3V}{\pi}\right)^{1/3}$.

25. **(b)** $x = 16.2$ **(c)** $L \approx 28$ cm

27. Radius $= \sqrt{2}$ m, height $= 1$ m, volume $= \dfrac{2\pi}{3}$ m^3

29. 1 **31.** $\dfrac{9b}{9 + \sqrt{3\pi}}$ m, triangle; $\dfrac{b\sqrt{3\pi}}{9 + \sqrt{3\pi}}$ m, circle

33. $\dfrac{3}{2} \times 2$ **35.** **(a)** 16 **(b)** -1

37. $r = \dfrac{2\sqrt{2}}{3}$ $h = \dfrac{4}{3}$ **39.** Area 8 when $a = 2$

41. **(a)** $v(0) = 29.4$ m/s **(b)** 78.4 m at $t = 3$ s
(c) Velocity when $s = 0$ is $v(7) = -39.2$ m/s.

43. ≈ 9.58 m **45.** 4 m

47. **(a)** $10\sqrt{3} \times 10\sqrt{6}$ cm

49. **(a)** $10\pi \approx 31.42$ cm/s; when $t = 0.5$ s, 1.5 s, 2.5 s, 3.5 s; $s = 0$, acceleration is 0.
(b) 10 cm from rest position; speed is 0.

51. **(a)** $s = ((12 - 12t)^2 + 64t^2)^{1/2}$
(b) -12 knots, 8 knots
(c) No
(d) $4\sqrt{13}$. This limit is the square root of the sums of the squares of the individual speeds.

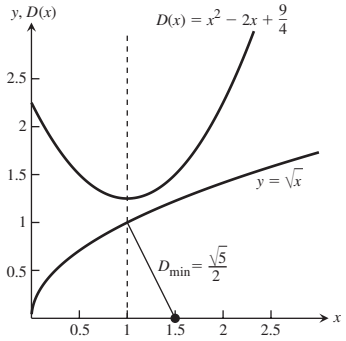
53. $x = \dfrac{a}{2}$, $v = \dfrac{ka^2}{4}$ **55.** $\dfrac{c}{2} + 50$

57. **(a)** $\sqrt{\dfrac{2km}{h}}$ **(b)** $\sqrt{\dfrac{2km}{h}}$ **61.** $2 \times 2 \times \dfrac{3}{2}$ m, \$720

63. $M = \dfrac{C}{2}$ **69.** **(a)** $y = -1$

71. **(a)** The minimum distance is $\frac{\sqrt{5}}{2}$.

(b) The minimum distance is from the point $(3/2, 0)$ to the point $(1, 1)$ on the graph of $y = \sqrt{x}$, and this occurs at the value $x = 1$, where $D(x)$, the distance squared, has its minimum value.

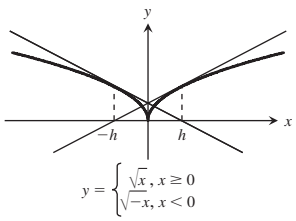


SECTION 4.6, pp. 247–249

1. $x_2 = -\frac{5}{3}, \frac{13}{21}$ **3.** $x_2 = -\frac{51}{31}, \frac{5763}{4945}$ **5.** $x_2 = \frac{2387}{2000}$

7. $x_2 = \frac{17}{14}$ **9.** x_1, and all later approximations will equal x_0.

11.



13. The points of intersection of $y = x^3$ and $y = 3x + 1$ or $y = x^3 - 3x$ and $y = 1$ have the same x-values as the roots of part (i) or the solutions of part (iv). **15.** 1.165561185

17. **(a)** Two **(b)** 0.35003501505249 and −1.0261731615301

19. ± 1.3065629648764, ± 0.5411961001462 **21.** $x \approx 0.45$

23. 0.8192 **25.** The root is 1.17951.

27. **(a)** For $x_0 = -2$ or $x_0 = -0.8$, $x_i \to -1$ as i gets large.

(b) For $x_0 = -0.5$ or $x_0 = 0.25$, $x_i \to 0$ as i gets large.

(c) For $x_0 = 0.8$ or $x_0 = 2$, $x_i \to 1$ as i gets large.

(d) For $x_0 = -\sqrt{21}/7$ or $x_0 = \sqrt{21}/7$, Newton's method does not converge. The values of x_i alternate between $-\sqrt{21}/7$ and $\sqrt{21}/7$ as i increases.

29. Answers will vary with machine speed.

SECTION 4.7, pp. 255–259

1. **(a)** x^2 **(b)** $\frac{x^3}{3}$ **(c)** $\frac{x^3}{3} - x^2 + x$

3. **(a)** x^{-3} **(b)** $-\frac{1}{3}x^{-3}$ **(c)** $-\frac{1}{3}x^{-3} + x^2 + 3x$

5. **(a)** $-\frac{1}{x}$ **(b)** $-\frac{5}{x}$ **(c)** $2x + \frac{5}{x}$

7. **(a)** $\sqrt{x^3}$ **(b)** $\sqrt{x}$ **(c)** $\frac{2\sqrt{x^3}}{3} + 2\sqrt{x}$

9. **(a)** $x^{2/3}$ **(b)** $x^{1/3}$ **(c)** $x^{-1/3}$

11. **(a)** $\cos(\pi x)$ **(b)** $-3\cos x$ **(c)** $-\frac{1}{\pi}\cos(\pi x) + \cos(3x)$

13. **(a)** $\frac{1}{2}\tan x$ **(b)** $2\tan\left(\frac{x}{3}\right)$ **(c)** $-\frac{2}{3}\tan\left(\frac{3x}{2}\right)$

15. **(a)** $-\csc x$ **(b)** $\frac{1}{5}\csc(5x)$ **(c)** $2\csc\left(\frac{\pi x}{2}\right)$

17. $\frac{x^2}{2} + x + C$ **19.** $t^3 + \frac{t^2}{4} + C$ **21.** $\frac{x^4}{2} - \frac{5x^2}{2} + 7x + C$

23. $-\frac{1}{x} - \frac{x^3}{3} - \frac{x}{3} + C$ **25.** $\frac{3}{2}x^{2/3} + C$

27. $\frac{2}{3}x^{3/2} + \frac{3}{4}x^{4/3} + C$ **29.** $4y^2 - \frac{8}{3}y^{3/4} + C$

31. $x^2 + \frac{2}{x} + C$ **33.** $2\sqrt{t} - \frac{2}{\sqrt{t}} + C$ **35.** $-2\sin t + C$

37. $-21\cos\frac{\theta}{3} + C$ **39.** $3\cot x + C$ **41.** $-\frac{1}{2}\csc\theta + C$

43. $4\sec x - 2\tan x + C$ **45.** $-\frac{1}{2}\cos 2x + \cot x + C$

47. $\frac{t}{2} + \frac{\sin 4t}{8} + C$ **49.** $\frac{3x^{(\sqrt{3}+1)}}{\sqrt{3}+1} + C$

51. $\tan\theta + C$ **53.** $-\cot x - x + C$ **55.** $-\cos\theta + \theta + C$

63. **(a)** Wrong: $\frac{d}{dx}\left(\frac{x^2}{2}\sin x + C\right) = \frac{2x}{2}\sin x + \frac{x^2}{2}\cos x = x\sin x + \frac{x^2}{2}\cos x$

(b) Wrong: $\frac{d}{dx}(-x\cos x + C) = -\cos x + x\sin x$

(c) Right: $\frac{d}{dx}(-x\cos x + \sin x + C) = -\cos x + x\sin x + \cos x = x\sin x$

65. **(a)** Wrong: $\frac{d}{dx}\left(\frac{(2x+1)^3}{3} + C\right) = \frac{3(2x+1)^2(2)}{3} = 2(2x+1)^2$

(b) Wrong: $\frac{d}{dx}((2x+1)^3 + C) = 3(2x+1)^2(2) = 6(2x+1)^2$

(c) Right: $\frac{d}{dx}((2x+1)^3 + C) = 6(2x+1)^2$

67. Right **69.** **(b)** **71.** $y = x^2 - 7x + 10$

73. $y = -\frac{1}{x} + \frac{x^2}{2} - \frac{1}{2}$ **75.** $y = 9x^{1/3} + 4$

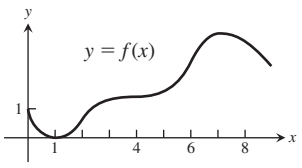
77. $s = t + \sin t + 4$ **79.** $r = \cos(\pi\theta) - 1$

81. $v = \frac{1}{2}\sec t + \frac{1}{2}$ **83.** $y = x^2 - x^3 + 4x + 1$

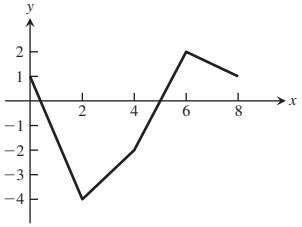
85. $r = \frac{1}{t} + 2t - 2$ **87.** $y = x^3 - 4x^2 + 5$

89. $y = -\sin t + \cos t + t^3 - 1$ **91.** $y = 2x^{3/2} - 50$

93.



95.



97. $y = x - x^{4/3} + \frac{1}{2}$ **99.** $y = -\sin x - \cos x - 2$

101. **(a)** **(i)** 33.2 units, **(ii)** 33.2 units, **(iii)** 33.2 units
(b) True

103. $t = 30/k, k = 6$

105. **(a)** $v = 10t^{3/2} - 6t^{1/2}$ **(b)** $s = 4t^{5/2} - 4t^{3/2}$

109. **(a)** $-\sqrt{x} + C$ **(b)** $x + C$ **(c)** $\sqrt{x} + C$
(d) $-x + C$ **(e)** $x - \sqrt{x} + C$ **(f)** $-x - \sqrt{x} + C$

PRACTICE EXERCISES, pp. 259–262

1. Minimum value is 1 at $x = 2$.

3. Local maximum at $(-2, 17)$; local minimum at $\left(\frac{4}{3}, -\frac{41}{27}\right)$

5. Minimum value is 0 at $x = -1$ and $x = 1$.

7. There is a local minimum at $(0, 1)$.

9. Maximum value is $\frac{1}{2}$ at $x = 1$; minimum value is $-\frac{1}{2}$ at $x = -1$.

11. No **13.** No minimum; absolute maximum: $f(1) = 16$; critical points: $x = 1$ and $11/3$

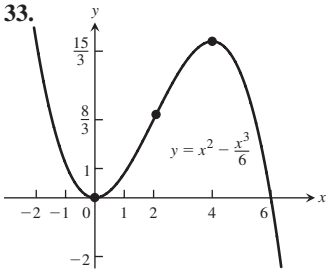
15. Yes, except at $x = 0$ **17.** No **21.** **(b)** one

23. **(b)** 0.8555 99677 2

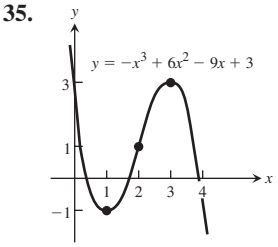
29. Global minimum value of $\frac{1}{2}$ at $x = 2$

31. **(a)** $t = 0, 6, 12$ **(b)** $t = 3, 9$ **(c)** $6 < t < 12$
(d) $0 < t < 6, 12 < t < 14$

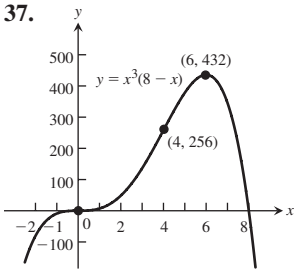
33.



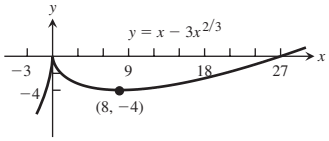
35.



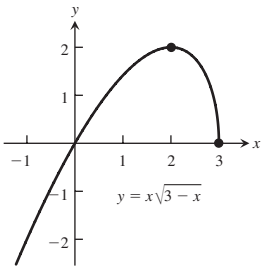
37.



39.

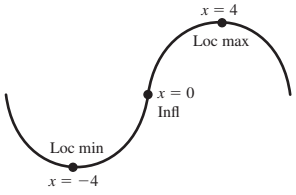


41.

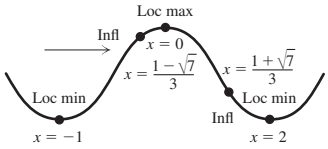


43. **(a)** Local maximum at $x = 4$, local minimum at $x = -4$, inflection point at $x = 0$

(b)

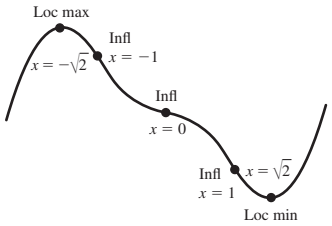


45. **(a)** Local maximum at $x = 0$, local minima at $x = -1$ and $x = 2$, inflection points at $x = \left(1 \pm \sqrt{7}\right)/3$

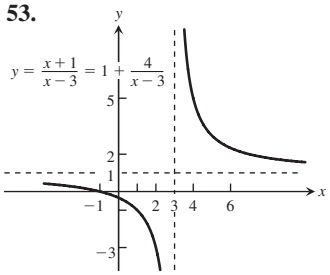


47. **(a)** Local maximum at $x = -\sqrt{2}$, local minimum at $x = \sqrt{2}$, inflection points at $x = \pm 1$ and 0

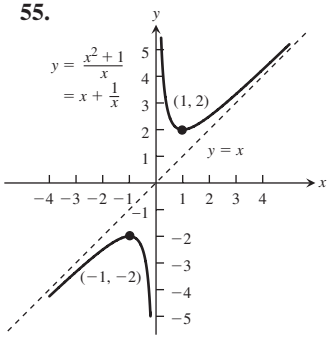
(b)



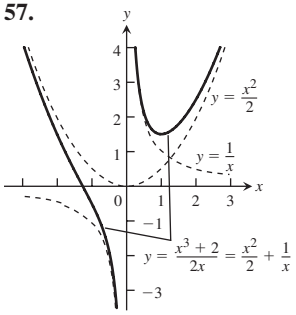
53.



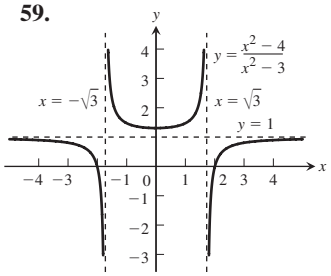
55.



57.



59.



61. **(a)** 0, 36 **(b)** 18, 18 **63.** 54 square units

65. height $= 2$, radius $= \sqrt{2}$

67. $x = 5 - \sqrt{5}$ hundred $\approx$ 276 tires,
$y = 2\left(5 - \sqrt{5}\right)$ hundred $\approx$ 553 tires

69. Dimensions: base is 15 cm by 30 cm, height $= 5$ cm; maximum volume $= 2250\ \text{cm}^3$

71. $x_5 = 2.1958\ 23345$ **73.** $\frac{x^4}{4} + \frac{5}{2}x^2 - 7x + C$

75. $2t^{3/2} - \frac{4}{t} + C$ **77.** $-\frac{1}{r + 5} + C$

79. $(\theta^2 + 1)^{3/2} + C$ **81.** $\frac{1}{3}(1 + x^4)^{3/4} + C$

83. $10 \tan \frac{s}{10} + C$ **85.** $-\frac{1}{\sqrt{2}} \csc \sqrt{2}\theta + C$

87. $\frac{1}{2}x - \sin \frac{x}{2} + C$ **89.** $y = x - \frac{1}{x} - 1$

91. $r = 4t^{5/2} + 4t^{3/2} - 8t$

ADDITIONAL AND ADVANCED EXERCISES, pp. 262–265

1. The function is constant on the interval.

3. The extreme points will not be at the end of an open interval.

5. (a) A local minimum at $x = -1$, points of inflection at $x = 0$ and $x = 2$

(b) A local maximum at $x = 0$ and local minima at $x = -1$ and $x = 2$, points of inflection at $x = \dfrac{1 \pm \sqrt{7}}{3}$

9. No **11.** $a = 1, b = 0, c = 1$ **13.** Yes

15. Drill the hole at $y = h/2$.

17. $r = \dfrac{RH}{2(H - R)}$ for $H > 2R$, $r = R$ if $H \le 2R$

19. $\dfrac{12}{5}$ and 5

21. (a) $\dfrac{c - b}{2e}$ **(b)** $\dfrac{c + b}{2}$ **(c)** $\dfrac{b^2 - 2bc + c^2 + 4ae}{4e}$

(d) $\dfrac{c + b + t}{2}$

23. $m_0 = 1 - \dfrac{1}{q}, m_1 = \dfrac{1}{q}$

25. (a) $k = 15$ **(b)** 7.5 m

27. Yes, $y = x + C$ **29.** $v_0 = \dfrac{2\sqrt{2}}{3} b^{3/4}$ **33.** 3

Chapter 5

SECTION 5.1, pp. 274–276

1. (a) 0.125 **(b)** 0.21875 **(c)** 0.625 **(d)** 0.46875

3. (a) 1.066667 **(b)** 1.283333 **(c)** 2.666667 **(d)** 2.083333

5. 0.3125, 0.328125 **7.** 1.5, 1.574603

9. (a) 245 cm **(b)** 245 cm **11. (a)** 1180 m **(b)** 1300 m

13. (a) 22.862 m/s **(b)** 13.867 m/s **(c)** 44.893 m

15. $\dfrac{31}{16}$ **17.** 1

19. (a) Upper = 758 L, lower = 543 L

(b) Upper = 2363 L, lower = 1693 L

(c) $\approx$ 31.4 h, $\approx$ 32.4 h

21. (a) 2 **(b)** $2\sqrt{2} \approx 2.828$ **(c)** $8 \sin\left(\dfrac{\pi}{8}\right) \approx 3.061$

(d) Each area is less than the area of the circle, π. As n increases, the polygon area approaches π.

SECTION 5.2, pp. 282–283

1. $\dfrac{6(1)}{1 + 1} + \dfrac{6(2)}{2 + 1} = 7$

3. $\cos(1)\pi + \cos(2)\pi + \cos(3)\pi + \cos(4)\pi = 0$

5. $\sin \pi - \sin \dfrac{\pi}{2} + \sin \dfrac{\pi}{3} = \dfrac{\sqrt{3} - 2}{2}$

7. All of them **9.** b

11. $\displaystyle\sum_{k=1}^{6} k$ **13.** $\displaystyle\sum_{k=1}^{4} \frac{1}{2^k}$ **15.** $\displaystyle\sum_{k=1}^{5} (-1)^{k+1} \frac{1}{k}$

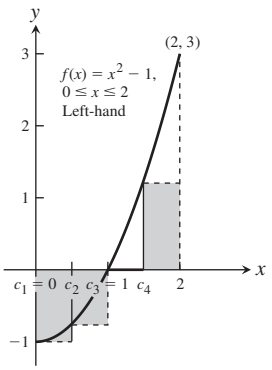
17. (a) -15 **(b)** 1 **(c)** 1 **(d)** -11 **(e)** 16

19. (a) 55 **(b)** 385 **(c)** 3025 **21.** -56 **23.** -73

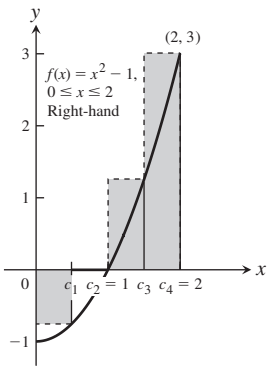
25. 240 **27.** 3376 **29. (a)** 21 **(b)** 3500 **(c)** 2620

31. (a) $4n$ **(b)** cn **(c)** $(n^2 - n)/2$ **33.** 2600 **35.** $-2\sqrt{3}$

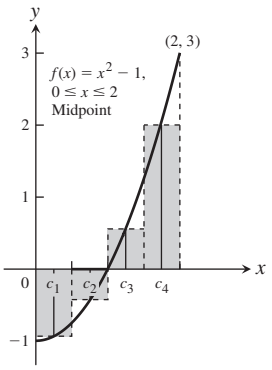
37. (a)



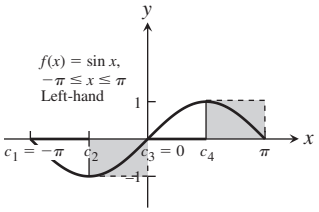
(b)



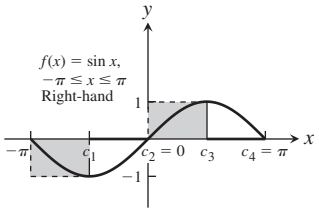
(c)



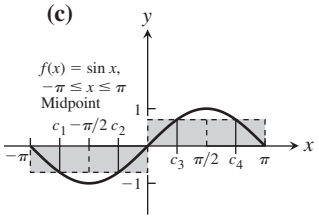
39. (a)



(b)



(c)



41. 1.2

43. $\dfrac{2}{3} - \dfrac{1}{2n} - \dfrac{1}{6n^2}, \dfrac{2}{3}$

45. $12 + \dfrac{27n + 9}{2n^2}, 12$

47. $\dfrac{5}{6} + \dfrac{6n + 1}{6n^2}, \dfrac{5}{6}$

49. $\dfrac{1}{2} + \dfrac{1}{n} + \dfrac{1}{2n^2}, \dfrac{1}{2}$

SECTION 5.3, pp. 292–295

1. $\displaystyle\int_0^2 x^2\,dx$ **3.** $\displaystyle\int_{-7}^{5} (x^2 - 3x)\,dx$ **5.** $\displaystyle\int_2^3 \frac{1}{1 - x}\,dx$

7. $\displaystyle\int_{-\pi/4}^{0} \sec x\,dx$

9. (a) 0 **(b)** -8 **(c)** -12 **(d)** 10 **(e)** -2 **(f)** 16

11. (a) 5 **(b)** $5\sqrt{3}$ **(c)** -5 **(d)** -5

13. (a) 4 **(b)** -4 **15.** Area = 21 square units

17. Area $= 9\pi/2$ square units **19.** Area $= 2.5$ square units
21. Area $= 3$ square units **23.** $b^2/4$ **25.** $b^2 - a^2$
27. **(a)** 2π **(b)** π **29.** $1/2$ **31.** $3\pi^2/2$ **33.** $7/3$
35. $1/24$ **37.** $3a^2/2$ **39.** $b/3$ **41.** -14
43. -2 **45.** $-7/4$ **47.** 7 **49.** 0
51. Using n subintervals of length $\Delta x = b/n$ and right-endpoint values:

$$\text{Area} = \int_0^b 3x^2\,dx = b^3$$

53. Using n subintervals of length $\Delta x = b/n$ and right-endpoint values:

$$\text{Area} = \int_0^b 2x\,dx = b^2$$

55. $\text{av}(f) = 0$ **57.** $\text{av}(f) = -2$ **59.** $\text{av}(f) = 1$
61. **(a)** $\text{av}(g) = -1/2$ **(b)** $\text{av}(g) = 1$ **(c)** $\text{av}(g) = 1/4$
63. $c(b - a)$ **65.** $b^3/3 - a^3/3$ **67.** 9
69. $b^4/4 - a^4/4$ **71.** $a = 0$ and $b = 1$ maximize the integral.
73. Upper bound $= 1$, lower bound $= 1/2$
75. For example, $\displaystyle\int_0^1 \sin(x^2)\,dx \le \int_0^1 dx = 1$
77. $\displaystyle\int_a^b f(x)\,dx \ge \int_a^b 0\,dx = 0$ **79.** Upper bound $= 1/2$

SECTION 5.4, pp. 305–307

1. $-10/3$ **3.** $124/125$ **5.** $753/16$ **7.** 1 **9.** $2\sqrt{3}$
11. 0 **13.** $-\pi/4$ **15.** $1 - \dfrac{\pi}{4}$ **17.** $\dfrac{2 - \sqrt{2}}{4}$ **19.** $-8/3$
21. $-3/4$ **23.** $\sqrt{2} - \sqrt[4]{8} + 1$ **25.** -1 **27.** 16
29. $1/2$ **31.** $\sqrt{26} - \sqrt{5}$ **33.** $(\cos\sqrt{x})\left(\dfrac{1}{2\sqrt{x}}\right)$
35. $4t^5$ **37.** 3 **39.** $\sqrt{1 + x^2}$ **41.** $-\dfrac{1}{2}x^{-1/2}\sin x$
43. $\mathbf{0}$ **45.** $\mathbf{1}$ **47.** $28/3$
49. $1/2$ **51.** π **53.** $\dfrac{\sqrt{2}\pi}{2}$
55. d, since $y' = \dfrac{1}{x}$ and $y(\pi) = \displaystyle\int_\pi^\pi \frac{1}{t}dt - 3 = -3$
57. b, since $y' = \sec x$ and $y(0) = \displaystyle\int_0^0 \sec t\,dt + 4 = 4$
59. $y = \displaystyle\int_2^x \sec t\,dt + 3$ **61.** $\dfrac{2}{3}bh$ **63.** \$9.00
65. **(a)** $T(0) = 20°\text{C}$, $T(16) = 24°\text{C}$, $T(25) = 30°\text{C}$
(b) $\text{av}(T) = 23.33°\text{C}$
67. $2x - 2$ **69.** $-3x + 5$
71. **(a)** True. Since f is continuous, g is differentiable by Part 1 of the Fundamental Theorem of Calculus.
(b) True: g is continuous because it is differentiable.
(c) True, since $g'(1) = f(1) = 0$.
(d) False, since $g''(1) = f'(1) > 0$.
(e) True, since $g'(1) = 0$ and $g''(1) = f'(1) > 0$.
(f) False: $g''(x) = f'(x) > 0$, so g'' never changes sign.
(g) True, since $g'(1) = f(1) = 0$ and $g'(x) = f(x)$ is an increasing function of x (because $f'(x) > 0$).
73. **(a)** $v = \dfrac{ds}{dt} = \dfrac{d}{dt}\displaystyle\int_0^t f(x)\,dx = f(t) \Rightarrow v(5) = f(5) = 2\ \text{m/s}$
(b) $a = df/dt$ is negative, since the slope of the tangent line at $t = 5$ is negative.
(c) $s = \displaystyle\int_0^3 f(x)\,dx = \frac{1}{2}(3)(3) = \frac{9}{2}\text{m}$, since the integral is the area of the triangle formed by $y = f(x)$, the x-axis, and $x = 3$.
(d) $t = 6$, since after $t = 6$ to $t = 9$, the region lies below the x-axis.
(e) At $t = 4$ and $t = 7$, since there are horizontal tangents there.
(f) Toward the origin between $t = 6$ and $t = 9$, since the velocity is negative on this interval. Away from the origin between $t = 0$ and $t = 6$, since the velocity is positive there.
(g) Right or positive side, because the integral of f from 0 to 9 is positive, there being more area above the x-axis than below.

SECTION 5.5, pp. 313–314

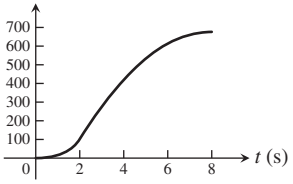
1. $\dfrac{1}{6}(2x + 4)^6 + C$ **3.** $-\dfrac{1}{3}(x^2 + 5)^{-3} + C$
5. $\dfrac{1}{10}(3x^2 + 4x)^5 + C$ **7.** $-\dfrac{1}{3}\cos 3x + C$
9. $\dfrac{1}{2}\sec 2t + C$ **11.** $-6(1 - r^3)^{1/2} + C$
13. $\dfrac{1}{3}(x^{3/2} - 1) - \dfrac{1}{6}\sin(2x^{3/2} - 2) + C$
15. **(a)** $-\dfrac{1}{4}(\cot^2 2\theta) + C$ **(b)** $-\dfrac{1}{4}(\csc^2 2\theta) + C$
17. $-\dfrac{1}{3}(3 - 2s)^{3/2} + C$ **19.** $-\dfrac{2}{5}(1 - \theta^2)^{5/4} + C$
21. $\left(-2/\left(1 + \sqrt{x}\right)\right) + C$ **23.** $\dfrac{1}{3}\tan(3x + 2) + C$
25. $\dfrac{1}{2}\sin^6\left(\dfrac{x}{3}\right) + C$ **27.** $\left(\dfrac{r^3}{18} - 1\right)^6 + C$
29. $-\dfrac{2}{3}\cos(x^{3/2} + 1) + C$ **31.** $\dfrac{1}{2\cos(2t + 1)} + C$
33. $-\sin\left(\dfrac{1}{t} - 1\right) + C$ **35.** $-\dfrac{\sin^2(1/\theta)}{2} + C$
37. $\dfrac{2}{3}(1 + x)^{3/2} - 2(1 + x)^{1/2} + C$ **39.** $\dfrac{2}{3}\left(2 - \dfrac{1}{x}\right)^{3/2} + C$
41. $\dfrac{2}{27}\left(1 - \dfrac{3}{x^3}\right)^{3/2} + C$ **43.** $\dfrac{1}{12}(x - 1)^{12} + \dfrac{1}{11}(x - 1)^{11} + C$
45. $-\dfrac{1}{8}(1 - x)^8 + \dfrac{4}{7}(1 - x)^7 - \dfrac{2}{3}(1 - x)^6 + C$
47. $\dfrac{1}{5}(x^2 + 1)^{5/2} - \dfrac{1}{3}(x^2 + 1)^{3/2} + C$ **49.** $\dfrac{-1}{4(x^2 - 4)^2} + C$
51. **(a)** $-\dfrac{6}{2 + \tan^3 x} + C$ **(b)** $-\dfrac{6}{2 + \tan^3 x} + C$
(c) $-\dfrac{6}{2 + \tan^3 x} + C$
53. $\dfrac{1}{6}\sin\sqrt{3(2r - 1)^2 + 6} + C$ **55.** $s = \dfrac{1}{2}(3t^2 - 1)^4 - 5$
57. $s = 4t - 2\sin\left(2t + \dfrac{\pi}{6}\right) + 9$
59. $s = \sin\left(2t - \dfrac{\pi}{2}\right) + 100t + 1$ **61.** 6 m

SECTION 5.6, pp. 321–324

1. (a) $14/3$ (b) $2/3$ **3.** (a) $1/2$ (b) $-1/2$
5. (a) $15/16$ (b) 0 **7.** (a) 0 (b) $1/8$ **9.** (a) 4 (b) 0
11. (a) $506/375$ (b) $86{,}744/375$ **13.** (a) 0 (b) 0
15. $2\sqrt{3}$ **17.** $3/4$ **19.** $3^{5/2} - 1$ **21.** 3 **23.** $\pi/3$
25. $16/3$ **27.** $2^{5/2}$ **29.** $\pi/2$ **31.** $128/15$ **33.** $4/3$
35. $5/6$ **37.** $38/3$ **39.** $49/6$ **41.** $32/3$ **43.** $48/5$
45. $8/3$ **47.** 8 **49.** $5/3$ (There are three intersection points.)
51. 18 **53.** $243/8$ **55.** $8/3$ **57.** 2 **59.** $104/15$
61. $56/15$ **63.** 4 **65.** $\frac{4}{3} - \frac{4}{\pi}$ **67.** $\pi/2$ **69.** 2
71. $1/2$ **73.** 1
75. (a) $\left(\pm\sqrt{c}, c\right)$ (b) $c = 4^{2/3}$ (c) $c = 4^{2/3}$
77. $11/3$ **79.** $3/4$ **81.** Neither **83.** $F(6) - F(2)$
85. (a) -3 (b) 3 **87.** $I = a/2$

PRACTICE EXERCISES, pp. 325–327

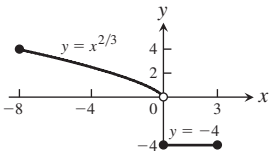
1. (a) About 680 m (b) h (meters)



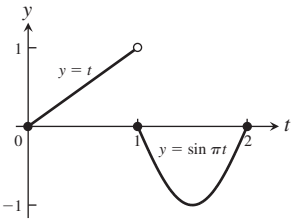
3. (a) $-1/2$ (b) 31 (c) 13 (d) 0
5. $\int_1^5 (2x-1)^{-1/2}\,dx = 2$ **7.** $\int_{-\pi}^{0} \cos\frac{x}{2}\,dx = 2$
9. (a) 4 (b) 2 (c) -2 (d) -2π (e) $8/5$
11. $8/3$ **13.** 62 **15.** 1 **17.** $1/6$ **19.** 18
21. $9/8$ **23.** $\frac{\pi^2}{32} + \frac{\sqrt{2}}{2} - 1$ **25.** 4 **27.** $\frac{8\sqrt{2} - 7}{6}$
29. Min: -4, max: 0, area: $27/4$ **31.** $6/5$
35. $y = \int_5^x \left(\frac{\sin t}{t}\right) dt - 3$ **37.** $-4(\cos x)^{1/2} + C$
39. $\theta^2 + \theta + \sin(2\theta + 1) + C$ **41.** $\frac{t^3}{3} + \frac{4}{t} + C$
43. $-\frac{1}{3}\cos\left(2t^{3/2}\right) + C$ **45.** $\frac{1}{4(\sin 2\theta + \cos 2\theta)^2} + C$
47. 16 **49.** 2 **51.** 1 **53.** 8 **55.** $27\sqrt{3}/160$
57. $\pi/2$ **59.** $\sqrt{3}$ **61.** $6\sqrt{3} - 2\pi$ **63.** -1 **65.** 2
67. 1 **69.** (a) b (b) b **73.** $-4°$ C
75. $\sqrt{2 + \cos^3 x}$ **77.** $\frac{-6}{3 + x^4}$
79. Yes **81.** $-\sqrt{1 + x^2}$
83. Cost $\approx$ 13,897.50 using a lower sum estimate

ADDITIONAL AND ADVANCED EXERCISES, pp. 328–330

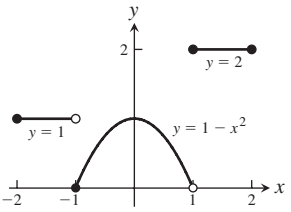
1. (a) Yes (b) No **5.** (a) $1/4$ (b) $\sqrt[3]{12}$
7. $f(x) = \frac{x}{\sqrt{x^2 + 1}}$ **9.** $y = x^3 + 2x - 4$
11. $36/5$



13. $\frac{1}{2} - \frac{2}{\pi}$



15. $13/3$



17. $1/2$ **19.** $1/6$
21. $\int_0^1 f(x)\,dx$ **23.** (b) πr^2
25. (a) 0 (b) -1
(c) $-\pi$ (d) $x = 1$
(e) $y = 2x + 2 - \pi$
(f) $x = -1, x = 2$
(g) $[-2\pi, 0]$
27. $2/x$ **29.** $\frac{\sin 4y}{\sqrt{y}} - \frac{\sin y}{2\sqrt{y}}$

Chapter 6

SECTION 6.1, pp. 339–343

1. 16 **3.** $16/3$ **5.** (a) $2\sqrt{3}$ (b) 8 **7.** (a) 60 (b) 36
9. 8π **11.** 10 **13.** (a) s^2h (b) s^2h **15.** $8/3$
17. $\frac{2\pi}{3}$ **19.** $4 - \pi$ **21.** $\frac{32\pi}{5}$ **23.** 36π **25.** π
27. $\pi\left(\frac{\pi}{2} + 2\sqrt{2} - \frac{11}{3}\right)$ **29.** 2π **31.** 2π
33. $4\pi \ln 4$ **35.** $\pi^2 - 2\pi$ **37.** $\frac{2\pi}{3}$ **39.** $\frac{117\pi}{5}$
41. $\pi(\pi - 2)$ **43.** $\frac{4\pi}{3}$ **45.** 8π **47.** $\frac{7\pi}{6}$
49. (a) 8π (b) $\frac{32\pi}{5}$ (c) $\frac{8\pi}{3}$ (d) $\frac{224\pi}{15}$
51. (a) $\frac{16\pi}{15}$ (b) $\frac{56\pi}{15}$ (c) $\frac{64\pi}{15}$ **53.** $V = 2a^2b\pi^2$
55. (a) $V = \frac{\pi h^2(3a - h)}{3}$ (b) $\frac{1}{120\pi}$ m/s
59. $V = 3308\ \text{cm}^3$ **61.** $\frac{4 - b + a}{2}$

SECTION 6.2, pp. 348–350

1. 6π **3.** 2π **5.** $14\pi/3$ **7.** 8π **9.** $5\pi/6$
11. $\frac{7\pi}{15}$ **13.** (b) 4π **15.** $\frac{16\pi}{15}\left(3\sqrt{2} + 5\right)$
17. $\frac{8\pi}{3}$ **19.** $\frac{4\pi}{3}$ **21.** $\frac{16\pi}{3}$
23. (a) 16π (b) 32π (c) 28π
(d) 24π (e) 60π (f) 48π
25. (a) $\frac{27\pi}{2}$ (b) $\frac{27\pi}{2}$ (c) $\frac{72\pi}{5}$ (d) $\frac{108\pi}{5}$

27. (a) $\frac{6\pi}{5}$ (b) $\frac{4\pi}{5}$ (c) 2π (d) 2π

29. (a) About the x-axis: $V = \frac{2\pi}{15}$; about the y-axis: $V = \frac{\pi}{6}$

(b) About the x-axis: $V = \frac{2\pi}{15}$; about the y-axis: $V = \frac{\pi}{6}$

31. (a) $\frac{5\pi}{3}$ (b) $\frac{4\pi}{3}$ (c) 2π (d) $\frac{2\pi}{3}$

33. (a) $\frac{4\pi}{15}$ (b) $\frac{7\pi}{30}$ **35.** (a) $\frac{24\pi}{5}$ (b) $\frac{48\pi}{5}$

37. (a) $\frac{9\pi}{16}$ (b) $\frac{9\pi}{16}$

39. Disk: 2 integrals; washer: 2 integrals; shell: 1 integral

41. (a) $\frac{256\pi}{3}$ (b) $\frac{244\pi}{3}$ **45.** 2

SECTION 6.3, pp. 355–356

1. 12 **3.** $\frac{53}{6}$ **5.** $\frac{123}{32}$ **7.** $\frac{99}{8}$ **9.** $\frac{53}{6}$ **11.** 2

13. (a) $\int_{-1}^{2} \sqrt{1 + 4x^2}\, dx$ (c) ≈ 6.13

15. (a) $\int_{0}^{\pi} \sqrt{1 + \cos^2 y}\, dy$ (c) ≈ 3.82

17. (a) $\int_{-1}^{3} \sqrt{1 + (y + 1)^2}\, dy$ (c) ≈ 9.29

19. (a) $\int_{0}^{\pi/6} \sec x\, dx$ (c) ≈ 0.55

21. (a) $y = \sqrt{x}$ from (1, 1) to (4, 2)
(b) Only one. We know the derivative of the function and the value of the function at one value of x.

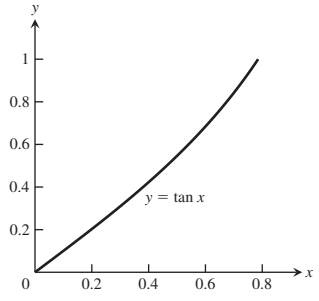
23. 1 **27.** Yes, $f(x) = \pm x + C$ where C is any real number.

33. $\int_{0}^{x} \sqrt{1 + 9t}\, dt, \frac{2}{27}(10^{3/2} - 1)$

SECTION 6.4, pp. 360–361

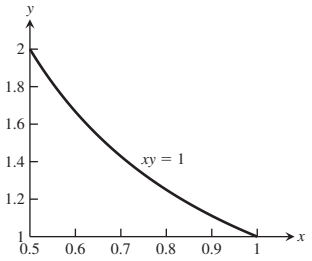
1. (a) $2\pi \int_{0}^{\pi/4} (\tan x)\sqrt{1 + \sec^4 x}\, dx$ (c) $S \approx 3.84$

(b)



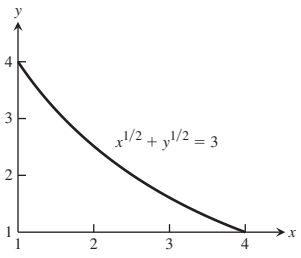
3. (a) $2\pi \int_{1}^{2} \frac{1}{y}\sqrt{1 + y^{-4}}\, dy$ (c) $S \approx 5.02$

(b)



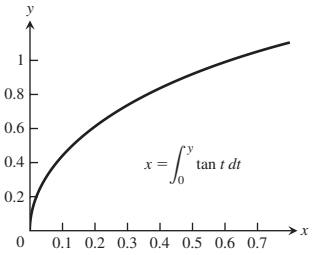
5. (a) $2\pi \int_{1}^{4} (3 - x^{1/2})^2 \sqrt{1 + (1 - 3x^{-1/2})^2}\, dx$ (c) $S \approx 63.37$

(b)



7. (a) $2\pi \int_{0}^{\pi/3} \left(\int_{0}^{y} \tan t\, dt \right) \sec y\, dy$ (c) $s \approx 2.08$

(b)



9. $4\pi\sqrt{5}$ **11.** $3\pi\sqrt{5}$ **13.** $98\pi/81$ **15.** 2π

17. $\pi\left(\sqrt{8} - 1\right)/9$ **19.** $35\pi\sqrt{5}/3$ **21.** $(2\pi/3)\left(2\sqrt{2} - 1\right)$

23. $253\pi/20$ **27.** Order 226.2 liters of each color.

SECTION 6.5, pp. 367–371

1. 116 J **3.** 400 N/m **5.** 4 cm, 0.08 J

7. (a) 12,000 N/cm (b) 6000 N · cm, 18,000 N · cm

9. 780 J **11.** 108,000 J **13.** 234 J

15. (a) 235,200,000 J (b) 17 h, 46 min (c) 266 min
(d) At 9780 N/m^3: a) 234,720,000 J b) 17 h, 44 min
At 9820 N/m^3: a) 235,680,000 J b) 17 h, 48 min

17. 5,772,676.5 J **19.** 8,977,666 J **21.** 385,369 J

23. 15,073,099.75 J **27.** 120 J **29.** 138.72 J

31. 2.175 J **33.** 5.144×10^{10} J **35.** 9146.7 N

37. (a) 26,989.2 N (b) 25,401.6 N **39.** (a) 182,933 N
(b) 187,600 N **41.** 5808 N

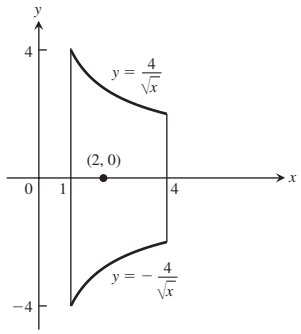
43. (a) 19,600 N (b) 14,700 N (c) 16,135 N

45. (a) 14,933 N (b) 2.94 m **47.** $\frac{wb}{2}$

49. No. The tank will overflow because the movable end will have moved only 2.18 m by the time the tank is full.

SECTION 6.6, pp. 381–383

1. $M = 14/3,\ \bar{x} = 93/35$ **3.** $M = \ln 4,\ \bar{x} = (3 - \ln 4)/(\ln 4)$
5. $M = 13,\ \bar{x} = 41/26$ **7.** $\bar{x} = 0, \bar{y} = 12/5$
9. $\bar{x} = 1, \bar{y} = -3/5$ **11.** $\bar{x} = 16/105, \bar{y} = 8/15$
13. $\bar{x} = 0, \bar{y} = \pi/8$ **15.** **(a)** $(4/\pi, 4/\pi)$ **(b)** $(0, 4/\pi)$
17. $\bar{x} = 7, \bar{y} = \dfrac{\ln 16}{12}$
19. $\bar{x} = 5/7, \bar{y} = 10/33$. $(\bar{x})^4 < \bar{y}$, so the center of mass is outside the region.
21. $\bar{x} = 3/2, \bar{y} = 1/2$
23. **(a)** $\dfrac{224\pi}{3}$ **(b)** $\bar{x} = 2, \bar{y} = 0$
(c)



27. $\bar{x} = \bar{y} = 1/3$ **29.** $\bar{x} = a/3, \bar{y} = b/3$ **31.** $13\delta/6$
33. $\bar{x} = 0, \bar{y} = \dfrac{a\pi}{4}$ **35.** $\bar{x} = 1/2, \bar{y} = 4$
37. $\bar{x} = 6/5, \bar{y} = 8/7$ **39.** $V = 32\pi, S = 32\sqrt{2}\pi$ **43.** $4\pi^2$
45. $\bar{x} = 0, \bar{y} = \dfrac{2a}{\pi}$ **47.** $\bar{x} = 0, \bar{y} = \dfrac{4b}{3\pi}$
49. $\sqrt{2}\pi a^3(4 + 3\pi)/6$ **51.** $\bar{x} = \dfrac{a}{3}, \bar{y} = \dfrac{b}{3}$

PRACTICE EXERCISES, pp. 384–386

1. $\dfrac{9\pi}{280}$ **3.** π^2 **5.** $\dfrac{72\pi}{35}$
7. **(a)** 2π **(b)** π **(c)** $12\pi/5$ **(d)** $26\pi/5$
9. **(a)** 8π **(b)** $1088\pi/15$ **(c)** $512\pi/15$
11. $\pi\left(3\sqrt{3} - \pi\right)/3$
13. **(a)** $16\pi/15$ **(b)** $8\pi/5$ **(c)** $8\pi/3$ **(d)** $32\pi/5$
15. $\dfrac{28\pi}{3}\ \text{m}^3$ **17.** $(\pi/3)(a^2 + ab + b^2)h$ **19.** $\dfrac{10}{3}$ **21.** $\dfrac{285}{5}$
23. $28\pi\sqrt{2}/3$ **25.** 4π **27.** 4640 J
29. $\dfrac{w}{2}(2ar - a^2)$ **31.** 65,680,230 J
33. $3{,}375{,}000\pi$ J, 257 s **35.** **(a)** 11.31 J **(b)** 29.18 J
37. $\bar{x} = 0, \bar{y} = 8/5$ **39.** $\bar{x} = 3/2, \bar{y} = 12/5$
41. $\bar{x} = 9/5, \bar{y} = 11/10$ **43.** 52,267 J **45.** 344,960 N

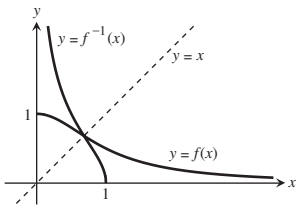
ADDITIONAL AND ADVANCED EXERCISES, pp. 386–387

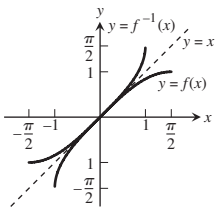
1. $f(x) = \sqrt{\dfrac{2x - a}{\pi}}$ **3.** $f(x) = \sqrt{C^2 - 1}\,x + a$, where $C \geq 1$
5. $\dfrac{\pi}{30\sqrt{2}}$ **7.** $28/3$ **9.** $\dfrac{4h\sqrt{3mh}}{3}$
11. $\bar{x} = 0, \bar{y} = \dfrac{n}{2n + 1}, (0, 1/2)$
15. **(a)** $\bar{x} = \bar{y} = 4(a^2 + ab + b^2)/(3\pi(a + b))$
(b) $(2a/\pi, 2a/\pi)$
17. $\approx$ 365,867 N

Chapter 7

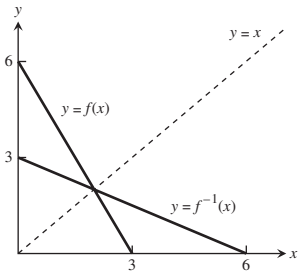
SECTION 7.1, pp. 394–396

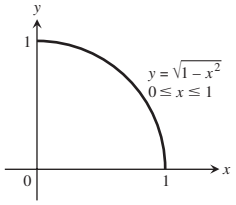
1. One-to-one **3.** Not one-to-one **5.** One-to-one
7. Not one-to-one **9.** One-to-one
11. D: $(0, 1]$ R: $[0, \infty)$ **13.** D: $[-1, 1]$ R: $[-\pi/2, \pi/2]$



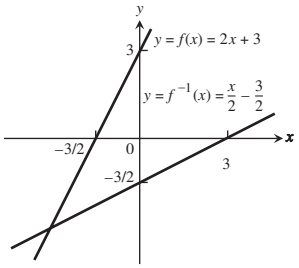


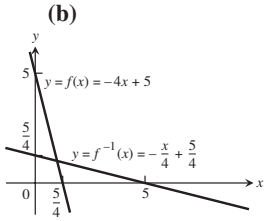
15. D: $[0, 6]$ R: $[0, 3]$ **17.** **(a)** Symmetric about the line $y = x$





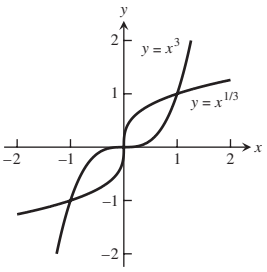
19. $f^{-1}(x) = \sqrt{x - 1}$ **21.** $f^{-1}(x) = \sqrt[3]{x + 1}$
23. $f^{-1}(x) = \sqrt{x} - 1$
25. $f^{-1}(x) = \sqrt[5]{x}$; domain: $-\infty < x < \infty$; range: $-\infty < y < \infty$
27. $f^{-1}(x) = 5\sqrt{x - 1}$; domain: $-\infty < x < \infty$; range: $-\infty < y < \infty$
29. $f^{-1}(x) = \dfrac{1}{\sqrt{x}}$; domain: $x > 0$; range: $y > 0$
31. $f^{-1}(x) = \dfrac{2x + 3}{x - 1}$; domain: $-\infty < x < \infty, x \neq 1$; range: $-\infty < y < \infty, y \neq 2$
33. $f^{-1}(x) = 1 - \sqrt{x + 1}$; domain: $-1 \leq x < \infty$; range: $-\infty < y \leq 1$
35. **(a)** $f^{-1}(x) = \dfrac{x}{2} - \dfrac{3}{2}$ **37.** **(a)** $f^{-1}(x) = -\dfrac{x}{4} + \dfrac{5}{4}$
(b) **(b)**





(c) 2, 1/2 **(c)** $-4, -1/4$

39. **(b)**



(c) Slope of f at $(1, 1)$: 3; slope of g at $(1, 1)$: $1/3$; slope of f at $(-1, -1)$: 3; slope of g at $(-1, -1)$: $1/3$

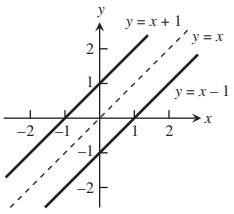
(d) $y = 0$ is tangent to $y = x^3$ at $x = 0$; $x = 0$ is tangent to $y = \sqrt[3]{x}$ at $x = 0$.

41. $1/9$ **43.** 3

45. **(a)** $f^{-1}(x) = \frac{1}{m}x$

(b) The graph of f^{-1} is the line through the origin with slope $1/m$.

47. **(a)** $f^{-1}(x) = x - 1$



(b) $f^{-1}(x) = x - b$. The graph of f^{-1} is a line parallel to the graph of f. The graphs of f and f^{-1} lie on opposite sides of the line $y = x$ and are equidistant from that line.

(c) Their graphs will be parallel to one another and lie on opposite sides of the line $y = x$ equidistant from that line.

51. Increasing, therefore one-to-one; $df^{-1}/dx = \frac{1}{9}x^{-2/3}$

53. Decreasing, therefore one-to-one; $df^{-1}/dx = -\frac{1}{3}x^{-2/3}$

SECTION 7.2, pp. 403–404

1. **(a)** $\ln 3 - 2\ln 2$ **(b)** $2(\ln 2 - \ln 3)$ **(c)** $-\ln 2$ **(d)** $\frac{2}{3}\ln 3$ **(e)** $\ln 3 + \frac{1}{2}\ln 2$ **(f)** $\frac{1}{2}(3\ln 3 - \ln 2)$

3. **(a)** $\ln 5$ **(b)** $\ln(x - 3)$ **(c)** $\ln(t^2)$

5. $t = e^2/(e^2 - 1)$ **7.** $1/x$ **9.** $2/t$ **11.** $-1/x$

13. $\frac{1}{\theta + 1}$ **15.** $3/x$ **17.** $2(\ln t) + (\ln t)^2$ **19.** $x^3 \ln x$

21. $\frac{1 - \ln t}{t^2}$ **23.** $\frac{1}{x(1 + \ln x)^2}$ **25.** $\frac{1}{x \ln x}$ **27.** $2\cos(\ln\theta)$

29. $-\frac{3x + 2}{2x(x + 1)}$ **31.** $\frac{2}{t(1 - \ln t)^2}$ **33.** $\frac{\tan(\ln\theta)}{\theta}$

35. $\frac{10x}{x^2 + 1} + \frac{1}{2(1 - x)}$ **37.** $2x\ln|x| - x\ln\frac{|x|}{\sqrt{2}}$

39. $\ln\left(\frac{2}{3}\right)$ **41.** $\ln|y^2 - 25| + C$ **43.** $\ln 3$

45. $(\ln 2)^2$ **47.** $\frac{1}{\ln 4}$ **49.** $\ln|6 + 3\tan t| + C$

51. $\ln 2$ **53.** $\ln 27$ **55.** $\ln(1 + \sqrt{x}) + C$

57. $\left(\frac{1}{2}\right)\sqrt{x(x + 1)}\left(\frac{1}{x} + \frac{1}{x + 1}\right) = \frac{2x + 1}{2\sqrt{x(x + 1)}}$

59. $\left(\frac{1}{2}\right)\sqrt{\frac{t}{t + 1}}\left(\frac{1}{t} - \frac{1}{t + 1}\right) = \frac{1}{2\sqrt{t}(t + 1)^{3/2}}$

61. $\sqrt{\theta + 3}(\sin\theta)\left(\frac{1}{2(\theta + 3)} + \cot\theta\right)$

63. $t(t + 1)(t + 2)\left[\frac{1}{t} + \frac{1}{t + 1} + \frac{1}{t + 2}\right] = 3t^2 + 6t + 2$

65. $\frac{\theta + 5}{\theta\cos\theta}\left[\frac{1}{\theta + 5} - \frac{1}{\theta} + \tan\theta\right]$

67. $\frac{x\sqrt{x^2 + 1}}{(x + 1)^{2/3}}\left[\frac{1}{x} + \frac{x}{x^2 + 1} - \frac{2}{3(x + 1)}\right]$

69. $\frac{1}{3}\sqrt[3]{\frac{x(x - 2)}{x^2 + 1}}\left(\frac{1}{x} + \frac{1}{x - 2} - \frac{2x}{x^2 + 1}\right)$

71. **(a)** Max $= 0$ at $x = 0$, min $= -\ln 2$ at $x = \pi/3$

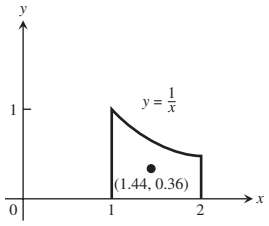
(b) Max $= 1$ at $x = 1$, min $= \cos(\ln 2)$ at $x = 1/2$ and $x = 2$

73. $\ln 16$ **75.** **(a)** Increasing on $(0, e^{-2})$ and $(1, \infty)$; decreasing on $(e^{-2}, 1)$ **(b)** local maximum is $4/e^2$ at $x = e^{-2}$; absolute minimum is 0 at $x = 1$; no absolute maximum **77.** $4\pi\ln 4$

79. $\pi\ln 16$ **81.** **(a)** $6 + \ln 2$ **(b)** $8 + \ln 9$

83. **(a)** $\bar{x} \approx 1.44$, $\bar{y} \approx 0.36$

(b)



87. $y = x + \ln|x| + 2$ **89.** **(b)** 0.00469

SECTION 7.3, pp. 412–415

1. **(a)** $t = -10\ln 3$ **(b)** $t = -\frac{\ln 2}{k}$ **(c)** $t = \frac{\ln .4}{\ln .2}$

3. $4(\ln x)^2$ **5.** $\ln 3$ **7.** $-5e^{-5x}$ **9.** $-7e^{(5-7x)}$ **11.** xe^x

13. x^2e^x **15.** $2e^\theta\cos\theta$ **17.** $2\theta e^{-\theta^2}\sin(e^{-\theta^2})$ **19.** $\frac{1 - t}{t}$

21. $1/(1 + e^\theta)$ **23.** $e^{\cos t}(1 - t\sin t)$ **25.** $(\sin x)/x$

27. $\frac{ye^y\cos x}{1 - ye^y\sin x}$ **29.** $\frac{2e^{2x} - \cos(x + 3y)}{3\cos(x + 3y)}$

31. $y' = \frac{3x^2}{1 - \cos y}$, $y'' = \frac{6x(1 - \cos y)^2 - 9x^4\sin y}{(1 - \cos y)^3}$

33. $\frac{1}{3}e^{3x} - 5e^{-x} + C$ **35.** 1 **37.** $8e^{(x+1)} + C$

39. 2 **41.** $2e^{\sqrt{r}} + C$ **43.** $-e^{-t^2} + C$ **45.** $-e^{1/x} + C$

47. e **49.** $\frac{1}{\pi}e^{\sec\pi t} + C$ **51.** 1 **53.** $\ln(1 + e^r) + C$

55. $y = 1 - \cos(e^t - 2)$ **57.** $y = 2(e^{-x} + x) - 1$

59. $2^x\ln 2$ **61.** $\left(\frac{\ln 5}{2\sqrt{s}}\right)5^{\sqrt{s}}$ **63.** $\pi x^{(\pi-1)}$

65. $-\sqrt{2}\cos\theta^{(\sqrt{2}-1)}\sin\theta$ **67.** $7^{\sec\theta}(\ln 7)^2(\sec\theta\tan\theta)$

69. $(3\cos 3t)(2^{\sin 3t})\ln 2$ **71.** $\frac{1}{\theta\ln 2}$ **73.** $\frac{3}{x\ln 4}$

75. $\frac{x^2}{\ln 10} + 3x^2 \log_{10} x$ **77.** $\frac{-2}{(x+1)(x-1)}$

79. $\sin(\log_7 \theta) + \frac{1}{\ln 7}\cos(\log_7 \theta)$ **81.** $\frac{1}{\ln 10}$

83. $\frac{1}{t}(\log_2 3)3^{\log_2 t}$ **85.** $\frac{1}{t}$ **87.** $\frac{5^x}{\ln 5} + C$ **89.** $\frac{1}{2\ln 2}$

91. $\frac{1}{\ln 2}$ **93.** $\frac{6}{\ln 7}$ **95.** 32760 **97.** $\frac{3x^{(\sqrt{3}+1)}}{\sqrt{3}+1} + C$

99. $3^{\sqrt{2}+1}$ **101.** $\frac{1}{\ln 10}\left(\frac{(\ln x)^2}{2}\right) + C$ **103.** $2(\ln 2)^2$

105. $\frac{3\ln 2}{2}$ **107.** $\ln 10$ **109.** $(\ln 10)\ln|\ln x| + C$

111. $\ln(\ln x)$, $x > 1$ **113.** $-\ln x$

115. $(x+1)^x\left(\frac{x}{x+1} + \ln(x+1)\right)$ **117.** $(\sqrt{t})^t\left(\frac{\ln t}{2} + \frac{1}{2}\right)$

119. $(\sin x)^x(\ln \sin x + x\cot x)$ **121.** $\cos x^x \cdot x^x(1 + \ln x)$

123. $\frac{3y - xy\ln y}{x^2 - x}$ **125.** $\frac{1 - xy\ln y}{x^2(1 + \ln y)}$ **127.** $(1 + \ln t)^2$

129. Maximum: 1 at $x = 0$, minimum: $2 - 2\ln 2$ at $x = \ln 2$

131. **(a)** Abs max: $\frac{1}{e}$ at $x = 1$ **(b)** $\left(2, \frac{2}{e^2}\right)$

133. Abs max of $1/(2e)$ assumed at $x = 1/\sqrt{e}$ **135.** 2

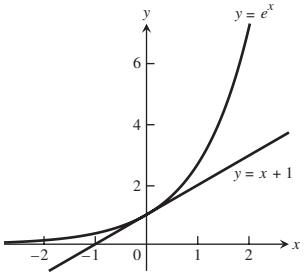
137. $y = e^{x/2} - 1$ **139.** $\frac{e^2 - 1}{2e}$ **141.** $\ln(\sqrt{2} + 1)$

143. **(a)** $\frac{d}{dx}(x\ln x - x + C) = x \cdot \frac{1}{x} + \ln x - 1 + 0 = \ln x$

(b) $\frac{1}{e - 1}$

145. **(b)** $|\text{error}| \approx 0.02140$

(c) $L(x) = x + 1$ never overestimates e^x.



147. $2\ln 5$ **149.** **(a)** $4 + \ln 2$ **(b)** $\left(\frac{1}{2}\right)(4 + \ln 2)$

151. $x \approx -0.76666$

153. **(a)** $L(x) = 1 + (\ln 2)x \approx 0.69x + 1$

SECTION 7.4, pp. 423–425

9. $\frac{2}{3}y^{3/2} - x^{1/2} = C$ **11.** $e^y - e^x = C$

13. $-x + 2\tan\sqrt{y} = C$ **15.** $e^{-y} + 2e^{\sqrt{x}} = C$

17. $y = \sin(x^2 + C)$ **19.** $\frac{1}{3}\ln|y^3 - 2| = x^3 + C$

21. $4\ln\left(\sqrt{y} + 2\right) = e^{x^2} + C$

23. **(a)** -0.00001 **(b)** 10,536 years **(c)** 82%

25. 54.88 g **27.** 19.9 m **29.** 2.8147498×10^{14}

31. **(a)** 8 years **(b)** 32.02 years **33.** Yes, $y(20) < 1$

35. 15.28 years **37.** 56,562 years

41. **(a)** 17.5 min **(b)** 13.26 min

43. $-3°C$ **45.** About 6693 years **47.** 54.62%

49. $\approx$15,683 years

SECTION 7.5, pp. 432–433

1. $-1/4$ **3.** $5/7$ **5.** $1/2$ **7.** $1/4$ **9.** $-23/7$

11. $5/7$ **13.** 0 **15.** -16 **17.** -2 **19.** $1/4$

21. 2 **23.** 3 **25.** -1 **27.** $\ln 3$ **29.** $\frac{1}{\ln 2}$ **31.** $\ln 2$

33. 1 **35.** $1/2$ **37.** $\ln 2$ **39.** $-\infty$ **41.** $-1/2$

43. -1 **45.** 1 **47.** 0 **49.** 2 **51.** $1/e$ **53.** 1

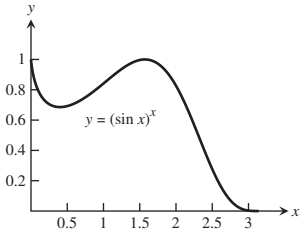
55. $1/e$ **57.** $e^{1/2}$ **59.** 1 **61.** e^3 **63.** 0 **65.** $+1$

67. 3 **69.** 1 **71.** 0 **73.** ∞ **75.** **(b)** is correct.

77. **(d)** is correct. **79.** $c = \frac{27}{10}$ **81.** **(b)** $\frac{-1}{2}$ **83.** -1

87. **(a)** $y = 1$ **(b)** $y = 0, y = \frac{3}{2}$

89. **(a)** We should assign the value 1 to $f(x) = (\sin x)^x$ to make it continuous at $x = 0$.



(c) The maximum value of $f(x)$ is close to 1 near the point $x \approx 1.55$ (see the graph in part (a)).

SECTION 7.6, pp. 442–446

1. **(a)** $\pi/4$ **(b)** $-\pi/3$ **(c)** $\pi/6$

3. **(a)** $-\pi/6$ **(b)** $\pi/4$ **(c)** $-\pi/3$

5. **(a)** $\pi/3$ **(b)** $3\pi/4$ **(c)** $\pi/6$

7. **(a)** $3\pi/4$ **(b)** $\pi/6$ **(c)** $2\pi/3$

9. $1/\sqrt{2}$ **11.** $-1/\sqrt{3}$ **13.** $\pi/2$ **15.** $\pi/2$ **17.** $\pi/2$

19. 0 **21.** $\frac{-2x}{\sqrt{1 - x^4}}$ **23.** $\frac{\sqrt{2}}{\sqrt{1 - 2t^2}}$

25. $\frac{1}{|2s + 1|\sqrt{s^2 + s}}$ **27.** $\frac{-2x}{(x^2 + 1)\sqrt{x^4 + 2x^2}}$

29. $\frac{-1}{\sqrt{1 - t^2}}$ **31.** $\frac{-1}{2\sqrt{t}(1 + t)}$ **33.** $\frac{1}{(\tan^{-1} x)(1 + x^2)}$

35. $\frac{-e^t}{|e^t|\sqrt{(e^t)^2 - 1}} = \frac{-1}{\sqrt{e^{2t} - 1}}$ **37.** $\frac{-2s^n}{\sqrt{1 - s^2}}$ **39.** 0

41. $\sin^{-1} x$ **43.** 0 **45.** $\frac{8\sqrt{2}}{4 + 3\pi}$ **47.** $\sin^{-1}\frac{x}{3} + C$

49. $\frac{1}{\sqrt{17}}\tan^{-1}\frac{x}{\sqrt{17}} + C$ **51.** $\frac{1}{\sqrt{2}}\sec^{-1}\left|\frac{5x}{\sqrt{2}}\right| + C$

53. $2\pi/3$ **55.** $\pi/16$ **57.** $-\pi/12$

59. $\frac{3}{2}\sin^{-1} 2(r - 1) + C$ **61.** $\frac{\sqrt{2}}{2}\tan^{-1}\left(\frac{x - 1}{\sqrt{2}}\right) + C$

63. $\frac{1}{4}\sec^{-1}\left|\frac{2x - 1}{2}\right| + C$ **65.** π **67.** $\pi/12$

69. $\frac{1}{2}\sin^{-1} y^2 + C$ **71.** $\sin^{-1}(x-2) + C$ **73.** π

75. $\frac{1}{2}\tan^{-1}\left(\frac{y-1}{2}\right) + C$ **77.** 2π

79. $\frac{1}{2}\ln(x^2+4) + 2\tan^{-1}\frac{x}{2} + C$

81. $x + \ln(x^2+9) - \frac{10}{3}\tan^{-1}\frac{x}{3} + C$

83. $\sec^{-1}|x+1| + C$ **85.** $e^{\sin^{-1} x} + C$

87. $\frac{1}{3}(\sin^{-1} x)^3 + C$ **89.** $\ln|\tan^{-1} y| + C$ **91.** $\sqrt{3} - 1$

93. $\frac{2}{3}\tan^{-1}\left(\frac{\tan^{-1}\sqrt{x}}{3}\right) + C$ **95.** $\pi^2/32$ **97.** 5

99. 2 **101.** 1 **103.** 1 **109.** $y = \sin^{-1} x$

111. $y = \sec^{-1} x + \frac{2\pi}{3}, x > 1$ **113. (b)** $x = \sqrt{5}$

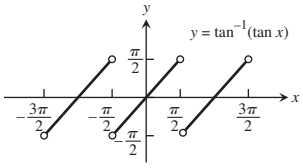
115. $\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 54.7°$

127. $\pi^2/2$ **129. (a)** $\pi^2/2$ **(b)** 2π

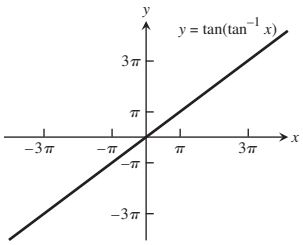
131. (a) 0.84107 **(b)** −0.72973 **(c)** 0.46365

133. (a) Domain: all real numbers except those having the form $\frac{\pi}{2} + k\pi$, where k is an integer

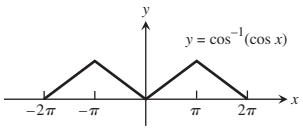
Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$



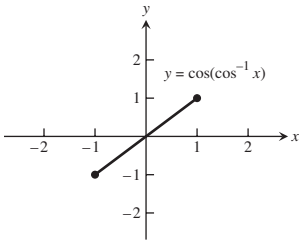
(b) Domain: $-\infty < x < \infty$; Range: $-\infty < y < \infty$



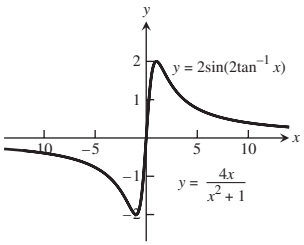
135. (a) Domain: $-\infty < x < \infty$;
Range: $0 \le y \le \pi$



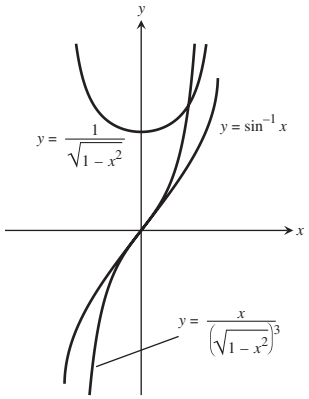
(b) Domain: $-1 \le x \le 1$;
Range: $-1 \le y \le 1$



137. The graphs are identical.



139.



SECTION 7.7, pp. 451–454

1. $\cosh x = 5/4$, $\tanh x = -3/5$, $\coth x = -5/3$, $\operatorname{sech} x = 4/5$, $\operatorname{csch} x = -4/3$

3. $\sinh x = 8/15$, $\tanh x = 8/17$, $\coth x = 17/8$, $\operatorname{sech} x = 15/17$, $\operatorname{csch} x = 15/8$

5. $x + \frac{1}{x}$ **7.** e^{5x} **9.** e^{4x} **13.** $2\cosh\frac{x}{3}$

15. $\operatorname{sech}^2\sqrt{t} + \frac{\tanh\sqrt{t}}{\sqrt{t}}$ **17.** $\coth z$

19. $(\ln\operatorname{sech}\theta)(\operatorname{sech}\theta\tanh\theta)$ **21.** $\tanh^3 v$ **23.** 2

25. $\frac{1}{2\sqrt{x(1+x)}}$ **27.** $\frac{1}{1+\theta} - \tanh^{-1}\theta$

29. $\frac{1}{2\sqrt{t}} - \coth^{-1}\sqrt{t}$ **31.** $-\operatorname{sech}^{-1} x$ **33.** $\frac{\ln 2}{\sqrt{1+\left(\frac{1}{2}\right)^{2\theta}}}$

35. $|\sec x|$ **41.** $\frac{\cosh 2x}{2} + C$ **43.** $12\sinh\left(\frac{x}{2} - \ln 3\right) + C$

45. $7\ln|e^{x/7} + e^{-x/7}| + C$ **47.** $\tanh\left(x - \frac{1}{2}\right) + C$

49. $-2\operatorname{sech}\sqrt{t} + C$ **51.** $\ln\frac{5}{2}$ **53.** $\frac{3}{32} + \ln 2$

55. $e - e^{-1}$ **57.** $3/4$ **59.** $\frac{3}{8} + \ln\sqrt{2}$

61. $\ln(2/3)$ **63.** $\frac{-\ln 3}{2}$ **65.** $\ln 3$

67. (a) $\sinh^{-1}(\sqrt{3})$ **(b)** $\ln(\sqrt{3}+2)$

69. (a) $\coth^{-1}(2) - \coth^{-1}(5/4)$ **(b)** $\left(\frac{1}{2}\right)\ln\left(\frac{1}{3}\right)$

71. (a) $-\operatorname{sech}^{-1}\left(\frac{12}{13}\right) + \operatorname{sech}^{-1}\left(\frac{4}{5}\right)$

(b) $-\ln\left(\frac{1+\sqrt{1-(12/13)^2}}{(12/13)}\right) + \ln\left(\frac{1+\sqrt{1-(4/5)^2}}{(4/5)}\right)$

$= -\ln\left(\frac{3}{2}\right) + \ln(2) = \ln(4/3)$

73. (a) 0 **(b)** 0

77. (b) $\sqrt{\frac{mg}{k}}$ **(c)** $70\sqrt{30/47} \approx 55.93$ m/s **79.** 2π **81.** $\frac{6}{5}$

SECTION 7.8, pp. 458–459

1. **(a)** Slower **(b)** Slower **(c)** Slower **(d)** Faster **(e)** Slower **(f)** Slower **(g)** Same **(h)** Slower

3. **(a)** Same **(b)** Faster **(c)** Same **(d)** Same **(e)** Slower **(f)** Faster **(g)** Slower **(h)** Same

5. **(a)** Same **(b)** Same **(c)** Same **(d)** Faster **(e)** Faster **(f)** Same **(g)** Slower **(h)** Faster

7. d, a, c, b

9. **(a)** False **(b)** False **(c)** True **(d)** True **(e)** True **(f)** True **(g)** False **(h)** True

13. When the degree of f is less than or equal to the degree of g.

15. 1, 1

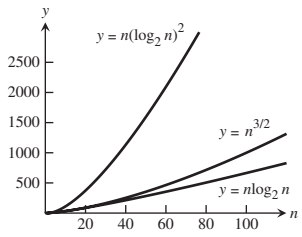
21. **(b)** $\ln(e^{17000000}) = 17{,}000{,}000 < (e^{17\times 10^6})^{1/10^6} = e^{17} \approx 24{,}154{,}952.75$

(c) $x \approx 3.4306311 \times 10^{15}$

(d) They cross at $x \approx 3.4306311 \times 10^{15}$.

23. **(a)** The algorithm that takes $O(n \log_2 n)$ steps

(b)



25. It could take one million for a sequential search; at most 20 steps for a binary search.

PRACTICE EXERCISES, pp. 460–462

1. $-2e^{-x/5}$ **3.** xe^{4x} **5.** $\dfrac{2\sin\theta\cos\theta}{\sin^2\theta} = 2\cot\theta$ **7.** $\dfrac{2}{(\ln 2)x}$

9. $-8^{-t}(\ln 8)$ **11.** $18x^{2.6}$

13. $(x+2)^{x+2}(\ln(x+2)+1)$ **15.** $-\dfrac{1}{\sqrt{1-u^2}}$

17. $\dfrac{-1}{\sqrt{1-x^2}\cos^{-1}x}$ **19.** $\tan^{-1}(t) + \dfrac{t}{1+t^2} - \dfrac{1}{2t}$

21. $\dfrac{1-z}{\sqrt{z^2-1}} + \sec^{-1}z$ **23.** -1

25. $\dfrac{2(x^2+1)}{\sqrt{\cos 2x}}\left[\dfrac{2x}{x^2+1} + \tan 2x\right]$

27. $5\left[\dfrac{(t+1)(t-1)}{(t-2)(t+3)}\right]^5\left[\dfrac{1}{t+1} + \dfrac{1}{t-1} - \dfrac{1}{t-2} - \dfrac{1}{t+3}\right]$

29. $\dfrac{1}{\sqrt{\theta}}(\sin\theta)^{\sqrt{\theta}}\left(\dfrac{\ln\sqrt{\sin\theta}}{2} + \theta\cot\theta\right)$ **31.** $-\cos e^x + C$

33. $\tan(e^x - 7) + C$ **35.** $e^{\tan x} + C$ **37.** $\dfrac{-\ln 7}{3}$

39. $\ln 8$ **41.** $\ln(9/25)$ **43.** $-[\ln|\cos(\ln v)|] + C$

45. $-\dfrac{1}{2}(\ln x)^{-2} + C$ **47.** $-\cot(1+\ln r) + C$

49. $\dfrac{1}{2\ln 3}\left(3^{x^2}\right) + C$ **51.** $3\ln 7$ **53.** $15/16 + \ln 2$

55. $e - 1$ **57.** $1/6$ **59.** $9/14$

61. $\dfrac{1}{3}[(\ln 4)^3 - (\ln 2)^3]$ or $\dfrac{7}{3}(\ln 2)^3$ **63.** $\dfrac{9\ln 2}{4}$ **65.** π

67. $\pi/\sqrt{3}$ **69.** $\sec^{-1}|2y| + C$ **71.** $\pi/12$

73. $\sin^{-1}(x+1) + C$ **75.** $\pi/2$ **77.** $\dfrac{1}{3}\sec^{-1}\left(\dfrac{t+1}{3}\right) + C$

79. $y = \dfrac{\ln 2}{\ln(3/2)}$ **81.** $y = \ln x - \ln 3$ **83.** $y = \dfrac{1}{1-e^x}$

85. 5 **87.** 0 **89.** 1 **91.** $3/7$ **93.** 0 **95.** 1

97. $\ln 10$ **99.** $\ln 2$ **101.** 5 **103.** $-\infty$ **105.** 1

107. 1 **109.** **(a)** Same rate **(b)** Same rate **(c)** Faster **(d)** Faster **(e)** Same rate **(f)** Same rate

111. **(a)** True **(b)** False **(c)** False **(d)** True **(e)** True **(f)** True

113. $1/3$

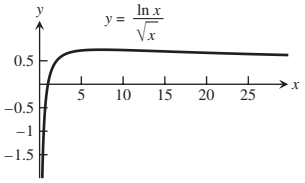
115. Absolute maximum $= 0$ at $x = e/2$, absolute minimum $= -0.5$ at $x = 0.5$

117. 1

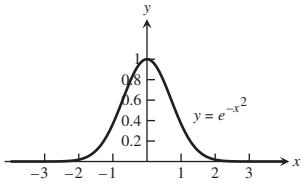
119. $1/e$ m/s

121. $1/\sqrt{2}$ units long by $1/\sqrt{e}$ units high, $A = 1/\sqrt{2e} \approx 0.43$ units^2

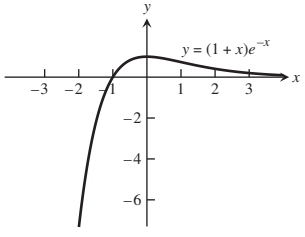
123. **(a)** Absolute maximum of $2/e$ at $x = e^2$; inflection point $(e^{8/3}, (8/3)e^{-4/3})$; concave up on $(e^{8/3}, \infty)$; concave down on $(0, e^{8/3})$



(b) Absolute maximum of 1 at $x = 0$; inflection points $(\pm 1/\sqrt{2}, 1/\sqrt{e})$; concave up on $(-\infty, -1/\sqrt{2}) \cup (1/\sqrt{2}, \infty)$; concave down on $(-1/\sqrt{2}, 1/\sqrt{2})$



(c) Absolute maximum of 1 at $x = 0$; inflection point $(1, 2/e)$; concave up on $(1, \infty)$; concave down on $(-\infty, 1)$



125. $y = \left(\tan^{-1}\left(\dfrac{x+C}{2}\right)\right)^2$ **127.** $y^2 = \sin^{-1}(2\tan x + C)$

129. $y = -2 + \ln(2 - e^{-x})$ **131.** $y = 4x - 4\sqrt{x} + 1$

133. 18,935 years **135.** $20(5 - \sqrt{17})$ m

ADDITIONAL AND ADVANCED EXERCISES, pp. 463–464

1. $\pi/2$ **3.** $1/\sqrt{e}$ **5.** $\ln 2$

7. **(a)** 1 **(b)** $\pi/2$ **(c)** π

9. $y' = \dfrac{x^{y-1}y^2 - ye^x(x^y+1)\ln y}{e^x(x^y+1) - x^y y\ln x}$ **11.** $\dfrac{1}{\ln 2}, \dfrac{1}{2\ln 2}, 2:1$

13. $x = 2$ **15.** $2/17$ **19.** $\bar{x} = \dfrac{\ln 4}{\pi}$, $\bar{y} = 0$ **21.** **(b)** 61°

Chapter 8

SECTION 8.1, pp. 469–470

1. $\ln 5$ **3.** $2\tan x - 2\sec x - x + C$
5. $\sin^{-1} x + \sqrt{1-x^2} + C$ **7.** $e^{-\cot z} + C$
9. $\tan^{-1}(e^z) + C$ **11.** π **13.** $t + \cot t + \csc t + C$
15. $\sqrt{2}$ **17.** $\frac{1}{8}\ln(1 + 4\ln^2 y) + C$
19. $\ln|1 + \sin\theta| + C$ **21.** $2t^2 - t + 2\tan^{-1}\left(\frac{t}{2}\right) + C$
23. $2\left(\sqrt{2} - 1\right) \approx 0.82843$ **25.** $\sec^{-1}(e^y) + C$
27. $\sin^{-1}(2\ln x) + C$ **29.** $\ln|\sin x| + \ln|\cos x| + C$
31. $7 + \ln 8$ **33.** $\left(\sin^{-1} y - \sqrt{1-y^2}\right]_{-1}^{0} = \frac{\pi}{2} - 1$
35. $\sec^{-1}\left|\frac{x-1}{7}\right| + C$ **37.** $\frac{\theta^3}{3} - \frac{\theta^2}{2} + \theta + \frac{5}{2}\ln|2\theta - 5| + C$
39. $x - \ln(1 + e^x) + C$ **41.** $(1/2)e^{2x} - e^x + \ln(1 + e^x) + C$
43. $2\arctan\left(\sqrt{x}\right) + C$ **45.** $2\sqrt{2} - \ln\left(3 + 2\sqrt{2}\right)$
47. $\ln\left(2 + \sqrt{3}\right)$ **49.** $\bar{x} = 0, \quad \bar{y} = \dfrac{1}{\ln\left(3 + 2\sqrt{2}\right)}$
51. $xe^{x^3} + C$ **53.** $\frac{1}{30}(x^4 + 1)^{3/2}(3x^4 - 2) + C$

SECTION 8.2, pp. 475–478

1. $-2x\cos(x/2) + 4\sin(x/2) + C$
3. $t^2\sin t + 2t\cos t - 2\sin t + C$
5. $\ln 4 - \frac{3}{4}$ **7.** $xe^x - e^x + C$
9. $-(x^2 + 2x + 2)e^{-x} + C$
11. $y\tan^{-1}(y) - \ln\sqrt{1+y^2} + C$
13. $x\tan x + \ln|\cos x| + C$
15. $(x^3 - 3x^2 + 6x - 6)e^x + C$ **17.** $(x^2 - 7x + 7)e^x + C$
19. $(x^5 - 5x^4 + 20x^3 - 60x^2 + 120x - 120)e^x + C$
21. $\frac{1}{2}(-e^\theta\cos\theta + e^\theta\sin\theta) + C$
23. $\frac{e^{2x}}{13}(3\sin 3x + 2\cos 3x) + C$
25. $\frac{2}{3}\left(\sqrt{3s+9}\,e^{\sqrt{3s+9}} - e^{\sqrt{3s+9}}\right) + C$
27. $\frac{\pi\sqrt{3}}{3} - \ln(2) - \frac{\pi^2}{18}$
29. $\frac{1}{2}[-x\cos(\ln x) + x\sin(\ln x)] + C$
31. $\frac{1}{2}\ln|\sec x^2 + \tan x^2| + C$
33. $\frac{1}{2}x^2(\ln x)^2 - \frac{1}{2}x^2\ln x + \frac{1}{4}x^2 + C$
35. $-\frac{1}{x}\ln x - \frac{1}{x} + C$ **37.** $\frac{1}{4}e^{x^4} + C$
39. $\frac{1}{3}x^2(x^2+1)^{3/2} - \frac{2}{15}(x^2+1)^{5/2} + C$
41. $-\frac{2}{5}\sin 3x\sin 2x - \frac{3}{5}\cos 3x\cos 2x + C$
43. $\frac{2}{9}x^{3/2}(3\ln x - 2) + C$
45. $2\sqrt{x}\sin\sqrt{x} + 2\cos\sqrt{x} + C$
47. $\frac{\pi^2 - 4}{8}$ **49.** $\frac{5\pi - 3\sqrt{3}}{9}$
51. $\frac{1}{2}(x^2+1)\tan^{-1}x - \frac{x}{2} + C$ **53.** $xe^{x^2} + C$
55. $(2/3)x^{3/2}\arcsin\left(\sqrt{x}\right) + (2/9)x\sqrt{1-x} + (4/9)\sqrt{1-x} + C$
57. **(a)** π **(b)** 3π **(c)** 5π **(d)** $(2n+1)\pi$
59. $2\pi(1 - \ln 2)$ **61.** **(a)** $\pi(\pi - 2)$ **(b)** 2π
63. **(a)** 1 **(b)** $(e-2)\pi$ **(c)** $\frac{\pi}{2}(e^2 + 9)$
(d) $\bar{x} = \frac{1}{4}(e^2+1), \bar{y} = \frac{1}{2}(e-2)$
65. $\frac{1}{2\pi}(1 - e^{-2\pi})$ **67.** $u = x^n, dv = \cos x\,dx$
69. $u = x^n, dv = e^{ax}\,dx$ **73.** $u = x^n, dv = (x+1)^{-(1/2)}\,dx$
77. $x\sin^{-1}x + \cos(\sin^{-1}x) + C$
79. $x\sec^{-1}x - \ln|x + \sqrt{x^2-1}| + C$ **81.** Yes
83. **(a)** $x\sinh^{-1}x - \cosh(\sinh^{-1}x) + C$
(b) $x\sinh^{-1}x - (1+x^2)^{1/2} + C$

SECTION 8.3, pp. 483–484

1. $\frac{1}{2}\sin 2x + C$ **3.** $-\frac{1}{4}\cos^4 x + C$
5. $\frac{1}{3}\cos^3 x - \cos x + C$ **7.** $-\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C$
9. $\sin x - \frac{1}{3}\sin^3 x + C$ **11.** $\frac{1}{4}\sin^4 x - \frac{1}{6}\sin^6 x + C$
13. $\frac{1}{2}x + \frac{1}{4}\sin 2x + C$ **15.** $16/35$ **17.** 3π
19. $-4\sin x\cos^3 x + 2\cos x\sin x + 2x + C$
21. $-\cos^4 2\theta + C$ **23.** 4 **25.** 2
27. $\sqrt{\frac{3}{2}} - \frac{2}{3}$ **29.** $\frac{4}{5}\left(\frac{3}{2}\right)^{5/2} - \frac{18}{35} - \frac{2}{7}\left(\frac{3}{2}\right)^{7/2}$ **31.** $\sqrt{2}$
33. $\frac{1}{2}\tan^2 x + C$ **35.** $\frac{1}{3}\sec^3 x + C$ **37.** $\frac{1}{3}\tan^3 x + C$
39. $2\sqrt{3} + \ln\left(2 + \sqrt{3}\right)$ **41.** $\frac{2}{3}\tan\theta + \frac{1}{3}\sec^2\theta\tan\theta + C$
43. $4/3$ **45.** $2\tan^2 x - 2\ln(1 + \tan^2 x) + C$
47. $\frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \ln|\sec x| + C$ **49.** $\frac{4}{3} - \ln\sqrt{3}$
51. $-\frac{1}{10}\cos 5x - \frac{1}{2}\cos x + C$ **53.** π
55. $\frac{1}{2}\sin x + \frac{1}{14}\sin 7x + C$
57. $\frac{1}{6}\sin 3\theta - \frac{1}{4}\sin\theta - \frac{1}{20}\sin 5\theta + C$
59. $-\frac{2}{5}\cos^5\theta + C$ **61.** $\frac{1}{4}\cos\theta - \frac{1}{20}\cos 5\theta + C$
63. $\sec x - \ln|\csc x + \cot x| + C$ **65.** $\cos x + \sec x + C$
67. $\frac{1}{4}x^2 - \frac{1}{4}x\sin 2x - \frac{1}{8}\cos 2x + C$ **69.** $\ln\left(2 + \sqrt{3}\right)$
71. $\pi^2/2$ **73.** $\bar{x} = \frac{4\pi}{3}, \bar{y} = \frac{8\pi^2 + 3}{12\pi}$ **75.** $(\pi/4)(4 - \pi)$

SECTION 8.4, pp. 488–489

1. $\ln|\sqrt{9 + x^2} + x| + C$ **3.** $\pi/4$ **5.** $\pi/6$
7. $\frac{25}{2}\sin^{-1}\left(\frac{t}{5}\right) + \frac{t\sqrt{25 - t^2}}{2} + C$
9. $\frac{1}{2}\ln\left|\frac{2x}{7} + \frac{\sqrt{4x^2 - 49}}{7}\right| + C$
11. $7\left[\frac{\sqrt{y^2 - 49}}{7} - \sec^{-1}\left(\frac{y}{7}\right)\right] + C$ **13.** $\frac{\sqrt{x^2 - 1}}{x} + C$
15. $-\sqrt{9 - x^2} + C$ **17.** $\frac{1}{3}(x^2 + 4)^{3/2} - 4\sqrt{x^2 + 4} + C$
19. $\frac{-2\sqrt{4 - w^2}}{w} + C$ **21.** $\sin^{-1}x - \sqrt{1 - x^2} + C$
23. $4\sqrt{3} - \frac{4\pi}{3}$ **25.** $-\frac{x}{\sqrt{x^2 - 1}} + C$

27. $-\frac{1}{5}\left(\frac{\sqrt{1-x^2}}{x}\right)^5 + C$ **29.** $2\tan^{-1}2x + \frac{4x}{(4x^2+1)} + C$

31. $\frac{1}{2}x^2 + \frac{1}{2}\ln|x^2-1| + C$ **33.** $\frac{1}{3}\left(\frac{v}{\sqrt{1-v^2}}\right)^3 + C$

35. $\ln 9 - \ln\left(1+\sqrt{10}\right)$ **37.** $\pi/6$ **39.** $\sec^{-1}|x| + C$

41. $\sqrt{x^2-1} + C$ **43.** $\frac{1}{2}\ln\left|\sqrt{1+x^4}+x^2\right| + C$

45. $4\sin^{-1}\frac{\sqrt{x}}{2} + \sqrt{x}\sqrt{4-x} + C$

47. $\frac{1}{4}\sin^{-1}\sqrt{x} - \frac{1}{4}\sqrt{x}\sqrt{1-x}(1-2x) + C$

49. $(9/2)\arcsin\left(\frac{x+1}{3}\right) + (1/2)(x+1)\sqrt{8-2x-x^2} + C$

51. $\sqrt{x^2+4x+3} - \operatorname{arcsec}(x+2) + C$

53. $y = 2\left[\frac{\sqrt{x^2-4}}{2} - \sec^{-1}\left(\frac{x}{2}\right)\right]$

55. $y = \frac{3}{2}\tan^{-1}\left(\frac{x}{2}\right) - \frac{3\pi}{8}$ **57.** $3\pi/4$

59. **(a)** $\frac{1}{12}(\pi + 6\sqrt{3} - 12)$

(b) $\bar{x} = \frac{3\sqrt{3}-\pi}{4(\pi+6\sqrt{3}-12)}, \bar{y} = \frac{\pi^2+12\sqrt{3}\pi-72}{12(\pi+6\sqrt{3}-12)}$

61. **(a)** $-\frac{1}{3}x^2(1-x^2)^{3/2} - \frac{2}{15}(1-x^2)^{5/2} + C$

(b) $-\frac{1}{3}(1-x^2)^{3/2} + \frac{1}{5}(1-x^2)^{5/2} + C$

(c) $\frac{1}{5}(1-x^2)^{5/2} - \frac{1}{3}(1-x^2)^{3/2} + C$

63. $\sqrt{3} - \frac{\sqrt{2}}{2} + \frac{1}{2}\ln\left(\frac{2+\sqrt{3}}{\sqrt{2}+1}\right)$

SECTION 8.5, pp. 495–497

1. $\frac{2}{x-3} + \frac{3}{x-2}$ **3.** $\frac{1}{x+1} + \frac{3}{(x+1)^2}$

5. $\frac{-2}{z} + \frac{-1}{z^2} + \frac{2}{z-1}$ **7.** $1 + \frac{17}{t-3} + \frac{-12}{t-2}$

9. $\frac{1}{2}[\ln|1+x| - \ln|1-x|] + C$

11. $\frac{1}{7}\ln|(x+6)^2(x-1)^5| + C$ **13.** $(\ln 15)/2$

15. $-\frac{1}{2}\ln|t| + \frac{1}{6}\ln|t+2| + \frac{1}{3}\ln|t-1| + C$ **17.** $3\ln 2 - 2$

19. $\frac{1}{4}\ln\left|\frac{x+1}{x-1}\right| - \frac{x}{2(x^2-1)} + C$ **21.** $(\pi + 2\ln 2)/8$

23. $\tan^{-1}y - \frac{1}{y^2+1} + C$

25. $-(s-1)^{-2} + (s-1)^{-1} + \tan^{-1}s + C$

27. $\frac{2}{3}\ln|x-1| + \frac{1}{6}\ln|x^2+x+1| - \sqrt{3}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$

29. $\frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| + \frac{1}{2}\tan^{-1}x + C$

31. $\frac{-1}{\theta^2+2\theta+2} + \ln(\theta^2+2\theta+2) - \tan^{-1}(\theta+1) + C$

33. $x^2 + \ln\left|\frac{x-1}{x}\right| + C$

35. $9x + 2\ln|x| + \frac{1}{x} + 7\ln|x-1| + C$

37. $\frac{y^2}{2} - \ln|y| + \frac{1}{2}\ln(1+y^2) + C$ **39.** $\ln\left(\frac{e^t+1}{e^t+2}\right) + C$

41. $\frac{1}{5}\ln\left|\frac{\sin y - 2}{\sin y + 3}\right| + C$

43. $\frac{(\tan^{-1}2x)^2}{4} - 3\ln|x-2| + \frac{6}{x-2} + C$

45. $\ln\left|\frac{\sqrt{x}-1}{\sqrt{x}+1}\right| + C$

47. $2\sqrt{1+x} + \ln\left|\frac{\sqrt{x+1}-1}{\sqrt{x+1}+1}\right| + C$

49. $\frac{1}{4}\ln\left|\frac{x^4}{x^4+1}\right| + C$

51. $\frac{1}{\sqrt{2}}\ln\left|\frac{\sqrt{2}\cos\theta+1}{\sqrt{2}\cos\theta-1}\right| + \frac{1}{2}\ln\left|\frac{1-\cos\theta}{1+\cos\theta}\right| + C$

53. $4\sqrt{1+\sqrt{x}} + 2\ln\left|\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}\right| + C$

55. $\frac{1}{3}x^3 - 2x^2 + 5x - 10\ln|x+2| + C$

57. $\frac{1}{\ln 2}\ln(2^x + 2^{-x}) + C$ **59.** $\frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| - \frac{1}{2}\arctan x + C$

61. $\frac{1}{2}\ln|(\ln x+1)(\ln x+3)| + C$

63. $\ln|x + \sqrt{x^2-1}| + C$

65. $\frac{2}{9}x^3(x^3+1)^{3/2} - \frac{4}{45}(x^3+1)^{5/2} + C$

67. $x = \ln|t-2| - \ln|t-1| + \ln 2$

69. $x = \frac{6t}{t+2} - 1$ **71.** $3\pi\ln 25$

73. $\ln(3) - \frac{1}{2}$ **75.** 1.10

77. **(a)** $x = \frac{1000e^{4t}}{499+e^{4t}}$ **(b)** 1.55 days

SECTION 8.6, pp. 501–503

1. $\frac{2}{\sqrt{3}}\left(\tan^{-1}\sqrt{\frac{x-3}{3}}\right) + C$

3. $\sqrt{x-2}\left(\frac{2(x-2)}{3}+4\right) + C$ **5.** $\frac{(2x-3)^{3/2}(x+1)}{5} + C$

7. $\frac{-\sqrt{9-4x}}{x} - \frac{2}{3}\ln\left|\frac{\sqrt{9-4x}-3}{\sqrt{9-4x}+3}\right| + C$

9. $\frac{(x+2)(2x-6)\sqrt{4x-x^2}}{6} + 4\sin^{-1}\left(\frac{x-2}{2}\right) + C$

11. $-\frac{1}{\sqrt{7}}\ln\left|\frac{\sqrt{7}+\sqrt{7+x^2}}{x}\right| + C$

13. $\sqrt{4-x^2} - 2\ln\left|\frac{2+\sqrt{4-x^2}}{x}\right| + C$

15. $\frac{e^{2t}}{13}(2\cos 3t + 3\sin 3t) + C$

17. $\frac{x^2}{2}\cos^{-1}x + \frac{1}{4}\sin^{-1}x - \frac{1}{4}x\sqrt{1-x^2} + C$

19. $\frac{x^3}{3}\tan^{-1}x - \frac{x^2}{6} + \frac{1}{6}\ln(1+x^2) + C$

21. $-\frac{\cos 5x}{10} - \frac{\cos x}{2} + C$

23. $8\left[\frac{\sin(7t/2)}{7} - \frac{\sin(9t/2)}{9}\right] + C$

25. $6\sin(\theta/12) + \frac{6}{7}\sin(7\theta/12) + C$

27. $\frac{1}{2}\ln(x^2+1) + \frac{x}{2(1+x^2)} + \frac{1}{2}\tan^{-1}x + C$

29. $\left(x - \frac{1}{2}\right)\sin^{-1}\sqrt{x} + \frac{1}{2}\sqrt{x - x^2} + C$

31. $\sin^{-1}\sqrt{x} - \sqrt{x - x^2} + C$

33. $\sqrt{1 - \sin^2 t} - \ln\left|\frac{1 + \sqrt{1 - \sin^2 t}}{\sin t}\right| + C$

35. $\ln\left|\ln y + \sqrt{3 + (\ln y)^2}\right| + C$

37. $\ln\left|x + 1 + \sqrt{x^2 + 2x + 5}\right| + C$

39. $\frac{x+2}{2}\sqrt{5 - 4x - x^2} + \frac{9}{2}\sin^{-1}\left(\frac{x+2}{3}\right) + C$

41. $-\frac{\sin^4 2x\cos 2x}{10} - \frac{2\sin^2 2x\cos 2x}{15} - \frac{4\cos 2x}{15} + C$

43. $\frac{\sin^3 2\theta\cos^2 2\theta}{10} + \frac{\sin^3 2\theta}{15} + C$

45. $\tan^2 2x - 2\ln|\sec 2x| + C$

47. $\frac{(\sec \pi x)(\tan \pi x)}{\pi} + \frac{1}{\pi}\ln|\sec \pi x + \tan \pi x| + C$

49. $\frac{-\csc^3 x\cot x}{4} - \frac{3\csc x\cot x}{8} - \frac{3}{8}\ln|\csc x + \cot x| + C$

51. $\frac{1}{2}[\sec(e^t - 1)\tan(e^t - 1) + \ln|\sec(e^t - 1) + \tan(e^t - 1)|] + C$

53. $\sqrt{2} + \ln\left(\sqrt{2} + 1\right)$ **55.** $\pi/3$

57. $2\pi\sqrt{3} + \pi\sqrt{2}\ln\left(\sqrt{2} + \sqrt{3}\right)$ **59.** $\bar{x} = 4/3, \bar{y} = \ln\sqrt{2}$

61. 8 **63.** $\pi/8$ **67.** $\pi/4$

SECTION 8.7, pp. 510–512

1. I: **(a)** 1.5, 0 **(b)** 1.5, 0 **(c)** 0%
II: **(a)** 1.5, 0 **(b)** 1.5, 0 **(c)** 0%

3. I: **(a)** 2.75, 0.08 **(b)** 2.67, 0.08 **(c)** 0.0312 ≈ 3%
II: **(a)** 2.67, 0 **(b)** 2.67, 0 **(c)** 0%

5. I: **(a)** 6.25, 0.5 **(b)** 6, 0.25 **(c)** 0.0417 ≈ 4%
II: **(a)** 6, 0 **(b)** 6, 0 **(c)** 0%

7. I: **(a)** 0.509, 0.03125 **(b)** 0.5, 0.009 **(c)** 0.018 ≈ 2%
II: **(a)** 0.5, 0.002604 **(b)** 0.5, 0.4794 **(c)** 0%

9. I: **(a)** 1.8961, 0.161 **(b)** 2, 0.1039 **(c)** 0.052 ≈ 5%
II: **(a)** 2.0045, 0.0066 **(b)** 2, 0.00454 **(c)** 0.2%

11. **(a)** 1 **(b)** 2 **13.** **(a)** 116 **(b)** 2
15. **(a)** 283 **(b)** 2 **17.** **(a)** 71 **(b)** 10
19. **(a)** 76 **(b)** 12 **21.** **(a)** 82 **(b)** 8
23. 106.6 m³ **25.** ≈2.55 m
27. **(a)** ≈0.00021 **(b)** ≈1.37079 **(c)** ≈0.015%
31. **(a)** ≈5.870 **(b)** $|E_T| \le 0.0032$
33. 21.07 cm **35.** 14.4 **39.** ≈28.7 mg

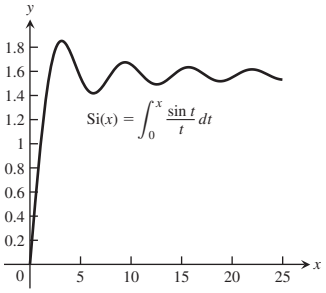
SECTION 8.8, pp. 521–523

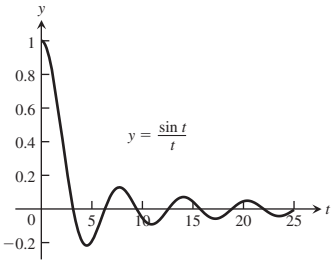
1. $\pi/2$ 3. 2 5. 6 7. $\pi/2$ 9. ln 3 **11.** ln 4

13. 0 **15.** $\sqrt{3}$ **17.** π **19.** $\ln\left(1 + \frac{\pi}{2}\right)$

21. −1 **23.** 1 **25.** −1/4 **27.** $\pi/2$ **29.** $\pi/3$

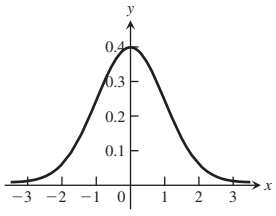
31. 6 **33.** ln 2 **35.** Diverges **37.** Diverges
39. Diverges **41.** Diverges **43.** Converges
45. Converges **47.** Diverges **49.** Converges
51. Converges **53.** Diverges **55.** Converges
57. Converges **59.** Diverges **61.** Converges
63. Diverges **65.** Converges **67.** Converges
69. **(a)** Converges when $p < 1$ **(b)** Converges when $p > 1$
71. 1 **73.** 2π **75.** ln 2
77. **(a)** 1 **(b)** $\pi/3$ **(c)** Diverges
79. **(a)** $\pi/2$ **(b)** π **81.** **(b)** ≈0.88621
83. **(a)**





(b) $\pi/2$

85. **(a)**



(b) ≈0.683, ≈0.954, ≈0.997

91. ≈0.16462

PRACTICE EXERCISES, pp. 524–526

1. $(x+1)(\ln(x+1)) - (x+1) + C$

3. $x\tan^{-1}(3x) - \frac{1}{6}\ln(1 + 9x^2) + C$

5. $(x+1)^2e^x - 2(x+1)e^x + 2e^x + C$

7. $\frac{2e^x\sin 2x}{5} + \frac{e^x\cos 2x}{5} + C$

9. $2\ln|x-2| - \ln|x-1| + C$

11. $\ln|x| - \ln|x+1| + \frac{1}{x+1} + C$

13. $-\frac{1}{3}\ln\left|\frac{\cos\theta - 1}{\cos\theta + 2}\right| + C$

15. $4\ln|x| - \frac{1}{2}\ln(x^2+1) + 4\tan^{-1}x + C$

17. $\frac{1}{16}\ln\left|\frac{(v-2)^5(v+2)}{v^6}\right| + C$

19. $\frac{1}{2}\tan^{-1}t - \frac{\sqrt{3}}{6}\tan^{-1}\frac{t}{\sqrt{3}} + C$

21. $\frac{x^2}{2} + \frac{4}{3}\ln|x+2| + \frac{2}{3}\ln|x-1| + C$

23. $\frac{x^2}{2} - \frac{9}{2}\ln|x+3| + \frac{3}{2}\ln|x+1| + C$

25. $\frac{1}{3}\ln\left|\frac{\sqrt{x+1} - 1}{\sqrt{x+1} + 1}\right| + C$ **27.** $\ln|1 - e^{-s}| + C$

29. $-\sqrt{16 - y^2} + C$ **31.** $-\frac{1}{2}\ln|4 - x^2| + C$

33. $\ln \dfrac{1}{\sqrt{9-x^2}} + C$ **35.** $\dfrac{1}{6}\ln\left|\dfrac{x+3}{x-3}\right| + C$

37. $-\dfrac{\cos^5 x}{5} + \dfrac{\cos^7 x}{7} + C$ **39.** $\dfrac{\tan^5 x}{5} + C$

41. $\dfrac{\cos\theta}{2} - \dfrac{\cos 11\theta}{22} + C$ **43.** $4\sqrt{1-\cos(t/2)} + C$

45. At least 16 **47.** $T = \pi, S = \pi$ **49.** $-4°\text{C}$

51. **(a)** ≈ 2.42 L **(b)** ≈ 24.83 km/L

53. $\pi/2$ **55.** 6 **57.** $\ln 3$ **59.** 2 **61.** $\pi/6$

63. Diverges **65.** Diverges **67.** Converges

69. $\dfrac{1}{2}xe^{2x} - \dfrac{1}{4}e^{2x} + C$ 71. $2\tan x - x + C$

73. $x\tan x - \ln|\sec x| + C$ **75.** $-\dfrac{1}{3}(\cos x)^3 + C$

77. $1 + \dfrac{1}{2}\ln\left(\dfrac{2}{1+e^2}\right)$ **79.** $2\ln\left|1-\dfrac{1}{x}\right| + \dfrac{4x+1}{2x^2} + C$

81. $\dfrac{e^{2x}-1}{e^x} + C$ **83.** $9/4$ **85.** $256/15$

87. $-\dfrac{1}{3}\csc^3 x + C$

89. $\dfrac{2x^{3/2}}{3} - x + 2\sqrt{x} - 2\ln\left(\sqrt{x}+1\right) + C$

91. $\dfrac{1}{2}\sin^{-1}(x-1) + \dfrac{1}{2}(x-1)\sqrt{2x-x^2} + C$

93. $-2\cot x - \ln|\csc x + \cot x| + \csc x + C$

95. $\dfrac{1}{12}\ln\left|\dfrac{3+v}{3-v}\right| + \dfrac{1}{6}\tan^{-1}\dfrac{v}{3} + C$

97. $\dfrac{\theta\sin(2\theta+1)}{2} + \dfrac{\cos(2\theta+1)}{4} + C$

99. $\dfrac{1}{4}\sec^2\theta + C$ **101.** $2\left(\dfrac{\left(\sqrt{2-x}\right)^3}{3} - 2\sqrt{2-x}\right) + C$

103. $\tan^{-1}(y-1) + C$

105. $\dfrac{1}{4}\ln|z| - \dfrac{1}{4z} - \dfrac{1}{4}\left[\dfrac{1}{2}\ln(z^2+4) + \dfrac{1}{2}\tan^{-1}\left(\dfrac{z}{2}\right)\right] + C$

107. $-\dfrac{1}{4}\sqrt{9-4t^2} + C$ **109.** $\ln\left(\dfrac{e^t+1}{e^t+2}\right) + C$

111. $1/4$ **113.** $\dfrac{2}{3}x^{3/2} + C$ **115.** $-\dfrac{1}{5}\tan^{-1}(\cos 5t) + C$

117. $2\sqrt{r} - 2\ln\left(1+\sqrt{r}\right) + C$

119. $\dfrac{1}{2}x^2 - \dfrac{1}{2}\ln(x^2+1) + C$

121. $\dfrac{2}{3}\ln|x+1| + \dfrac{1}{6}\ln|x^2-x+1| + \dfrac{1}{\sqrt{3}}\tan^{-1}\left(\dfrac{2x-1}{\sqrt{3}}\right) + C$

123. $\dfrac{4}{7}\left(1+\sqrt{x}\right)^{7/2} - \dfrac{8}{5}\left(1+\sqrt{x}\right)^{5/2} + \dfrac{4}{3}\left(1+\sqrt{x}\right)^{3/2} + C$

125. $2\ln\left|\sqrt{x} + \sqrt{1+x}\right| + C$

127. $\ln x - \ln|1 + \ln x| + C$

129. $\dfrac{1}{2}x^{\ln x} + C$ **131.** $\dfrac{1}{2}\ln\left|\dfrac{1-\sqrt{1-x^4}}{x^2}\right| + C$

133. **(b)** $\dfrac{\pi}{4}$ **135.** $x - \dfrac{1}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\tan x\right) + C$

ADDITIONAL AND ADVANCED EXERCISES, pp. 526–529

1. $x(\sin^{-1}x)^2 + 2(\sin^{-1}x)\sqrt{1-x^2} - 2x + C$

3. $\dfrac{x^2\sin^{-1}x}{2} + \dfrac{x\sqrt{1-x^2} - \sin^{-1}x}{4} + C$

5. $\dfrac{1}{2}\left(\ln\left(t - \sqrt{1-t^2}\right) - \sin^{-1}t\right) + C$ **7.** 0

9. $\ln(4) - 1$ **11.** 1 **13.** $32\pi/35$ **15.** 2π

17. **(a)** π **(b)** $\pi(2e-5)$

19. **(b)** $\pi\left(\dfrac{8(\ln 2)^2}{3} - \dfrac{16(\ln 2)}{9} + \dfrac{16}{27}\right)$

21. $\left(\dfrac{e^2+1}{4}, \dfrac{e-2}{2}\right)$

23. $\sqrt{1+e^2} - \ln\left(\dfrac{\sqrt{1+e^2}}{e} + \dfrac{1}{e}\right) - \sqrt{2} + \ln\left(1+\sqrt{2}\right)$

25. $\dfrac{12\pi}{5}$ **27.** $a = \dfrac{1}{2}, -\dfrac{\ln 2}{4}$ **29.** $\dfrac{1}{2} < p \le 1$

33. $\dfrac{2}{1-\tan(x/2)} + C$ **35.** 1 **37.** $\dfrac{\sqrt{3}\pi}{9}$

39. $\dfrac{1}{\sqrt{2}}\ln\left|\dfrac{\tan(t/2)+1-\sqrt{2}}{\tan(t/2)+1+\sqrt{2}}\right| + C$

41. $\ln\left|\dfrac{1+\tan(\theta/2)}{1-\tan(\theta/2)}\right| + C$

Chapter 9

SECTION 9.1, pp. 539–543

1. $a_1 = 0, a_2 = -1/4, a_3 = -2/9, a_4 = -3/16$

3. $a_1 = 1, a_2 = -1/3, a_3 = 1/5, a_4 = -1/7$

5. $a_1 = 1/2, a_2 = 1/2, a_3 = 1/2, a_4 = 1/2$

7. $1, \dfrac{3}{2}, \dfrac{7}{4}, \dfrac{15}{8}, \dfrac{31}{16}, \dfrac{63}{32}, \dfrac{127}{64}, \dfrac{255}{128}, \dfrac{511}{256}, \dfrac{1023}{512}$

9. $2, 1, -\dfrac{1}{2}, -\dfrac{1}{4}, \dfrac{1}{8}, \dfrac{1}{16}, -\dfrac{1}{32}, -\dfrac{1}{64}, \dfrac{1}{128}, \dfrac{1}{256}$

11. 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

13. $a_n = (-1)^{n+1}, n \ge 1$

15. $a_n = (-1)^{n+1}(n)^2, n \ge 1$ **17.** $a_n = \dfrac{2^{n-1}}{3(n+2)}, n \ge 1$

19. $a_n = n^2 - 1, n \ge 1$ **21.** $a_n = 4n - 3, n \ge 1$

23. $a_n = \dfrac{3n+2}{n!}, n \ge 1$ **25.** $a_n = \dfrac{1+(-1)^{n+1}}{2}, n \ge 1$

27. $a_n = \dfrac{1}{(n+1)(n+2)}$ **29.** $a_n = \sin\left(\dfrac{\sqrt{n+1}}{1+(n+1)^2}\right)$

31. Converges, 2 **33.** Converges, -1 **35.** Converges, -5

37. Diverges **39.** Diverges **41.** Converges, $1/2$

43. Converges, 0 **45.** Converges, $\sqrt{2}$ **47.** Converges, 1

49. Converges, 0 **51.** Converges, 0 **53.** Converges, 0

55. Converges, 1 **57.** Converges, e^7 **59.** Converges, 1

61. Converges, 1 **63.** Diverges **65.** Converges, 4

67. Converges, 0 **69.** Diverges **71.** Converges, e^{-1}

73. Diverges **75.** Converges, 0 **77.** Diverges

79. Converges, $e^{2/3}$ **81.** Converges, x $(x > 0)$

83. Converges, 0 **85.** Converges, 1 **87.** Converges, $1/2$

89. Converges, 1 **91.** Converges, $\pi/2$ **93.** Converges, 0
95. Converges, 0 **97.** Converges, $1/2$ **99.** Converges, 0

101. 8 **103.** 4 **105.** 5 **107.** $1 + \sqrt{2}$ **109.** $x_n = 2^{n-2}$
111. **(a)** $f(x) = x^2 - 2$, $1.414213562 \approx \sqrt{2}$
(b) $f(x) = \tan(x) - 1$, $0.7853981635 \approx \pi/4$
(c) $f(x) = e^x$, diverges
113. 1
121. Nondecreasing, bounded
123. Not nondecreasing, bounded
125. Converges, nondecreasing sequence theorem
127. Converges, nondecreasing sequence theorem
129. Diverges, definition of divergence
131. Converges
133. Converges
145. **(b)** $\sqrt{3}$

SECTION 9.2, pp. 550–552

1. $s_n = \dfrac{2(1 - (1/3)^n)}{1 - (1/3)}$, 3 **3.** $s_n = \dfrac{1 - (-1/2)^n}{1 - (-1/2)}$, $2/3$

5. $s_n = \dfrac{1}{2} - \dfrac{1}{n+2}$, $\dfrac{1}{2}$ **7.** $1 - \dfrac{1}{4} + \dfrac{1}{16} - \dfrac{1}{64} + \cdots$, $\dfrac{4}{5}$

9. $-\dfrac{3}{4} + \dfrac{9}{16} + \dfrac{57}{64} + \dfrac{249}{256} + \cdots$, diverges.

11. $(5 + 1) + \left(\dfrac{5}{2} + \dfrac{1}{3}\right) + \left(\dfrac{5}{4} + \dfrac{1}{9}\right) + \left(\dfrac{5}{8} + \dfrac{1}{27}\right) + \cdots$, $\dfrac{23}{2}$

13. $(1 + 1) + \left(\dfrac{1}{2} - \dfrac{1}{5}\right) + \left(\dfrac{1}{4} + \dfrac{1}{25}\right) + \left(\dfrac{1}{8} - \dfrac{1}{125}\right) + \cdots$, $\dfrac{17}{6}$

15. Converges, $5/3$ **17.** Converges, $1/7$

19. Converges, $\dfrac{e}{e+2}$ **21.** Diverges **23.** $23/99$

25. $7/9$ **27.** $1/15$ **29.** $41333/33300$ **31.** Diverges
33. Inconclusive **35.** Diverges **37.** Diverges

39. $s_n = 1 - \dfrac{1}{n+1}$; converges, 1

41. $s_n = \ln\sqrt{n+1}$; diverges

43. $s_n = \dfrac{\pi}{3} - \cos^{-1}\left(\dfrac{1}{n+2}\right)$; converges, $-\dfrac{\pi}{6}$ **45.** 1 **47.** 5

49. 1 **51.** $-\dfrac{1}{\ln 2}$ **53.** Converges, $2 + \sqrt{2}$
55. Converges, 1 **57.** Diverges
59. Converges, $\dfrac{e^2}{e^2 - 1}$
61. Converges, $2/9$ **63.** Converges, $3/2$ **65.** Diverges
67. Converges, 4 **69.** Diverges **71.** Converges, $\dfrac{\pi}{\pi - e}$
73. Converges, $-5/6$ **75.** Diverges
77. $a = 1$, $r = -x$; converges to $1/(1 + x)$ for $|x| < 1$
79. $a = 3$, $r = (x - 1)/2$; converges to $6/(3 - x)$ for x in $(-1, 3)$

81. $|x| < \dfrac{1}{2}$, $\dfrac{1}{1 - 2x}$ **83.** $-2 < x < 0$, $\dfrac{1}{2 + x}$

85. $x \neq (2k + 1)\dfrac{\pi}{2}$, k an integer; $\dfrac{1}{1 - \sin x}$

87. **(a)** $\displaystyle\sum_{n=-2}^{\infty} \frac{1}{(n+4)(n+5)}$ **(b)** $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(n+2)(n+3)}$
(c) $\displaystyle\sum_{n=5}^{\infty} \frac{1}{(n-3)(n-2)}$

97. **(a)** $r = 3/5$ **(b)** $r = -3/10$ **99.** $|r| < 1$, $\dfrac{1 + 2r}{1 - r^2}$

101. **(a)** 16.84 mg, 17.79 mg **(b)** 17.84 mg

103. **(a)** $0, \dfrac{1}{27}, \dfrac{2}{27}, \dfrac{1}{9}, \dfrac{2}{9}, \dfrac{7}{27}, \dfrac{8}{27}, \dfrac{1}{3}, \dfrac{2}{3}, \dfrac{7}{9}, \dfrac{8}{9}, 1$
(b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{2}{3}\right)^{n-1} = 1$ **105.** $(4/3)\pi$

SECTION 9.3, pp. 557–559

1. Converges **3.** Converges **5.** Converges **7.** Diverges

9. Converges **11.** Diverges

13. Diverges; $\displaystyle\lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$

15. Diverges; p-series, $p < 1$

17. Converges; geometric series, $r = \dfrac{1}{8} < 1$

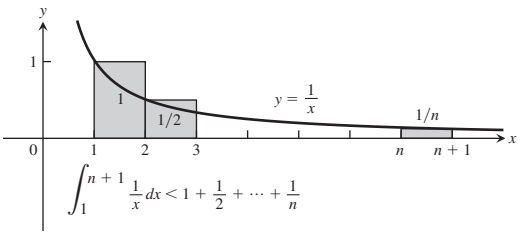
19. Converges; geometric series, $r = \dfrac{1}{8} < 1$
21. Diverges; Integral Test
23. Converges; geometric series, $r = 2/3 < 1$
25. Diverges; Integral Test

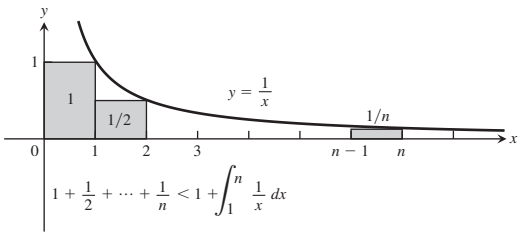
27. Diverges; $\displaystyle\lim_{n\to\infty} \frac{2^n}{n+1} \neq 0$

29. Diverges; $\lim_{n\to\infty}\left(\sqrt{n}/\ln n\right) \neq 0$

31. Diverges; geometric series, $r = \dfrac{1}{\ln 2} > 1$
33. Converges; Integral Test
35. Diverges; nth-Term Test
37. Converges; Integral Test
39. Diverges; nth-Term Test
41. Converges; by taking limit of partial sums
43. Converges; Integral Test
45. Converges; Integral Test **47.** $a = 1$

49. **(a)**





(b) ≈ 41.55
51. True **53.** $n \geq 251{,}415$

55. $s_8 = \displaystyle\sum_{n=1}^{8} \frac{1}{n^3} \approx 1.195$ **57.** 10^{60}

65. **(a)** $1.20166 \leq S \leq 1.20253$
(b) $S \approx 1.2021$, error < 0.0005

67. $\left(\dfrac{\pi^2}{6} - 1\right) \approx 0.64493$

SECTION 9.4, pp. 563–564

1. Converges; compare with $\sum(1/n^2)$
3. Diverges; compare with $\sum\left(1/\sqrt{n}\right)$
5. Converges; compare with $\sum(1/n^{3/2})$
7. Converges; compare with $\sum\sqrt{\dfrac{n+4n}{n^4+0}} = \sqrt{5}\,\sum\dfrac{1}{n^{3/2}}$
9. Converges
11. Diverges; limit comparison with $\sum(1/n)$
13. Diverges; limit comparison with $\sum\left(1/\sqrt{n}\right)$
15. Diverges
17. Diverges; limit comparison with $\sum\left(1/\sqrt{n}\right)$
19. Converges; compare with $\sum(1/2^n)$
21. Diverges; nth-Term Test
23. Converges; compare with $\sum(1/n^2)$
25. Converges; $\left(\dfrac{n}{3n+1}\right)^n < \left(\dfrac{n}{3n}\right)^n = \left(\dfrac{1}{3}\right)^n$
27. Diverges; direct comparison with $\sum(1/n)$
29. Diverges; limit comparison with $\sum(1/n)$
31. Diverges; limit comparison with $\sum(1/n)$
33. Converges; compare with $\sum(1/n^{3/2})$
35. Converges; $\dfrac{1}{n2^n} \le \dfrac{1}{2^n}$ **37.** Converges; $\dfrac{1}{3^{n-1}+1} < \dfrac{1}{3^{n-1}}$
39. Converges; comparison with $\sum(1/5n^2)$
41. Diverges; comparison with $\sum(1/n)$
43. Converges; comparison with $\sum\dfrac{1}{n(n-1)}$ or limit comparison with $\sum(1/n^2)$
45. Diverges; limit comparison with $\sum(1/n)$
47. Converges; $\dfrac{\tan^{-1}n}{n^{1.1}} < \dfrac{\pi/2}{n^{1.1}}$
49. Converges; compare with $\sum(1/n^2)$
51. Diverges; limit comparison with $\sum(1/n)$
53. Converges; limit comparison with $\sum(1/n^2)$
55. Diverges nth-Term Test
67. Converges **69.** Converges **71.** Converges

SECTION 9.5, pp. 569–570

1. Converges **3.** Diverges **5.** Converges
7. Converges **9.** Converges **11.** Diverges
13. Converges **15.** Converges
17. Converges; Ratio Test **19.** Diverges; Ratio Test
21. Converges; Ratio Test
23. Converges; compare with $\sum(3/(1.25)^n)$
25. Diverges; $\lim\limits_{n\to\infty}\left(1-\dfrac{3}{n}\right)^n = e^{-3} \ne 0$
27. Converges; compare with $\sum(1/n^2)$
29. Diverges; compare with $\sum(1/(2n))$ **31.** Diverges; $a_n \not\to 0$
33. Converges; Ratio Test **35.** Converges; Ratio Test
37. Converges; Ratio Test **39.** Converges; Root Test
41. Converges; compare with $\sum(1/n^2)$
43. Converges; Ratio Test **45.** Diverges; Ratio Test
47. Converges; Ratio Test **49.** Diverges; Ratio Test
51. Converges; Ratio Test **53.** Converges; Ratio Test
55. Diverges; $a_n = \left(\dfrac{1}{3}\right)^{(1/n!)} \to 1$ **57.** Converges; Ratio Test
59. Diverges; Root Test **61.** Converges; Root Test
63. Converges; Ratio Test
65. **(a)** Diverges; nth-Term Test
(b) Diverges; Root Test
(c) Converges; Root Test
(d) Converges; Ratio Test
69. Yes

SECTION 9.6, pp. 575–577

1. Converges by Alternating Series Test
3. Converges; Alternating Series Test
5. Converges; Alternating Series Test
7. Diverges; $a_n \not\to 0$
9. Diverges; $a_n \not\to 0$
11. Converges; Alternating Series Test
13. Converges by Alternating Series Test
15. Converges absolutely. Series of absolute s is a convergent geometric series.
17. Converges conditionally; $1/\sqrt{n} \to 0$ but $\sum_{n=1}^{\infty}\dfrac{1}{\sqrt{n}}$ diverges.
19. Converges absolutely; compare with $\sum_{n=1}^{\infty}(1/n^2)$.
21. Converges conditionally; $1/(n+3) \to 0$ but $\sum_{n=1}^{\infty}\dfrac{1}{n+3}$ diverges (compare with $\sum_{n=1}^{\infty}(1/n)$).
23. Diverges; $\dfrac{3+n}{5+n} \to 1$
25. Converges conditionally; $\left(\dfrac{1}{n^2}+\dfrac{1}{n}\right) \to 0$ but $(1+n)/n^2 > 1/n$
27. Converges absolutely; Ratio Test
29. Converges absolutely by Integral Test
31. Diverges; $a_n \not\to 0$
33. Converges absolutely by Ratio Test
35. Converges absolutely, since $\left|\dfrac{\cos n\pi}{n\sqrt{n}}\right| = \left|\dfrac{(-1)^{n+1}}{n^{3/2}}\right| = \dfrac{1}{n^{3/2}}$ (convergent p-series)
37. Converges absolutely by Root Test
39. Diverges; $a_n \to \infty$
41. Converges conditionally; $\sqrt{n+1}-\sqrt{n} = 1/\left(\sqrt{n}+\sqrt{n+1}\right) \to 0$, but series of absolute values diverges $\left(\text{compare with } \sum\left(1/\sqrt{n}\right)\right)$.
43. Diverges, $a_n \to 1/2 \ne 0$
45. Converges absolutely; $\operatorname{sech} n = \dfrac{2}{e^n+e^{-n}} = \dfrac{2e^n}{e^{2n}+1} < \dfrac{2e^n}{e^{2n}} = \dfrac{2}{e^n}$, a term from a convergent geometric series.
47. Converges conditionally; $\sum(-1)^{n+1}\dfrac{1}{2(n+1)}$ converges by Alternating Series Test; $\sum\dfrac{1}{2(n+1)}$ diverges by limit comparison with $\sum(1/n)$.
49. $|\text{Error}| < 0.2$ **51.** $|\text{Error}| < 2\times10^{-11}$
53. $n \ge 31$ **55.** $n \ge 4$ **57.** Converges; Root Test
59. Converges; Limit of Partial Sums
61. Converges; Ratio Test **63.** Diverges; p-series Test
65. Converges; Root Test **67.** Converges; Limit Comparison Test
69. Diverges; Limit of Partial Sums
71. Diverges; Limit Comparison Test
73. Diverges; nth-Term Test **75.** Diverges; Limit of Partial Sums
77. Converges; Limit Comparison Test
79. Converges; Limit Comparison Test

81. Converges; Ratio Test
83. 0.54030 **85.** **(a)** $a_n \geq a_{n+1}$ **(b)** $-1/2$

SECTION 9.7, pp. 586–589

1. **(a)** $1, -1 < x < 1$ **(b)** $-1 < x < 1$ **(c)** none
3. **(a)** $1/4, -1/2 < x < 0$ **(b)** $-1/2 < x < 0$ **(c)** none
5. **(a)** $10, -8 < x < 12$ **(b)** $-8 < x < 12$ **(c)** none
7. **(a)** $1, -1 < x < 1$ **(b)** $-1 < x < 1$ **(c)** none
9. **(a)** $3, -3 \leq x \leq 3$ **(b)** $-3 \leq x \leq 3$ **(c)** none
11. **(a)** ∞, for all x **(b)** for all x **(c)** none
13. **(a)** $1/2, -1/2 < x < 1/2$ **(b)** $-1/2 < x < 1/2$ **(c)** none
15. **(a)** $1, -1 \leq x < 1$ **(b)** $-1 < x < 1$ **(c)** $x = -1$
17. **(a)** $5, -8 < x < 2$ **(b)** $-8 < x < 2$ **(c)** none
19. **(a)** $3, -3 < x < 3$ **(b)** $-3 < x < 3$ **(c)** none
21. **(a)** $1, -2 < x < 0$ **(b)** $-2 < x < 0$ **(c)** none
23. **(a)** $1, -1 < x < 1$ **(b)** $-1 < x < 1$ **(c)** none
25. **(a)** $0, x = 0$ **(b)** $x = 0$ **(c)** none
27. **(a)** $2, -4 < x \leq 0$ **(b)** $-4 < x < 0$ **(c)** $x = 0$
29. **(a)** $1, -1 \leq x \leq 1$ **(b)** $-1 \leq x \leq 1$ **(c)** none
31. **(a)** $1/4, 1 \leq x \leq 3/2$ **(b)** $1 \leq x \leq 3/2$ **(c)** none
33. **(a)** ∞, for all x **(b)** for all x **(c)** none
35. **(a)** $1, -1 \leq x < 1$ **(b)** $-1 < x < 1$ **(c)** -1
37. 3 **39.** 8 **41.** $-1/3 < x < 1/3, 1/(1 - 3x)$
43. $-1 < x < 3, 4/(3 + 2x - x^2)$
45. $0 < x < 16, 2/\left(4 - \sqrt{x}\right)$
47. $-\sqrt{2} < x < \sqrt{2}, 3/(2 - x^2)$
49. $\dfrac{2}{x} = \displaystyle\sum_{n=0}^{\infty} 2(-1)^n (x-1)^n, \ 0 < x < 2$

51. $\displaystyle\sum_{n=0}^{\infty} \left(-\tfrac{1}{3}\right)^n (x-5)^n, \ 2 < x < 8$

53. $1 < x < 5, \ 2/(x-1), \ \displaystyle\sum_{n=1}^{\infty} \left(-\tfrac{1}{2}\right)^n n(x-3)^{n-1},$
$1 < x < 5, \ -2/(x-1)^2$

55. **(a)** $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \dfrac{x^8}{8!} - \dfrac{x^{10}}{10!} + \cdots$; converges for all x
(b) Same answer as part (c)
(c) $2x - \dfrac{2^3x^3}{3!} + \dfrac{2^5x^5}{5!} - \dfrac{2^7x^7}{7!} + \dfrac{2^9x^9}{9!} - \dfrac{2^{11}x^{11}}{11!} + \cdots$

57. **(a)** $\dfrac{x^2}{2} + \dfrac{x^4}{12} + \dfrac{x^6}{45} + \dfrac{17x^8}{2520} + \dfrac{31x^{10}}{14175}, \ -\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$
(b) $1 + x^2 + \dfrac{2x^4}{3} + \dfrac{17x^6}{45} + \dfrac{62x^8}{315} + \cdots, \ -\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$

63. **(a)** T **(b)** T **(c)** F **(d)** T **(e)** N **(f)** F **(g)** N **(h)** T

SECTION 9.8, pp. 593–594

1. $P_0(x) = 1, P_1(x) = 1 + 2x, P_2(x) = 1 + 2x + 2x^2,$
$P_3(x) = 1 + 2x + 2x^2 + \dfrac{4}{3}x^3$

3. $P_0(x) = 0, P_1(x) = x - 1, P_2(x) = (x-1) - \dfrac{1}{2}(x-1)^2,$
$P_3(x) = (x-1) - \dfrac{1}{2}(x-1)^2 + \dfrac{1}{3}(x-1)^3$

5. $P_0(x) = \dfrac{1}{2}, P_1(x) = \dfrac{1}{2} - \dfrac{1}{4}(x-2),$
$P_2(x) = \dfrac{1}{2} - \dfrac{1}{4}(x-2) + \dfrac{1}{8}(x-2)^2,$
$P_3(x) = \dfrac{1}{2} - \dfrac{1}{4}(x-2) + \dfrac{1}{8}(x-2)^2 - \dfrac{1}{16}(x-2)^3$

7. $P_0(x) = \dfrac{\sqrt{2}}{2}, P_1(x) = \dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}\left(x - \dfrac{\pi}{4}\right),$
$P_2(x) = \dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}\left(x - \dfrac{\pi}{4}\right) - \dfrac{\sqrt{2}}{4}\left(x - \dfrac{\pi}{4}\right)^2,$
$P_3(x) = \dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}\left(x - \dfrac{\pi}{4}\right) - \dfrac{\sqrt{2}}{4}\left(x - \dfrac{\pi}{4}\right)^2$
$- \dfrac{\sqrt{2}}{12}\left(x - \dfrac{\pi}{4}\right)^3$

9. $P_0(x) = 2, P_1(x) = 2 + \dfrac{1}{4}(x-4),$
$P_2(x) = 2 + \dfrac{1}{4}(x-4) - \dfrac{1}{64}(x-4)^2,$
$P_3(x) = 2 + \dfrac{1}{4}(x-4) - \dfrac{1}{64}(x-4)^2 + \dfrac{1}{512}(x-4)^3$

11. $\displaystyle\sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \cdots$

13. $\displaystyle\sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots$

15. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{2n+1}}{(2n+1)!}$ **17.** $7\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ **19.** $\displaystyle\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

21. $x^4 - 2x^3 - 5x + 4$ **23.** $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n}}{(2n-1)!}$

25. $8 + 10(x-2) + 6(x-2)^2 + (x-2)^3$
27. $21 - 36(x+2) + 25(x+2)^2 - 8(x+2)^3 + (x+2)^4$

29. $\displaystyle\sum_{n=0}^{\infty} (-1)^n (n+1)(x-1)^n$ **31.** $\displaystyle\sum_{n=0}^{\infty} \frac{e^2}{n!}(x-2)^n$

33. $\displaystyle\sum_{n=0}^{\infty} (-1)^{n+1} \frac{2^{2n}}{(2n)!}\left(x - \frac{\pi}{4}\right)^{2n}$

35. $-1 - 2x - \dfrac{5}{2}x^2 - \cdots, -1 < x < 1$

37. $x^2 - \dfrac{1}{2}x^3 + \dfrac{1}{6}x^4 + \cdots, -1 < x < 1$

39. $x^4 + x^6 + \dfrac{x^8}{2} + \cdots, (-\infty, \infty)$

45. $L(x) = 0, Q(x) = -x^2/2$ **47.** $L(x) = 1, Q(x) = 1 + x^2/2$
49. $L(x) = x, Q(x) = x$

SECTION 9.9, pp. 600–601

1. $\displaystyle\sum_{n=0}^{\infty} \frac{(-5x)^n}{n!} = 1 - 5x + \frac{5^2x^2}{2!} - \frac{5^3x^3}{3!} + \cdots$

3. $\displaystyle\sum_{n=0}^{\infty} \frac{5(-1)^n(-x)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{5(-1)^{n+1}x^{2n+1}}{(2n+1)!}$
$= -5x + \dfrac{5x^3}{3!} - \dfrac{5x^5}{5!} + \dfrac{5x^7}{7!} + \cdots$

5. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n(5x^2)^{2n}}{(2n)!} = 1 - \frac{25x^4}{2!} + \frac{625x^8}{4!} - \cdots$

7. $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n}}{n} = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} + \cdots$

9. $\displaystyle\sum_{n=0}^{\infty} (-1)^n \left(\frac{3}{4}\right)^n x^{3n} = 1 - \frac{3}{4}x^3 + \frac{3^2}{4^2}x^6 - \frac{3^3}{4^3}x^9 + \cdots$

11. $\ln 3 + \displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^n x^n}{n} = \ln 3 + 2x - 2x^2 + \frac{8}{3}x^3 - \cdots$

13. $\displaystyle\sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} = x + x^2 + \frac{x^3}{2!} + \frac{x^4}{3!} + \frac{x^5}{4!} + \cdots$

15. $\sum_{n=2}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \cdots$

17. $x - \frac{\pi^2 x^3}{2!} + \frac{\pi^4 x^5}{4!} - \frac{\pi^6 x^7}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n} x^{2n+1}}{(2n)!}$

19. $1 + \sum_{n=1}^{\infty} \frac{(-1)^n (2x)^{2n}}{2 \cdot (2n)!} =$

$1 - \frac{(2x)^2}{2 \cdot 2!} + \frac{(2x)^4}{2 \cdot 4!} - \frac{(2x)^6}{2 \cdot 6!} + \frac{(2x)^8}{2 \cdot 8!} - \cdots$

21. $x^2 \sum_{n=0}^{\infty} (2x)^n = x^2 + 2x^3 + 4x^4 + \cdots$

23. $\sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + 4x^3 + \cdots$

25. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{4n-1}}{2n-1} = x^3 - \frac{x^7}{3} + \frac{x^{11}}{5} - \frac{x^{15}}{7} + \cdots$

27. $\sum_{n=0}^{\infty} \left(\frac{1}{n!} + (-1)^n\right) x^n = 2 + \frac{3}{2}x^2 - \frac{5}{6}x^3 + \frac{25}{24}x^4 - \cdots$

29. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{3n} = \frac{x^3}{3} - \frac{x^5}{6} + \frac{x^7}{9} - \cdots$

31. $x + x^2 + \frac{x^3}{3} - \frac{x^5}{30} + \cdots$

33. $x^2 - \frac{2}{3}x^4 + \frac{23}{45}x^6 - \frac{44}{105}x^8 + \cdots$

35. $1 + x + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \cdots$ **37.** $1 - \frac{x^2}{2} - \frac{x^3}{2} - \frac{x^4}{4} - \cdots$

39. $|\text{Error}| \le \frac{1}{10^4 \cdot 4!} < 4.2 \times 10^{-6}$

41. $|x| < (0.06)^{1/5} < 0.56968$

43. $|\text{Error}| < (10^{-3})^3/6 < 1.67 \times 10^{-10}, \quad -10^{-3} < x < 0$

45. $|\text{Error}| < (3^{0.1})(0.1)^3/6 < 1.87 \times 10^{-4}$

53. (a) $Q(x) = 1 + kx + \frac{k(k-1)}{2}x^2$ **(b)** $0 \le x < 100^{-1/3}$

SECTION 9.10, pp. 608–610

1. $1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$ **3.** $1 + 3x + 6x^2 + 10x^3$

5. $1 - x + \frac{3x^2}{4} - \frac{x^3}{2}$ **7.** $1 - \frac{x^3}{2} + \frac{3x^6}{8} - \frac{5x^9}{16}$

9. $1 + \frac{1}{2x} - \frac{1}{8x^2} + \frac{1}{16x^3}$

11. $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

13. $(1 - 2x)^3 = 1 - 6x + 12x^2 - 8x^3$

15. 0.0713362 **17.** 0.4969536 **19.** 0.0999445 **21.** 0.10000

23. $\frac{1}{13 \cdot 6!} \approx 0.00011$ **25.** $\frac{x^3}{3} - \frac{x^7}{7 \cdot 3!} + \frac{x^{11}}{11 \cdot 5!}$

27. (a) $\frac{x^2}{2} - \frac{x^4}{12}$

(b) $\frac{x^2}{2} - \frac{x^4}{3 \cdot 4} + \frac{x^6}{5 \cdot 6} - \frac{x^8}{7 \cdot 8} + \cdots + (-1)^{15} \frac{x^{32}}{31 \cdot 32}$

29. $1/2$ **31.** $-1/24$ **33.** $1/3$ **35.** -1 **37.** 2

39. $3/2$ **41.** e **43.** $\cos \frac{3}{4}$ **45.** $\frac{\sqrt{3}}{2}$ **47.** $\frac{x^3}{1-x}$

49. $\frac{x^3}{1+x^2}$ **51.** $\frac{-1}{(1+x)^2}$ **55.** 500 terms **57.** 4 terms

59. (a) $x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112}$, radius of convergence $= 1$

(b) $\frac{\pi}{2} - x - \frac{x^3}{6} - \frac{3x^5}{40} - \frac{5x^7}{112}$

61. $1 - 2x + 3x^2 - 4x^3 + \cdots$

67. (a) -1 **(b)** $(1/\sqrt{2})(1 + i)$ **(c)** $-i$

71. $x + x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5 + \cdots$, for all x

PRACTICE EXERCISES, pp. 611–613

1. Converges to 1 **3.** Converges to -1 **5.** Diverges
7. Converges to 0 **9.** Converges to 1 **11.** Converges o e^{-5}
13. Converges to 3 **15.** Converges to ln 2 **17.** Diverges
19. $1/6$ **21.** $3/2$ **23.** $e/(e-1)$ **25.** Diverges
27. Converges conditionally **29.** Converges conditionally
31. Converges absolutely **33.** Converges absolutely
35. Converges absolutely **37.** Converges absolutely
39. Converges absolutely **41.** Converges absolutely
43. Diverges
45. (a) $3, -7 \le x < -1$ **(b)** $-7 < x < -1$ **(c)** $x = -7$
47. (a) $1/3, 0 \le x \le 2/3$ **(b)** $0 \le x \le 2/3$ **(c)** None
49. (a) ∞, for all x **(b)** For all x **(c)** None
51. (a) $\sqrt{3}, -\sqrt{3} < x < \sqrt{3}$ **(b)** $-\sqrt{3} < x < \sqrt{3}$ **(c)** None
53. (a) $e, -e < x < e$ **(b)** $-e < x < e$ **(c)** Empty set

55. $\frac{1}{1+x}, \frac{1}{4}, \frac{4}{5}$ **57.** $\sin x, \pi, 0$ **59.** e^x, ln 2, 2 **61.** $\sum_{n=0}^{\infty} 2^n x^n$

63. $\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1} x^{2n+1}}{(2n+1)!}$ **65.** $\sum_{n=0}^{\infty} \frac{(-1)^n x^{10n/3}}{(2n)!}$ **67.** $\sum_{n=0}^{\infty} \frac{((\pi x)/2)^n}{n!}$

69. $2 - \frac{(x+1)}{2 \cdot 1!} + \frac{3(x+1)^2}{2^3 \cdot 2!} + \frac{9(x+1)^3}{2^5 \cdot 3!} + \cdots$

71. $\frac{1}{4} - \frac{1}{4^2}(x-3) + \frac{1}{4^3}(x-3)^2 - \frac{1}{4^4}(x-3)^3$

73. 0.4849171431 **75.** 0.4872223583 **77.** $7/2$ **79.** $1/12$
81. -2 **83.** $r = -3, s = 9/2$ **85.** $2/3$
87. $\ln\left(\frac{n+1}{2n}\right)$; the series converges to $\ln\left(\frac{1}{2}\right)$.
89. (a) ∞ **(b)** $a = 1, b = 0$ **91.** It converges.
99. (a) Converges; Limit Comparison Test
(b) Converges; Direct Comparison Test
(c) Diverges; nth-Term Test
101. 2

ADDITIONAL AND ADVANCED EXERCISES, pp. 613–615

1. Converges; Comparison Test
3. Diverges; nth-Term Test
5. Converges; Comparison Test
7. Diverges; nth-Term Test

9. With $a = \pi/3$, $\cos x = \frac{1}{2} - \frac{\sqrt{3}}{2}(x - \pi/3) - \frac{1}{4}(x - \pi/3)^2 + \frac{\sqrt{3}}{12}(x - \pi/3)^3 + \cdots$

11. With $a = 0$, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

13. With $a = 22\pi$, $\cos x = 1 - \frac{1}{2}(x - 22\pi)^2 + \frac{1}{4!}(x - 22\pi)^4 - \frac{1}{6!}(x - 22\pi)^6 + \cdots$

15. Converges, limit $= b$ **17.** $\pi/2$ **21.** $b = \pm\frac{1}{5}$
23. $a = 2, L = -7/6$ **27. (b)** Yes

31. (a) $\sum_{n=1}^{\infty} n x^{n-1}$ **(b)** 6 **(c)** $\frac{1}{q}$

33. (a) $R_n = C_0 e^{-kt_0}(1 - e^{-nkt_0})/(1 - e^{-kt_0})$,

$R = C_0(e^{-kt_0})/(1 - e^{-kt_0}) = C_0/(e^{kt_0} - 1)$

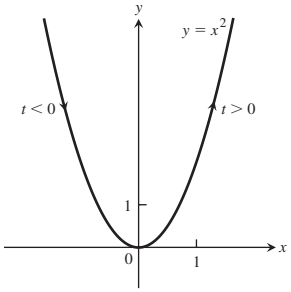
(b) $R_1 = 1/e \approx 0.368$,
$R_{10} = R(1 - e^{-10}) \approx R(0.9999546) \approx 0.58195$;
$R \approx 0.58198$; $0 < (R - R_{10})/R < 0.0001$

(c) 7

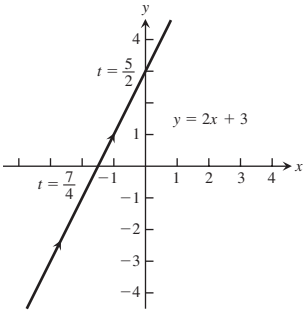
Chapter 10

SECTION 10.1, pp. 622–625

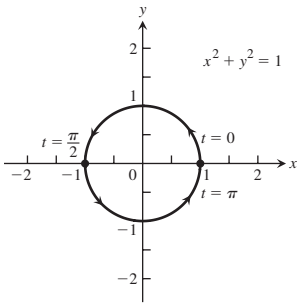
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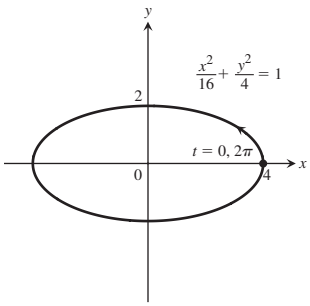
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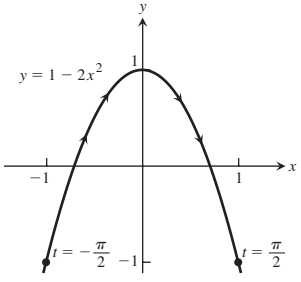
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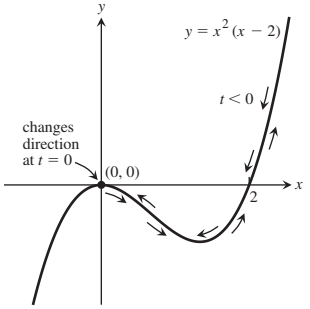
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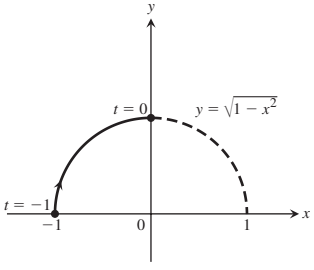
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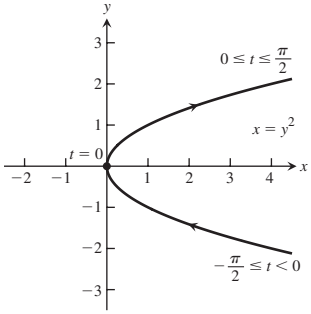
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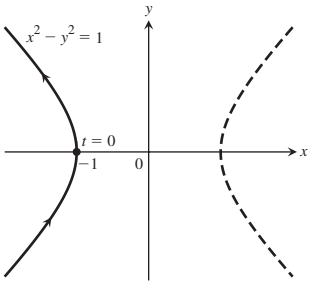
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15.

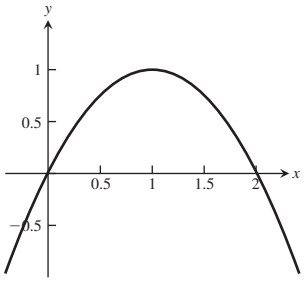


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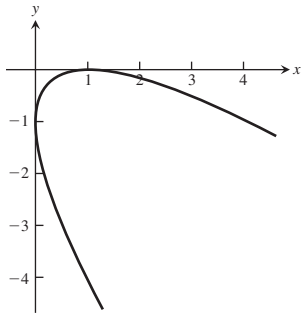


19. D **21.** E **23.** C

25.



27.



29. **(a)** $x = a\cos t, \quad y = -a\sin t, \quad 0 \le t \le 2\pi$
(b) $x = a\cos t, \quad y = a\sin t, \quad 0 \le t \le 2\pi$
(c) $x = a\cos t, \quad y = -a\sin t, \quad 0 \le t \le 4\pi$
(d) $x = a\cos t, \quad y = a\sin t, \quad 0 \le t \le 4\pi$

31. Possible answer: $x = -1 + 5t, \quad y = -3 + 4t, \quad 0 \le t \le 1$

33. Possible answer: $x = t^2 + 1, \quad y = t, \quad t \le 0$

35. Possible answer: $x = 2 - 3t, \quad y = 3 - 4t, \quad t \ge 0$

37. Possible answer: $x = 2\cos t, \quad y = 2\,|\sin t\,|, \quad 0 \le t \le 4\pi$

39. Possible answer: $x = \dfrac{-at}{\sqrt{1+t^2}}, \quad y = \dfrac{a}{\sqrt{1+t^2}}, -\infty < t < \infty$

41. Possible answer: $x = \dfrac{4}{1 + 2\tan\theta}, \quad y = \dfrac{4\tan\theta}{1 + 2\tan\theta}$,
$0 \le \theta < \pi/2$ and $x = 0, y = 2$ if $\theta = \pi/2$

43. Possible answer: $x = 2 - \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi$

45. $x = 2\cot t, \quad y = 2\sin^2 t, \quad 0 < t < \pi$

47. $x = a\sin^2 t\tan t, \quad y = a\sin^2 t, \quad 0 \le t < \pi/2$ **49.** (1, 1)

SECTION 10.2, pp. 633–634

1. $y = -x + 2\sqrt{2}, \quad \dfrac{d^2y}{dx^2} = -\sqrt{2}$

3. $y = -\dfrac{1}{2}x + 2\sqrt{2}, \quad \dfrac{d^2y}{dx^2} = -\dfrac{\sqrt{2}}{4}$

5. $y = x + \dfrac{1}{4}, \quad \dfrac{d^2y}{dx^2} = -2$ **7.** $y = 2x - \sqrt{3}, \quad \dfrac{d^2y}{dx^2} = -3\sqrt{3}$

9. $y = x - 4, \quad \dfrac{d^2y}{dx^2} = \dfrac{1}{2}$

11. $y = \sqrt{3}x - \dfrac{\pi\sqrt{3}}{3} + 2, \quad \dfrac{d^2y}{dx^2} = -4$

13. $y = 9x - 1, \quad \dfrac{d^2y}{dx^2} = 108$ **15.** $-\dfrac{3}{16}$ **17.** -6

19. 1 **21.** $3a^2\pi$ **23.** $|ab|\pi$ **25.** 4 **27.** 12

29. π^2 **31.** $8\pi^2$ **33.** $\dfrac{52\pi}{3}$ **35.** $3\pi\sqrt{5}$

37. $(\bar{x}, \bar{y}) = \left(\dfrac{12}{\pi} - \dfrac{24}{\pi^2}, \dfrac{24}{\pi^2} - 2\right)$

39. $(\bar{x}, \bar{y}) = \left(\dfrac{1}{3}, \pi - \dfrac{4}{3}\right)$ **41.** **(a)** π **(b)** π

43. **(a)** $x = 1, \quad y = 0, \quad \dfrac{dy}{dx} = \dfrac{1}{2}$ **(b)** $x = 0, \quad y = 3, \quad \dfrac{dy}{dx} = 0$

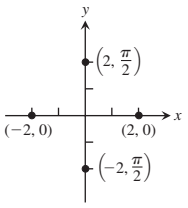
(c) $x = \dfrac{\sqrt{3} - 1}{2}, \quad y = \dfrac{3 - \sqrt{3}}{2}, \quad \dfrac{dy}{dx} = \dfrac{2\sqrt{3} - 1}{\sqrt{3} - 2}$

45. $\left(\dfrac{\sqrt{2}}{2}, 1\right), \quad y = 2x$ at $t = 0$, $y = -2x$ at $t = \pi$

47. **(a)** 8a **(b)** $\dfrac{64\pi}{3}$ **49.** $32\pi/15$

SECTION 10.3, pp. 637–638

1. a, e; b, g; c, h; d, f **3.**



(a) $\left(2, \dfrac{\pi}{2} + 2n\pi\right)$ and $\left(-2, \dfrac{\pi}{2} + (2n + 1)\pi\right)$, n an integer

(b) $(2, 2n\pi)$ and $(-2, (2n + 1)\pi)$, n an integer

(c) $\left(2, \dfrac{3\pi}{2} + 2n\pi\right)$ and $\left(-2, \dfrac{3\pi}{2} + (2n + 1)\pi\right)$, n an integer

(d) $(2, (2n + 1)\pi)$ and $(-2, 2n\pi)$, n an integer

5. **(a)** $(3, 0)$ **(b)** $(-3, 0)$ **(c)** $\left(-1, \sqrt{3}\right)$ **(d)** $\left(1, \sqrt{3}\right)$
(e) $(3, 0)$ **(f)** $\left(1, \sqrt{3}\right)$ **(g)** $(-3, 0)$ **(h)** $\left(-1, \sqrt{3}\right)$

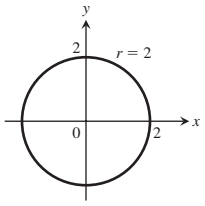
7. **(a)** $\left(\sqrt{2}, \dfrac{\pi}{4}\right)$ **(b)** $(3, \pi)$ **(c)** $\left(2, \dfrac{11\pi}{6}\right)$

(d) $\left(5, \pi - \tan^{-1}\dfrac{4}{3}\right)$

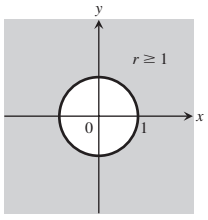
9. **(a)** $\left(-3\sqrt{2}, \dfrac{5\pi}{4}\right)$ **(b)** $(-1, 0)$ **(c)** $\left(-2, \dfrac{5\pi}{3}\right)$

(d) $\left(-5, \pi - \tan^{-1}\dfrac{3}{4}\right)$

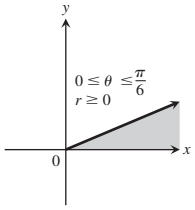
11.



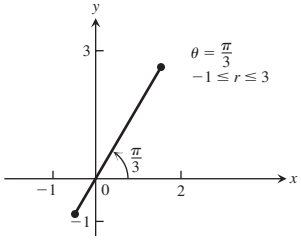
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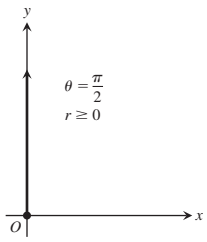
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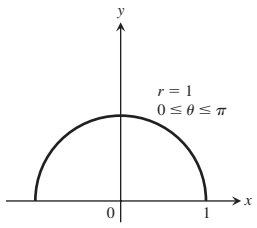
17.



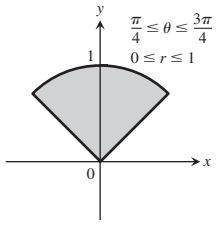
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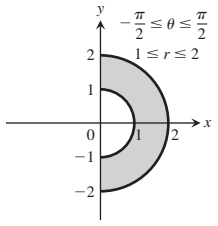
21.



23.



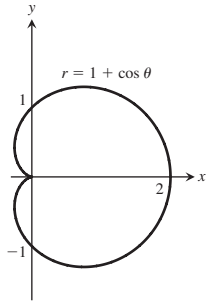
25.



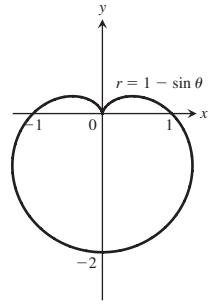
27. $x = 2$, vertical line through $(2, 0)$ **29.** $y = 0$, the x-axis
31. $y = 4$, horizontal line through $(0, 4)$
33. $x + y = 1$, line, $m = -1, b = 1$
35. $x^2 + y^2 = 1$, circle, $C(0, 0)$, radius 1
37. $y - 2x = 5$, line, $m = 2, b = 5$
39. $y^2 = x$, parabola, vertex $(0, 0)$, opens right
41. $y = e^x$, graph of natural exponential function
43. $x + y = \pm 1$, two straight lines of slope -1, y-intercepts $b = \pm 1$
45. $(x + 2)^2 + y^2 = 4$, circle, $C(-2, 0)$, radius 2
47. $x^2 + (y - 4)^2 = 16$, circle, $C(0, 4)$, radius 4
49. $(x - 1)^2 + (y - 1)^2 = 2$, circle, $C(1, 1)$, radius $\sqrt{2}$
51. $\sqrt{3}y + x = 4$ **53.** $r \cos\theta = 7$ **55.** $\theta = \pi/4$
57. $r = 2$ or $r = -2$ **59.** $4r^2\cos^2\theta + 9r^2\sin^2\theta = 36$
61. $r\sin^2\theta = 4\cos\theta$ **63.** $r = 4\sin\theta$
65. $r^2 = 6r\cos\theta - 2r\sin\theta - 6$
67. $(0, \theta)$, where θ is any angle

SECTION 10.4, pp. 641–642

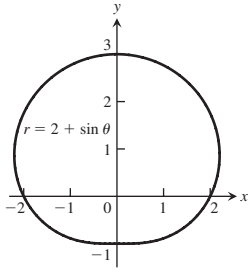
1. x-axis



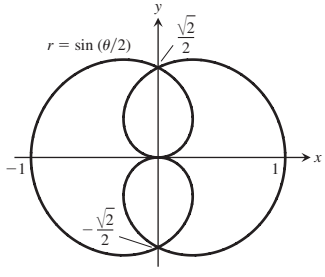
3. y-axis



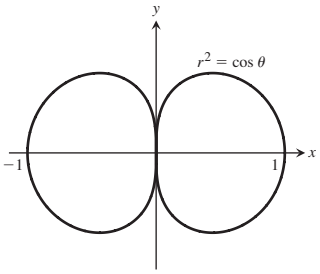
5. y-axis



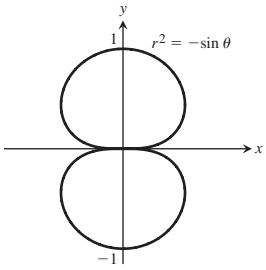
7. x-axis, y-axis, origin



9. x-axis, y-axis, origin

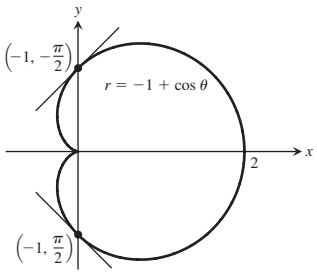


11. y-axis, x-axis, origin

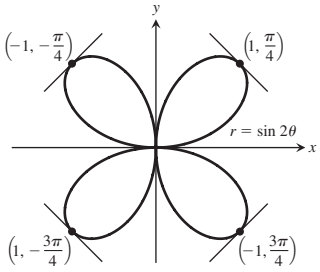


13. x-axis, y-axis, origin **15.** Origin

17. The slope at $(-1, \pi/2)$ is -1, at $(-1, -\pi/2)$ is 1.



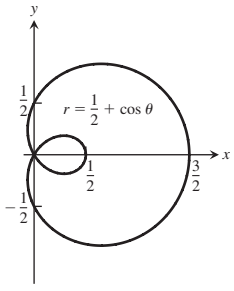
19. The slope at $(1, \pi/4)$ is -1, at $(-1, -\pi/4)$ is 1, at $(-1, 3\pi/4)$ is 1, at $(1, -3\pi/4)$ is -1.



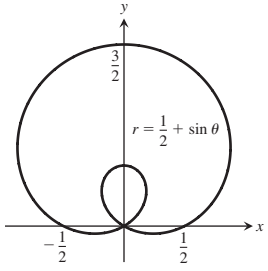
21. At $\pi/6$: slope $\sqrt{3}$, concavity 16 (concave up); at $\pi/3$: slope $-\sqrt{3}$, concavity -16 (concave down).

23. At 0: slope 0, concavity 2 (concave up); at $\pi/2$: slope $-2/\pi$, concavity $-2(8 + \pi^2)/\pi^3$ (concave down).

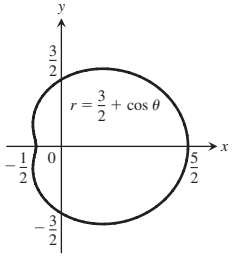
25. (a)



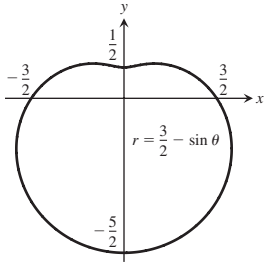
(b)



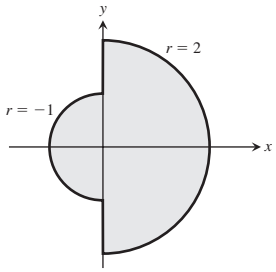
27. (a)



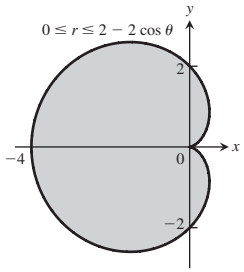
(b)



29.



31.



33. Equation (a)

SECTION 10.5, pp. 646–647

1. $\frac{1}{6}\pi^3$ **3.** 18π **5.** $\frac{\pi}{8}$ **7.** 2 **9.** $\frac{\pi}{2} - 1$

11. $5\pi - 8$ **13.** $3\sqrt{3} - \pi$ **15.** $\frac{\pi}{3} + \frac{\sqrt{3}}{2}$

17. $\frac{8\pi}{3} + \sqrt{3}$ **19. (a)** $\frac{3}{2} - \frac{\pi}{4}$ **21.** $19/3$ **23.** 8

25. $3\left(\sqrt{2} + \ln\left(1 + \sqrt{2}\right)\right)$ **27.** $\frac{\pi}{8} + \frac{3}{8}$

31. (a) a **(b)** a **(c)** $2a/\pi$

SECTION 10.6, pp. 653–655

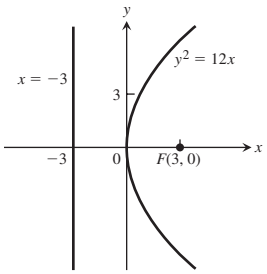
1. $y^2 = 8x$, $F(2, 0)$, directrix: $x = -2$

3. $x^2 = -6y$, $F(0, -3/2)$, directrix: $y = 3/2$

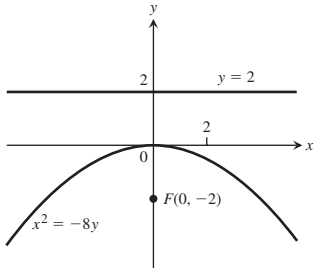
5. $\frac{x^2}{4} - \frac{y^2}{9} = 1$, $F\left(\pm\sqrt{13}, 0\right)$, $V(\pm 2, 0)$, asymptotes: $y = \pm\frac{3}{2}x$

7. $\frac{x^2}{2} + y^2 = 1$, $F(\pm 1, 0)$, $V\left(\pm\sqrt{2}, 0\right)$

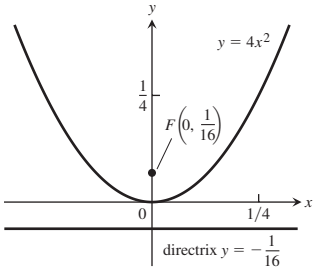
9.



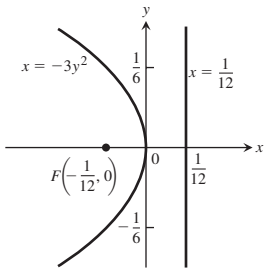
11.



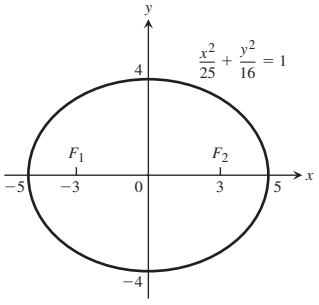
13.



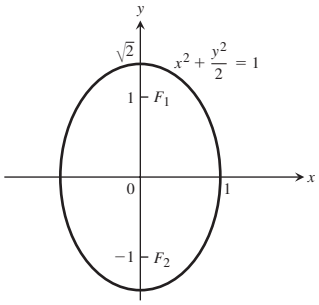
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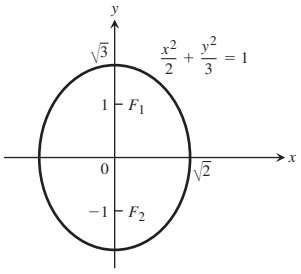
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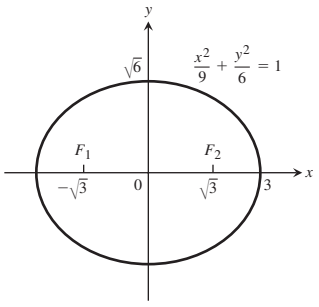
19.



21.

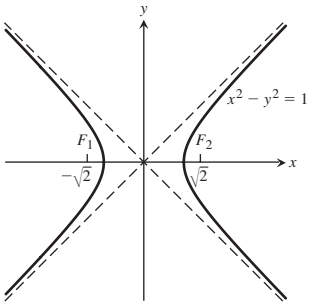


23.

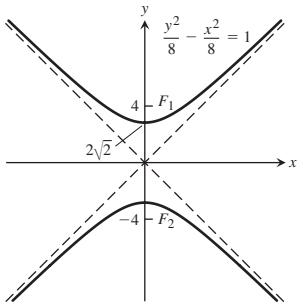


25. $\dfrac{x^2}{4} + \dfrac{y^2}{2} = 1$

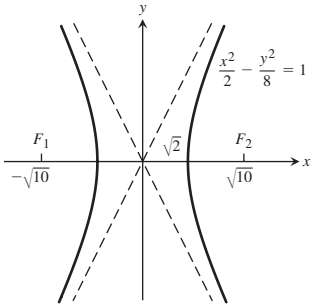
27. Asymptotes: $y = \pm x$



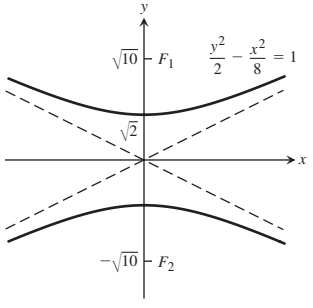
29. Asymptotes: $y = \pm x$



31. Asymptotes: $y = \pm 2x$



33. Asymptotes: $y = \pm x/2$

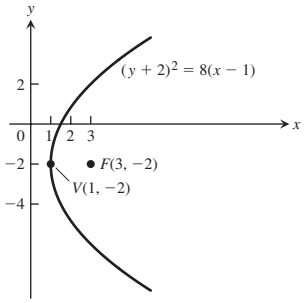


35. $y^2 - x^2 = 1$

37. $\dfrac{x^2}{9} - \dfrac{y^2}{16} = 1$

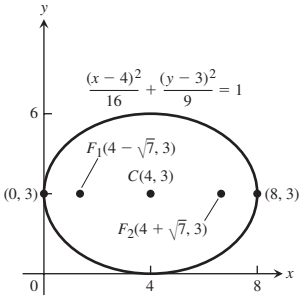
39. **(a)** Vertex: $(1, -2)$; focus: $(3, -2)$; directrix: $x = -1$

(b)



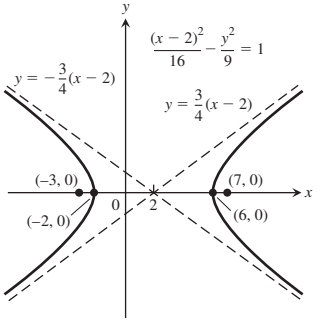
41. **(a)** Foci: $\left(4 \pm \sqrt{7}, 3\right)$; vertices: (8, 3) and (0, 3); center: (4, 3)

(b)



43. **(a)** Center: (2, 0); foci: (7, 0) and $(-3, 0)$; vertices: (6, 0) and $(-2, 0)$; asymptotes: $y = \pm\dfrac{3}{4}(x - 2)$

(b)



45. $(y + 3)^2 = 4(x + 2)$, $V(-2, -3)$, $F(-1, -3)$, directrix: $x = -3$

47. $(x - 1)^2 = 8(y + 7)$, $V(1, -7)$, $F(1, -5)$, directrix: $y = -9$

49. $\dfrac{(x + 2)^2}{6} + \dfrac{(y + 1)^2}{9} = 1$, $F\left(-2, \pm\sqrt{3} - 1\right)$, $V(-2, \pm 3 - 1)$, $C(-2, -1)$

51. $\dfrac{(x - 2)^2}{3} + \dfrac{(y - 3)^2}{2} = 1$, $F(3, 3)$ and $F(1, 3)$, $V\left(\pm\sqrt{3} + 2, 3\right)$, $C(2, 3)$

53. $\dfrac{(x - 2)^2}{4} - \dfrac{(y - 2)^2}{5} = 1$, $C(2, 2)$, $F(5, 2)$ and $F(-1, 2)$, $V(4, 2)$ and $V(0, 2)$; asymptotes: $(y - 2) = \pm\dfrac{\sqrt{5}}{2}(x - 2)$

55. $(y + 1)^2 - (x + 1)^2 = 1$, $C(-1, -1)$, $F\left(-1, \sqrt{2} - 1\right)$ and $F\left(-1, -\sqrt{2} - 1\right)$, $V(-1, 0)$ and $V(-1, -2)$; asymptotes $(y + 1) = \pm(x + 1)$

57. $C(-2, 0)$, $a = 4$ **59.** $V(-1, 1)$, $F(-1, 0)$

61. Ellipse: $\dfrac{(x+2)^2}{5} + y^2 = 1$, $C(-2, 0)$, $F(0, 0)$ and $F(-4, 0)$, $V\left(\sqrt{5}-2, 0\right)$ and $V\left(-\sqrt{5}-2, 0\right)$

63. Ellipse: $\dfrac{(x-1)^2}{2} + (y-1)^2 = 1$, $C(1, 1)$, $F(2, 1)$ and $F(0, 1)$, $V\left(\sqrt{2}+1, 1\right)$ and $V\left(-\sqrt{2}+1, 1\right)$

65. Hyperbola: $(x-1)^2 - (y-2)^2 = 1$, $C(1, 2)$, $F\left(1+\sqrt{2}, 2\right)$ and $F\left(1-\sqrt{2}, 2\right)$, $V(2, 2)$ and $V(0, 2)$; asymptotes: $(y-2) = \pm(x-1)$

67. Hyperbola: $\dfrac{(y-3)^2}{6} - \dfrac{x^2}{3} = 1$, $C(0, 3)$, $F(0, 6)$ and $F(0, 0)$, $V\left(0, \sqrt{6}+3\right)$ and $V\left(0, -\sqrt{6}+3\right)$; asymptotes: $y = \sqrt{2}x + 3$ or $y = -\sqrt{2}x + 3$

69. **(b)** 1:1 **73.** Length $= 2\sqrt{2}$, width $= \sqrt{2}$, area $= 4$

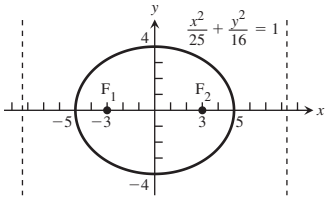
75. 24π

77. $x = 0, y = 0$: $y = -2x$; $x = 0, y = 2$: $y = 2x + 2$; $x = 4, y = 0$: $y = 2x - 8$

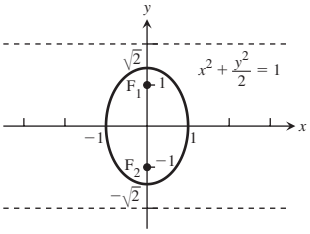
79. $\bar{x} = 0$, $\bar{y} = \dfrac{16}{3\pi}$

SECTION 10.7, pp. 660–661

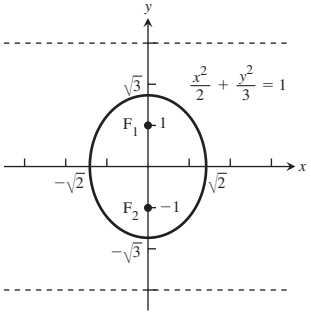
1. $e = \dfrac{3}{5}$, $F(\pm 3, 0)$; directrices are $x = \pm\dfrac{25}{3}$.



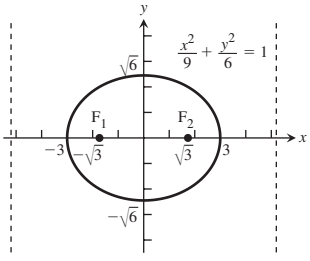
3. $e = \dfrac{1}{\sqrt{2}}$; $F(0, \pm 1)$; directrices are $y = \pm 2$.



5. $e = \dfrac{1}{\sqrt{3}}$; $F(0, \pm 1)$; directrices are $y = \pm 3$.



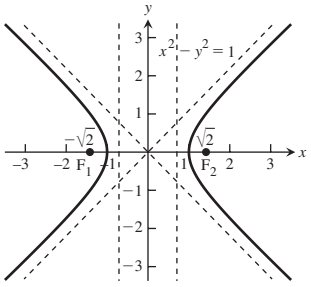
7. $e = \dfrac{\sqrt{3}}{3}$; $F\left(\pm\sqrt{3}, 0\right)$; directrices are $x = \pm 3\sqrt{3}$.



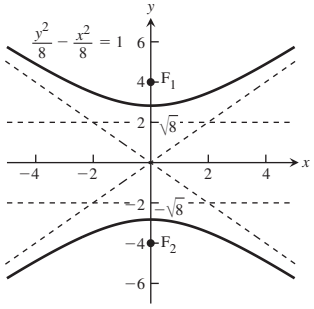
9. $\dfrac{x^2}{27} + \dfrac{y^2}{36} = 1$ **11.** $\dfrac{x^2}{4851} + \dfrac{y^2}{4900} = 1$

13. $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ **15.** $\dfrac{x^2}{64} + \dfrac{y^2}{48} = 1$

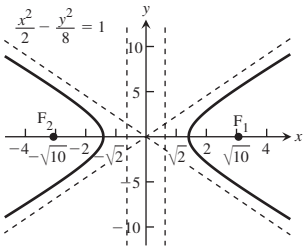
17. $e = \sqrt{2}$; $F\left(\pm\sqrt{2}, 0\right)$; directrices are $x = \pm\dfrac{1}{\sqrt{2}}$.



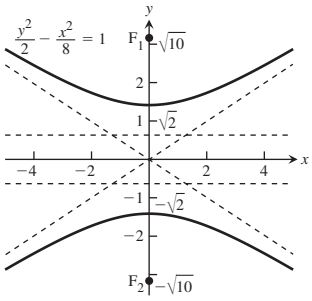
19. $e = \sqrt{2}$; $F(0, \pm 4)$; directrices are $y = \pm 2$.



21. $e = \sqrt{5}$; $F\left(\pm\sqrt{10}, 0\right)$; directrices are $x = \pm\dfrac{2}{\sqrt{10}}$.



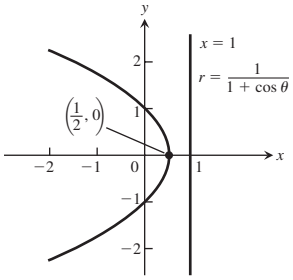
23. $e = \sqrt{5}$; $F\left(0, \pm\sqrt{10}\right)$; directrices are $y = \pm\dfrac{2}{\sqrt{10}}$.



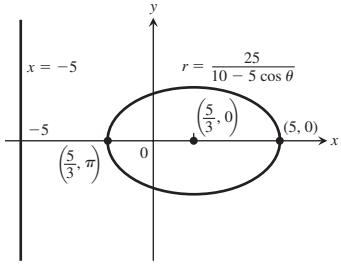
25. $y^2 - \dfrac{x^2}{8} = 1$ **27.** $x^2 - \dfrac{y^2}{8} = 1$ **29.** $r = \dfrac{2}{1+\cos\theta}$

31. $r = \dfrac{30}{1-5\sin\theta}$ **33.** $r = \dfrac{1}{2+\cos\theta}$ **35.** $r = \dfrac{10}{5-\sin\theta}$

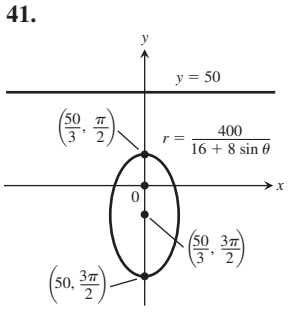
37.



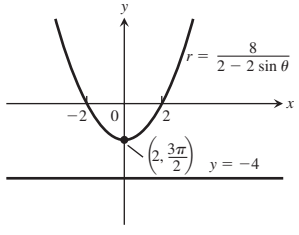
39.



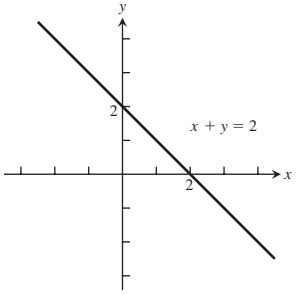
41.



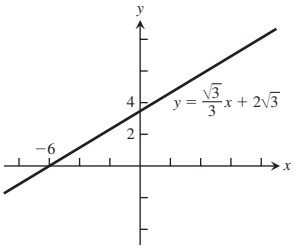
43.



45. $y = 2 - x$



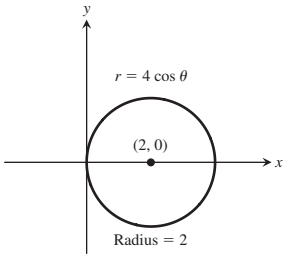
47. $y = \frac{\sqrt{3}}{3}x + 2\sqrt{3}$



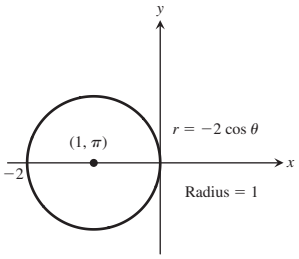
49. $r\cos\left(\theta - \frac{\pi}{4}\right) = 3$

51. $r\cos\left(\theta + \frac{\pi}{2}\right) = 5$

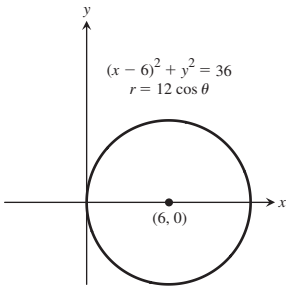
53.



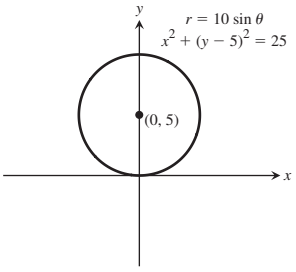
55.



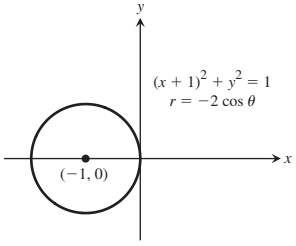
57. $r = 12\cos\theta$



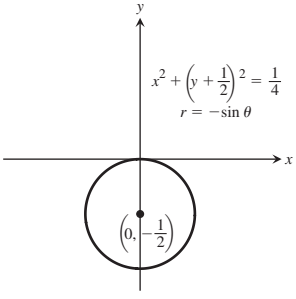
59. $r = 10\sin\theta$



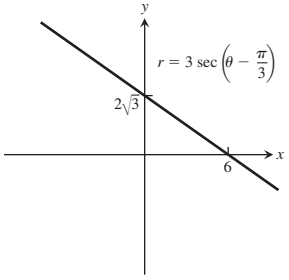
61. $r = -2\cos\theta$



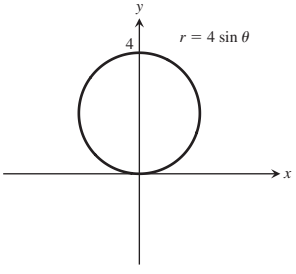
63. $r = -\sin\theta$



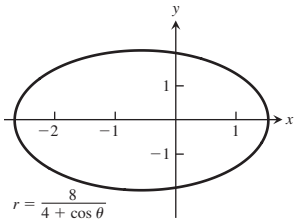
65.



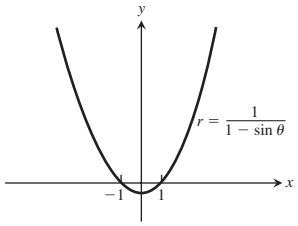
67.



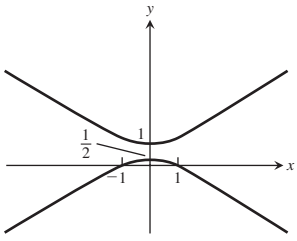
69.



71.



73.

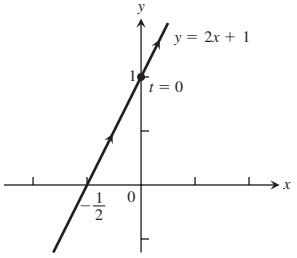


75. (b)

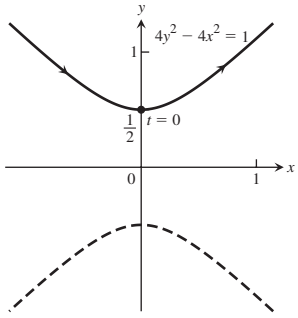
Planet	Perihelion	Aphelion
Mercury	0.3075 AU	0.4667 AU
Venus	0.7184 AU	0.7282 AU
Earth	0.9833 AU	1.0167 AU
Mars	1.3817 AU	1.6663 AU
Jupiter	4.9512 AU	5.4548 AU
Saturn	9.0210 AU	10.0570 AU
Uranus	18.2977 AU	20.0623 AU
Neptune	29.8135 AU	30.3065 AU

PRACTICE EXERCISES, pp. 662–664

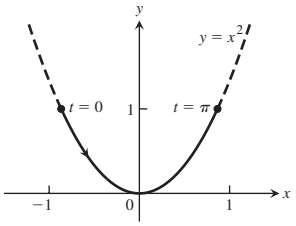
1.



3.



5.



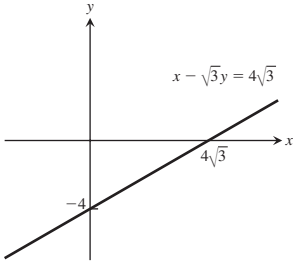
7. $x = 3\cos t, \quad y = 4\sin t, \quad 0 \le t \le 2\pi$

9. $y = \dfrac{\sqrt{3}}{2}x + \dfrac{1}{4}, \dfrac{1}{4}$

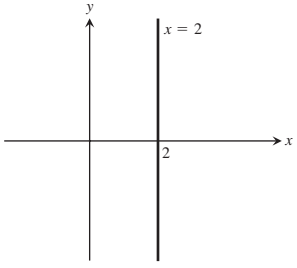
11. (a) $y = \dfrac{\pm|x|^{3/2}}{8} - 1$ **(b)** $y = \dfrac{\pm\sqrt{1-x^2}}{x}$

13. $\dfrac{10}{3}$ **15.** $\dfrac{285}{8}$ **17.** 10 **19.** $\dfrac{9\pi}{2}$ **21.** $\dfrac{76\pi}{3}$

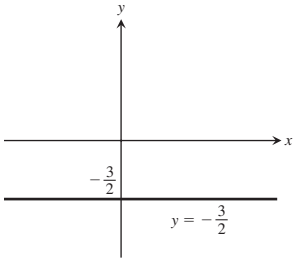
23. $y = \dfrac{\sqrt{3}}{3}x - 4$



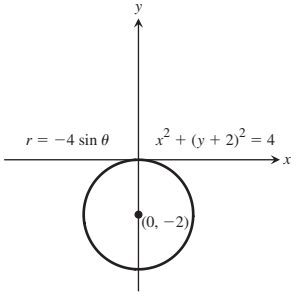
25. $x = 2$



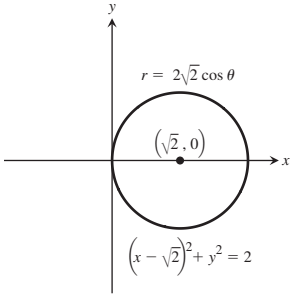
27. $y = -\dfrac{3}{2}$



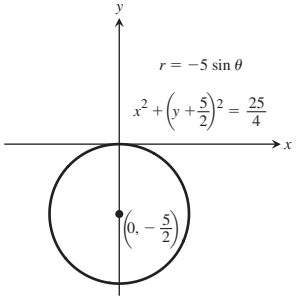
29. $x^2 + (y+2)^2 = 4$



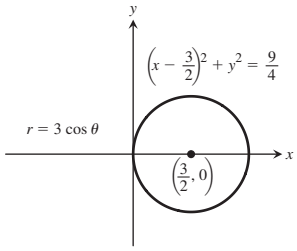
31. $\left(x - \sqrt{2}\right)^2 + y^2 = 2$



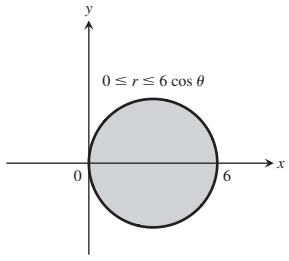
33. $r = -5\sin\theta$



35. $r = 3\cos\theta$



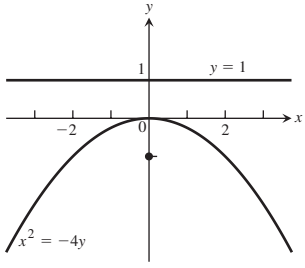
37.



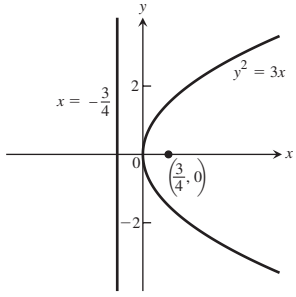
39. d **41.** l **43.** k **45.** i **47.** $\dfrac{9}{2}\pi$ **49.** $2 + \dfrac{\pi}{4}$

51. 8 **53.** $\pi - 3$

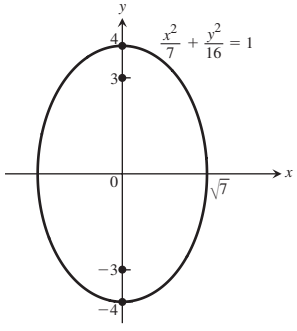
55. Focus is $(0, -1)$, directrix is $y = 1$.



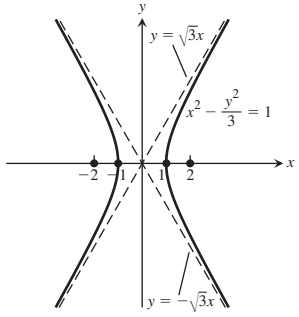
57. Focus is $\left(\dfrac{3}{4}, 0\right)$, directrix is $x = -\dfrac{3}{4}$.



59. $e = \dfrac{3}{4}$



61. $e = 2$; the asymptotes are $y = \pm\sqrt{3}\,x$.



63. $(x-2)^2 = -12(y-3)$, $V(2, 3)$, $F(2, 0)$, directrix is $y = 6$.

65. $\dfrac{(x+3)^2}{9} + \dfrac{(y+5)^2}{25} = 1$, $C(-3, -5)$, $F(-3, -1)$ and $F(-3, -9)$, $V(-3, -10)$ and $V(-3, 0)$.

67. $\dfrac{\left(y - 2\sqrt{2}\right)^2}{8} - \dfrac{(x-2)^2}{2} = 1$, $C\left(2, 2\sqrt{2}\right)$, $F\left(2, 2\sqrt{2} \pm \sqrt{10}\right)$, $V\left(2, 4\sqrt{2}\right)$ and $V(2, 0)$, the asymptotes are $y = 2x - 4 + 2\sqrt{2}$ and $y = -2x + 4 + 2\sqrt{2}$.

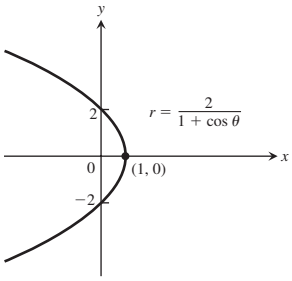
69. Hyperbola: $C(2, 0)$, $V(0, 0)$ and $V(4, 0)$, the foci are $F\left(2 \pm \sqrt{5}, 0\right)$, and the asymptotes are $y = \pm\dfrac{x-2}{2}$.

71. Parabola: $V(-3, 1)$, $F(-7, 1)$, and the directrix is $x = 1$.

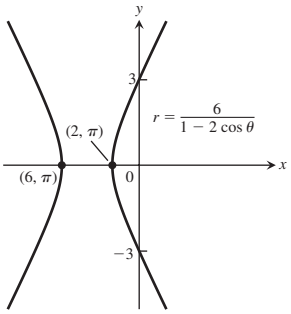
73. Ellipse: $C(-3, 2)$, $F\left(-3 \pm \sqrt{7}, 2\right)$, $V(1, 2)$ and $V(-7, 2)$

75. Circle: $C(1, 1)$ and radius $= \sqrt{2}$

77. $V(1, 0)$



79. $V(2, \pi)$ and $V(6, \pi)$

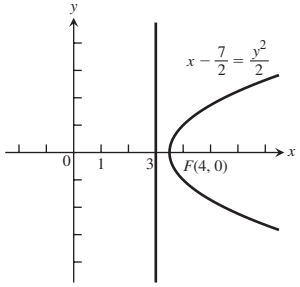


81. $r = \dfrac{4}{1 + 2\cos\theta}$ **83.** $r = \dfrac{2}{2 + \sin\theta}$

85. **(a)** 24π **(b)** 16π

ADDITIONAL AND ADVANCED EXERCISES, pp. 664–666

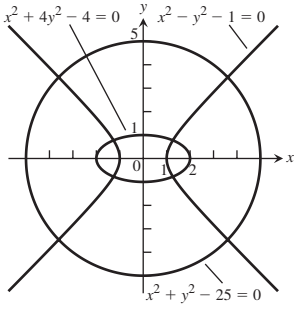
1. $x - \dfrac{7}{2} = \dfrac{y^2}{2}$



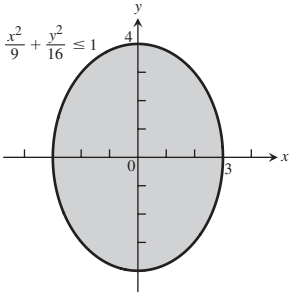
3. $3x^2 + 3y^2 - 8y + 4 = 0$ **5.** $F(0, \pm 1)$

7. **(a)** $\dfrac{(y - 1)^2}{16} - \dfrac{x^2}{48} = 1$ **(b)** $\dfrac{\left(y + \frac{3}{4}\right)^2}{\left(\frac{25}{16}\right)} - \dfrac{x^2}{\left(\frac{75}{2}\right)} = 1$

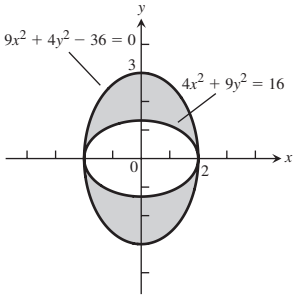
11.



13.



15.



17. **(a)** $r = e^{2\theta}$ **(b)** $\dfrac{\sqrt{5}}{2}(e^{4\pi} - 1)$

19. $r = \dfrac{4}{1 + 2\cos\theta}$ **21.** $r = \dfrac{2}{2 + \sin\theta}$

23. $x = (a + b)\cos\theta - b\cos\left(\dfrac{a + b}{b}\theta\right)$,

$y = (a + b)\sin\theta - b\sin\left(\dfrac{a + b}{b}\theta\right)$

27. $\dfrac{\pi}{2}$

Chapter 11

SECTION 11.1, pp. 670–672

1. The line through the point $(2, 3, 0)$ parallel to the z-axis
3. The x-axis
5. The circle $x^2 + y^2 = 4$ in the xy-plane
7. The circle $x^2 + z^2 = 4$ in the xz-plane
9. The circle $y^2 + z^2 = 1$ in the yz-plane
11. The circle $x^2 + y^2 = 16$ in the xy-plane
13. The ellipse formed by the intersection of the cylinder $x^2 + y^2 = 4$ and the plane $z = y$
15. The parabola $y = x^2$ in the xy-plane
17. **(a)** The first quadrant of the xy-plane
(b) The fourth quadrant of the xy-plane
19. **(a)** The ball of radius 1 centered at the origin
(b) All points more than 1 unit from the origin
21. **(a)** The ball of radius 2 centered at the origin with the interior of the ball of radius 1 centered at the origin removed
(b) The solid upper hemisphere of radius 1 centered at the origin
23. **(a)** The region on or inside the parabola $y = x^2$ in the xy-plane and all points above this region
(b) The region on or to the left of the parabola $x = y^2$ in the xy-plane and all points above it that are 2 units or less away from the xy-plane
25. 3 **27.** 7 **29.** $2\sqrt{3}$ **31.** **(a)** 2 **(b)** 3 **(c)** 4
33. **(a)** 3 **(b)** 4 **(c)** 5
35. **(a)** $x = 3$ **(b)** $y = -1$ **(c)** $z = -2$
37. **(a)** $z = 1$ **(b)** $x = 3$ **(c)** $y = -1$
39. **(a)** $x^2 + (y - 2)^2 = 4, z = 0$
(b) $(y - 2)^2 + z^2 = 4, x = 0$ **(c)** $x^2 + z^2 = 4, y = 2$
41. **(a)** $y = 3, z = -1$ **(b)** $x = 1, z = -1$ **(c)** $x = 1, y = 3$
43. $x^2 + y^2 + z^2 = 25, z = 3$ **45.** $0 \le z \le 1$ **47.** $z \le 0$
49. **(a)** $(x - 1)^2 + (y - 1)^2 + (z - 1)^2 < 1$
(b) $(x - 1)^2 + (y - 1)^2 + (z - 1)^2 > 1$
51. $C(-2, 0, 2), a = 2\sqrt{2}$ **53.** $C\left(\sqrt{2}, \sqrt{2}, -\sqrt{2}\right), a = \sqrt{2}$
55. $C(-2, 0, 2), a = \sqrt{8}$ **57.** $C\left(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right), a = \dfrac{5\sqrt{3}}{4}$
59. $C(2, -3, 5), a = 7$
61. $(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 14$
63. $(x + 1)^2 + \left(y - \dfrac{1}{2}\right)^2 + \left(z + \dfrac{2}{3}\right)^2 = \dfrac{16}{81}$
65. **(a)** $\sqrt{y^2 + z^2}$ **(b)** $\sqrt{x^2 + z^2}$ **(c)** $\sqrt{x^2 + y^2}$
67. $\sqrt{17} + \sqrt{33} + 6$ **69.** $y = 1$
71. **(a)** $(0, 3, -3)$ **(b)** $(0, 5, -5)$
73. $z = x^2/4 + 1$ **75.** **(a)** $z^2 = x^2$ **(b)** $y^2 = x^2$

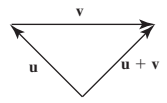
SECTION 11.2, pp. 679–681

1. (a) $\langle 9, -6 \rangle$ (b) $3\sqrt{13}$ **3.** (a) $\langle 1, 3 \rangle$ (b) $\sqrt{10}$
5. (a) $\langle 12, -19 \rangle$ (b) $\sqrt{505}$
7. (a) $\left\langle \frac{1}{5}, \frac{14}{5} \right\rangle$ (b) $\frac{\sqrt{197}}{5}$ **9.** $\langle 1, -4 \rangle$

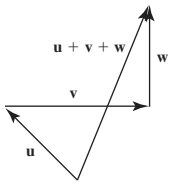
11. $\langle -2, -3 \rangle$ **13.** $\left\langle -\frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$ **15.** $\left\langle -\frac{\sqrt{3}}{2}, -\frac{1}{2} \right\rangle$

17. $-3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ **19.** $-3\mathbf{i} + 16\mathbf{j}$
21. $3\mathbf{i} + 5\mathbf{j} - 8\mathbf{k}$
23. The vector v is horizontal and 25 mm long. The vectors **u** and **w** are 17 mm long. **w** is vertical and **u** makes a 45° angle with the horizontal. All vectors must be drawn to scale.

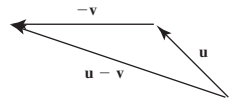
(a)



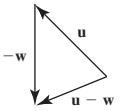
(b)



(c)



(d)



25. $3\left(\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}\right)$ **27.** $5(\mathbf{k})$

29. $\sqrt{\frac{1}{2}}\left(\frac{1}{\sqrt{3}}\mathbf{i} - \frac{1}{\sqrt{3}}\mathbf{j} - \frac{1}{\sqrt{3}}\mathbf{k}\right)$

31. (a) $2\mathbf{i}$ (b) $-\sqrt{3}\mathbf{k}$ (c) $\frac{3}{10}\mathbf{j} + \frac{2}{5}\mathbf{k}$ (d) $6\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

33. $\frac{7}{13}(12\mathbf{i} - 5\mathbf{k})$

35. (a) $\frac{3}{5\sqrt{2}}\mathbf{i} + \frac{4}{5\sqrt{2}}\mathbf{j} - \frac{1}{\sqrt{2}}\mathbf{k}$ (b) $(1/2, 3, 5/2)$

37. (a) $-\frac{1}{\sqrt{3}}\mathbf{i} - \frac{1}{\sqrt{3}}\mathbf{j} - \frac{1}{\sqrt{3}}\mathbf{k}$ (b) $\left(\frac{5}{2}, \frac{7}{2}, \frac{9}{2}\right)$

39. $A(4, -3, 5)$ **41.** $a = \frac{3}{2}, b = \frac{1}{2}$

43. $a = -1, b = 2, c = 1$ **45.** $\approx \langle -338.095, 725.046 \rangle$

47. $|\mathbf{F}_1| = \dfrac{100 \cos 45°}{\sin 75°} \approx 73.205$ N,

$|\mathbf{F}_2| = \dfrac{100 \cos 30°}{\sin 75°} \approx 89.658$ N,

$\mathbf{F}_1 = \langle -|\mathbf{F}_1| \cos 30°, |\mathbf{F}_1| \sin 30° \rangle \approx \langle -63.397, 36.603 \rangle$,
$\mathbf{F}_2 = \langle |\mathbf{F}_2| \cos 45°, |\mathbf{F}_2| \sin 45° \rangle \approx \langle 63.397, 63.397 \rangle$

49. $w = \dfrac{100 \sin 75°}{\cos 40°} \approx 126.093$ N,

$|\mathbf{F}_1| = \dfrac{w \cos 35°}{\sin 75°} \approx 106.933$ N

51. (a) $(5 \cos 60°, 5 \sin 60°) = \left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$

(b) $(5 \cos 60° + 10 \cos 315°, 5 \sin 60° + 10 \sin 315°) = \left(\frac{5 + 10\sqrt{2}}{2}, \frac{5\sqrt{3} - 10\sqrt{2}}{2}\right)$

53. (a) $\frac{3}{2}\mathbf{i} + \frac{3}{2}\mathbf{j} - 3\mathbf{k}$ (b) $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ (c) $(2, 2, 1)$

59. (a) $\langle 0, 0, 0 \rangle$ (b) $\langle 0, 0, 0 \rangle$

SECTION 11.3, pp. 687–689

1. (a) $-25, 5, 5$ (b) -1 (c) -5 (d) $-2\mathbf{i} + 4\mathbf{j} - \sqrt{5}\mathbf{k}$

3. (a) $25, 15, 5$ (b) $\frac{1}{3}$ (c) $\frac{5}{3}$ (d) $\frac{1}{9}(10\mathbf{i} + 11\mathbf{j} - 2\mathbf{k})$

5. (a) $2, \sqrt{34}, \sqrt{3}$ (b) $\dfrac{2}{\sqrt{3}\sqrt{34}}$ (c) $\dfrac{2}{\sqrt{34}}$

(d) $\frac{1}{17}(5\mathbf{j} - 3\mathbf{k})$

7. (a) $10 + \sqrt{17}, \sqrt{26}, \sqrt{21}$ (b) $\dfrac{10 + \sqrt{17}}{\sqrt{546}}$ (c) $\dfrac{10 + \sqrt{17}}{\sqrt{26}}$

(d) $\dfrac{10 + \sqrt{17}}{26}(5\mathbf{i} + \mathbf{j})$

9. 0.75 rad **11.** 1.77 rad

13. Angle at $A = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right) \approx 63.435$ degrees, angle at

$B = \cos^{-1}\left(\frac{3}{5}\right) \approx 53.130$ degrees, angle at

$C = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right) \approx 63.435$ degrees.

17. $\cos^{-1}\left(\frac{3}{\sqrt{10}}\right) \approx 0.322$ radian or 18.43 degrees

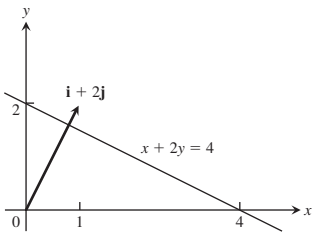
25. Horizontal component: ≈ 396 m/s, vertical component: ≈ 55.7 m/s

27. (a) Since $|\cos\theta| \le 1$, we have $|\mathbf{u} \cdot \mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\,|\cos\theta| \le |\mathbf{u}|\,|\mathbf{v}|\,(1) = |\mathbf{u}|\,|\mathbf{v}|$.

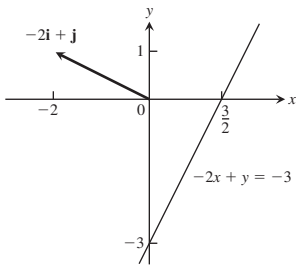
(b) We have equality precisely when $|\cos\theta| = 1$ or when one or both of **u** and v are **0**. In the case of nonzero vectors, we have equality when $\theta = 0$ or π, that is, when the vectors are parallel.

29. a

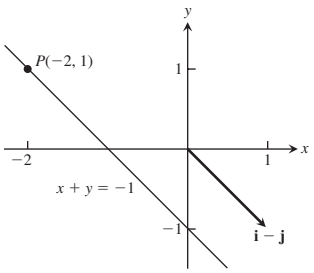
35. $x + 2y = 4$



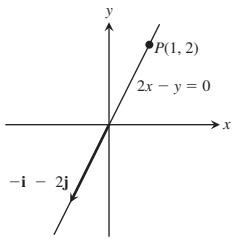
37. $-2x + y = -3$



39. $x + y = -1$



41. $2x - y = 0$



43. 5 J **45.** 3464 J **47.** $\frac{\pi}{4}$ **49.** $\frac{\pi}{6}$ **51.** 0.14

SECTION 11.4, pp. 694–695

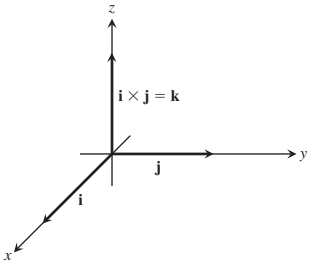
1. $|\mathbf{u} \times \mathbf{v}| = 3$, direction is $\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$; $|\mathbf{v} \times \mathbf{u}| = 3$, direction is $-\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

3. $|\mathbf{u} \times \mathbf{v}| = 0$, no direction; $|\mathbf{v} \times \mathbf{u}| = 0$, no direction

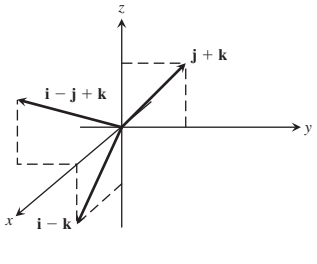
5. $|\mathbf{u} \times \mathbf{v}| = 6$, direction is $-\mathbf{k}$; $|\mathbf{v} \times \mathbf{u}| = 6$, direction is $\mathbf{k}$

7. $|\mathbf{u} \times \mathbf{v}| = 6\sqrt{5}$, direction is $\frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{k}$; $|\mathbf{v} \times \mathbf{u}| = 6\sqrt{5}$, direction is $-\frac{1}{\sqrt{5}}\mathbf{i} + \frac{2}{\sqrt{5}}\mathbf{k}$

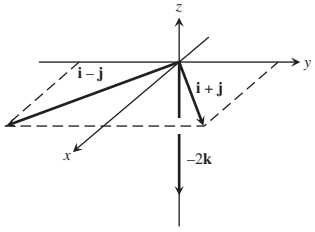
9.



11.



13.



15. (a) $2\sqrt{6}$ (b) $\pm\frac{1}{\sqrt{6}}(2\mathbf{i} + \mathbf{j} + \mathbf{k})$

17. (a) $\frac{\sqrt{2}}{2}$ (b) $\pm\frac{1}{\sqrt{2}}(\mathbf{i} - \mathbf{j})$

19. 8 **21.** 7 **23.** (a) None (b) **u** and **w**

25. $1.5\sqrt{3}$ N-m

27. (a) True (b) Not always true (c) True (d) True (e) Not always true (f) True (g) True (h) True

29. (a) $\text{proj}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}$ (b) $\pm\,\mathbf{u} \times \mathbf{v}$ (c) $\pm\,(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ (d) $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$ (e) $(\mathbf{u} \times \mathbf{v}) \times (\mathbf{u} \times \mathbf{w})$ (f) $|\mathbf{u}|\,\frac{\mathbf{v}}{|\mathbf{v}|}$

31. (a) Yes (b) No (c) Yes (d) No

33. No, v need not equal **w**. For example, $\mathbf{i} + \mathbf{j} \neq -\mathbf{i} + \mathbf{j}$, but $\mathbf{i} \times (\mathbf{i} + \mathbf{j}) = \mathbf{i} \times \mathbf{i} + \mathbf{i} \times \mathbf{j} = \mathbf{0} + \mathbf{k} = \mathbf{k}$ and $\mathbf{i} \times (-\mathbf{i} + \mathbf{j}) = -\mathbf{i} \times \mathbf{i} + \mathbf{i} \times \mathbf{j} = \mathbf{0} + \mathbf{k} = \mathbf{k}$.

35. 2 **37.** 13 **39.** $\sqrt{129}$ **41.** $\frac{11}{2}$ **43.** $\frac{25}{2}$

45. $\frac{3}{2}$ **47.** $\frac{\sqrt{21}}{2}$

49. If $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{B} = b_1\mathbf{i} + b_2\mathbf{j}$, then

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & 0 \\ b_1 & b_2 & 0 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}\mathbf{k}$$

and the triangle's area is

$$\frac{1}{2}\left|\mathbf{A} \times \mathbf{B}\right| = \pm\frac{1}{2}\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}.$$

The applicable sign is (+) if the acute angle from **A** to **B** runs counterclockwise in the xy-plane, and (−) if it runs clockwise.

51. 4 **53.** $44/3$ **55.** Coplanar **57.** Not coplanar

SECTION 11.5, pp. 702–704

1. $x = 3 + t,\quad y = -4 + t,\quad z = -1 + t$

3. $x = -2 + 5t,\quad y = 5t,\quad z = 3 - 5t$

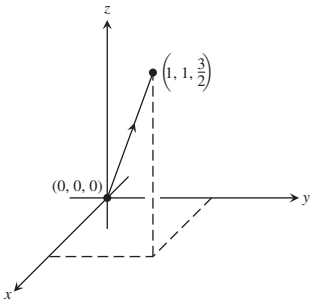
5. $x = 0,\quad y = 2t,\quad z = t$

7. $x = 1,\quad y = 1,\quad z = 1 + t$

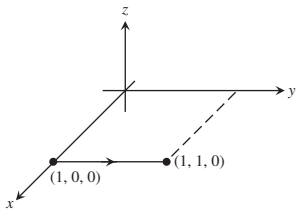
9. $x = t,\quad y = -7 + 2t,\quad z = 2t$

11. $x = t,\quad y = 0,\quad z = 0$

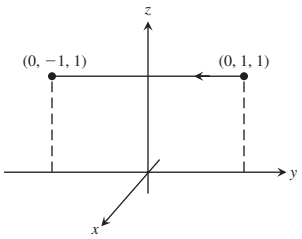
13. $x = t,\quad y = t,\quad z = \frac{3}{2}t,\quad 0 \le t \le 1$



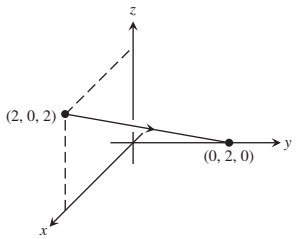
15. $x = 1,\quad y = 1 + t,\quad z = 0,\quad -1 \le t \le 0$



17. $x = 0,\quad y = 1 - 2t,\quad z = 1,\quad 0 \le t \le 1$



19. $x = 2 - 2t,\quad y = 2t,\quad z = 2 - 2t,\quad 0 \le t \le 1$



21. $3x - 2y - z = -3$ **23.** $7x - 5y - 4z = 6$

25. $x + 3y + 4z = 34$ **27.** $(1, 2, 3), -20x + 12y + z = 7$

29. $y + z = 3$ **31.** $x - y + z = 0$ **33.** $2\sqrt{30}$ **35.** 0

37. $\frac{9\sqrt{42}}{7}$ **39.** 3 **41.** $19/5$ **43.** $5/3$ **45.** $9/\sqrt{41}$

47. $\pi/4$ **49.** $\arccos(-1/6) \approx 1.738$ radians

51. $\arcsin\left(2/\sqrt{154}\right) \approx 0.161$ radians **53.** 1.38 rad

55. 0.82 rad **57.** $\left(\frac{3}{2}, -\frac{3}{2}, \frac{1}{2}\right)$ **59.** (1, 1, 0)

61. $x = 1 - t,\quad y = 1 + t,\quad z = -1$

63. $x = 4,\quad y = 3 + 6t,\quad z = 1 + 3t$

65. $L1$ intersects $L2$; $L2$ is parallel to $L3$, $\sqrt{5}/3$; $L1$ and $L3$ are skew, $10\sqrt{2}/3$

67. $x = 2 + 2t,\quad y = -4 - t,\quad z = 7 + 3t;\quad x = -2 - t,\quad y = -2 + (1/2)t,\quad z = 1 - (3/2)t$

69. $\left(0, -\frac{1}{2}, -\frac{3}{2}\right), (-1, 0, -3), (1, -1, 0)$

73. Many possible answers. One possibility: $x + y = 3$ and $2y + z = 7$.

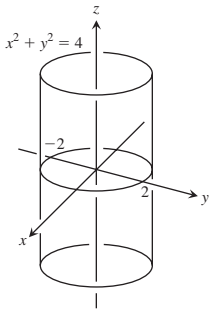
75. $(x/a) + (y/b) + (z/c) = 1$ describes all planes *except* those through the origin or parallel to a coordinate axis.

SECTION 11.6, pp. 708–709

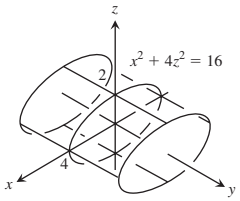
1. (d), ellipsoid **3.** (a), cylinder **5.** (l), hyperbolic paraboloid

7. (b), cylinder **9.** (k), hyperbolic paraboloid **11.** (h), cone

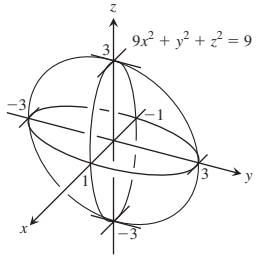
13.



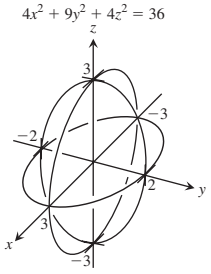
15.



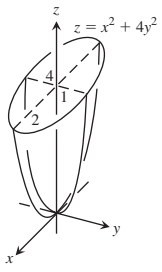
17.



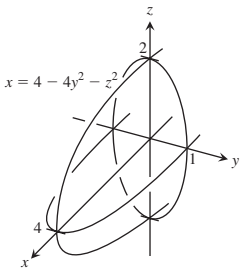
19.



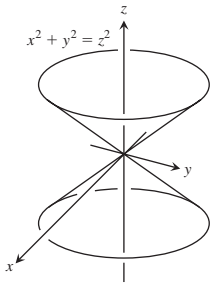
21.



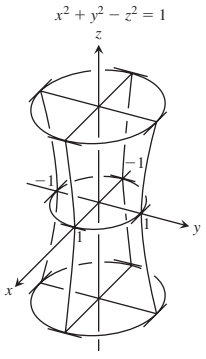
23.



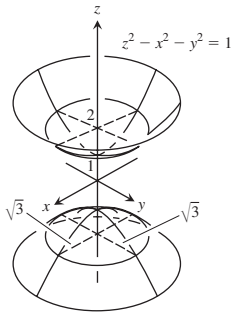
25.



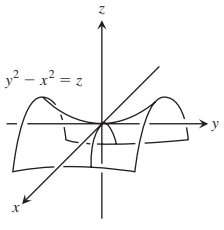
27.



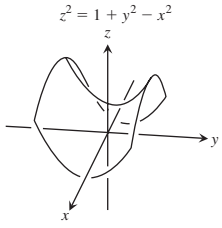
29.



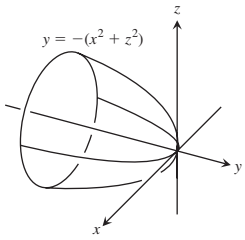
31.



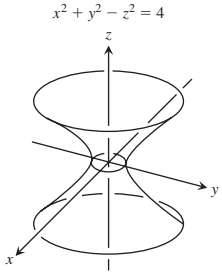
33.



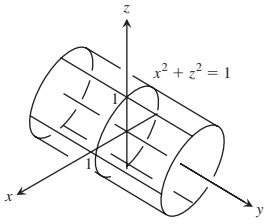
35.



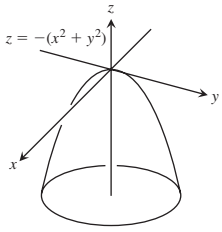
37.



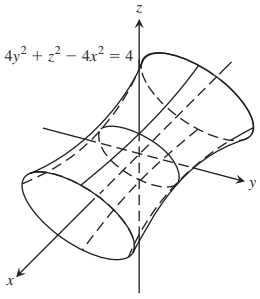
39.



41.



43.



45. **(a)** $\dfrac{2\pi(9 - c^2)}{9}$ **(b)** 8π **(c)** $\dfrac{4\pi abc}{3}$

PRACTICE EXERCISES, pp. 710–712

1. **(a)** $\langle -17, 32 \rangle$ **(b)** $\sqrt{1313}$

3. **(a)** $\langle 6, -8 \rangle$ **(b)** 10

5. $\left\langle -\dfrac{\sqrt{3}}{2}, -\dfrac{1}{2} \right\rangle$ [assuming counterclockwise]

7. $\left\langle \dfrac{8}{\sqrt{17}}, -\dfrac{2}{\sqrt{17}} \right\rangle$

9. Length $= 2$, direction is $\dfrac{1}{\sqrt{2}}\mathbf{i} + \dfrac{1}{\sqrt{2}}\mathbf{j}$.

11. $\mathbf{v}(\pi/2) = 2(-\mathbf{i})$

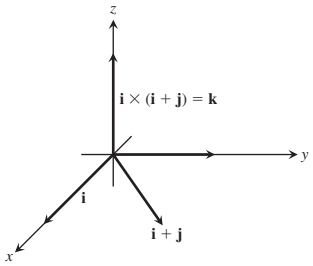
13. Length $= 7$, direction is $\dfrac{2}{7}\mathbf{i} - \dfrac{3}{7}\mathbf{j} + \dfrac{6}{7}\mathbf{k}$.

15. $\dfrac{8}{\sqrt{33}}\mathbf{i} - \dfrac{2}{\sqrt{33}}\mathbf{j} + \dfrac{8}{\sqrt{33}}\mathbf{k}$

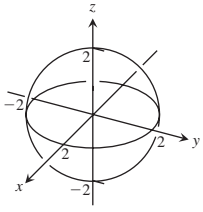
17. $|\mathbf{v}| = \sqrt{2}$, $|\mathbf{u}| = 3$, $\mathbf{v} \cdot \mathbf{u} = \mathbf{u} \cdot \mathbf{v} = 3$, $\mathbf{v} \times \mathbf{u} = -2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{u} \times \mathbf{v} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $|\mathbf{v} \times \mathbf{u}| = 3$, $\theta = \cos^{-1}\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi}{4}$, $|\mathbf{u}| \cos\theta = \dfrac{3}{\sqrt{2}}$, $\text{proj}_{\mathbf{v}}\,\mathbf{u} = \dfrac{3}{2}(\mathbf{i} + \mathbf{j})$

19. $\dfrac{4}{3}(2\mathbf{i} + \mathbf{j} - \mathbf{k})$

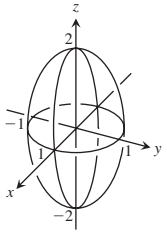
21. $\mathbf{u} \times \mathbf{v} = \mathbf{k}$



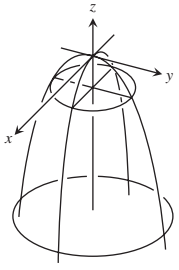
23. $2\sqrt{7}$ **25.** (a) $\sqrt{14}$ (b) 1 **29.** $\sqrt{78}/3$
31. $x = 1 - 3t, \quad y = 2, \quad z = 3 + 7t$ **33.** $\sqrt{2}$
35. $2x + y + z = 5$ **37.** $-9x + y + 7z = 4$
39. $\left(0, -\frac{1}{2}, -\frac{3}{2}\right), (-1, 0, -3), (1, -1, 0)$ **41.** $\pi/3$
43. $x = -5 + 5t, \quad y = 3 - t, \quad z = -3t$
45. (b) $x = -12t, \quad y = 19/12 + 15t, \quad z = 1/6 + 6t$
47. Yes; v is parallel to the plane.
49. 3 **51.** $-3\mathbf{j} + 3\mathbf{k}$
53. $\frac{2}{\sqrt{35}}(5\mathbf{i} - \mathbf{j} - 3\mathbf{k})$ **55.** $\left(\frac{11}{9}, \frac{26}{9}, -\frac{7}{9}\right)$
57. $(1, -2, -1); \; x = 1 - 5t, \quad y = -2 + 3t, \quad z = -1 + 4t$
59. $2x + 7y + 2z + 10 = 0$
61. (a) No (b) No (c) No (d) No (e) Yes
63. $11/\sqrt{107}$
65. $x^2 + y^2 + z^2 = 4$



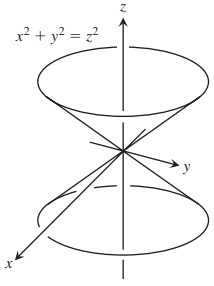
67. $4x^2 + 4y^2 + z^2 = 4$



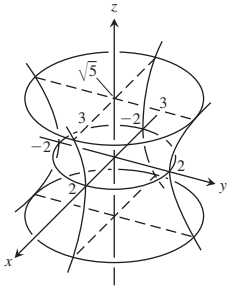
69. $z = -(x^2 + y^2)$



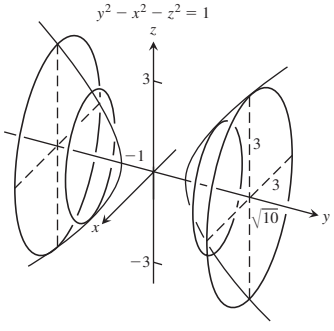
71.



73. $x^2 + y^2 - z^2 = 4$



75. $y^2 - x^2 - z^2 = 1$



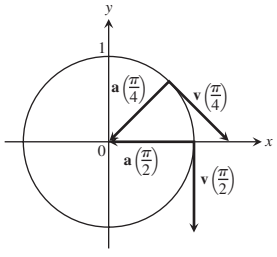
ADDITIONAL AND ADVANCED EXERCISES, pp. 712–714

1. $(26, 23, -1/3)$ **3.** $|\mathbf{F}| = 88.7$ N
5. (a) $|\mathbf{F}_1| = 80$ N, $|\mathbf{F}_2| = 60$ N, $\mathbf{F}_1 = \langle -48, 64 \rangle$,
$\mathbf{F}_2 = \langle 48, 36 \rangle$, $\alpha = \tan^{-1}\frac{4}{3}$, $\beta = \tan^{-1}\frac{3}{4}$
(b) $|\mathbf{F}_1| = \frac{2400}{13} \approx 184.615$ N, $|\mathbf{F}_2| = \frac{1000}{13} \approx 76.923$ N,
$\mathbf{F}_1 = \left\langle \frac{-12{,}000}{169}, \frac{28{,}800}{169} \right\rangle \approx \langle -71.006, 170.414 \rangle$,
$\mathbf{F}_2 = \left\langle \frac{12{,}000}{169}, \frac{5000}{169} \right\rangle \approx \langle 71.006, 29.586 \rangle$,
$\alpha = \tan^{-1}\frac{12}{5}, \qquad \beta = \tan^{-1}\frac{5}{12}$
9. (a) $\theta = \tan^{-1}\sqrt{2} \approx 54.74°$ (b) $\theta = \tan^{-1} 2\sqrt{2} \approx 70.53°$
13. (a) $\frac{6}{\sqrt{14}}$ (b) $2x - y + 2z = 8$
(c) $x - 2y + z = 3 + 5\sqrt{6}$ and $x - 2y + z = 3 - 5\sqrt{6}$
15. $\frac{32}{41}\mathbf{i} + \frac{23}{41}\mathbf{j} - \frac{13}{41}\mathbf{k}$
17. (a) $\mathbf{0}, \mathbf{0}$ (b) $-10\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}, -9\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$
(c) $-4\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}, \mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$
(d) $-10\mathbf{i} - 10\mathbf{k}, -12\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}$
19. The formula is always true.

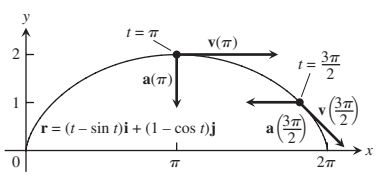
Chapter 12

SECTION 12.1, pp. 723–725

1. $\mathbf{i} - \frac{1}{2}\mathbf{j} + \mathbf{k}$ **3.** $2\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{\pi}{4}\mathbf{k}$
5. $y = x^2 - 2x, \quad \mathbf{v} = \mathbf{i} + 2\mathbf{j}, \quad \mathbf{a} = 2\mathbf{j}$
7. $y = \frac{2}{9}x^2, \quad \mathbf{v} = 3\mathbf{i} + 4\mathbf{j}, \quad \mathbf{a} = 3\mathbf{i} + 8\mathbf{j}$
9. $t = \frac{\pi}{4}$: $\mathbf{v} = \frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j}, \quad \mathbf{a} = \frac{-\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j}$;
$t = \pi/2$: $\mathbf{v} = -\mathbf{j}, \quad \mathbf{a} = -\mathbf{i}$



11. $t = \pi$: $\mathbf{v} = 2\mathbf{i}, \quad \mathbf{a} = -\mathbf{j}; \quad t = \frac{3\pi}{2}$: $\mathbf{v} = \mathbf{i} - \mathbf{j}, \quad \mathbf{a} = -\mathbf{i}$



13. $\mathbf{v} = \mathbf{i} + 2t\mathbf{j} + 2\mathbf{k}; \; \mathbf{a} = 2\mathbf{j}$; speed: 3; direction: $\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$;

$\mathbf{v}(1) = 3\left(\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}\right)$

15. $\mathbf{v} = (-2\sin t)\mathbf{i} + (3\cos t)\mathbf{j} + 4\mathbf{k}$;
$\mathbf{a} = (-2\cos t)\mathbf{i} - (3\sin t)\mathbf{j}$; speed: $2\sqrt{5}$;
direction: $\left(-1/\sqrt{5}\right)\mathbf{i} + \left(2/\sqrt{5}\right)\mathbf{k}$;
$\mathbf{v}(\pi/2) = 2\sqrt{5}\left[\left(-1/\sqrt{5}\right)\mathbf{i} + \left(2/\sqrt{5}\right)\mathbf{k}\right]$

17. $\mathbf{v} = \left(\frac{2}{t+1}\right)\mathbf{i} + 2t\mathbf{j} + t\mathbf{k}$; $\mathbf{a} = \left(\frac{-2}{(t+1)^2}\right)\mathbf{i} + 2\mathbf{j} + \mathbf{k}$;
speed: $\sqrt{6}$; direction: $\frac{1}{\sqrt{6}}\mathbf{i} + \frac{2}{\sqrt{6}}\mathbf{j} + \frac{1}{\sqrt{6}}\mathbf{k}$;
$\mathbf{v}(1) = \sqrt{6}\left(\frac{1}{\sqrt{6}}\mathbf{i} + \frac{2}{\sqrt{6}}\mathbf{j} + \frac{1}{\sqrt{6}}\mathbf{k}\right)$

19. $\pi/2$ **21.** $\pi/2$ **23.** $x = t, \quad y = -1, \quad z = 1 + t$

25. $x = t, \quad y = \frac{1}{3}t, \quad z = t$ **27.** $4, -2$ **29.** $2, -2$

31. E **33.** D **35.** C

37. **(a)** **(i):** It has constant speed 1. **(ii):** Yes
(iii): Counterclockwise **(iv):** Yes
(b) **(i):** It has constant speed 2. **(ii):** Yes
(iii): Counterclockwise **(iv):** Yes
(c) **(i):** It has constant speed 1. **(ii):** Yes
(iii): Counterclockwise
(iv): It starts at $(0, -1)$ instead of $(1, 0)$.
(d) **(i):** It has constant speed 1. **(ii):** Yes
(iii): Clockwise **(iv):** Yes
(i): It has variable speed. **(ii):** No
(iii): Counterclockwise **(iv):** Yes

39. $\mathbf{v} = 2\sqrt{5}\mathbf{i} + \sqrt{5}\mathbf{j}$

SECTION 12.2, pp. 730–734

1. $(1/4)\mathbf{i} + 7\mathbf{j} + (3/2)\mathbf{k}$ **3.** $\left(\frac{\pi + 2\sqrt{2}}{2}\right)\mathbf{j} + 2\mathbf{k}$

5. $(\ln 4)\mathbf{i} + (\ln 4)\mathbf{j} + (\ln 2)\mathbf{k}$

7. $\frac{e-1}{2}\mathbf{i} + \frac{e-1}{e}\mathbf{j} + \mathbf{k}$ **9.** $\mathbf{i} - \mathbf{j} + \frac{\pi}{4}\mathbf{k}$

11. $\mathbf{r}(t) = \left(\frac{-t^2}{2} + 1\right)\mathbf{i} + \left(\frac{-t^2}{2} + 2\right)\mathbf{j} + \left(\frac{-t^2}{2} + 3\right)\mathbf{k}$

13. $\mathbf{r}(t) = ((t+1)^{3/2} - 1)\mathbf{i} + (-e^{-t} + 1)\mathbf{j} + (\ln(t+1) + 1)\mathbf{k}$

15. $\mathbf{r}(t) = (3 + \ln|\sec t|)\mathbf{i} + (-2 + 2\sin(t/2))\mathbf{j}$
$+ (1 - (1/2)\ln|\sec 2t + \tan 2t|)\mathbf{k}$

17. $\mathbf{r}(t) = 8t\mathbf{i} + 8t\mathbf{j} + (-16t^2 + 100)\mathbf{k}$

19. $\mathbf{r}(t) = (e^t - 2t + 2)\mathbf{i} + (-e^{-t} + 3t + 2)\mathbf{j} + (e^{2t} - 2t + 1)\mathbf{k}$

21. $\mathbf{r}(t) = \left(\frac{3}{2}t^2 + \frac{6}{\sqrt{11}}t + 1\right)\mathbf{i} - \left(\frac{1}{2}t^2 + \frac{2}{\sqrt{11}}t - 2\right)\mathbf{j}$
$+\left(\frac{1}{2}t^2 + \frac{2}{\sqrt{11}}t + 3\right)\mathbf{k} = \left(\frac{1}{2}t^2 + \frac{2t}{\sqrt{11}}\right)(3\mathbf{i} - \mathbf{j} + \mathbf{k})$
$+(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$

23. 50 s

25. **(a)** 72.2 s; 25,510 m **(b)** 4020 m **(c)** 6378 m

27. **(a)** $v_0 \approx 9.9$ m/s **(b)** $\alpha \approx 18.4°$ or $71.6°$

29. $39.3°$ or $50.7°$ **35.** **(b)** $\mathbf{v}_0$ would bisect $\angle AOR$.

37. **(a)** (Assuming that "x" is zero at the point of impact)
$\mathbf{r}(t) = (x(t))\mathbf{i} + (y(t))\mathbf{j}$, where $x(t) = (12\cos 27°)t$ and
$y(t) = 1.3 + (12\sin 27°)t - 4.9t^2$.
(b) At $t \approx 0.556$ s, it reaches its maximum height of about 2.814 m.
(c) Range $\approx$ 14 m; flight time $\approx$ 1.31 s
(d) At $t \approx 0.232$ and $t \approx 0.880$ s, when it is $\approx$11.52 and $\approx$4.59 m from where it will land
(e) Yes. It changes things because the ball won't clear the net.

39. 1.225 m, 5.45 m/s

47. **(a)** $\mathbf{r}(t) = (x(t))\mathbf{i} + (y(t))\mathbf{j}$; where

$$x(t) = \left(\frac{1}{0.08}\right)(1 - e^{-0.08t})(50\cos 20° - 5) \text{ and}$$

$$y(t) = 1 + \left(\frac{50}{0.08}\right)(1 - e^{-0.08t})(\sin 20°)$$
$$+\left(\frac{9.8}{0.08^2}\right)(1 - 0.08t - e^{-0.08t})$$

(b) At $t \approx 1.633$ s it reaches a maximum height of about 14.66 m.
(c) Range $\approx$ 125.11 m; flight time $\approx$ 3.404 s
(d) At $t \approx 0.670$ and 2.622 s, when it is about 27.39 and 99.30 m from home plate
(e) No

SECTION 12.3, pp. 737–738

1. $\mathbf{T} = \left(-\frac{2}{3}\sin t\right)\mathbf{i} + \left(\frac{2}{3}\cos t\right)\mathbf{j} + \frac{\sqrt{5}}{3}\mathbf{k}$, 3π

3. $\mathbf{T} = \frac{1}{\sqrt{1+t}}\mathbf{i} + \frac{\sqrt{t}}{\sqrt{1+t}}\mathbf{k}$, $\frac{52}{3}$

5. $\mathbf{T} = -\cos t\mathbf{j} + \sin t\mathbf{k}$, $\frac{3}{2}$

7. $\mathbf{T} = \left(\frac{\cos t - t\sin t}{t+1}\right)\mathbf{i} + \left(\frac{\sin t + t\cos t}{t+1}\right)\mathbf{j}$
$+\left(\frac{\sqrt{2}t^{1/2}}{t+1}\right)\mathbf{k}$, $\frac{\pi^2}{2} + \pi$

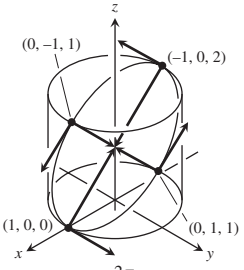
9. $(0, 5, 24\pi)$

11. $s(t) = 5t, \quad L = \frac{5\pi}{2}$

13. $s(t) = \sqrt{3}e^t - \sqrt{3}, \quad L = \frac{3\sqrt{3}}{4}$

15. $\sqrt{2} + \ln\left(1 + \sqrt{2}\right)$

17. **(a)** Cylinder is $x^2 + y^2 = 1$; plane is $x + z = 1$.
(b) and **(c)**



(d) $L = \int_0^{2\pi} \sqrt{1 + \sin^2 t}\, dt$ **(e)** $L \approx 7.64$

SECTION 12.4, pp. 743–744

1. $\mathbf{T} = (\cos t)\mathbf{i} - (\sin t)\mathbf{j}, \quad \mathbf{N} = (-\sin t)\mathbf{i} - (\cos t)\mathbf{j}, \quad \kappa = \cos t$

3. **(a)** $\mathbf{T} = \frac{1}{\sqrt{1+t^2}}\mathbf{i} - \frac{t}{\sqrt{1+t^2}}\mathbf{j}$, $\mathbf{N} = \frac{-t}{\sqrt{1+t^2}}\mathbf{i} -$
$\frac{1}{\sqrt{1+t^2}}\mathbf{j}$, $\kappa = \frac{1}{2\left(\sqrt{1+t^2}\right)^3}$

5. **(b)** $\cos x$

7. **(b)** $\mathbf{N} = \dfrac{-2e^{2t}}{\sqrt{1+4e^{4t}}}\mathbf{i} + \dfrac{1}{\sqrt{1+4e^{4t}}}\mathbf{j}$

(c) $\mathbf{N} = -\dfrac{1}{2}\left(\sqrt{4-t^2}\,\mathbf{i} + t\mathbf{j}\right)$

9. $\mathbf{T} = \dfrac{3\cos t}{5}\mathbf{i} - \dfrac{3\sin t}{5}\mathbf{j} + \dfrac{4}{5}\mathbf{k}$,

$\mathbf{N} = (-\sin t)\mathbf{i} - (\cos t)\mathbf{j}$, $\kappa = \dfrac{3}{25}$

11. $\mathbf{T} = \left(\dfrac{\cos t - \sin t}{\sqrt{2}}\right)\mathbf{i} + \left(\dfrac{\cos t + \sin t}{\sqrt{2}}\right)\mathbf{j}$,

$\mathbf{N} = \left(\dfrac{-\cos t - \sin t}{\sqrt{2}}\right)\mathbf{i} + \left(\dfrac{-\sin t + \cos t}{\sqrt{2}}\right)\mathbf{j}$, $\kappa = \dfrac{1}{e^t\sqrt{2}}$

13. $\mathbf{T} = \dfrac{t}{\sqrt{t^2+1}}\mathbf{i} + \dfrac{1}{\sqrt{t^2+1}}\mathbf{j}$,

$\mathbf{N} = \dfrac{\mathbf{i}}{\sqrt{t^2+1}} - \dfrac{t\mathbf{j}}{\sqrt{t^2+1}}$, $\kappa = \dfrac{1}{t(t^2+1)^{3/2}}$

15. $\mathbf{T} = \left(\operatorname{sech}\dfrac{t}{a}\right)\mathbf{i} + \left(\tanh\dfrac{t}{a}\right)\mathbf{j}$,

$\mathbf{N} = \left(-\tanh\dfrac{t}{a}\right)\mathbf{i} + \left(\operatorname{sech}\dfrac{t}{a}\right)\mathbf{j}$,

$\kappa = \dfrac{1}{a}\operatorname{sech}^2\dfrac{t}{a}$

19. $1/(2b)$

21. $\left(x - \dfrac{\pi}{2}\right)^2 + y^2 = 1$

23. $\kappa(x) = 2/(1+4x^2)^{3/2}$

25. $\kappa(x) = |\sin x|/(1+\cos^2 x)^{3/2}$

27. maximum curvature $2/\left(3\sqrt{3}\right)$ at $x = 1/\sqrt{2}$

SECTION 12.5, p. 750

1. $\mathbf{a} = |a|\mathbf{N}$ **3.** $\mathbf{a}(1) = \dfrac{4}{3}\mathbf{T} + \dfrac{2\sqrt{5}}{3}\mathbf{N}$ **5.** $\mathbf{a}(0) = 2\mathbf{N}$

7. $\mathbf{r}\left(\dfrac{\pi}{4}\right) = \dfrac{\sqrt{2}}{2}\mathbf{i} + \dfrac{\sqrt{2}}{2}\mathbf{j} - \mathbf{k}$, $\mathbf{T}\left(\dfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2}\mathbf{i} + \dfrac{\sqrt{2}}{2}\mathbf{j}$,

$\mathbf{N}\left(\dfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2}\mathbf{i} - \dfrac{\sqrt{2}}{2}\mathbf{j}$, $\mathbf{B}\left(\dfrac{\pi}{4}\right) = \mathbf{k}$; osculating plane:

$z = -1$; normal plane: $-x + y = 0$; rectifying plane:
$x + y = \sqrt{2}$

9. $\mathbf{B} = \left(\dfrac{4}{5}\cos t\right)\mathbf{i} - \left(\dfrac{4}{5}\sin t\right)\mathbf{j} - \dfrac{3}{5}\mathbf{k}$, $\tau = -\dfrac{4}{25}$

11. $\mathbf{B} = \mathbf{k}$, $\tau = 0$ **13.** $\mathbf{B} = -\mathbf{k}$, $\tau = 0$ **15.** $\mathbf{B} = \mathbf{k}$, $\tau = 0$

17. Yes. If the car is moving on a curved path ($\kappa \neq 0$), then $a_N = \kappa|\mathbf{v}|^2 \neq 0$ and $\mathbf{a} \neq \mathbf{0}$.

23. $\kappa = \dfrac{1}{t}$, $\rho = t$

27. Components of $\mathbf{v}$: -1.8701, 0.7089, 1.0000
Components of $\mathbf{a}$: -1.6960, -2.0307, 0
Speed: 2.2361; Components of $\mathbf{T}$: -0.8364, 0.3170, 0.4472
Components of $\mathbf{N}$: -0.4143, -0.8998, -0.1369
Components of $\mathbf{B}$: 0.3590, -0.2998, 0.8839; Curvature: 0.5060
Torsion: 0.2813; Tangential component of acceleration: 0.7746
Normal component of acceleration: 2.5298

29. Components of $\mathbf{v}$: 2.0000, 0, -0.1629
Components of $\mathbf{a}$: 0, -1.0000, -0.0086; Speed: 2.0066
Components of $\mathbf{T}$: 0.9967, 0, -0.0812
Components of $\mathbf{N}$: -0.0007, -1.0000, -0.0086
Components of $\mathbf{B}$: -0.0812, 0.0086, 0.9967;
Curvature: 0.2484
Torsion: 0.0411; Tangential component of acceleration: 0.0007
Normal component of acceleration: 1.0000

SECTION 12.6, p. 754

1. $\mathbf{v} = 2\mathbf{u}_r + 2\theta\mathbf{u}_\theta$
$\mathbf{a} = -4\theta\mathbf{u}_r + 8\mathbf{u}_\theta$

3. $\mathbf{v} = (3a\sin\theta)\mathbf{u}_r + 3a(1-\cos\theta)\mathbf{u}_\theta$
$\mathbf{a} = 9a(2\cos\theta - 1)\mathbf{u}_r + (18a\sin\theta)\mathbf{u}_\theta$

5. $\mathbf{v} = 2ae^{a\theta}\mathbf{u}_r + 2e^{a\theta}\mathbf{u}_\theta$
$\mathbf{a} = 4e^{a\theta}(a^2-1)\mathbf{u}_r + 8ae^{a\theta}\mathbf{u}_\theta$

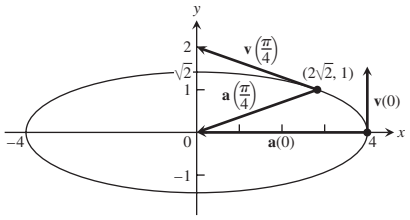
7. $\mathbf{v} = (-8\sin 4t)\mathbf{u}_r + (4\cos 4t)\mathbf{u}_\theta$
$\mathbf{a} = (-40\cos 4t)\mathbf{u}_r - (32\sin 4t)\mathbf{u}_\theta$

13. $\approx 29.93 \times 10^{10}$ m **15.** $\approx 2.25 \times 10^9$ km^2/s

17. $\approx 1.876 \times 10^{27}$ kg

PRACTICE EXERCISES, pp. 755–756

1. $\dfrac{x^2}{16} + \dfrac{y^2}{2} = 1$



At $t = 0$: $a_T = 0$, $a_N = 4$, $\kappa = 2$;
At $t = \dfrac{\pi}{4}$: $a_T = \dfrac{7}{3}$, $a_N = \dfrac{4\sqrt{2}}{3}$, $\kappa = \dfrac{4\sqrt{2}}{27}$

3. $|\mathbf{v}|_{\max} = 1$ **5.** $\kappa = 1/5$ **7.** $dy/dt = -x$; clockwise

11. Shot put is on the ground, about 21.88 m from the stopboard.

15. Length $= \dfrac{\pi}{4}\sqrt{1+\dfrac{\pi^2}{16}} + \ln\left(\dfrac{\pi}{4} + \sqrt{1+\dfrac{\pi^2}{16}}\right)$

17. $\mathbf{T}(0) = \dfrac{2}{3}\mathbf{i} - \dfrac{2}{3}\mathbf{j} + \dfrac{1}{3}\mathbf{k}$; $\mathbf{N}(0) = \dfrac{1}{\sqrt{2}}\mathbf{i} + \dfrac{1}{\sqrt{2}}\mathbf{j}$;

$\mathbf{B}(0) = -\dfrac{1}{3\sqrt{2}}\mathbf{i} + \dfrac{1}{3\sqrt{2}}\mathbf{j} + \dfrac{4}{3\sqrt{2}}\mathbf{k}$; $\kappa = \dfrac{\sqrt{2}}{3}$; $\tau = \dfrac{1}{6}$

19. $\mathbf{T}(\ln 2) = \dfrac{1}{\sqrt{17}}\mathbf{i} + \dfrac{4}{\sqrt{17}}\mathbf{j}$; $\mathbf{N}(\ln 2) = -\dfrac{4}{\sqrt{17}}\mathbf{i} + \dfrac{1}{\sqrt{17}}\mathbf{j}$;

$\mathbf{B}(\ln 2) = \mathbf{k}$; $\kappa = \dfrac{8}{17\sqrt{17}}$; $\tau = 0$

21. $\mathbf{a}(0) = 10\mathbf{T} + 6\mathbf{N}$

23. $\mathbf{T} = \left(\dfrac{1}{\sqrt{2}}\cos t\right)\mathbf{i} - (\sin t)\mathbf{j} + \left(\dfrac{1}{\sqrt{2}}\cos t\right)\mathbf{k}$;

$\mathbf{N} = \left(-\dfrac{1}{\sqrt{2}}\sin t\right)\mathbf{i} - (\cos t)\mathbf{j} - \left(\dfrac{1}{\sqrt{2}}\sin t\right)\mathbf{k}$;

$\mathbf{B} = \dfrac{1}{\sqrt{2}}\mathbf{i} - \dfrac{1}{\sqrt{2}}\mathbf{k}$; $\kappa = \dfrac{1}{\sqrt{2}}$; $\tau = 0$

25. $\dfrac{\pi}{3}$ **27.** $x = 1 + t$, $y = t$, $z = -t$ **31.** $\kappa = \dfrac{1}{a}$

ADDITIONAL AND ADVANCED EXERCISES, pp. 757–758

1. **(a)** $\left.\dfrac{d\theta}{dt}\right|_{\theta=2\pi} = 2\sqrt{\dfrac{\pi gb}{a^2+b^2}}$

(b) $\theta = \dfrac{gbt^2}{2(a^2+b^2)}$, $z = \dfrac{gb^2t^2}{2(a^2+b^2)}$

(c) $\mathbf{v}(t) = \dfrac{gbt}{\sqrt{a^2 + b^2}}\mathbf{T}$;

$$\frac{d^2\mathbf{r}}{dt^2} = \frac{bg}{\sqrt{a^2 + b^2}}\mathbf{T} + a\left(\frac{bgt}{a^2 + b^2}\right)^2\mathbf{N}$$

There is no component in the direction of $\mathbf{B}$.

5. (a) $\dfrac{dx}{dt} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta, \dfrac{dy}{dt} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

(b) $\dfrac{dr}{dt} = \dot{x}\cos\theta + \dot{y}\sin\theta, r\dfrac{d\theta}{dt} = -\dot{x}\sin\theta + \dot{y}\cos\theta$

7. (a) $\mathbf{a}(1) = -9\mathbf{u}_r - 6\mathbf{u}_\theta, \mathbf{v}(1) = -\mathbf{u}_r + 3\mathbf{u}_\theta$

(b) 6.5 cm

9. **(c)** $\mathbf{v} = \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta + \dot{z}\mathbf{k}, \mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{u}_\theta + \ddot{z}\mathbf{k}$

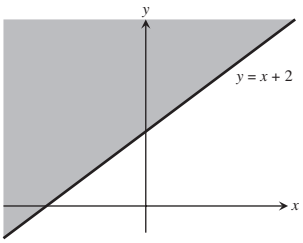
Chapter 13

SECTION 13.1, pp. 765–767

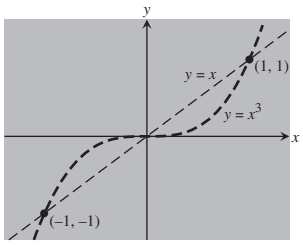
1. **(a)** 0 **(b)** 0 **(c)** 58 **(d)** 33

3. **(a)** $4/5$ **(b)** $8/5$ **(c)** 3 **(d)** 0

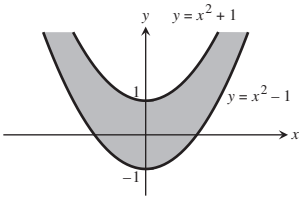
5. Domain: all points (x, y) on or above line $y = x + 2$



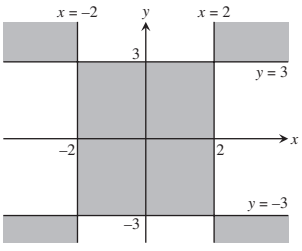
7. Domain: all points (x, y) not lying on the graph of $y = x$ or $y = x^3$



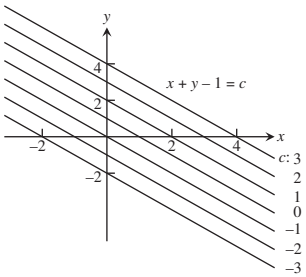
9. Domain: all points (x, y) satisfying $x^2 - 1 \le y \le x^2 + 1$



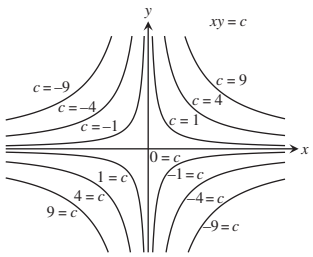
11. Domain: all points (x, y) for which $(x - 2)(x + 2)(y - 3)(y + 3) \ge 0$



13.



15.



17. **(a)** All points in the xy-plane **(b)** All reals
(c) The lines $y - x = c$ **(d)** No boundary points
(e) Both open and closed **(f)** Unbounded

19. **(a)** All points in the xy-plane **(b)** $z \ge 0$
(c) For $f(x, y) = 0$, the origin; for $f(x, y) \ne 0$, ellipses with the center $(0, 0)$, and major and minor axes along the x- and y-axes, respectively
(d) No boundary points **(e)** Both open and closed
(f) Unbounded

21. **(a)** All points in the xy-plane **(b)** All reals
(c) For $f(x, y) = 0$, the x- and y-axes; for $f(x, y) \ne 0$, hyperbolas with the x- and y-axes as asymptotes
(d) No boundary points **(e)** Both open and closed
(f) Unbounded

23. **(a)** All (x, y) satisfying $x^2 + y^2 < 16$ **(b)** $z \ge 1/4$
(c) Circles centered at the origin with radii $r < 4$
(d) Boundary is the circle $x^2 + y^2 = 16$
(e) Open **(f)** Bounded

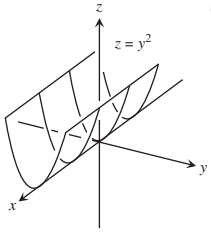
25. **(a)** $(x, y) \ne (0, 0)$ **(b)** All reals
(c) The circles with center $(0, 0)$ and radii $r > 0$
(d) Boundary is the single point $(0, 0)$
(e) Open **(f)** Unbounded

27. **(a)** All (x, y) satisfying $-1 \le y - x \le 1$
(b) $-\pi/2 \le z \le \pi/2$
(c) Straight lines of the form $y - x = c$ where $-1 \le c \le 1$
(d) Boundary is two straight lines $y = 1 + x$ and $y = -1 + x$
(e) Closed **(f)** Unbounded

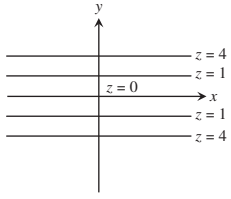
29. **(a)** Domain: all points (x, y) outside the circle $x^2 + y^2 = 1$
(b) Range: all reals
(c) Circles centered at the origin with radii $r > 1$
(d) Boundary: $x^2 + y^2 = 1$
(e) Open **(f)** Unbounded

31. (f), (h) **33.** (a), (i) **35.** (d), (j)

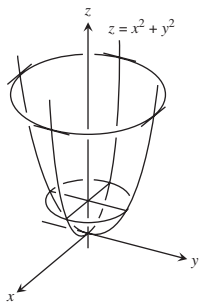
37. **(a)**



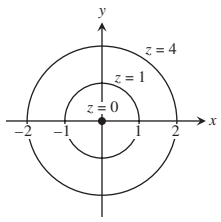
(b)



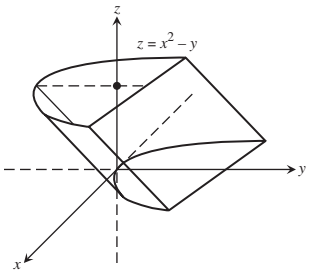
39. (a)



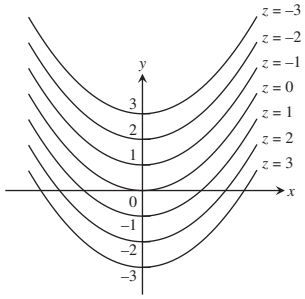
(b)



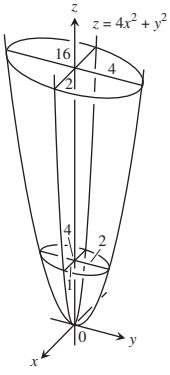
41. (a)



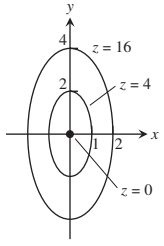
(b)



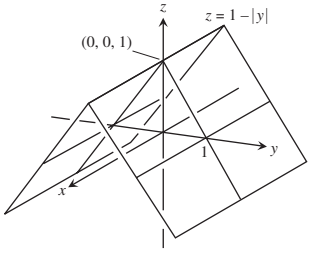
43. (a)



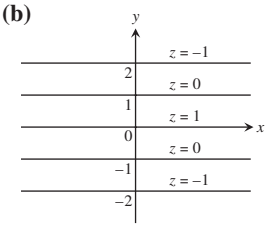
(b)



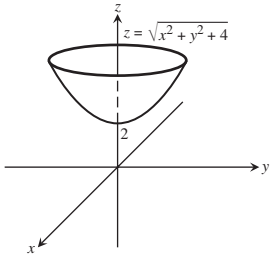
45. (a)



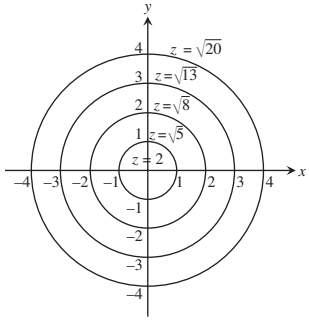
(b)



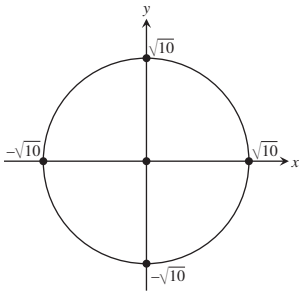
47. (a)



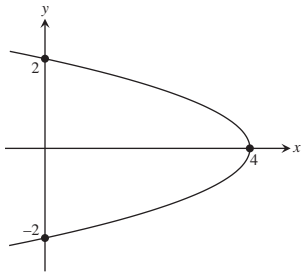
(b)



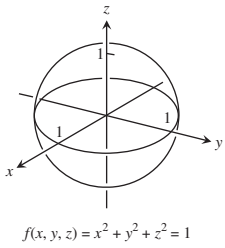
49. $x^2 + y^2 = 10$



51. $x + y^2 = 4$

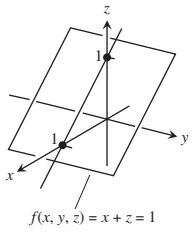


53.



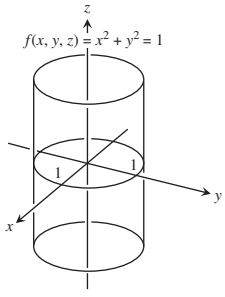
$f(x, y, z) = x^2 + y^2 + z^2 = 1$

55.

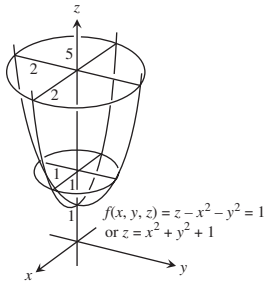


$f(x, y, z) = x + z = 1$

57.



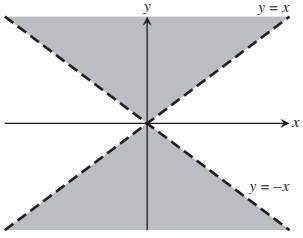
59.



61. $\sqrt{x - y} - \ln z = 2$

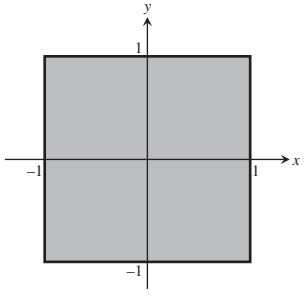
63. $x^2 + y^2 + z^2 = 4$

65. Domain: all points (x, y) satisfying $|x| < |y|$



level curve: $y = 2x$

67. Domain: all points (x, y) satisfying $-1 \le x \le 1$ and $-1 \le y \le 1$



level curve: $\sin^{-1} y - \sin^{-1} x = \dfrac{\pi}{2}$

SECTION 13.2, pp. 773–776

1. $5/2$ **3.** $2\sqrt{6}$ **5.** 1 **7.** $1/2$ **9.** 1
11. $1/4$ **13.** 0 **15.** -1 **17.** 2 **19.** $1/4$
21. 1 **23.** 3 **25.** $19/12$ **27.** 2 **29.** 3
31. (a) All (x, y) **(b)** All (x, y) except $(0, 0)$
33. (a) All (x, y) except where $x = 0$ or $y = 0$ **(b)** All (x, y)
35. (a) All (x, y, z)
(b) All (x, y, z) except the interior of the cylinder $x^2 + y^2 = 1$
37. (a) All (x, y, z) with $z \neq 0$ **(b)** All (x, y, z) with $x^2 + z^2 \neq 1$
39. (a) All points (x, y, z) satisfying $z > x^2 + y^2 + 1$
(b) All points (x, y, z) satisfying $z \neq \sqrt{x^2 + y^2}$
41. Consider paths along $y = x, x > 0$, and along $y = x, x < 0$.
43. Consider the paths $y = kx^2$, k a constant.
45. Consider the paths $y = mx$, m a constant, $m \neq -1$.
47. Consider the paths $y = kx^2$, k a constant, $k \neq 0$.
49. Consider the paths $x = 1$ and $y = x$.
51. Along $y = 1$ the limit is 0; along $y = e^x$ the limit is $1/2$.
53. Along $y = 0$ the limit is 1; along $y = -\sin x$ the limit is 0.
55. (a) 1 **(b)** 0 **(c)** Does not exist
59. The limit is 1. **61.** The limit is 0.
63. (a) $f(x, y)\big|_{y=mx} = \sin 2\theta$ where $\tan\theta = m$ **65.** 0
67. Does not exist **69.** $\pi/2$ **71.** $f(0, 0) = \ln 3$
73. $\delta = 0.1$ **75.** $\delta = 0.005$ **77.** $\delta = 0.04$
79. $\delta = \sqrt{0.015}$ **81.** $\delta = 0.005$

SECTION 13.3, pp. 785–788

1. $\dfrac{\partial f}{\partial x} = 4x, \dfrac{\partial f}{\partial y} = -3$ **3.** $\dfrac{\partial f}{\partial x} = 2x(y + 2), \dfrac{\partial f}{\partial y} = x^2 - 1$

5. $\dfrac{\partial f}{\partial x} = 2y(xy - 1), \dfrac{\partial f}{\partial y} = 2x(xy - 1)$

7. $\dfrac{\partial f}{\partial x} = \dfrac{x}{\sqrt{x^2 + y^2}}, \dfrac{\partial f}{\partial y} = \dfrac{y}{\sqrt{x^2 + y^2}}$

9. $\dfrac{\partial f}{\partial x} = \dfrac{-1}{(x + y)^2}, \dfrac{\partial f}{\partial y} = \dfrac{-1}{(x + y)^2}$

11. $\dfrac{\partial f}{\partial x} = \dfrac{-y^2 - 1}{(xy - 1)^2}, \dfrac{\partial f}{\partial y} = \dfrac{-x^2 - 1}{(xy - 1)^2}$

13. $\dfrac{\partial f}{\partial x} = e^{x+y+1}, \dfrac{\partial f}{\partial y} = e^{x+y+1}$ **15.** $\dfrac{\partial f}{\partial x} = \dfrac{1}{x + y}, \dfrac{\partial f}{\partial y} = \dfrac{1}{x + y}$

17. $\dfrac{\partial f}{\partial x} = 2\sin(x - 3y)\cos(x - 3y)$,
$\dfrac{\partial f}{\partial y} = -6\sin(x - 3y)\cos(x - 3y)$

19. $\dfrac{\partial f}{\partial x} = yx^{y-1}, \dfrac{\partial f}{\partial y} = x^y \ln x$ **21.** $\dfrac{\partial f}{\partial x} = -g(x), \dfrac{\partial f}{\partial y} = g(y)$

23. $f_x = y^2, f_y = 2xy, f_z = -4z$

25. $f_x = 1, f_y = -y(y^2 + z^2)^{-1/2}, f_z = -z(y^2 + z^2)^{-1/2}$

27. $f_x = \dfrac{yz}{\sqrt{1 - x^2y^2z^2}}, f_y = \dfrac{xz}{\sqrt{1 - x^2y^2z^2}}, f_z = \dfrac{xy}{\sqrt{1 - x^2y^2z^2}}$

29. $f_x = \dfrac{1}{x + 2y + 3z}, f_y = \dfrac{2}{x + 2y + 3z}, f_z = \dfrac{3}{x + 2y + 3z}$

31. $f_x = -2xe^{-(x^2+y^2+z^2)}, f_y = -2ye^{-(x^2+y^2+z^2)}, f_z = -2ze^{-(x^2+y^2+z^2)}$

33. $f_x = \operatorname{sech}^2(x + 2y + 3z), f_y = 2\operatorname{sech}^2(x + 2y + 3z)$,
$f_z = 3\operatorname{sech}^2(x + 2y + 3z)$

35. $\dfrac{\partial f}{\partial t} = -2\pi\sin(2\pi t - \alpha), \dfrac{\partial f}{\partial \alpha} = \sin(2\pi t - \alpha)$

37. $\dfrac{\partial h}{\partial \rho} = \sin\phi\cos\theta, \dfrac{\partial h}{\partial \phi} = \rho\cos\phi\cos\theta, \dfrac{\partial h}{\partial \theta} = -\rho\sin\phi\sin\theta$

39. $W_P(P, V, \delta, v, g) = V, W_V(P, V, \delta, v, g) = P + \dfrac{\delta v^2}{2g}$,
$W_\delta(P, V, \delta, v, g) = \dfrac{Vv^2}{2g}, W_v(P, V, \delta, v, g) = \dfrac{V\delta v}{g}$,
$W_g(P, V, \delta, v, g) = -\dfrac{V\delta v^2}{2g^2}$

41. $\dfrac{\partial f}{\partial x} = 1 + y, \dfrac{\partial f}{\partial y} = 1 + x, \dfrac{\partial^2 f}{\partial x^2} = 0, \dfrac{\partial^2 f}{\partial y^2} = 0, \dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y} = 1$

43. $\dfrac{\partial g}{\partial x} = 2xy + y\cos x, \dfrac{\partial g}{\partial y} = x^2 - \sin y + \sin x$,
$\dfrac{\partial^2 g}{\partial x^2} = 2y - y\sin x, \dfrac{\partial^2 g}{\partial y^2} = -\cos y$,
$\dfrac{\partial^2 g}{\partial y\,\partial x} = \dfrac{\partial^2 g}{\partial x\,\partial y} = 2x + \cos x$

45. $\dfrac{\partial r}{\partial x} = \dfrac{1}{x + y}, \dfrac{\partial r}{\partial y} = \dfrac{1}{x + y}, \dfrac{\partial^2 r}{\partial x^2} = \dfrac{-1}{(x + y)^2}, \dfrac{\partial^2 r}{\partial y^2} = \dfrac{-1}{(x + y)^2}$,
$\dfrac{\partial^2 r}{\partial y\,\partial x} = \dfrac{\partial^2 r}{\partial x\,\partial y} = \dfrac{-1}{(x + y)^2}$

47. $\dfrac{\partial w}{\partial x} = x^2y\sec^2(xy) + 2x\tan(xy), \dfrac{\partial w}{\partial y} = x^3\sec^2(xy)$,
$\dfrac{\partial^2 w}{\partial y\,\partial x} = \dfrac{\partial^2 w}{\partial x\,\partial y} = 2x^3y\sec^2(xy)\tan(xy) + 3x^2\sec^2(xy)$
$\dfrac{\partial^2 w}{\partial x^2} = 4xy\sec^2(xy) + 2x^2y^2\sec^2(xy)\tan(xy) + 2\tan(xy)$
$\dfrac{\partial^2 w}{\partial y^2} = 2x^4\sec^2(xy)\tan(xy)$

49. $\dfrac{\partial w}{\partial x} = \sin(x^2y) + 2x^2y\cos(x^2y), \dfrac{\partial w}{\partial y} = x^3\cos(x^2y)$,
$\dfrac{\partial^2 w}{\partial y\,\partial x} = \dfrac{\partial^2 w}{\partial x\,\partial y} = 3x^2\cos(x^2y) - 2x^4y\sin(x^2y)$
$\dfrac{\partial^2 w}{\partial x^2} = 6xy\cos(x^2y) - 4x^3y^2\sin(x^2y)$
$\dfrac{\partial^2 w}{\partial y^2} = -x^5\sin(x^2y)$

51. $\dfrac{\partial f}{\partial x} = 2xy^3 - 4x^3, \dfrac{\partial f}{\partial y} = 3x^2y^2 + 5y^4$,
$\dfrac{\partial^2 f}{\partial x^2} = 2y^3 - 12x^2, \dfrac{\partial^2 f}{\partial y^2} = 6x^2y + 20y^3$,
$\dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y} = 6xy^2$

53. $\dfrac{\partial z}{\partial x} = 2x\cos(2x - y^2) + \sin(2x - y^2)$,
$\dfrac{\partial z}{\partial y} = -2xy\cos(2x - y^2)$,
$\dfrac{\partial^2 z}{\partial x^2} = 4\cos(2x - y^2) - 4x\sin(2x - y^2)$,
$\dfrac{\partial^2 z}{\partial y^2} = -4xy^2\sin(2x - y^2) - 2x\cos(2x - y^2)$,
$\dfrac{\partial^2 z}{\partial x\,\partial y} = \dfrac{\partial^2 z}{\partial y\,\partial x} = 4xy\sin(2x - y^2) - 2y\cos(2x - y^2)$

55. $\dfrac{\partial w}{\partial x} = \dfrac{2}{2x + 3y}, \dfrac{\partial w}{\partial y} = \dfrac{3}{2x + 3y}, \dfrac{\partial^2 w}{\partial y\,\partial x} = \dfrac{\partial^2 w}{\partial x\,\partial y} = \dfrac{-6}{(2x + 3y)^2}$

57. $\dfrac{\partial w}{\partial x} = y^2 + 2xy^3 + 3x^2y^4, \dfrac{\partial w}{\partial y} = 2xy + 3x^2y^2 + 4x^3y^3$,

$\dfrac{\partial^2 w}{\partial y\,\partial x} = \dfrac{\partial^2 w}{\partial x\,\partial y} = 2y + 6xy^2 + 12x^2y^3$

59. $\dfrac{\partial \omega}{\partial x} = \dfrac{2x}{y^3}, \dfrac{\partial \omega}{\partial y} = \dfrac{-3x^2}{y^4}$

$\dfrac{\partial^2 \omega}{\partial y\,\partial x} = \dfrac{-6x}{y^4}, \dfrac{\partial^2 \omega}{\partial x\,\partial y} = \dfrac{-6x}{y^4}$

61. **(a)** x first **(b)** y first **(c)** x first
(d) x first **(e)** y first **(f)** y first

63. $f_x(1, 2) = -13, f_y(1, 2) = -2$

65. $f_x(-2, 3) = 1/2, f_y(-2, 3) = 3/4$

67. **(a)** 3 **(b)** 2 **69.** 12

71. $\dfrac{\partial f}{\partial x} = 3x^2y^2 - 2x \Rightarrow$

$f(x, y) = x^3y^2 - x^2 + g(y) \Rightarrow$

$\dfrac{\partial f}{\partial y} = 2x^3y + g'(y) = 2x^3y + 64 \Rightarrow$

$g'(y) = 6y \Rightarrow g(y) = 3y^2$ works $\Rightarrow$

$f(x, y) = x^3y^2 - x^2 + 3y^2$ works

73. $\dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{2x - 2y}{(x + y)^3} \neq \dfrac{\partial^2 f}{\partial x\,\partial y} = \dfrac{2y - 2x}{(x + y)^3}$ so impossible **75.** -2

77. $\dfrac{\partial A}{\partial a} = \dfrac{a}{bc\sin A}, \dfrac{\partial A}{\partial b} = \dfrac{c\cos A - b}{bc\sin A}$

79. $v_x = \dfrac{\ln v}{(\ln u)(\ln v) - 1}$

81. $f_x(x, y) = 0$ for all points (x, y),

$f_y(x, y) = \begin{cases} 3y^2, & y \geq 0 \\ -2y, & y < 0 \end{cases}$,

$f_{xy}(x, y) = f_{yx}(x, y) = 0$ for all points (x, y)

99. Yes

SECTION 13.4, pp. 795–797

1. **(a)** $\dfrac{dw}{dt} = 0$, **(b)** $\dfrac{dw}{dt}(\pi) = 0$

3. **(a)** $\dfrac{dw}{dt} = 1$, **(b)** $\dfrac{dw}{dt}(3) = 1$

5. **(a)** $\dfrac{dw}{dt} = 4t\tan^{-1}t + 1$, **(b)** $\dfrac{dw}{dt}(1) = \pi + 1$

7. **(a)** $\dfrac{\partial z}{\partial u} = 4\cos v \ln(u\sin v) + 4\cos v$,

$\dfrac{\partial z}{\partial v} = -4u\sin v \ln(u\sin v) + \dfrac{4u\cos^2 v}{\sin v}$

(b) $\dfrac{\partial z}{\partial u} = \sqrt{2}(\ln 2 + 2), \dfrac{\partial z}{\partial v} = -2\sqrt{2}(\ln 2 - 2)$

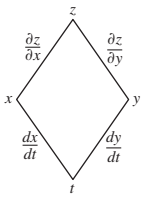
9. **(a)** $\dfrac{\partial w}{\partial u} = 2u + 4uv, \dfrac{\partial w}{\partial v} = -2v + 2u^2$

(b) $\dfrac{\partial w}{\partial u} = 3, \dfrac{\partial w}{\partial v} = -\dfrac{3}{2}$

11. **(a)** $\dfrac{\partial u}{\partial x} = 0, \dfrac{\partial u}{\partial y} = \dfrac{z}{(z - y)^2}, \dfrac{\partial u}{\partial z} = \dfrac{-y}{(z - y)^2}$

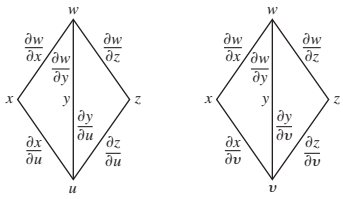
(b) $\dfrac{\partial u}{\partial x} = 0, \dfrac{\partial u}{\partial y} = 1, \dfrac{\partial u}{\partial z} = -2$

13. $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$

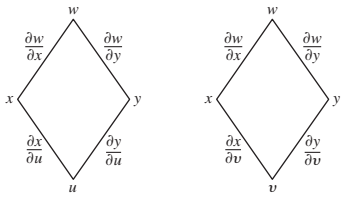


15. $\dfrac{\partial w}{\partial u} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial u} + \dfrac{\partial w}{\partial z}\dfrac{\partial z}{\partial u}$,

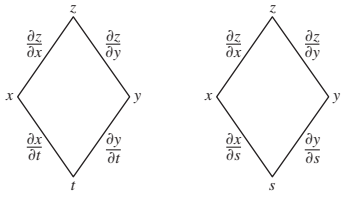
$\dfrac{\partial w}{\partial v} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial v} + \dfrac{\partial w}{\partial z}\dfrac{\partial z}{\partial v}$



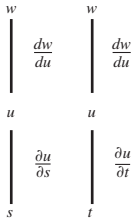
17. $\dfrac{\partial w}{\partial u} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial u}, \dfrac{\partial w}{\partial v} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial v}$.



19. $\dfrac{\partial z}{\partial t} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial t} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial t}, \dfrac{\partial z}{\partial s} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}$

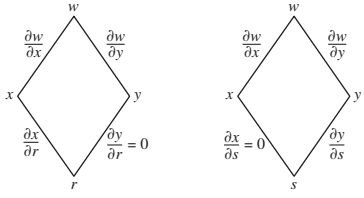


21. $\dfrac{\partial w}{\partial s} = \dfrac{dw}{du}\dfrac{\partial u}{\partial s}, \dfrac{\partial w}{\partial t} = \dfrac{dw}{du}\dfrac{\partial u}{\partial t}$



23. $\dfrac{\partial w}{\partial r} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial r} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial r} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial r}$ since $\dfrac{\partial y}{\partial r} = 0$,

$\dfrac{\partial w}{\partial s} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial s} = \dfrac{\partial w}{\partial y}\dfrac{\partial y}{\partial s}$ since $\dfrac{\partial x}{\partial s} = 0$



25. $4/3$ **27.** $-4/5$ **29.** 20 **31.** $\dfrac{\partial z}{\partial x} = \dfrac{1}{4}, \dfrac{\partial z}{\partial y} = -\dfrac{3}{4}$

33. $\dfrac{\partial z}{\partial x} = -1, \dfrac{\partial z}{\partial y} = -1$ **35.** 12 **37.** -7

39. $\dfrac{\partial z}{\partial u} = 2, \dfrac{\partial z}{\partial v} = 1$ **41.** $\dfrac{\partial w}{\partial t} = 2t\,e^{s^3+t^2}, \dfrac{\partial w}{\partial s} = 3s^2\,e^{s^3+t^2}$

43. 23 **45.** $-16, 2$ **47.** -0.00005 amp/s

53. $(\cos 1, \sin 1, 1)$ and $(\cos(-2), \sin(-2), -2)$

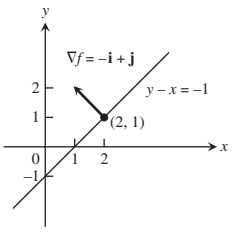
55. **(a)** Maximum at $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$; minimum at $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

(b) Max $= 6$, min $= 2$

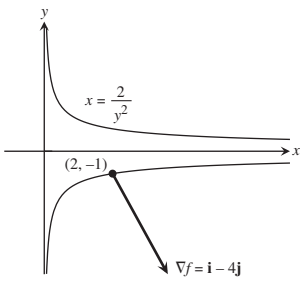
57. $5°\text{C/s}$ **59.** $2x\sqrt{x^8 + x^3} + \displaystyle\int_0^{x^2} \frac{3x^2}{2\sqrt{t^4 + x^3}}\,dt$

SECTION 13.5, pp. 805–806

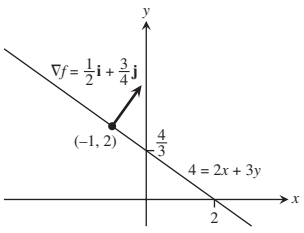
1.



3.



5.



7. $\nabla f = 3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ **9.** $\nabla f = -\frac{26}{27}\mathbf{i} + \frac{23}{54}\mathbf{j} - \frac{23}{54}\mathbf{k}$

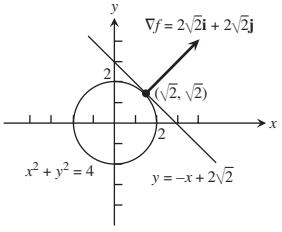
11. -4 **13.** $21/13$ **15.** 3 **17.** 2

19. $\mathbf{u} = -\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$, $(D_{\mathbf{u}}f)_{P_0} = \sqrt{2}$; $-\mathbf{u} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}$, $(D_{-\mathbf{u}}f)_{P_0} = -\sqrt{2}$

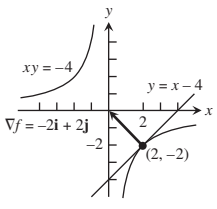
21. $\mathbf{u} = \frac{1}{3\sqrt{3}}\mathbf{i} - \frac{5}{3\sqrt{3}}\mathbf{j} - \frac{1}{3\sqrt{3}}\mathbf{k}$, $(D_{\mathbf{u}}f)_{P_0} = 3\sqrt{3}$; $-\mathbf{u} = -\frac{1}{3\sqrt{3}}\mathbf{i} + \frac{5}{3\sqrt{3}}\mathbf{j} + \frac{1}{3\sqrt{3}}\mathbf{k}$, $(D_{-\mathbf{u}}f)_{P_0} = -3\sqrt{3}$

23. $\mathbf{u} = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $(D_{\mathbf{u}}f)_{P_0} = 2\sqrt{3}$; $-\mathbf{u} = -\frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $(D_{-\mathbf{u}}f)_{P_0} = -2\sqrt{3}$

25.



27.



29. **(a)** $\mathbf{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$, $D_{\mathbf{u}}f(1, -1) = 5$

(b) $\mathbf{u} = -\frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$, $D_{\mathbf{u}}f(1, -1) = -5$

(c) $\mathbf{u} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$, $\mathbf{u} = -\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$

(d) $\mathbf{u} = -\mathbf{j}$, $\mathbf{u} = \frac{24}{25}\mathbf{i} - \frac{7}{25}\mathbf{j}$

(e) $\mathbf{u} = -\mathbf{i}$, $\mathbf{u} = \frac{7}{25}\mathbf{i} + \frac{24}{25}\mathbf{j}$

31. $\mathbf{u} = \frac{7}{\sqrt{53}}\mathbf{i} - \frac{2}{\sqrt{53}}\mathbf{j}$, $-\mathbf{u} = -\frac{7}{\sqrt{53}}\mathbf{i} + \frac{2}{\sqrt{53}}\mathbf{j}$

33. No, the maximum rate of change is $\sqrt{185} < 14$.

35. $-7/\sqrt{5}$ **41.** $r(t) = (-3 - 6t)\mathbf{i} + (4 + 8t)\mathbf{j}$, $-\infty < t < \infty$

43. $r(t) = (3 + 6t)\mathbf{i} + (-2 - 4t)\mathbf{j} + (1 + 2t)\mathbf{k}$, $-\infty < t < \infty$

SECTION 13.6, pp. 813–816

1. **(a)** $x + y + z = 3$

(b) $x = 1 + 2t, y = 1 + 2t, z = 1 + 2t$

3. **(a)** $2x - z - 2 = 0$

(b) $x = 2 - 4t, y = 0, z = 2 + 2t$

5. **(a)** $2x + 2y + z - 4 = 0$

(b) $x = 2t, y = 1 + 2t, z = 2 + t$

7. **(a)** $x + y + z - 1 = 0$

(b) $x = t, y = 1 + t, z = t$

9. **(a)** $-x + 3y + z/e = 2$

(b) $x = 2 - t, y = 1 + 3t, z = e + (1/e)t$

11. $2x - z - 2 = 0$

13. $x - y + 2z - 1 = 0$

15. $x = 1, y = 1 + 2t, z = 1 - 2t$

17. $x = 1 - 2t, y = 1, z = \frac{1}{2} + 2t$

19. $x = 1 + 90t, y = 1 - 90t, z = 3$

21. $df = \frac{9}{11{,}830} \approx 0.0008$ **23.** $dg = 0$

25. **(a)** $\frac{\sqrt{3}}{2}\sin\sqrt{3} - \frac{1}{2}\cos\sqrt{3} \approx 0.935°\text{C/m}$

(b) $\sqrt{3}\sin\sqrt{3} - \cos\sqrt{3} \approx 1.87°\text{C/s}$

27. **(a)** $L(x, y) = 1$ **(b)** $L(x, y) = 2x + 2y - 1$

29. **(a)** $L(x, y) = 3x - 4y + 5$ **(b)** $L(x, y) = 3x - 4y + 5$

31. **(a)** $L(x, y) = 1 + x$ **(b)** $L(x, y) = -y + \frac{\pi}{2}$

33. **(a)** $W(30, -5) = -13°\text{C}$, $W(50, -25) = -42.2°\text{C}$, $W(30, -10) = -19.5°\text{C}$

(b) $W(15, -40) \approx -53.7°\text{C}$, $W(80, -40) \approx -66.6°\text{C}$, $W(90, 0) \approx -10.2°\text{C}$

(c) $L(v, T) \approx 1.336\,T - 0.11v - 2.952$

(d) **i)** $L(39, 9) \approx -19.3°\text{C}$

ii) $L(42, -12) \approx -23.6°\text{C}$

iii) $L(10, -25) \approx -37.5°\text{C}$

35. $L(x, y) = 7 + x - 6y$; 0.06 **37.** $L(x, y) = x + y + 1$; 0.08

39. $L(x, y) = 1 + x$; 0.0222

41. **(a)** $L(x, y, z) = 2x + 2y + 2z - 3$ **(b)** $L(x, y, z) = y + z$

(c) $L(x, y, z) = 0$

43. **(a)** $L(x, y, z) = x$

(b) $L(x, y, z) = \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y$

(c) $L(x, y, z) = \frac{1}{3}x + \frac{2}{3}y + \frac{2}{3}z$

45. **(a)** $L(x, y, z) = 2 + x$

(b) $L(x, y, z) = x - y - z + \frac{\pi}{2} + 1$

(c) $L(x, y, z) = x - y - z + \frac{\pi}{2} + 1$

47. $L(x, y, z) = 2x - 6y - 2z + 6$, 0.0024

49. $L(x, y, z) = x + y - z - 1$, 0.00135

51. Maximum error (estimate) ≤ 0.31 in magnitude

53. Pay more attention to the smaller of the two dimensions. It will generate the larger partial derivative.

55. f is most sensitive to a change in d.

61. **(a)** 1.75% **(b)** 1.75%

SECTION 13.7, pp. 823–825

1. $f(-3, 3) = -5$, local minimum **3.** $f(-2, 1)$, saddle point

5. $f\left(3, \frac{3}{2}\right) = \frac{17}{2}$, local maximum

7. $f(2, -1) = -6$, local minimum **9.** $f(1, 2)$, saddle point

11. $f\left(\frac{16}{7}, 0\right) = -\frac{16}{7}$, local maximum

13. $f(0, 0)$, saddle point; $f\left(-\frac{2}{3}, \frac{2}{3}\right) = \frac{170}{27}$, local maximum

15. $f(0, 0) = 0$, local minimum; $f(1, -1)$, saddle point

17. $f(0, \pm\sqrt{5})$, saddle points; $f(-2, -1) = 30$, local maximum; $f(2, 1) = -30$, local minimum

19. $f(0, 0)$, saddle point; $f(1, 1) = 2$, $f(-1, -1) = 2$, local maxima

21. $f(0, 0) = -1$, local maximum

23. $f(n\pi, 0)$, saddle points, for every integer n

25. $f(2, 0) = e^{-4}$, local minimum

27. $f(0, 0) = 0$, local minimum; $f(0, 2)$, saddle point

29. $f\left(\frac{1}{2}, 1\right) = \ln\left(\frac{1}{4}\right) - 3$, local maximum

31. Absolute maximum: 1 at (0, 0); absolute minimum: -5 at (1, 2)

33. Absolute maximum: 4 at (0, 2); absolute minimum: 0 at (0, 0)

35. Absolute maximum: 11 at $(0, -3)$; absolute minimum: -10 at $(4, -2)$

37. Absolute maximum: 4 at (2, 0); absolute minimum: $\frac{3\sqrt{2}}{2}$ at $\left(3, -\frac{\pi}{4}\right)$, $\left(3, \frac{\pi}{4}\right)$, $\left(1, -\frac{\pi}{4}\right)$, and $\left(1, \frac{\pi}{4}\right)$

39. $a = -3$, $b = 2$

41. Hottest is $2\frac{1}{4}^\circ$ at $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ and $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$; coldest is $-\frac{1}{4}^\circ$ at $\left(\frac{1}{2}, 0\right)$.

43. **(a)** $f(0, 0)$, saddle point **(b)** $f(1, 2)$, local minimum
(c) $f(1, -2)$, local minimum; $f(-1, -2)$, saddle point

49. $\left(\frac{1}{6}, \frac{1}{3}, \frac{355}{36}\right)$ **51.** $\left(\frac{9}{7}, \frac{6}{7}, \frac{3}{7}\right)$ **53.** 3, 3, 3 **55.** 12

57. $\frac{4}{\sqrt{3}} \times \frac{4}{\sqrt{3}} \times \frac{4}{\sqrt{3}}$ **59.** 2 m × 2 m × 1 m

61. Points (0, 2, 0) and $(0, -2, 0)$ have distance 2 from the origin.

63. **(a)** On the semicircle, max $f = 2\sqrt{2}$ at $t = \pi/4$, min $f = -2$ at $t = \pi$. On the quarter circle, max $f = 2\sqrt{2}$ at $t = \pi/4$, min $f = 2$ at $t = 0, \pi/2$.
(b) On the semicircle, max $g = 2$ at $t = \pi/4$, min $g = -2$ at $t = 3\pi/4$. On the quarter circle, max $g = 2$ at $t = \pi/4$, min $g = 0$ at $t = 0, \pi/2$.
(c) On the semicircle, max $h = 8$ at $t = 0, \pi$; min $h = 4$ at $t = \pi/2$. On the quarter circle, max $h = 8$ at $t = 0$, min $h = 4$ at $t = \pi/2$.

65. **i)** min $f = -1/2$ at $t = -1/2$; no max
ii) max $f = 0$ at $t = -1, 0$; min $f = -1/2$ at $t = -1/2$
iii) max $f = 4$ at $t = 1$; min $f = 0$ at $t = 0$

69. $y = -\frac{20}{13}x + \frac{9}{13}$, $y|_{x=4} = -\frac{71}{13}$

SECTION 13.8, pp. 832–835

1. $\left(\pm\frac{1}{\sqrt{2}}, \frac{1}{2}\right), \left(\pm\frac{1}{\sqrt{2}}, -\frac{1}{2}\right)$ **3.** 39 **5.** $(3, \pm 3\sqrt{2})$

7. **(a)** 8 **(b)** 64

9. $r = 2$ cm, $h = 4$ cm

11. Length $= 4\sqrt{2}$, width $= 3\sqrt{2}$

13. $f(0, 0) = 0$ is minimum; $f(2, 4) = 20$ is maximum.

15. Lowest $= 0°$, highest $= 125°$

17. $\left(\frac{3}{2}, 2, \frac{5}{2}\right)$ **19.** 1 **21.** $(0, 0, 2)$, $(0, 0, -2)$

23. $f(1, -2, 5) = 30$ is maximum; $f(-1, 2, -5) = -30$ is minimum.

25. 3, 3, 3 **27.** $\frac{2}{\sqrt{3}}$ by $\frac{2}{\sqrt{3}}$ by $\frac{2}{\sqrt{3}}$ units

29. $(\pm 4/3, -4/3, -4/3)$ **31.** ≈24,322 units

33. $U(8, 14) = \$128$ **37.** $f(2/3, 4/3, -4/3) = \frac{4}{3}$

39. (2, 4, 4) **41.** Maximum is $1 + 6\sqrt{3}$ at $\left(\pm\sqrt{6}, \sqrt{3}, 1\right)$; minimum is $1 - 6\sqrt{3}$ at $\left(\pm\sqrt{6}, -\sqrt{3}, 1\right)$.

43. Maximum is 4 at $(0, 0, \pm 2)$; minimum is 2 at $\left(\pm\sqrt{2}, \pm\sqrt{2}, 0\right)$.

SECTION 13.9, p. 839

1. Quadratic: $x + xy$; cubic: $x + xy + \frac{1}{2}xy^2$

3. Quadratic: xy; cubic: xy

5. Quadratic: $y + \frac{1}{2}(2xy - y^2)$;
cubic: $y + \frac{1}{2}(2xy - y^2) + \frac{1}{6}(3x^2y - 3xy^2 + 2y^3)$

7. Quadratic: $\frac{1}{2}(2x^2 + 2y^2) = x^2 + y^2$; cubic: $x^2 + y^2$

9. Quadratic: $1 + (x + y) + (x + y)^2$;
cubic: $1 + (x + y) + (x + y)^2 + (x + y)^3$

11. Quadratic: $1 - \frac{1}{2}x^2 - \frac{1}{2}y^2$; $E(x, y) \le 0.00134$

SECTION 13.10, p. 843

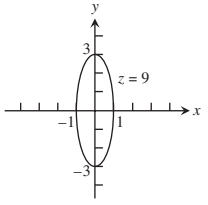
1. **(a)** 0 **(b)** $1 + 2z$ **(c)** $1 + 2z$

3. **(a)** $\frac{\partial U}{\partial P} + \frac{\partial U}{\partial T}\left(\frac{V}{nR}\right)$ **(b)** $\frac{\partial U}{\partial P}\left(\frac{nR}{V}\right) + \frac{\partial U}{\partial T}$

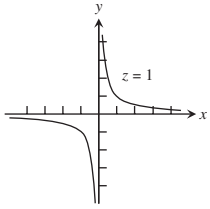
5. **(a)** 5 **(b)** 5 **7.** $\left(\frac{\partial x}{\partial r}\right)_\theta = \cos\theta \left(\frac{\partial r}{\partial x}\right)_y = \frac{x}{\sqrt{x^2 + y^2}}$

PRACTICE EXERCISES, pp. 844–847

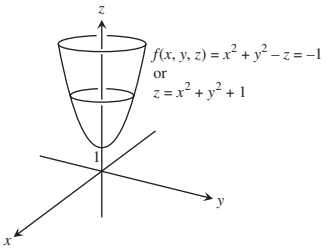
1. Domain: all points in the xy-plane; range: $z \ge 0$. Level curves are ellipses with major axis along the y-axis and minor axis along the x-axis.



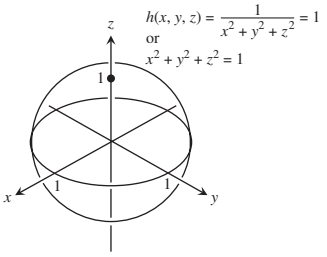
3. Domain: all (x, y) such that $x \ne 0$ and $y \ne 0$; range: $z \ne 0$. Level curves are hyperbolas with the x- and y-axes as asymptotes.



5. Domain: all points in xyz-space; range: all real numbers. Level surfaces are paraboloids of revolution with the z-axis as axis.



7. Domain: all (x, y, z) such that $(x, y, z) \neq (0, 0, 0)$; range: positive real numbers. Level surfaces are spheres with center $(0, 0, 0)$ and radius $r > 0$.



9. -2 **11.** $1/2$ **13.** 1

15. Let $y = kx^2, k \neq 1$

17. No; $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist.

19. $\dfrac{\partial g}{\partial r} = \cos\theta + \sin\theta, \dfrac{\partial g}{\partial \theta} = -r\sin\theta + r\cos\theta$

21. $\dfrac{\partial f}{\partial R_1} = -\dfrac{1}{{R_1}^2}, \dfrac{\partial f}{\partial R_2} = -\dfrac{1}{{R_2}^2}, \dfrac{\partial f}{\partial R_3} = -\dfrac{1}{{R_3}^2}$

23. $\dfrac{\partial P}{\partial n} = \dfrac{RT}{V}, \dfrac{\partial P}{\partial R} = \dfrac{nT}{V}, \dfrac{\partial P}{\partial T} = \dfrac{nR}{V}, \dfrac{\partial P}{\partial V} = -\dfrac{nRT}{V^2}$

25. $\dfrac{\partial^2 g}{\partial x^2} = 0, \dfrac{\partial^2 g}{\partial y^2} = \dfrac{2x}{y^3}, \dfrac{\partial^2 g}{\partial y\,\partial x} = \dfrac{\partial^2 g}{\partial x\,\partial y} = -\dfrac{1}{y^2}$

27. $\dfrac{\partial^2 f}{\partial x^2} = -30x + \dfrac{2 - 2x^2}{(x^2 + 1)^2}, \dfrac{\partial^2 f}{\partial y^2} = 0, \dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y} = 1$

29. $\left.\dfrac{dw}{dt}\right|_{t=0} = -1$

31. $\left.\dfrac{\partial w}{\partial r}\right|_{(r,s)=(\pi,0)} = 2, \left.\dfrac{\partial w}{\partial s}\right|_{(r,s)=(\pi,0)} = 2 - \pi$

33. $\left.\dfrac{df}{dt}\right|_{t=1} = -(\sin 1 + \cos 2)(\sin 1) + (\cos 1 + \cos 2)(\cos 1)$
$-2(\sin 1 + \cos 1)(\sin 2)$

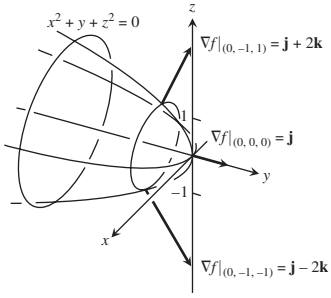
35. $\left.\dfrac{dy}{dx}\right|_{(x,y)=(0,1)} = -1$

37. Increases most rapidly in the direction $\mathbf{u} = -\dfrac{\sqrt{2}}{2}\mathbf{i} - \dfrac{\sqrt{2}}{2}\mathbf{j}$;
decreases most rapidly in the direction $-\mathbf{u} = \dfrac{\sqrt{2}}{2}\mathbf{i} + \dfrac{\sqrt{2}}{2}\mathbf{j}$;
$D_{\mathbf{u}} f = \dfrac{\sqrt{2}}{2}; D_{-\mathbf{u}} f = -\dfrac{\sqrt{2}}{2}; D_{\mathbf{u}_1} f = -\dfrac{7}{10}$ where $\mathbf{u}_1 = \dfrac{\mathbf{v}}{|\mathbf{v}|}$

39. Increases most rapidly in the direction $\mathbf{u} = \dfrac{2}{7}\mathbf{i} + \dfrac{3}{7}\mathbf{j} + \dfrac{6}{7}\mathbf{k}$;
decreases most rapidly in the direction $-\mathbf{u} = -\dfrac{2}{7}\mathbf{i} - \dfrac{3}{7}\mathbf{j} - \dfrac{6}{7}\mathbf{k}$;
$D_{\mathbf{u}} f = 7; D_{-\mathbf{u}} f = -7; D_{\mathbf{u}_1} f = 7$ where $\mathbf{u}_1 = \dfrac{\mathbf{v}}{|\mathbf{v}|}$

41. $\pi/\sqrt{2}$ **43.** **(a)** $f_x(1, 2) = f_y(1, 2) = 2$ **(b)** $14/5$

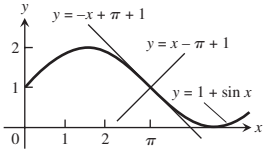
45.



47. Tangent: $4x - y - 5z = 4$; normal line: $x = 2 + 4t, y = -1 - t, z = 1 - 5t$

49. $2y - z - 2 = 0$

51. Tangent: $x + y = \pi + 1$; normal line: $y = x - \pi + 1$



53. $x = 1 - 2t, y = 1, z = 1/2 + 2t$

55. Answers will depend on the upper bound used for $|f_{xx}|, |f_{xy}|, |f_{yy}|$. With $M = \sqrt{2}/2$, $|E| \leq 0.0142$. With $M = 1$, $|E| \leq 0.02$.

57. $L(x, y, z) = y - 3z, L(x, y, z) = x + y - z - 1$

59. Be more careful with the diameter.

61. $dI = 0.038$, % change in $I = 15.83\%$, more sensitive to voltage change

63. **(a)** 5% **65.** Local minimum of -8 at $(-2, -2)$

67. Saddle point at $(0, 0)$, $f(0, 0) = 0$; local maximum of $1/4$ at $(-1/2, -1/2)$

69. Saddle point at $(0, 0)$, $f(0, 0) = 0$; local minimum of -4 at $(0, 2)$; local maximum of 4 at $(-2, 0)$; saddle point at $(-2, 2)$, $f(-2, 2) = 0$

71. Absolute maximum: 28 at $(0, 4)$; absolute minimum: $-9/4$ at $(3/2, 0)$

73. Absolute maximum: 18 at $(2, -2)$; absolute minimum: $-17/4$ at $(-2, 1/2)$

75. Absolute maximum: 8 at $(-2, 0)$; absolute minimum: -1 at $(1, 0)$

77. Absolute maximum: 4 at $(1, 0)$; absolute minimum: -4 at $(0, -1)$

79. Absolute maximum: 1 at $(0, \pm 1)$ and $(1, 0)$; absolute minimum: -1 at $(-1, 0)$

81. Maximum: 5 at $(0, 1)$; minimum: $-1/3$ at $(0, -1/3)$

83. Maximum: $\sqrt{3}$ at $\left(\dfrac{1}{\sqrt{3}}, -\dfrac{1}{\sqrt{3}}, \dfrac{1}{\sqrt{3}}\right)$; minimum: $-\sqrt{3}$ at $\left(-\dfrac{1}{\sqrt{3}}, \dfrac{1}{\sqrt{3}}, -\dfrac{1}{\sqrt{3}}\right)$

85. Width $= \left(\dfrac{c^2 V}{ab}\right)^{1/3}$, depth $= \left(\dfrac{b^2 V}{ac}\right)^{1/3}$, height $= \left(\dfrac{a^2 V}{bc}\right)^{1/3}$

87. Maximum: $\dfrac{3}{2}$ at $\left(\dfrac{1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}, \sqrt{2}\right)$ and $\left(-\dfrac{1}{\sqrt{2}}, -\dfrac{1}{\sqrt{2}}, -\sqrt{2}\right)$;
minimum: $\dfrac{1}{2}$ at $\left(-\dfrac{1}{\sqrt{2}}, \dfrac{1}{\sqrt{2}}, -\sqrt{2}\right)$ and $\left(\dfrac{1}{\sqrt{2}}, -\dfrac{1}{\sqrt{2}}, \sqrt{2}\right)$

89. $\dfrac{\partial w}{\partial x} = \cos\theta\dfrac{\partial w}{\partial r} - \dfrac{\sin\theta}{r}\dfrac{\partial w}{\partial \theta}, \dfrac{\partial w}{\partial y} = \sin\theta\dfrac{\partial w}{\partial r} + \dfrac{\cos\theta}{r}\dfrac{\partial w}{\partial \theta}$

95. $(t, -t \pm 4, t)$, t a real number

101. **(a)** $(2y + x^2z)e^{yz}$ **(b)** $x^2e^{yz}\left(y - \dfrac{z}{2y}\right)$ **(c)** $(1 + x^2y)e^{yz}$

ADDITIONAL AND ADVANCED EXERCISES, pp. 847–849

1. $f_{xy}(0, 0) = -1$, $f_{yx}(0, 0) = 1$

7. **(c)** $\dfrac{r^2}{2} = \dfrac{1}{2}(x^2 + y^2 + z^2)$ **13.** $V = \dfrac{\sqrt{3}abc}{2}$

17. $f(x, y) = \dfrac{y}{2} + 4$, $g(x, y) = \dfrac{x}{2} + \dfrac{9}{2}$

19. $y = 2\ln|\sin x| + \ln 2$

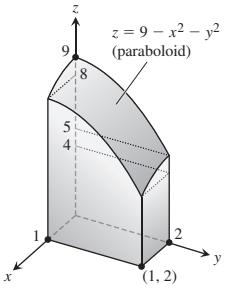
21. **(a)** $\dfrac{1}{\sqrt{53}}(2\mathbf{i} + 7\mathbf{j})$ **(b)** $\dfrac{-1}{\sqrt{29{,}097}}(98\mathbf{i} - 127\mathbf{j} + 58\mathbf{k})$

23. $w = e^{-c^2\pi^2 t}\sin \pi x$

Chapter 14

SECTION 14.1, pp. 854–855

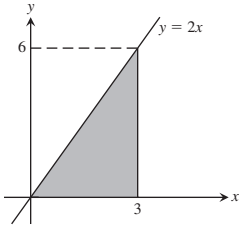
1. 24 **3.** 1 **5.** 16 **7.** $2\ln 2 - 1$ **9.** $(3/2)(5 - e)$
11. $3/2$ **13.** $\ln 2$ **15.** $3/2, -2$ **17.** 14 **19.** 0
21. $1/2$ **23.** $2\ln 2$ **25.** $(\ln 2)^2$
27.



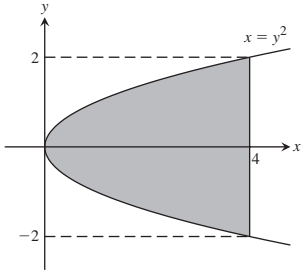
29. $8/3$ **31.** 1 **33.** $\sqrt{2}$ **35.** $2/27$
37. $\dfrac{3}{2}\ln 3 - 1$ **39.** **(a)** $1/3$ **(b)** $2/3$

SECTION 14.2, pp. 862–864

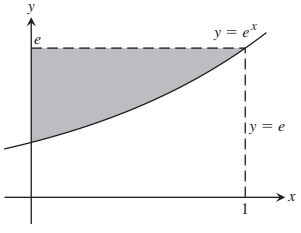
1.



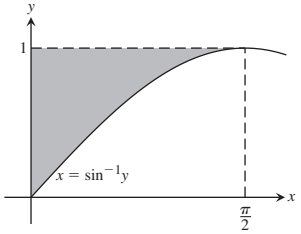
3.



5.



7.



9. **(a)** $0 \le x \le 2, x^3 \le y \le 8$
(b) $0 \le y \le 8, 0 \le x \le y^{1/3}$

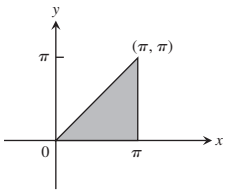
11. **(a)** $0 \le x \le 3, x^2 \le y \le 3x$
(b) $0 \le y \le 9, \dfrac{y}{3} \le x \le \sqrt{y}$

13. **(a)** $0 \le x \le 9, 0 \le y \le \sqrt{x}$
(b) $0 \le y \le 3, y^2 \le x \le 9$

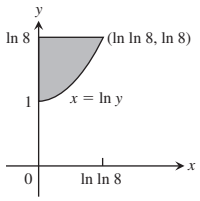
15. **(a)** $0 \le x \le \ln 3, e^{-x} \le y \le 1$
(b) $\dfrac{1}{3} \le y \le 1, -\ln y \le x \le \ln 3$

17. **(a)** $0 \le x \le 1, x \le y \le 3 - 2x$
(b) $0 \le y \le 1, 0 \le x \le y \cup 1 \le y \le 3, 0 \le x \le \dfrac{3 - y}{2}$

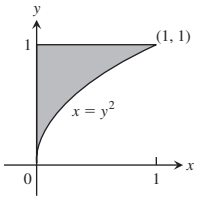
19. $\dfrac{\pi^2}{2} + 2$



21. $8\ln 8 - 16 + e$

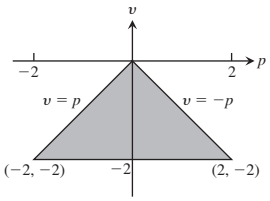


23. $e - 2$

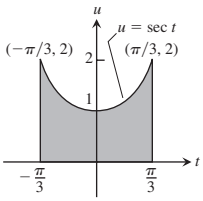


25. $\dfrac{3}{2}\ln 2$ **27.** $-1/10$

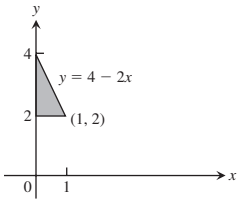
29. 8



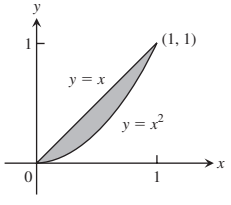
31. 2π



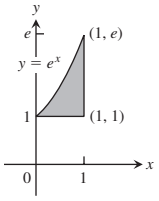
33. $\displaystyle\int_2^4 \int_0^{(4-y)/2} dx\, dy$



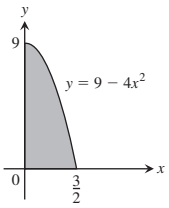
35. $\displaystyle\int_0^1 \int_{x^2}^{x} dy\, dx$



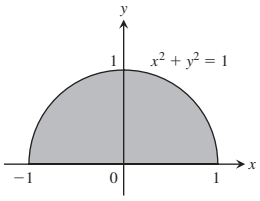
37. $\int_1^e \int_{\ln y}^1 dx\,dy$



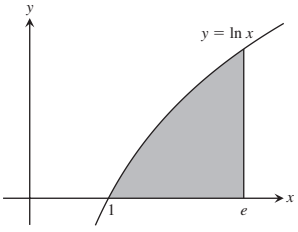
39. $\int_0^9 \int_0^{(\sqrt{9-y})/2} 16x\,dx\,dy$



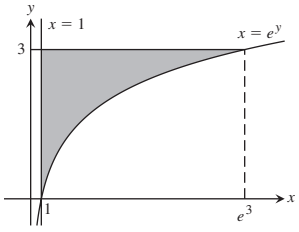
41. $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} 3y\,dy\,dx$



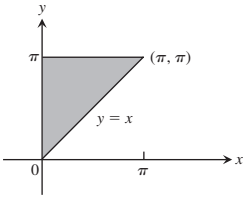
43. $\int_0^1 \int_{e^y}^e xy\,dx\,dy$



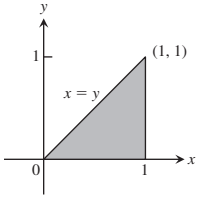
45. $\int_1^{e^3} \int_{\ln x}^3 (x + y)\,dy\,dx$



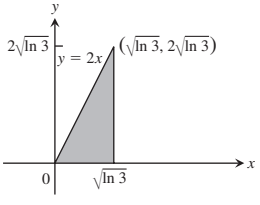
47. 2



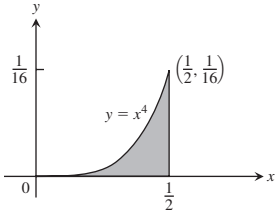
49. $\dfrac{e-2}{2}$



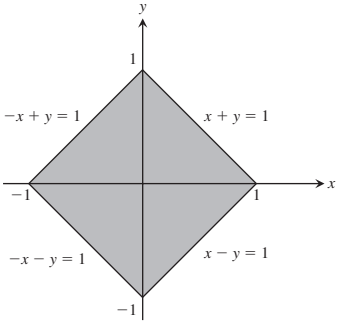
51. 2



53. $1/(80\pi)$

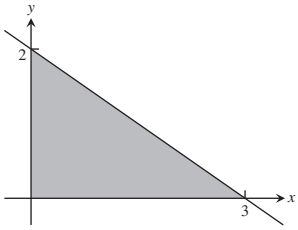


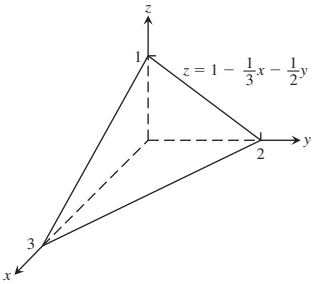
55. $-2/3$



57. $4/3$ **59.** $625/12$ **61.** 16 **63.** 20 **65.** $2(1 + \ln 2)$

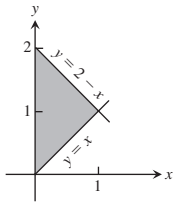
67.





69. 1 **71.** π^2 **73.** $-\dfrac{3}{32}$ **75.** $\dfrac{20\sqrt{3}}{9}$

77. $\int_0^1 \int_x^{2-x} (x^2 + y^2)\,dy\,dx = \dfrac{4}{3}$



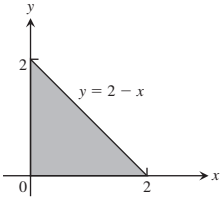
79. R is the set of points (x, y) such that $x^2 + 2y^2 < 4$.

81. No, by Fubini's Theorem, the two orders of integration must give the same result.

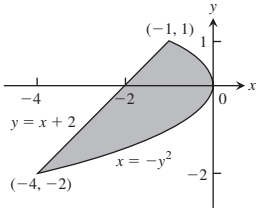
85. 0.603 **87.** 0.233

SECTION 14.3, p. 867

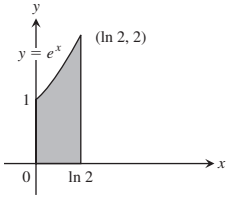
1. $\int_0^2 \int_0^{2-x} dy\,dx = 2$ or $\int_0^2 \int_0^{2-y} dx\,dy = 2$



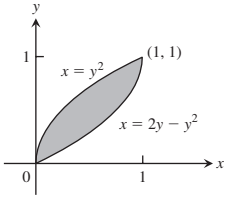
3. $\int_{-2}^1 \int_{y-2}^{-y^2} dx\,dy = \dfrac{9}{2}$



5. $\int_0^{\ln 2} \int_0^{e^x} dy\,dx = 1$

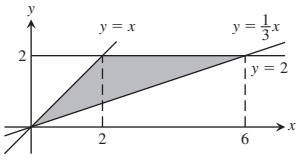


7. $\int_0^1 \int_{y^2}^{2y-y^2} dx\,dy = \dfrac{1}{3}$



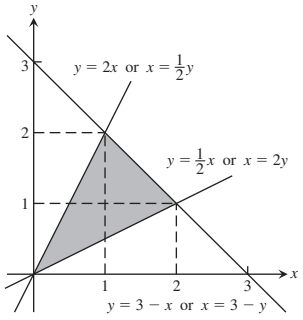
9. $\displaystyle\int_0^2\int_y^{3y} 1\,dx\,dy = 4$ or

$$\int_0^2\int_{x/3}^{x} 1\,dy\,dx + \int_2^6\int_{x/3}^{2} 1\,dy\,dx = 4$$

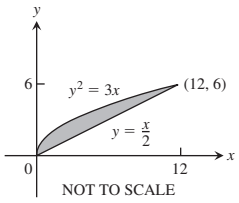


11. $\displaystyle\int_0^1\int_{x/2}^{2x} 1\,dy\,dx + \int_1^2\int_{x/2}^{3-x} 1\,dy\,dx = \frac{3}{2}$ or

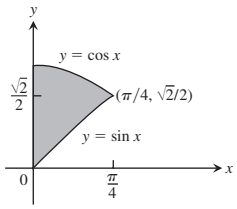
$$\int_0^1\int_{y/2}^{2y} 1\,dx\,dy + \int_1^2\int_{y/2}^{3-y} 1\,dx\,dy = \frac{3}{2}$$



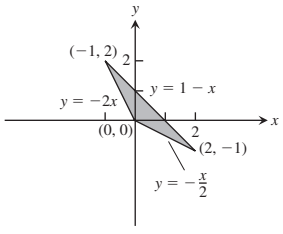
13. 12



15. $\sqrt{2} - 1$



17. $\frac{3}{2}$



19. **(a)** 0 **(b)** $4/\pi^2$ **21.** $8/3$ **23.** $\pi - 2$
25. $40{,}000(1 - e^{-2})\ln(7/2) \approx 43{,}329$

SECTION 14.4, pp. 872–874

1. $\frac{\pi}{2} \le \theta \le 2\pi, 0 \le r \le 9$ **3.** $\frac{\pi}{4} \le \theta \le \frac{3\pi}{4}, 0 \le r \le \csc\theta$

5. $0 \le \theta \le \frac{\pi}{6}, 1 \le r \le 2\sqrt{3}\sec\theta$;

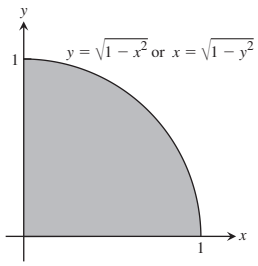
$$\frac{\pi}{6} \le \theta \le \frac{\pi}{2}, 1 \le r \le 2\csc\theta$$

7. $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}, 0 \le r \le 2\cos\theta$ **9.** $\frac{\pi}{2}$

11. 2π **13.** 36 **15.** $2 - \sqrt{3}$ **17.** $(1 - \ln 2)\,\pi$

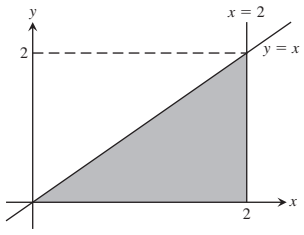
19. $(2\ln 2 - 1)(\pi/2)$ **21.** $\dfrac{2\left(1 + \sqrt{2}\right)}{3}$

23.



$$\int_0^1\int_0^{\sqrt{1-x^2}} xy\,dy\,dx \quad \text{or} \quad \int_0^1\int_0^{\sqrt{1-y^2}} xy\,dx\,dy$$

25.



$$\int_0^2\int_0^x y^2\left(x^2 + y^2\right)dy\,dx \quad \text{or} \quad \int_0^2\int_y^2 y^2\left(x^2 + y^2\right)dx\,dy$$

27. $2(\pi - 2)$ **29.** 12π **31.** $(3\pi/8) + 1$ **33.** $\frac{2a}{3}$

35. $\frac{2a}{3}$ **37.** $2\pi\left(2 - \sqrt{e}\right)$ **39.** $\frac{4}{3} + \frac{5\pi}{8}$

41. **(a)** $\dfrac{\sqrt{\pi}}{2}$ **(b)** 1 **43.** $\pi\ln 4$, no **45.** $\frac{1}{2}\left(a^2 + 2h^2\right)$

47. $\frac{8}{9}(3\pi - 4)$

SECTION 14.5, pp. 882–884

1. $1/6$

3. $\displaystyle\int_0^1\int_0^{2-2x}\int_0^{3-3x-3y/2} dz\,dy\,dx, \int_0^2\int_0^{1-y/2}\int_0^{3-3x-3y/2} dz\,dx\,dy,$

$$\int_0^1\int_0^{3-3x}\int_0^{2-2x-2z/3} dy\,dz\,dx, \int_0^3\int_0^{1-z/3}\int_0^{2-2x-2z/3} dy\,dx\,dz,$$

$$\int_0^2\int_0^{3-3y/2}\int_0^{1-y/2-z/3} dx\,dz\,dy, \int_0^3\int_0^{2-2z/3}\int_0^{1-y/2-z/3} dx\,dy\,dz.$$

The value of all six integrals is 1.

5. $\displaystyle\int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{x^2+y^2}^{8-x^2-y^2} 1\,dz\,dx\,dy, \int_{-2}^2\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}}\int_{x^2+y^2}^{8-x^2-y^2} 1\,dz\,dx\,dy,$

$$\int_{-2}^2\int_4^{8-y^2}\int_{-\sqrt{8-z-y^2}}^{\sqrt{8-z-y^2}} 1\,dx\,dz\,dy + \int_{-2}^2\int_{y^2}^4\int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} 1\,dx\,dz\,dy,$$

$$\int_4^8\int_{-\sqrt{8-z}}^{\sqrt{8-z}}\int_{-\sqrt{8-z-y^2}}^{\sqrt{8-z-y^2}} 1\,dx\,dy\,dz + \int_0^4\int_{-\sqrt{z}}^{\sqrt{z}}\int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} 1\,dx\,dy\,dz,$$

$$\int_{-2}^2\int_4^{8-x^2}\int_{-\sqrt{8-z-x^2}}^{\sqrt{8-z-x^2}} 1\,dy\,dz\,dx + \int_{-2}^2\int_{x^2}^4\int_{-\sqrt{z-x^2}}^{\sqrt{z-x^2}} 1\,dy\,dz\,dx,$$

$$\int_4^8 \int_{-\sqrt{8-z}}^{\sqrt{8-z}} \int_{-\sqrt{8-z-x^2}}^{\sqrt{8-z-x^2}} 1\, dy\, dx\, dz + \int_0^4 \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-x^2}}^{\sqrt{z-x^2}} 1\, dy\, dx\, dz.$$

The value of all six integrals is 16π.

7. 1 **9.** 6 **11.** $\dfrac{5(2-\sqrt{3})}{4}$ **13.** 18

15. 7/6 **17.** 0 **19.** $\dfrac{1}{2} - \dfrac{\pi}{8}$

21. **(a)** $\displaystyle\int_{-1}^{1}\int_{0}^{1-x^2}\int_{x^2}^{1-z} dy\, dz\, dx$ **(b)** $\displaystyle\int_{0}^{1}\int_{-\sqrt{1-z}}^{\sqrt{1-z}}\int_{x^2}^{1-z} dy\, dx\, dz$

(c) $\displaystyle\int_{0}^{1}\int_{0}^{1-z}\int_{-\sqrt{y}}^{\sqrt{y}} dx\, dy\, dz$ **(d)** $\displaystyle\int_{0}^{1}\int_{0}^{1-y}\int_{-\sqrt{y}}^{\sqrt{y}} dx\, dz\, dy$

(e) $\displaystyle\int_{0}^{1}\int_{-\sqrt{y}}^{\sqrt{y}}\int_{0}^{1-y} dz\, dx\, dy$

23. 2/3 **25.** 20/3 **27.** 1 **29.** 16/3 **31.** $8\pi - \dfrac{32}{3}$
33. 2 **35.** 4π **37.** 31/3 **39.** 1 **41.** 2 sin 4
43. 4 **45.** $a = 3$ or $a = 13/3$
47. The domain is the set of all points (x, y, z) such that $4x^2 + 4y^2 + z^2 \le 4$.

SECTION 14.6, pp. 892–894

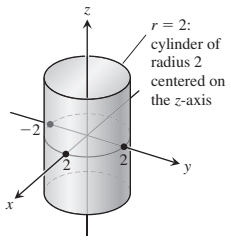
1. $\bar{x} = 5/14, \bar{y} = 38/35$ **3.** $\bar{x} = 64/35, \bar{y} = 5/7$
5. $\bar{x} = \bar{y} = 4a/(3\pi)$
7. $I_x = I_y = 4\pi$ gm/cm^2, $I_0 = 8\pi$ gm/cm^2
9. $\bar{x} = -1, \bar{y} = 1/4$ **11.** $I_x = 64/105$
13. $\bar{x} = 3/8, \bar{y} = 17/16$ **15.** $\bar{x} = 11/3, \bar{y} = 14/27, I_y = 432$
17. $\bar{x} = 0, \bar{y} = 13/31, I_y = 7/5$
19. $\bar{x} = 0, \bar{y} = 7/10$; $I_x = 9/10$ kg/m^2, $I_y = 3/10$ kg/m^2, $I_0 = 6/5$ kg/m^2
21. $I_x = \dfrac{M}{3}(b^2 + c^2), I_y = \dfrac{M}{3}(a^2 + c^2), I_z = \dfrac{M}{3}(a^2 + b^2)$
23. $\bar{x} = \bar{y} = 0, \bar{z} = 12/5$, $I_x = 7904/105 \approx 75.28$, $I_y = 4832/63 \approx 76.70$, $I_z = 256/45 \approx 5.69$
25. **(a)** $\bar{x} = \bar{y} = 0, \bar{z} = 8/3$ **(b)** $c = 2\sqrt{2}$
27. $I_L = 1386$
29. **(a)** 4/3 gm **(b)** $\bar{x} = 4/5$ cm, $\bar{y} = \bar{z} = 2/5$ cm
31. **(a)** 5/2 **(b)** $\bar{x} = \bar{y} = \bar{z} = 8/15$ **(c)** $I_x = I_y = I_z = 11/6$
33. 3 kg
37. **(a)** $I_{\text{c.m.}} = \dfrac{abc(a^2 + b^2)}{12}, R_{\text{c.m.}} = \sqrt{\dfrac{a^2 + b^2}{12}}$

(b) $I_L = \dfrac{abc(a^2 + 7b^2)}{3}, R_L = \sqrt{\dfrac{a^2 + 7b^2}{3}}$

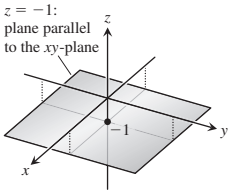
39. $\mu_x = \mu_y = 7/12$
41. $\mu_x = 3/4, \mu_y = 2/3$
43. $f(x, y) = 1/6, P(X < Y) = 2/3$

SECTION 14.7, pp. 902–906

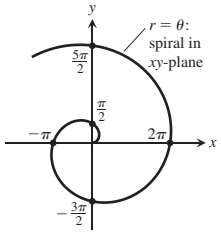
1.

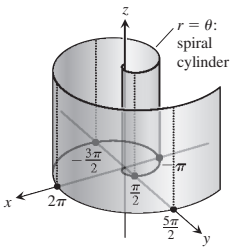


3.

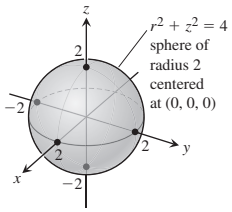


5.

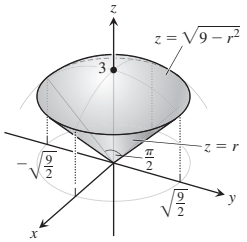




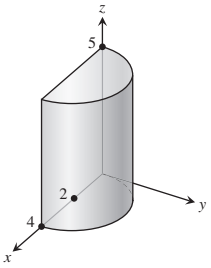
7.



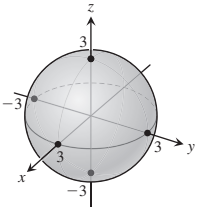
9. $r \le z \le \sqrt{9 - r^2}$: cone with vertex angle $\dfrac{\pi}{2}$ below a sphere of radius 3 centered at (0, 0, 0), and its interior



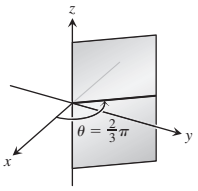
11. $0 \le r \le 4\cos\theta, 0 \le \theta \le \dfrac{\pi}{2}, 0 \le z \le 5$: half-cylinder of height 5, radius 2, and tangent to the z -axis, and its interior



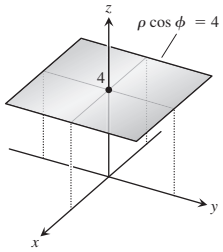
13. $\rho = 3$: sphere of radius 3 centered at (0, 0, 0)



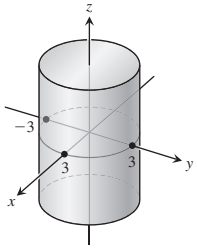
15. $\theta = \frac{2}{3}\pi$: closed half-plane along the z-axis



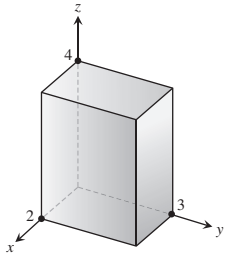
17. $\rho\cos\phi = 4$: plane with z-intercept 4 and parallel to the xy-plane



19. $0 \le \rho \le 3\csc\phi \Rightarrow 0 \le \rho\sin\phi \le 3$: cylinder of radius 3 centered on the z-axis, and its interior



21. $0 \le \rho\cos\theta\sin\phi \le 2$, $0 \le \rho\sin\theta\sin\phi \le 3$, $0 \le \rho\cos\phi \le 4$: rectangular box $2 \times 3 \times 4$, and its interior



23. $\dfrac{4\pi\left(\sqrt{2}-1\right)}{3}$ **25.** $\dfrac{17\pi}{5}$ **27.** $\pi\left(6\sqrt{2}-8\right)$ **29.** $\dfrac{3\pi}{10}$

31. $\pi/3$

33. (a) $\displaystyle\int_0^{2\pi}\int_0^{1}\int_0^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$

(b) $\displaystyle\int_0^{2\pi}\int_0^{\sqrt{3}}\int_0^{1} r\,dr\,dz\,d\theta + \int_0^{2\pi}\int_{\sqrt{3}}^{2}\int_0^{\sqrt{4-z^2}} r\,dr\,dz\,d\theta$

(c) $\displaystyle\int_0^{1}\int_0^{\sqrt{4-r^2}}\int_0^{2\pi} r\,d\theta\,dz\,dr$

35. $\displaystyle\int_{-\pi/2}^{\pi/2}\int_0^{\cos\theta}\int_0^{3r^2} f(r,\theta,z)\,dz\,r\,dr\,d\theta$

37. $\displaystyle\int_0^{\pi}\int_0^{2\sin\theta}\int_0^{4-r\sin\theta} f(r,\theta,z)\,dz\,r\,dr\,d\theta$

39. $\displaystyle\int_{-\pi/2}^{\pi/2}\int_1^{1+\cos\theta}\int_0^{4} f(r,\theta,z)\,dz\,r\,dr\,d\theta$

41. $\displaystyle\int_0^{\pi/4}\int_0^{\sec\theta}\int_0^{2-r\sin\theta} f(r,\theta,z)\,dz\,r\,dr\,d\theta$ **43.** π^2

45. $\pi/3$ **47.** 5π **49.** 2π **51.** $\left(\dfrac{8-5\sqrt{2}}{2}\right)\pi$

53. (a) $\displaystyle\int_0^{2\pi}\int_0^{\pi/6}\int_0^{2}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta + \int_0^{2\pi}\int_{\pi/6}^{\pi/2}\int_0^{\csc\phi}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

(b) $\displaystyle\int_0^{2\pi}\int_1^{2}\int_{\pi/6}^{\sin^{-1}(1/\rho)}\rho^2\sin\phi\,d\phi\,d\rho\,d\theta + \int_0^{2\pi}\int_0^{2}\int_0^{\pi/6}\rho^2\sin\phi\,d\phi\,d\rho\,d\theta + \int_0^{2\pi}\int_0^{1}\int_{\pi/6}^{\pi/2}\rho^2\sin\phi\,d\phi\,d\rho\,d\theta$

55. $\displaystyle\int_0^{2\pi}\int_0^{\pi/2}\int_{\cos\phi}^{2}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \frac{31\pi}{6}$

57. $\displaystyle\int_0^{2\pi}\int_0^{\pi}\int_0^{1-\cos\phi}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \frac{8\pi}{3}$

59. $\displaystyle\int_0^{2\pi}\int_{\pi/4}^{\pi/2}\int_0^{2\cos\phi}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \frac{\pi}{3}$

61. (a) $\displaystyle 8\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{2}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

(b) $\displaystyle 8\int_0^{\pi/2}\int_0^{2}\int_0^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$

(c) $\displaystyle 8\int_0^{2}\int_0^{\sqrt{4-x^2}}\int_0^{\sqrt{4-x^2-y^2}} dz\,dy\,dx$

63. (a) $\displaystyle\int_0^{2\pi}\int_0^{\pi/3}\int_{\sec\phi}^{2}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

(b) $\displaystyle\int_0^{2\pi}\int_0^{\sqrt{3}}\int_1^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$

(c) $\displaystyle\int_{-\sqrt{3}}^{\sqrt{3}}\int_{-\sqrt{3-x^2}}^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} dz\,dy\,dx$ **(d)** $5\pi/3$

65. $8\pi/3$ **67.** $9/4$ **69.** $\dfrac{3\pi-4}{18}$ **71.** $\dfrac{2\pi a^3}{3}$

73. $5\pi/3$ **75.** $\pi/2$ **77.** $\dfrac{4\left(2\sqrt{2}-1\right)\pi}{3}$ **79.** 16π

81. $5\pi/2$ **83.** $\dfrac{4\pi\left(8-3\sqrt{3}\right)}{3}$ **85.** $2/3$ **87.** $3/4$

89. $\bar{x} = \bar{y} = 0, \bar{z} = 3/8$ **91.** $(\bar{x},\bar{y},\bar{z}) = (0,0,3/8)$

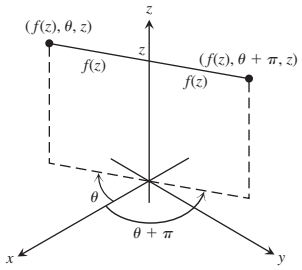
93. $\bar{x} = \bar{y} = 0, \bar{z} = 5/6$ **95.** $I_x = \pi/4$ **97.** $\dfrac{a^4 h\pi}{10}$

99. (a) $(\bar{x},\bar{y},\bar{z}) = \left(0,0,\dfrac{4}{5}\right), I_z = \dfrac{\pi}{12}$

(b) $(\bar{x},\bar{y},\bar{z}) = \left(0,0,\dfrac{5}{6}\right), I_z = \dfrac{\pi}{14}$

101. $\dfrac{3M}{\pi R^3}$

103. The surface's equation $r = f(z)$ tells us that the point $(r, \theta, z) = (f(z), \theta, z)$ will lie on the surface for all θ. In particular, $(f(z), \theta + \pi, z)$ lies on the surface whenever $(f(z), \theta, z)$ lies on the surface, so the surface is symmetric with respect to the z-axis.



SECTION 14.8, pp. 914–916

1. (a) $x = \dfrac{u+v}{3}, y = \dfrac{v-2u}{3}; \dfrac{1}{3}$

(b) Triangular region with boundaries $u = 0$, $v = 0$, and $u + v = 3$

3. (a) $x = \dfrac{1}{5}(2u - v), y = \dfrac{1}{10}(3v - u); \dfrac{1}{10}$

(b) Triangular region with boundaries $3v = u$, $v = 2u$, and $3u + v = 10$

7. $64/5$ **9.** $\displaystyle\int_1^2\int_1^3 (u+v)\frac{2u}{v}\,du\,dv = 8 + \frac{52}{3}\ln 2$

11. $\dfrac{\pi ab(a^2+b^2)}{4}$ **13.** $\dfrac{1}{3}\left(1 + \dfrac{3}{e^2}\right) \approx 0.4687$

15. $\dfrac{225}{16}$ **17.** 12 **19.** $\dfrac{a^2b^2c^2}{6}$

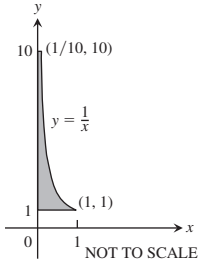
21. (a) $\begin{vmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{vmatrix} = u\cos^2 v + u\sin^2 v = u$

(b) $\begin{vmatrix} \sin v & u\cos v \\ \cos v & -u\sin v \end{vmatrix} = -u\sin^2 v - u\cos^2 v = -u$

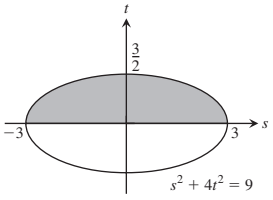
27. $\dfrac{3}{2}\ln 2$

PRACTICE EXERCISES, pp. 916–918

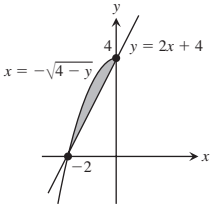
1. $9e - 9$



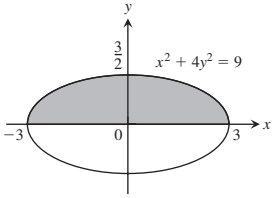
3. $9/2$



5. $\displaystyle\int_{-2}^{0}\int_{2x+4}^{4-x^2} dy\,dx = \frac{4}{3}$



7. $\displaystyle\int_{-3}^{3}\int_{0}^{(1/2)\sqrt{9-x^2}} y\,dy\,dx = \frac{9}{2}$



9. $\sin 4$ **11.** $\dfrac{\ln 17}{4}$ **13.** $4/3$ **15.** $4/3$ **17.** $1/4$

19. π **21.** $\dfrac{\pi - 2}{4}$ **23.** 0 **25.** $8/35$ **27.** $\pi/2$

29. $\dfrac{2(31 - 3^{5/2})}{3}$

31. (a) $\displaystyle\int_{-\sqrt{2}}^{\sqrt{2}}\int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{4-x^2-y^2}} 3\,dz\,dx\,dy$

(b) $\displaystyle\int_0^{2\pi}\int_0^{\pi/4}\int_0^{2} 3\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$ **(c)** $2\pi\left(8 - 4\sqrt{2}\right)$

33. $\displaystyle\int_0^{2\pi}\int_0^{\pi/4}\int_0^{\sec\phi} \rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \frac{\pi}{3}$

35. $\displaystyle\int_0^1\int_{\sqrt{1-x^2}}^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} z^2xy\,dz\,dy\,dx$

$\displaystyle + \int_1^{\sqrt{3}}\int_0^{\sqrt{3-x^2}}\int_1^{\sqrt{4-x^2-y^2}} z^2xy\,dz\,dy\,dx$

37. (a) $\dfrac{8\pi\left(4\sqrt{2} - 5\right)}{3}$ **(b)** $\dfrac{8\pi\left(4\sqrt{2} - 5\right)}{3}$

39. $I_z = \dfrac{8\pi\delta(b^5 - a^5)}{15}$

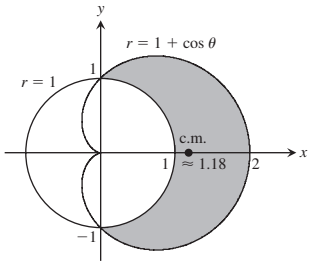
41. $\bar{x} = \bar{y} = \dfrac{1}{2 - \ln 4}$ **43.** $I_0 = 104$ **45.** $I_x = 2\delta$

47. $M = 4, M_x = 0, M_y = 0$

49. $\bar{x} = \dfrac{3\sqrt{3}}{\pi}, \bar{y} = 0$

51. (a) $\bar{x} = \dfrac{15\pi + 32}{6\pi + 48}, \bar{y} = 0$

(b)



ADDITIONAL AND ADVANCED EXERCISES, pp. 919–920

1. (a) $\displaystyle\int_{-3}^{2}\int_{x}^{6-x^2} x^2\,dy\,dx$ **(b)** $\displaystyle\int_{-3}^{2}\int_{x}^{6-x^2}\int_{0}^{x^2} dz\,dy\,dx$

(c) $125/4$

3. 2π **5.** $3\pi/2$

7. (a) Hole radius $= 1$, sphere radius $= 2$ **(b)** $4\sqrt{3}\pi$

9. $\pi/4$ **11.** $\ln\left(\dfrac{b}{a}\right)$ **15.** $1/\sqrt[4]{3}$

17. Mass $= a^2\cos^{-1}\left(\dfrac{b}{a}\right) - b\sqrt{a^2 - b^2}$,

$I_0 = \dfrac{a^4}{2}\cos^{-1}\left(\dfrac{b}{a}\right) - \dfrac{b^3}{2}\sqrt{a^2 - b^2} - \dfrac{b^3}{6}(a^2 - b^2)^{3/2}$

19. $\dfrac{1}{ab}(e^{a^2b^2} - 1)$ **21. (b)** 1 **(c)** 0

25. $h = \sqrt{180}$ cm, $h = \sqrt{540}$ cm **27.** $2\pi\left[\dfrac{1}{3} - \left(\dfrac{1}{3}\right)\dfrac{\sqrt{2}}{2}\right]$

Chapter 15

SECTION 15.1, pp. 927–929

1. Graph (c) **3.** Graph (g) **5.** Graph (d) **7.** Graph (f)

9. $\sqrt{2}$ **11.** $\frac{13}{2}$ **13.** $3\sqrt{14}$ **15.** $\frac{1}{6}\left(5\sqrt{5}+9\right)$

17. $\sqrt{3}\ln\left(\frac{b}{a}\right)$ **19.** **(a)** $4\sqrt{5}$ **(b)** $\frac{1}{12}\left(17^{3/2}-1\right)$

21. $\frac{15}{32}\left(e^{16}-e^{64}\right)$ **23.** $\frac{1}{27}\left(40^{3/2}-13^{3/2}\right)$

25. $\frac{1}{6}\left(5^{3/2}+7\sqrt{2}-1\right)$ **27.** $\frac{10\sqrt{5}-2}{3}$ **29.** 8

31. $\frac{1}{6}\left(17^{3/2}-1\right)$ **33.** $2\sqrt{2}-1$

35. **(a)** $4\sqrt{2}-2$ **(b)** $\sqrt{2}+\ln\left(1+\sqrt{2}\right)$ **37.** $I_z = 2\pi\delta a^3$

39. **(a)** $I_z = 2\pi\sqrt{2}\delta$ **(b)** $I_z = 4\pi\sqrt{2}\delta$ **41.** $I_x = 2\pi - 2$

SECTION 15.2, pp. 939–942

1. $\nabla f = -(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})(x^2+y^2+z^2)^{-3/2}$

3. $\nabla g = -\left(\frac{2x}{x^2+y^2}\right)\mathbf{i} - \left(\frac{2y}{x^2+y^2}\right)\mathbf{j} + e^z\mathbf{k}$

5. $\mathbf{F} = -\frac{kx}{(x^2+y^2)^{3/2}}\mathbf{i} - \frac{ky}{(x^2+y^2)^{3/2}}\mathbf{j}$, any $k > 0$

7. **(a)** 9/2 **(b)** 13/3 **(c)** 9/2

9. **(a)** 1/3 **(b)** −1/5 **(c)** 0

11. **(a)** 2 **(b)** 3/2 **(c)** 1/2

13. −15/2 **15.** 36 **17.** **(a)** −5/6 **(b)** 0 **(c)** −7/12

19. 1/2 **21.** $-\pi$ **23.** 69/4 **25.** −39/2 **27.** 25/6

29. **(a)** $\text{Circ}_1 = 0$, $\text{circ}_2 = 2\pi$, $\text{flux}_1 = 2\pi$, $\text{flux}_2 = 0$
(b) $\text{Circ}_1 = 0$, $\text{circ}_2 = 8\pi$, $\text{flux}_1 = 8\pi$, $\text{flux}_2 = 0$

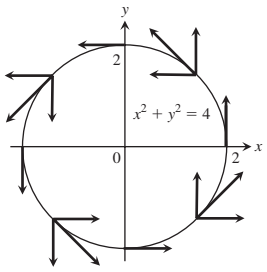
31. Circ = 0, flux $= a^2\pi$ **33.** Circ $= a^2\pi$, flux = 0

35. **(a)** $-\frac{\pi}{2}$ **(b)** 0 **(c)** 1 **37.** $(.0001)\pi$ kg/s

39. **(a)** 32 **(b)** 32 **(c)** 32 **41.** 115.2 J

43. $5/3 - (3/2)\ln 2$ m^2/s **45.** 5/3 g/s

47.



49. **(a)** $\mathbf{G} = -y\mathbf{i} + x\mathbf{j}$ **(b)** $\mathbf{G} = \sqrt{x^2+y^2}\,\mathbf{F}$

51. $\mathbf{F} = -\frac{x\mathbf{i}+y\mathbf{j}}{\sqrt{x^2+y^2}}$ **55.** 48 **57.** π **59.** 0 **61.** $\frac{1}{2}$

SECTION 15.3, pp. 951–953

1. Conservative **3.** Not conservative **5.** Not conservative

7. $f(x,y,z) = x^2 + \frac{3y^2}{2} + 2z^2 + C$ **9.** $f(x,y,z) = xe^{y+2z} + C$

11. $f(x,y,z) = x\ln x - x + \tan(x+y) + \frac{1}{2}\ln\left(y^2+z^2\right) + C$

13. 49 **15.** −16 **17.** 1 **19.** $9\ln 2$ **21.** 0 **23.** −3

27. $\mathbf{F} = \nabla\left(\frac{x^2-1}{y}\right)$ **29.** **(a)** 1 **(b)** 1 **(c)** 1

31. **(a)** 2 **(b)** 2 **33.** **(a)** $c = b = 2a$ **(b)** $c = b = 2$

35. It does not matter what path you use. The work will be the same on any path because the field is conservative.

37. The force $\mathbf{F}$ is conservative because all partial derivatives of M, N, and P are zero. $f(x,y,z) = ax + by + cz + C$; $A = (xa, ya, za)$ and $B = (xb, yb, zb)$. Therefore, $\int \mathbf{F}\cdot d\mathbf{r} = f(B) - f(A) = a(xb - xa) + b(yb - ya) + c(zb - za) = \mathbf{F}\cdot\overrightarrow{AB}$.

SECTION 15.4, pp. 963–965

1. $2y - 1$ **3.** $ye^x - xe^y$ **5.** $\sin y - \sin x$

7. Flux = 0, circ $= 2\pi a^2$ **9.** Flux $= -\pi a^2$, circ = 0

11. Flux = 2, circ = 0 **13.** Flux = −9, circ = 9

15. Flux = −11/60, circ = −7/60

17. Flux = 64/9, circ = 0 **19.** Flux = 1/2, circ = 1/2

21. Flux = 1/5, circ = −1/12 **23.** 0 **25.** 2/33 **27.** 0

29. -16π **31.** πa^2 **33.** $3\pi/8$

35. **(a)** 0 if C is traversed counterclockwise
(b) $(h-k)$(area of the region)

45. **(a)** 4π **(b)** 4π if (0, 0) lies inside K, 0 otherwise

SECTION 15.5, pp. 973–975

1. $\mathbf{r}(r,\theta) = (r\cos\theta)\mathbf{i} + (r\sin\theta)\mathbf{j} + r^2\mathbf{k}$, $0 \le r \le 2$, $0 \le \theta \le 2\pi$

3. $\mathbf{r}(r,\theta) = (r\cos\theta)\mathbf{i} + (r\sin\theta)\mathbf{j} + (r/2)\mathbf{k}$, $0 \le r \le 6$, $0 \le \theta \le \pi/2$

5. $\mathbf{r}(r,\theta) = (r\cos\theta)\mathbf{i} + (r\sin\theta)\mathbf{j} + \sqrt{9-r^2}\,\mathbf{k}$, $0 \le r \le 3\sqrt{2}/2$, $0 \le \theta \le 2\pi$; Also: $\mathbf{r}(\phi,\theta) = (3\sin\phi\cos\theta)\mathbf{i} + (3\sin\phi\sin\theta)\mathbf{j} + (3\cos\phi)\mathbf{k}$, $0 \le \phi \le \pi/4$, $0 \le \theta \le 2\pi$

7. $\mathbf{r}(\phi,\theta) = \left(\sqrt{3}\sin\phi\cos\theta\right)\mathbf{i} + \left(\sqrt{3}\sin\phi\sin\theta\right)\mathbf{j} + \left(\sqrt{3}\cos\phi\right)\mathbf{k}$, $\pi/3 \le \phi \le 2\pi/3$, $0 \le \theta \le 2\pi$

9. $\mathbf{r}(x,y) = x\mathbf{i} + y\mathbf{j} + (4-y^2)\mathbf{k}$, $0 \le x \le 2$, $-2 \le y \le 2$

11. $\mathbf{r}(u,v) = u\mathbf{i} + (3\cos v)\mathbf{j} + (3\sin v)\mathbf{k}$, $0 \le u \le 3$, $0 \le v \le 2\pi$

13. **(a)** $\mathbf{r}(r,\theta) = (r\cos\theta)\mathbf{i} + (r\sin\theta)\mathbf{j} + (1 - r\cos\theta - r\sin\theta)\mathbf{k}$, $0 \le r \le 3$, $0 \le \theta \le 2\pi$
(b) $\mathbf{r}(u,v) = (1 - u\cos v - u\sin v)\mathbf{i} + (u\cos v)\mathbf{j} + (u\sin v)\mathbf{k}$, $0 \le u \le 3$, $0 \le v \le 2\pi$

15. $\mathbf{r}(u,v) = (4\cos^2 v)\mathbf{i} + u\mathbf{j} + (4\cos v\sin v)\mathbf{k}$, $0 \le u \le 3$, $-(\pi/2) \le v \le (\pi/2)$; Another way: $\mathbf{r}(u,v) = (2 + 2\cos v)\mathbf{i} + u\mathbf{j} + (2\sin v)\mathbf{k}$, $0 \le u \le 3$, $0 \le v \le 2\pi$

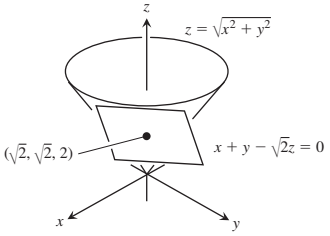
17. $\int_0^{2\pi}\int_0^1 \frac{\sqrt{5}}{2} r\,dr\,d\theta = \frac{\pi\sqrt{5}}{2}$

19. $\int_0^{2\pi}\int_1^3 r\sqrt{5}\,dr\,d\theta = 8\pi\sqrt{5}$ **21.** $\int_0^{2\pi}\int_1^4 1\,du\,dv = 6\pi$

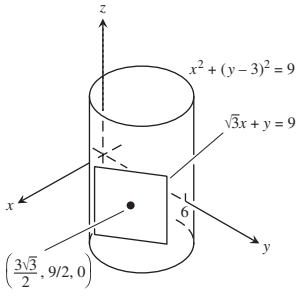
23. $\int_0^{2\pi}\int_0^1 u\sqrt{4u^2+1}\,du\,dv = \frac{\left(5\sqrt{5}-1\right)}{6}\pi$

25. $\int_0^{2\pi}\int_{\pi/4}^{\pi} 2\sin\phi\,d\phi\,d\theta = \left(4+2\sqrt{2}\right)\pi$

27.



29.



33. (b) $A = \int_0^{2\pi}\int_0^{\pi} [\,a^2b^2\sin^2\phi\cos^2\phi + b^2c^2\cos^4\phi\cos^2\theta + a^2c^2\cos^4\phi\sin^2\theta\,]^{1/2}\,d\phi\,d\theta$

35. $x_0x + y_0y = 25$ **37.** $13\pi/3$ **39.** 4

41. $6\sqrt{6} - 2\sqrt{2}$ **43.** $\pi\sqrt{c^2+1}$

45. $\frac{\pi}{6}\left(17\sqrt{17} - 5\sqrt{5}\right)$ **47.** $3 + 2\ln 2$

49. $\frac{\pi}{6}\left(13\sqrt{13} - 1\right)$ **51.** $5\pi\sqrt{2}$ **53.** $\frac{2}{3}\left(5\sqrt{5} - 1\right)$

SECTION 15.6, pp. 983–985

1. $\iint_S x\,d\sigma = \int_0^3\int_0^2 u\sqrt{4u^2+1}\,du\,dv = \frac{17\sqrt{17}-1}{4}$

3. $\iint_S x^2\,d\sigma = \int_0^{2\pi}\int_0^{\pi}\sin^3\phi\cos^2\theta\,d\phi\,d\theta = \frac{4\pi}{3}$

5. $\iint_S z\,d\sigma = \int_0^1\int_0^1 (4-u-v)\sqrt{3}\,dv\,du = 3\sqrt{3}$ (for $x = u$, $y = v$)

7. $\iint_S x^2\sqrt{5-4z}\,d\sigma = \int_0^1\int_0^{2\pi} u^2\cos^2 v\cdot\sqrt{4u^2+1}\cdot u\sqrt{4u^2+1}\,dv\,du = \int_0^1\int_0^{2\pi} u^3(4u^2+1)\cos^2 v\,dv\,du = \frac{11\pi}{12}$

9. $9a^3$ **11.** $\frac{abc}{4}(ab + ac + bc)$ **13.** 2

15. $\frac{1}{30}\left(\sqrt{2} + 6\sqrt{6}\right)$ **17.** $\sqrt{6}/30$ **19.** -32 **21.** $\frac{\pi a^3}{6}$

23. $13a^4/6$ **25.** $2\pi/3$ **27.** $-73\pi/6$ **29.** 18

31. $\frac{\pi a^3}{6}$ **33.** $\frac{\pi a^2}{4}$ **35.** $\frac{\pi a^3}{2}$ **37.** -32 **39.** -4

41. $3a^4$ **43.** $\left(\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)$

45. $(\bar{x},\bar{y},\bar{z}) = \left(0,0,\frac{14}{9}\right)$, $I_z = \frac{15\pi\sqrt{2}}{2}\delta$

47. (a) $\frac{8\pi}{3}a^4\delta$ **(b)** $\frac{20\pi}{3}a^4\delta$ **49.** $70/3$ mg

SECTION 15.7, pp. 996–998

1. $-\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ **3.** $(1-y)\mathbf{i} + (1-z)\mathbf{j} + (1-x)\mathbf{k}$

5. $x(z^2-y^2)\mathbf{i} + y(x^2-z^2)\mathbf{j} + z(y^2-x^2)\mathbf{k}$ **7.** 4π

9. $-5/6$ **11.** 0 **13.** -6π **15.** $2\pi a^2$ **17.** $-\pi$

19. 12π **21.** $-\pi/4$ **23.** -15π **25.** -8π

33. $16I_y + 16I_x$

SECTION 15.8, pp. 1009–1010

1. 0 **3.** $(y^2z + xz^2 + x^2y)e^{xyz}$ **5.** 0 **7.** 0

9. -16 11. -8π **13.** 3π **15.** $-40/3$ **17.** 12π

19. $12\pi\left(4\sqrt{2} - 1\right)$ **23.** No

25. The integral's value never exceeds the surface area of S.

27. $184/35$

PRACTICE EXERCISES, pp. 1011–1013

1. Path 1: $2\sqrt{3}$; path 2: $1 + 3\sqrt{2}$ **3.** $4a^2$ **5.** 0

7. $8\pi\sin(1)$ **9.** 0 **11.** $\pi\sqrt{3}$

13. $2\pi\left(1 - \frac{1}{\sqrt{2}}\right)$ **15.** $\frac{abc}{2}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}$ **17.** 50

19. $\mathbf{r}(\phi,\theta) = (6\sin\phi\cos\theta)\mathbf{i} + (6\sin\phi\sin\theta)\mathbf{j} + (6\cos\phi)\mathbf{k}$, $\frac{\pi}{6} \le \phi \le \frac{2\pi}{3}$, $0 \le \theta \le 2\pi$

21. $\mathbf{r}(r,\theta) = (r\cos\theta)\mathbf{i} + (r\sin\theta)\mathbf{j} + (1+r)\mathbf{k}$, $0 \le r \le 2$, $0 \le \theta \le 2\pi$

23. $\mathbf{r}(u,v) = (u\cos v)\mathbf{i} + 2u^2\mathbf{j} + (u\sin v)\mathbf{k}$, $0 \le u \le 1$, $0 \le v \le \pi$

25. $\sqrt{6}$ **27.** $\pi\left[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\right]$ **29.** Conservative

31. Not conservative **33.** $f(x,y,z) = y^2 + yz + 2x + z$

35. Path 1: 2; path 2: $8/3$ **37. (a)** $1 - e^{-2\pi}$ **(b)** $1 - e^{-2\pi}$

39. 0 **41. (a)** $4\sqrt{2} - 2$ **(b)** $\sqrt{2} + \ln\left(1+\sqrt{2}\right)$

43. $(\bar{x},\bar{y},\bar{z}) = \left(1,\frac{16}{15},\frac{2}{3}\right)$; $I_x = \frac{232}{45}$, $I_y = \frac{64}{15}$, $I_z = \frac{56}{9}$

45. $\bar{z} = \frac{3}{2}$, $I_z = \frac{7\sqrt{3}}{3}$ **47.** $(\bar{x},\bar{y},\bar{z}) = (0,0,49/12)$, $I_z = 640\pi$

49. Flux: $3/2$; circ: $-1/2$ **53.** 3

55. $\frac{2\pi}{3}\left(7 - 8\sqrt{2}\right)$ **57.** 0 **59.** π

ADDITIONAL AND ADVANCED EXERCISES, pp. 1014–1015

1. 6π **3.** $2/3$

5. (a) $\mathbf{F}(x,y,z) = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$ **(b)** $\mathbf{F}(x,y,z) = z\mathbf{i} + y\mathbf{k}$ **(c)** $\mathbf{F}(x,y,z) = z\mathbf{i}$

7. $\frac{16\pi R^3}{3}$ **9.** $a = 2$, $b = 1$. The minimum flux is -4.

11. (b) $\frac{16}{3}g$ **(c)** Work $= \left(\int_C gxy\,ds\right)\bar{y} = g\int_C xy^2\,ds = \frac{16}{3}g$

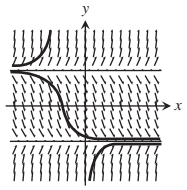
13. (c) $\frac{4}{3}\pi w$ **19.** False if $\mathbf{F} = y\mathbf{i} + x\mathbf{j}$

Chapter 16

SECTION 16.1, pp. 1022–1024

1. (d) **3.** (a)

5.



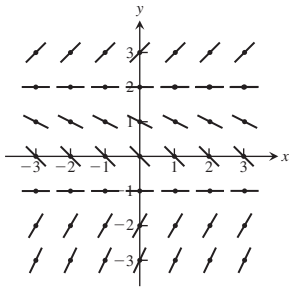
7. $y' = x - y$; $y(1) = -1$

9. $y' = -(1+y)\sin x$; $y(0) = 2$

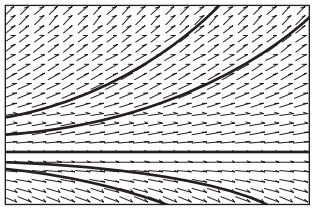
11. $y' = 1 + xe^y$; $y(-2) = 2$

13.

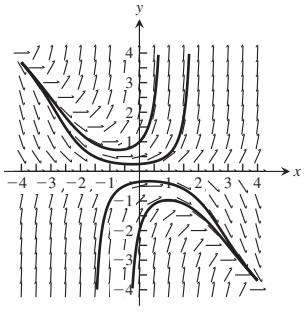
y	$f(y) = \frac{dy}{dx}$
-3	2
-2	1.5
-1	0
0	-1
1	-0.75
2	0
3	1



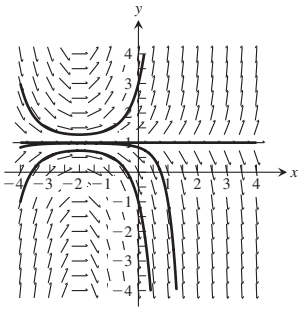
15. $y(\text{exact}) = -x^2,\ y_1 = -2,\ y_2 = -3.3333,\ y_3 = -5$
17. $y(\text{exact}) = 3e^{x(x+2)},\ y_1 = 4.2,\ y_2 = 6.216,\ y_3 = 9.697$
19. $y(\text{exact}) = e^{x^2} + 1,\ y_1 = 2.0,\ y_2 = 2.0202,\ y_3 = 2.0618$
21. $y \approx 2.48832$, exact value is e.
23. $y \approx -0.2272$, exact value is $1/\left(1 - 2\sqrt{5}\right) \approx -0.2880$.
27.



29.



31.



39. Euler's method gives $y \approx 3.45835$; the exact solution is $y = 1 + e \approx 3.71828$.
41. $y \approx 1.5000$; exact value is 1.5275.

SECTION 16.2, pp. 1028–1030

1. $y = \dfrac{e^x + C}{x}, \quad x > 0$

3. $y = \dfrac{C - \cos x}{x^3}, \quad x > 0$

5. $y = \dfrac{1}{2} - \dfrac{1}{x} + \dfrac{C}{x^2}, \quad x > 0$

7. $y = \dfrac{1}{2}xe^{x/2} + Ce^{x/2}$

9. $y = x(\ln x)^2 + Cx$

11. $s = \dfrac{t^3}{3(t-1)^4} - \dfrac{t}{(t-1)^4} + \dfrac{C}{(t-1)^4}$

13. $r = (\csc\theta)(\ln|\sec\theta| + C), \quad 0 < \theta < \pi/2$

15. $y = \dfrac{3}{2} - \dfrac{1}{2}e^{-2t}$

17. $y = -\dfrac{1}{\theta}\cos\theta + \dfrac{\pi}{2\theta}$

19. $y = 6e^{x^2} - \dfrac{e^{x^2}}{x+1}$

21. $y = y_0e^{kt}$

23. (b) is correct, but (a) is not.

25. $t = \dfrac{L}{R}\ln 2$ s

27. **(a)** $i = \dfrac{V}{R} - \dfrac{V}{R}e^{-3} = \dfrac{V}{R}(1 - e^{-3}) \approx 0.95\dfrac{V}{R}$ amp **(b)** 86%

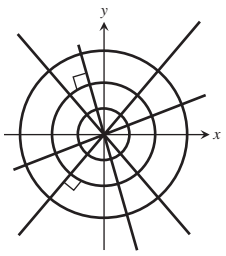
29. $y = \dfrac{1}{1 + Ce^{-x}}$

31. $y^3 = 1 + Cx^{-3}$

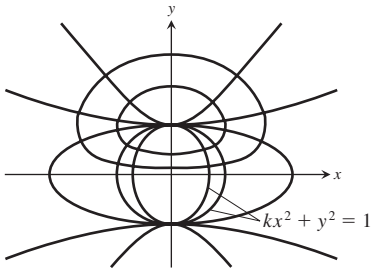
SECTION 16.3, pp. 1035–1036

1. **(a)** 168.5 m **(b)** 41.13 s
3. $s(t) = 4.91\left(1 - e^{-(22.36/39.92)t}\right)$

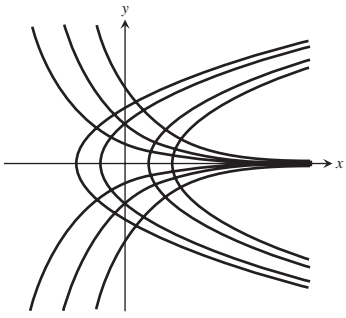
5. $x^2 + y^2 = C$



7. $\ln|y| - \dfrac{1}{2}y^2 = \dfrac{1}{2}x^2 + C$



9. $y = \pm\sqrt{2x + C}$



13. **(a)** 4 kg/min **(b)** $(400 + 4t)$ L **(c)** $4\left(\dfrac{y}{100 + t}\right)$ kg/min

(d) $\dfrac{dy}{dt} = 4 - \dfrac{4y}{100 + t}, \quad y(0) = 20,$

$$y = 0.8(100 + t) - \frac{60}{\left(1 + \dfrac{t}{100}\right)^4}$$

(e) Concentration $= \dfrac{y(25)}{\text{amt. brine in tank}} = \dfrac{75.424}{500} \approx 0.151$ kg/L

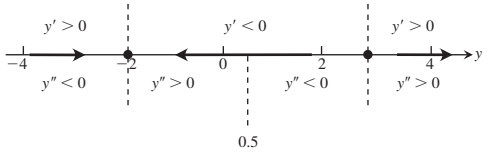
15. $y(27.8) \approx 5.93$ kg, $t \approx 27.8$ min

SECTION 16.4, pp. 1042–1043

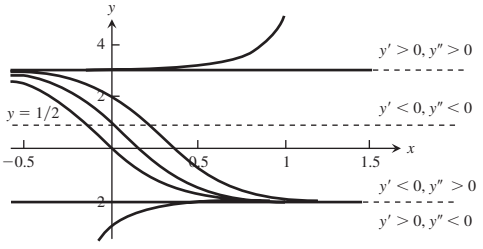
1. $y' = (y + 2)(y - 3)$

(a) $y = -2$ is a stable equilibrium value and $y = 3$ is an unstable equilibrium.

(b) $y'' = 2(y + 2)\left(y - \dfrac{1}{2}\right)(y - 3)$



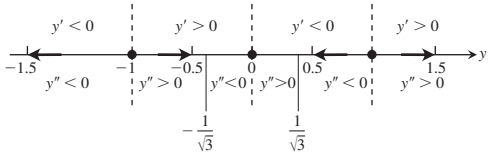
(c)



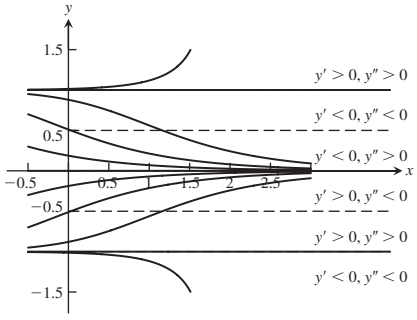
3. $y' = y^3 - y = (y + 1)y(y - 1)$

(a) $y = -1$ and $y = 1$ are unstable equilibria and $y = 0$ is a stable equilibrium.

(b) $y'' = (3y^2 - 1)y'$
$= 3(y + 1)\left(y + 1/\sqrt{3}\right)y\left(y - 1/\sqrt{3}\right)(y - 1)$



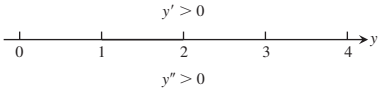
(c)



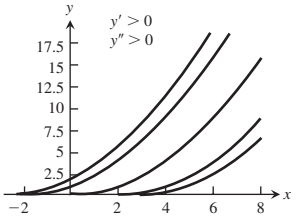
5. $y' = \sqrt{y}, y > 0$

(a) There are no equilibrium values.

(b) $y'' = \dfrac{1}{2}$



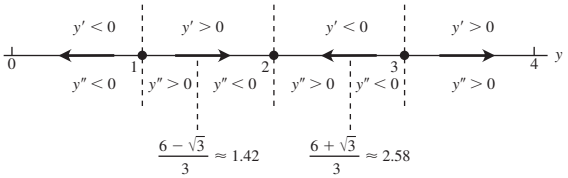
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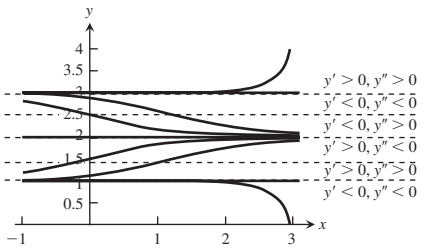
7. $y' = (y - 1)(y - 2)(y - 3)$

(a) $y = 1$ and $y = 3$ are unstable equilibria and $y = 2$ is a stable equilibrium.

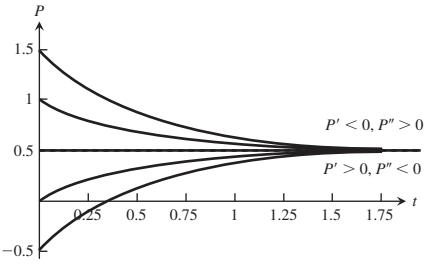
(b) $y'' = (3y^2 - 12y + 11)(y - 1)(y - 2)(y - 3) =$
$3(y - 1)\left(y - \dfrac{6 - \sqrt{3}}{3}\right)(y - 2)\left(y - \dfrac{6 + \sqrt{3}}{3}\right)(y - 3)$



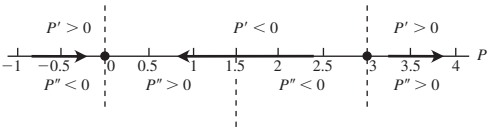
(c)

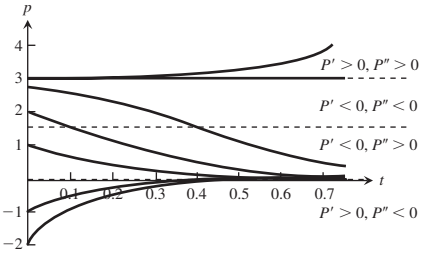


9. $\dfrac{dP}{dt} = 1 - 2P$ has a stable equilibrium at $P = \dfrac{1}{2}$;
$\dfrac{d^2P}{dt^2} = -2\dfrac{dP}{dt} = -2(1 - 2P)$.

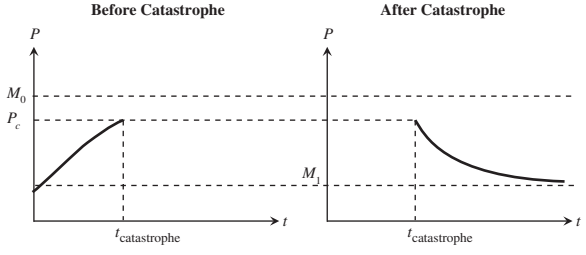


11. $\dfrac{dP}{dt} = 2P(P - 3)$ has a stable equilibrium at $P = 0$ and an unstable equilibrium at $P = 3$; $\dfrac{d^2P}{dt^2} = 2(2P - 3)\dfrac{dP}{dt} = 4P(2P - 3)(P - 3)$.





13. Before the catastrophe, the population exhibits logistic growth and $P(t)$ increases toward M_0, the stable equilibrium. After the catastrophe, the population declines logistically and $P(t)$ decreases toward M_1, the new stable equilibrium.

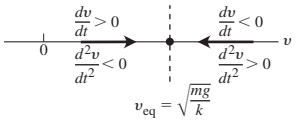


15. $\dfrac{dv}{dt} = g - \dfrac{k}{m}v^2, \quad g, k, m > 0$ and $v(t) \geq 0$

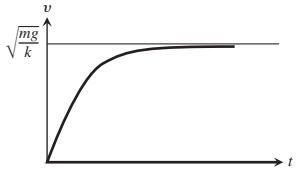
Equilibrium: $\dfrac{dv}{dt} = g - \dfrac{k}{m}v^2 = 0 \Rightarrow v = \sqrt{\dfrac{mg}{k}}$

Concavity: $\dfrac{d^2v}{dt^2} = -2\left(\dfrac{k}{m}v\right)\dfrac{dv}{dt} = -2\left(\dfrac{k}{m}v\right)\left(g - \dfrac{k}{m}v^2\right)$

(a)



(b)



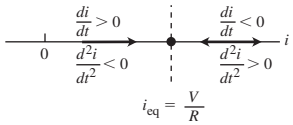
(c) $v_{\text{terminal}} = \sqrt{\dfrac{441}{0.2}} = 46.96 \text{ m/s} = 169 \text{ km/h}$ for the 45-kg skydiver

$v_{\text{terminal}} = \sqrt{\dfrac{7.84}{0.2}} = 62.6 \text{ m/s} = 225 \text{ km/h}$ for the 80-kg skydiver

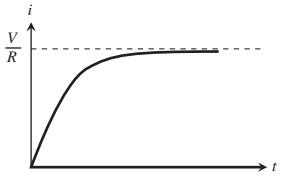
17. $F = F_p - F_r$; $ma = 200 - 50|v|$; $\dfrac{dv}{dt} = \dfrac{1}{m}(200 - 50|v|)$.

The maximum velocity occurs when $\dfrac{dv}{dt} = 0$ or $v = 4$ m/s.

19. Phase line:



If the switch is closed at $t = 0$, then $i(0) = 0$, and the graph of the solution looks like this:



As $t \to \infty$, $i(t) \to i_{\text{steady state}} = \dfrac{V}{R}$.

SECTION 16.5, pp. 1046–1049

1. Seasonal variations, nonconformity of the environments, effects of other interactions, unexpected disasters, etc.

3. This model assumes that the number of interactions is proportional to the product of x and y:

$\dfrac{dx}{dt} = (a - by)x, \quad a < 0,$

$\dfrac{dy}{dt} = m\left(1 - \dfrac{y}{M}\right)y - nxy = y\left(m - \dfrac{m}{M}y - nx\right).$

Rest points are $(0, 0)$, unstable, and $(0, M)$, stable.

5. (a) Logistic growth occurs in the absence of the competitor, and involves a simple interaction between the species: Growth dominates the competition when either population is small, so it is difficult to drive either species to extinction.

(b) a: per capita growth rate for trout
m: per capita growth rate for bass
b: intensity of competition to the trout
n: intensity of competition to the bass
k_1: environmental carrying capacity for the trout
k_2: environmental carrying capacity for the bass
$\dfrac{a}{b}$: growth versus competition or net growth of trout
$\dfrac{m}{n}$: relative survival of bass

(c) $\dfrac{dx}{dt} = 0$ when $x = 0$ or $y = \dfrac{a}{b} - \dfrac{a}{bk_1}x$,

$\dfrac{dy}{dt} = 0$ when $y = 0$ or $y = k_2 - \dfrac{k_2 n}{m}x$.

By picking $a/b > k_2$ and $m/n > k_1$, we ensure that an equilibrium point exists inside the first quadrant.

PRACTICE EXERCISES, pp. 1049–1051

1. $y = -\ln\left(C - \dfrac{2}{5}(x - 2)^{5/2} - \dfrac{4}{3}(x - 2)^{3/2}\right)$

3. $\tan y = -x \sin x - \cos x + C$

5. $(y + 1)e^{-y} = -\ln|x| + C$

7. $y = C\dfrac{x - 1}{x}$ **9.** $y = \dfrac{x^2}{4}e^{x/2} + Ce^{x/2}$

11. $y = \dfrac{x^2 - 2x + C}{2x^2}$ **13.** $y = \dfrac{e^{-x} + C}{1 + e^x}$ **15.** $xy + y^3 = C$

17. $\tan y = -\cos x + (1/3)\cos^3 x + C$

19. $e^{x^2}y = (x - 1)e^x + C$

21. $\ln|\ln y| = (1/2)x^2 \ln x - (1/4)x^2 + C$

23. $y = \dfrac{2x^3 + 3x^2 + 6}{6(x + 1)^2}$ **25.** $y = \dfrac{1}{3}(1 - 4e^{-x^3})$

27. $y = e^{-x}(3x^3 - 3x^2)$

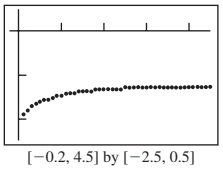
29.

x	y
0	0
0.1	0.1000
0.2	0.2095
0.3	0.3285
0.4	0.4568
0.5	0.5946
0.6	0.7418
0.7	0.8986
0.8	1.0649
0.9	1.2411
1.0	1.4273

x	y
1.1	1.6241
1.2	1.8319
1.3	2.0513
1.4	2.2832
1.5	2.5285
1.6	2.7884
1.7	3.0643
1.8	3.3579
1.9	3.6709
2.0	4.0057

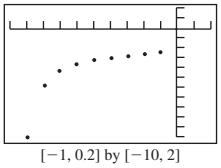
31. $y(3) \approx 0.9131$

33. (a)



[−0.2, 4.5] by [−2.5, 0.5]

(b) Note that we choose a small interval of x-values because the y-values decrease very rapidly and our calculator cannot handle the calculations for $x \leq -1$. (This occurs because the analytic solution is $y = -2 + \ln(2 - e^{-x})$, which has an asymptote at $x = -\ln 2 \approx -0.69$. Obviously, the Euler approximations are misleading for $x \leq -0.7$.)



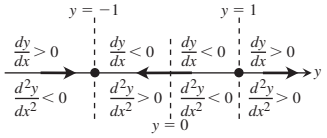
[−1, 0.2] by [−10, 2]

35. $y(\text{exact}) = \frac{1}{2}x^2 - \frac{3}{2}$; $y(2) \approx 0.4$; exact value is $\frac{1}{2}$.

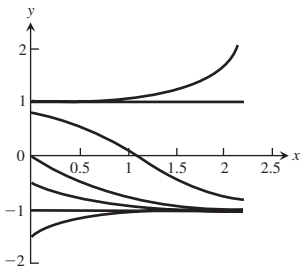
37. $y(\text{exact}) = -e^{(x^2-1)/2}$; $y(2) \approx -3.4192$; exact value is $-e^{3/2} \approx -4.4817$.

41. (a) $y = -1$ is stable and $y = 1$ is unstable.

(b) $\dfrac{d^2y}{dx^2} = 2y\dfrac{dy}{dx} = 2y(y^2 - 1)$



(c)



43. (a) 1 min: 608 L; 10 min: 680 L; 60 min: 1080 L
(b) $S(1) \approx 11.76$ kg, $S(10) \approx 57.89$ kg, $S(60) \approx 225.56$ kg

ADDITIONAL AND ADVANCED EXERCISES, p. 1051

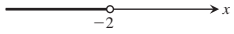
1. **(a)** $y = c + (y_0 - c)e^{-k(A/V)t}$
(b) Steady-state solution: $y_\infty = c$
5. $x^2(x^2 + 2y^2) = C$
7. $\ln|x| + e^{-y/x} = C$
9. $\ln|x| - \ln|\sec(y/x - 1) + \tan(y/x - 1)| = C$

Appendices

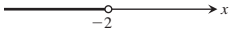
APPENDIX 1, p. AP-6

1. $0.\overline{1}$, $0.\overline{2}$, $0.\overline{3}$, $0.\overline{8}$, $0.\overline{9}$ or 1

3. $x < -2$



5. $x \le -\frac{1}{3}$



7. $3, -3$ **9.** $7/6, 25/6$

11. $-2 \le t \le 4$



13. $0 \le z \le 10$



15. $(-\infty, -2] \cup [2, \infty)$

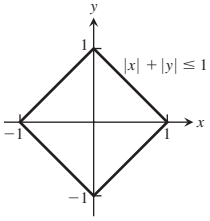


17. $(-\infty, -3] \cup [1, \infty)$



19. $(-3, -2) \cup (2, 3)$ **21.** $(0, 1)$ **23.** $(-\infty, 1]$

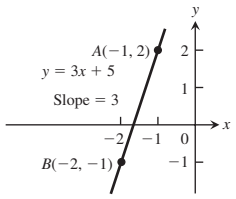
27. The graph of $|x| + |y| \le 1$ is the interior and boundary of the "diamond-shaped" region.



APPENDIX 3, pp. AP-17–AP-18

1. $2, -4$; $2\sqrt{5}$ **3.** Unit circle

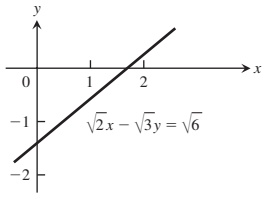
5. $m_\perp = -\frac{1}{3}$



7. (a) $x = -1$ **(b)** $y = 4/3$ **9.** $y = -x$

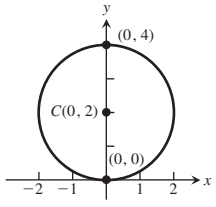
11. $y = -\frac{5}{4}x + 6$ **13.** $y = 4x + 4$ **15.** $y = -\frac{x}{2} + 12$

17. x-intercept $= \sqrt{3}$, y-intercept $= -\sqrt{2}$

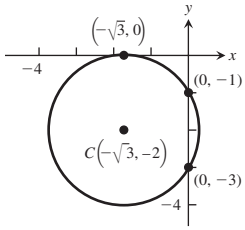


19. $(3, -3)$

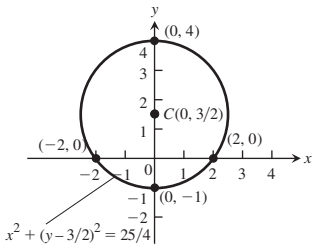
21. $x^2 + (y - 2)^2 = 4$



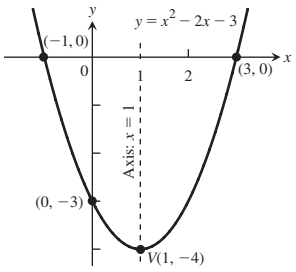
23. $(x + \sqrt{3})^2 + (y + 2)^2 = 4$



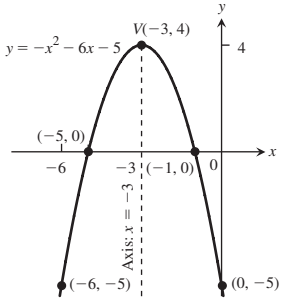
25. $x^2 + (y - 3/2)^2 = 25/4$



27.



29.



31. Exterior points of a circle of radius $\sqrt{7}$, centered at the origin

33. The washer between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ (points with distance from the origin between 1 and 2)

35. $(x + 2)^2 + (y - 1)^2 < 6$

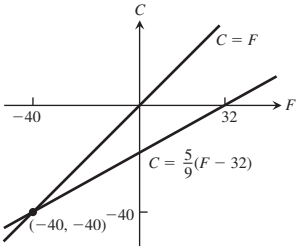
37. $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right), \left(-\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right)$

39. $\left(-\frac{1}{\sqrt{3}}, -\frac{1}{3}\right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{3}\right)$

41. **(a)** $\approx -0.1°\text{C/mm}$ **(b)** $\approx -0.31°\text{C/mm}$ **(c)** $\approx -0.15°\text{C/mm}$

43. 603.9 kPa

45. Yes: $C = F = -40°$



51. $k = -8, \quad k = 1/2$

APPENDIX 7, pp. AP-33–AP-34

1. **(a)** $14 + 8i$ **(b)** $-7 - 4i$ **(c)** $-5i$

3. **(a)** By reflecting z across the real axis
(b) By reflecting z across the imaginary axis
(c) By reflecting z across the real axis and then multiplying the length of the vector by $1/|z|^2$

5. **(a)** Points on the circle $x^2 + y^2 = 4$
(b) Points inside the circle $x^2 + y^2 = 4$
(c) Points outside the circle $x^2 + y^2 = 4$

7. Points on a circle of radius 1, center $(-1, 0)$

9. Points on the line $y = -x$ **11.** $4e^{2\pi i/3}$ **13.** $1e^{2\pi i/3}$

15. $\cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$

17. $1, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ **19.** $2i, -\sqrt{3} - i, \sqrt{3} - i$

21. $\frac{\sqrt{6}}{2} \pm \frac{\sqrt{2}}{2}i, -\frac{\sqrt{6}}{2} \pm \frac{\sqrt{2}}{2}i$ **23.** $1 \pm \sqrt{3}i, -1 \pm \sqrt{3}i$

APPENDIX 8, pp. AP-45–AP-47

1. No **3.** Yes **5.** Yes **7.** Yes **11.** ≈ 0.537

13. ≈ 0.688 **15.** ≈ 0.0502 **17.** $\sqrt{21}$ **19.** $\frac{1}{2}\ln 2$

21. $\frac{1}{\pi}, \frac{1}{\pi}\left(\tan^{-1} 2 - \frac{\pi}{4}\right) \approx 0.10242$

25. mean $= \frac{8}{3} \approx 2.67$, median $= \sqrt{8} \approx 2.83$

27. mean $= 2$, median $= \sqrt{2} \approx 1.41$

29. $P\left(X < \frac{1}{2}\right) \approx 0.3935$

31. **(a)** ≈ 0.57, so about 57 in every 100 bulbs will fail.
(b) ≈ 832 hours

33. ≈ 60 hydra **35.** **(a)** ≈ 0.393 **(b)** ≈ 0.135 **(c)** 0
(d) The probability that any customer waits longer than 3 minutes is $1 - (0.997521)^{200} \approx 0.391 < 1/2$. So the most likely outcome is that all 200 would be served within 3 minutes.

37. \$10, 256 **39.** ≈ 323, ≈ 262 **41.** ≈ 0.89435

43. **(a)** $\approx 16\%$ **(b)** ≈ 0.23832 **45.** ≈ 618 females

47. ≈ 61 adults **49.** ≈ 289 shafts

51. **(a)** ≈ 0.977 **(b)** ≈ 0.159 **(c)** ≈ 0.838

55. **(a)** {LLL, LLD, LDL, DLL, LLU, LUL, ULL, LDD, LDU, LUD, LUU, DLD, DLU, ULD, ULU, DDL, DUL, UDL, UUL, DDD, DDU, DUD, UDD, DUU, UDU, UUD, UUU}
(c) $7/27 \approx 0.26$ **(d)** $20/27 \approx 0.74$

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